Dual curvature and stability in Chambolle Pock dynamics

Abstract

We study the literal primal-first Chambolle–Pock iteration with extrapolation 1/8 and a quadratic dual objective. Our main extension allows arbitrary bounded anisotropic dual curvature and nonseparable convex primal objectives on real Hilbert spaces. For identity coupling, $\sigma = 1$, $Q \succeq (w^{-1} - 1)I$ and $47/50 \leq w < 1$, the sharp class-wide strict cap is $8(1 + w + w^2)/(w(9w - 1))$. A single asymmetric three-tap scalar graph multiplier works across all spectral modes; a dimension-free soft-IQC continuation proves actual strong convergence, square-summable errors and residuals, summable anchored gaps, and uniform true-restart bounds yielding anchored geometric convergence. Consequently $0 < r \leq 3$ is the included sharp curvature-independent cap over all bounded coercive Q: $r = 3$ lies strictly below each instance's cap. Sharp failure uses an admissible scalar subclass, not every fixed anisotropic operator. No rate uniform in vanishing coercivity or extension to $Q = 0$, unbounded Q, general coupling or incremental contraction is asserted.

For identity coupling, scalar resolvent parameter $0 < q < 1$, and every proper lower-semicontinuous convex primal objective on a real Hilbert space, we prove the complete class-universal sharp product range. Its open cap is $8(1 + q)/(5q)$ for $q \leq (\sqrt{105} - 3)/8$ and $8(1 + q + q^2)/(q(9q - 1))$ above that switch. Every strictly admissible product has strong paired convergence, error and port square summability, and a uniform true-restart tail bound in the ordinary physical pair norm. This gives geometric convergence and positive-output objective and residual estimates. At and above the cap, actual linear or fixed-objective period-three orbits provide counterexamples. For general bounded coupling we distinguish four sufficient certificate regimes: the exact three-history ansatz radius, an analytic mixed two-tap cap, a rational closed cap 337/100, and a three-tap closed cap 159/50 only for $\sigma = 1$ and $Q = I/19$. Their conclusions are weak primal and strong dual and coupling-image convergence, with their stated summability. The three-history curvature-dependent cap and its quantitative consequences remain separate; no new general-coupling rate is inferred from the mixed or three-tap certificates. Explicit scalar constants, affine boundary dynamics, fixed metrics, restricted robustness, multiplier comparisons, and actual period-three, four and five classifications retain their individual scopes. The storage and multiplier methodology is established prior art; theorem-level comparison with Wu–Ma remains unresolved.

Keywords: Chambolle–Pock iteration; dual curvature; general linear coupling; Hilbert spaces; graph-corrected storage; geometric convergence; integral quadratic constraints

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1 Problem sharp phase and scope

Let X, Y be real Hilbert spaces, $L : X \to Y$ bounded linear, and $f : X \to (-\infty, +\infty]$ proper lower-semicontinuous convex. For the quadratic-dual problem let $Q = Q^*$ satisfy

$0 < mI \preceq Q \preceq MI$ with $M < \infty$, and let $c \in Y$. Let $\tau, \sigma > 0$. The primal composite, KKT equations and iteration are fixed throughout:

$$\Phi(x) = f(x) + \frac{1}{2}\langle Q^{-1}(Lx - c), Lx - c\rangle. \tag{1}$$

Suppose a KKT pair exists, that is, some $(x^*, y^*) \in X \times Y$ satisfies

$$-L^*y^* \in \partial f(x^*), \qquad Lx^* = Qy^* + c. \tag{2}$$

Starting from arbitrary $(x_0, y_0) \in X \times Y$, consider the literal primal-first iteration

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^* y_{n-1}), \\ y_n &= (I + \sigma Q)^{-1}\left(y_{n-1} + \frac{\sigma L(9x_n - x_{n-1})}{8} - \sigma c\right), \qquad n \geq 1. \end{aligned} \tag{3}$$

For a general dual function h, the common update template is

$$x_{n+1} = \operatorname{prox}_{\tau f}(x_n - \tau L^* y_n), \qquad y_{n+1} = \operatorname{prox}_{\sigma h}\big(y_n + \sigma L(9x_{n+1} - x_n)/8\big),$$

with $-L^*y^* \in \partial f(x^*)$ and $Lx^* \in \partial h(y^*)$. The template itself asserts no convergence theorem for arbitrary two-function problems; the quadratic, affine and restricted nonlinear branches specify their own assumptions.

The initial pair is arbitrary. In particular, no initial function value or invented subgradient is required. The order is literal: one primal proximal step, followed by the dual step with $(9x_n - x_{n-1})/8$. The affine and restricted nonlinear-dual branches state their different hypotheses explicitly.

Put $d = \tau\sigma\|L\|^2$ and $K = [(1 + \sigma M)^{-1}, (1 + \sigma m)^{-1}]$. For identity coupling, $Q = \mu I$, $\sigma = 1$, $\tau = r$, write $q = (1 + \mu)^{-1}$ and

$$q_\star = \frac{\sqrt{161} - 7}{8}, \qquad q^\dagger = \frac{\sqrt{105} - 3}{8}, \qquad r_{\text{lin}}(q) = \frac{8(1 + q)}{5q}.$$

The identity KKT pair exists uniquely for every such objective: $x^* = \operatorname{prox}_{\mu f}(c)$ and $y^* = (x^* - c)/\mu$. General-coupling convergence instead assumes an attained KKT pair.

1.1 A common multiplier for nonseparable anisotropic curvature

The new central theorem is Theorem 1. With identity coupling and $\sigma = 1$, it extends the weak-curvature sharp boundary from scalar Q to the entire class of bounded $Q \succeq (w^{-1} - 1)I$, $47/50 \leq w < 1$. Its convergence proof also gives uniform physical-pair restart bounds and anchored geometric convergence. Corollary 2.1 identifies the included class-wide cap 3 over all bounded coercive Q. The cap's inclusion follows from a strictly larger instancewise radius, and does not assert convergence on a critical three-cycle boundary.

The obstruction to a scalar-to-operator transfer is nonlinear: spectral calculus for the linear filter does not split a nonseparable convex graph. Section 2 therefore constructs one scalar multiplier for all spectral modes. An asymmetric third lag repairs the failure of a particular two-tap transfer and its symmetric correction; that certificate failure is not an actual unstable orbit. Exact parameter positivity is followed by a self-contained causal-truncation homotopy argument, including the free physical initialization. This is the step that turns the soft frequency inequality into a trajectory theorem.

The complete scalar-curvature phase below remains a separate result and covers more scalar curvature values. The four general-coupling regimes also retain their individual hypotheses. Neither the anisotropic theorem nor its rate is claimed for arbitrary L.

1.2 The complete scalar phase and the operator regimes

Theorem 2 identifies the complete scalar cap (30); it equals r_{lin} up to $q^{\dagger}$ and the period-three threshold above it. The curvature switch is included in the first branch; the convergence interval always excludes its product endpoint. For each product at or above the cap, there exists an admissible fixed convex objective and an actual initialization that fails to converge. This is a class-universal threshold, not a failure claim for every objective or initialization. The affine point $q = 1$ has its own conclusions in Section 7.

The exact ansatz radius $\mathcal{R}(q)$ of (70) is a separate certificate quantity. It equals r_{lin} for $q \le q_{\star}$ and provides general-coupling results, restart bounds and growth-dependent rates on its own parameter region. The scalar rate in Theorem 9 extends that existing rate coverage to the full strict sharp cap. Its constant is uniform over the objective, Hilbert space, initial pair and every true restart for fixed parameters and the chosen certificate; it is not asserted uniform at endpoints.

Section 4 compares four precise sufficient operator regimes. The first three admit general bounded coercive self-adjoint Q and noncommuting Q, LL^{*}. The fourth fixes $\sigma = 1$ and $Q = I/19$. Their distinct curvature restrictions, open or closed product endpoints, and quantitative scopes are displayed together. Neither constant mixed cap replaces the old strong-curvature cap $(8/5)(2+\sigma m)$, which grows with the lower curvature. No general strong-primal or growth-free rate claim is deduced from the operator mixed and three-tap results.

1.3 Proof structure and established tools

Section 3 assembles the sharp scalar phase from actual graph energies and actual counterexamples. Appendix A keeps the low-q mixed, bridge three-square and cross-Gram storages separate, including their graph budgets, positive matrices, finite startup and DC closure. Section 4 and Appendix B develop the four general-coupling regimes. The common two-tap and the three-tap energies have different histories and first descent

indices. Coercivity on actual graph histories precedes all telescoping; frequency inequalities alone do not replace it. Section 5 separates exact ansatz radii, support families and alternative causal certificates. Section 6 and Appendix C prove the full-cap scalar rate and retain every older quantitative specialization. Restricted perturbation bounds keep their original hypotheses in Appendix D. Section 7 and Appendix E classify actual periodic dynamics and the affine boundary. The comparison proofs and alternative certificates occupy Appendix F. Exact finite source data supplement the analytic arguments in the manuscript.

The distinction between the two curvature regimes is already visible in (103): positive curvature determines the KKT pair by one proximal map. The question here is the stability of the prescribed iteration, rather than the existence of another way to compute its limit. At zero curvature the dual KKT value need not be unique.

The relevant comparison parameter is the universal one-eighth product

$$r_* = (88 + 16\sqrt{91})/81 \approx 2.970744. \tag{4}$$

The strict universal Hilbert theorem and its exact certificate are given in the prerequisite [1]. Scalar order and matched countable separability also permit convergence at equality [2, Theorem 1]. Theorems 11 and 10 treat a different, structured class: arbitrary, potentially nonseparable primal objectives, identity coupling, and an affine or scalar-quadratic dual objective. They reach beyond r_* without resolving the general two-function gap $(r_*, 3)$. The number 3 is sharp for the strict affine class, by the cycles in Section C.13; the complete positive-curvature scalar phase is Theorem 2, and the growing strong-curvature operator cap remains Corollary 4.1.

The independent explicit-constant proof is developed in Sections C.5 and C.6. The affine strict branch uses a shifted-dual distance argument, whereas positive curvature uses a stable dual filter and a restartable error-sum bound. The lower products retain one fixed exact certificate from [1]; no qualitative convergence statement is substituted for the quantitative bound needed for the rate. The scalar affine endpoint requires an order barrier, so its conclusion is not promoted to an arbitrary-Hilbert endpoint statement. No publication priority or exhaustive comparison with existing primal-dual linear-rate results is claimed.

Section 4.1 develops the general quadratic-dual argument using graph-corrected storage and continuous functional calculus. Its full nonlinear correction is a sum of causal primal history potentials and an anticausal conjugate potential. This is an application of established convex-potential storage templates, particularly [9, Theorem 5, Corollary 6 and Theorem 9]. The Hilbert-space, nonsmooth graph-consistent energy framework of [6, Section 4 and Theorem 2] is also directly relevant. The specific compact-band certificate and its convergence proof, rather than the graph-energy principle or Hilbert setting alone, are the objects established here. An indefinite quadratic coefficient is not a novelty marker: positivity must hold for the complete graph-corrected energy on actual histories.

Independent finite-dimensional soft-IQC estimates in Section 6 use [3, Theorem 2] and retain explicit gains for perturbation analysis. The static special case uses [3, Lemma 12] and a separate proof of positive quadratic storage. Section D.3 then treats only its identity and growth-based rectangular baselines, by a Hilbert finite-block comparison and a finite-dimensional small-gain estimate. Related finite-step Lyapunov machinery appears in [8]. The scalar classification in Section 7.2 has sufficient onset exactly equal to the period-three specialization of [4, Proposition 1]. Allowing the positive curvature to depend on the product gives Corollary E.1; it does not give a fixed-curvature converse.

For the exact primal-first map on general convex Hilbert problems, [5, Theorem 1.2], version 2 of 20 August 2026, gives the strict product range $\tau\sigma\|L\|^2 < 120/113$ at $\theta = 1/8$. Its Propositions 3.2–3.3 and proof of the main theorem also give summable anchored saddle gaps and squared increments. In the coercive quadratic-dual specialization, those conclusions already imply strong dual and coupling-image convergence: the dual error is square summable by coercivity, and the exact dual recurrence expresses $L(x_n - x^*)$ as a combination of that error and square-summable increments. This is a derived consequence of the cited estimates, not an additional theorem attributed to the source. The distinction here is the larger structured curvature-dependent range, not those qualitative features in isolation.

Automated Lyapunov analysis [6] includes quadratic-plus-value energies, graph-consistent positivity and finite-history extensions. Its published and newer preprint CP examples do not themselves supply the particular general-coupling certificate proved in Section 4.1; conversely, representability in that framework is not a reason to exclude an alternative derivation there. Classical PDHG–Douglas–Rachford equivalence [7] uses a different extrapolation and outer-relaxation map, so it requires an exact map identification before transfer. Growth and error-bound routes to geometric convergence are established ideas as well; see, for example, [10]. Here the growth assumption is global and relative to one minimizer, and the bound concerns the original unrestarted orbit. These bounded comparisons make no publication-priority claim and are not an exhaustive exclusion of existing certificates or frameworks.

2 Common multipliers and sharp anisotropic convergence

2.1 The sharp class theorem and the nonlinear compatibility problem

Scalar curvature allows a separate multiplier at each value of the resolvent parameter. For anisotropic curvature, such a family cannot simply be assembled by spectral calculus: the original convex graph need not split along the spectral subspaces. The linear plant

can be diagonalized, but an arbitrary nonseparable primal potential cannot. The result below uses one scalar temporal multiplier for every spectral mode and closes its soft integral quadratic constraint by a causal continuation argument. This is the substantive distinction from the scalar phase in Section 3 and the four general-coupling regimes in Section 4.

For this section alone set $L = I$, $\sigma = 1$, $\tau = r$, and let $Q = Q^*$ be bounded. For $47/50 \le w < 1$, assume

$$Q \succeq (w^{-1} - 1)I, \qquad T = (I + Q)^{-1}, \qquad 0 < r < R_3(w) = \frac{8(1 + w + w^2)}{w(9w - 1)}. \tag{5}$$

The operator T has spectrum in $[q_-, w]$, where $q_- = (1 + \|Q\|)^{-1} > 0$. The literal update is

$$\begin{aligned} x_n &= \operatorname{prox}_{rf}(x_{n-1} - r y_{n-1}), \\ y_n &= T\left[y_{n-1} + \frac{9x_n - x_{n-1}}{8} - c\right], \qquad n \ge 1. \end{aligned} \tag{6}$$

Theorem 1 (Sharp weak-curvature anisotropic class)**.** *Let $\mathcal{H}$ be a real Hilbert space, $f : \mathcal{H} \to (-\infty, +\infty]$ proper lower-semicontinuous convex, and $c \in \mathcal{H}$. Under* (5)*, there is a unique KKT pair (x_*, y_*). For every $x_0, y_0 \in \mathcal{H}$, the actual orbit* (6) *converges strongly to this pair. With*

$$\begin{aligned} p_n &= x_n - x_*, & Y_n &= y_n - y_*, \\ F(p) &= r\big[f(x_* + p) - f(x_*) + \langle y_*, p\rangle\big], & v_n &= p_{n-1} - p_n - rY_{n-1} \in \partial F(p_n), \end{aligned} \tag{7}$$

one has

$$\sum_{n\ge1} \big(\|p_n\|^2 + \|Y_n\|^2 + \|v_n\|^2 + F(p_n)\big) < \infty. \tag{8}$$

The paired error $z_n = (p_n, Y_n)$ also satisfies a uniform true-restart bound and an anchored geometric estimate:

$$\sum_{n\ge0} \|z_n\|^2 \le C\|z_0\|^2, \qquad \|z_n\| \le \sqrt{C}\,(1 - C^{-1})^{n/2}\|z_0\|. \tag{9}$$

Here $C > 1$ depends on the fixed parameters and spectral bounds, but not on f, $\mathcal{H}$ or the initial pair; an explicit choice is given in (27)*. No finite $f(x_0)$ is required.*

For each fixed $w \in [47/50, 1)$, the cap $R_3(w)$ is sharp over the whole class $Q \succeq (w^{-1} - 1)I$: at every $r \ge R_3(w)$ an admissible scalar operator $Q = (w^{-1} - 1)I$, a fixed scalar convex objective and an actual initial pair produce a nonconstant three-cycle. This does not assert endpoint failure for every fixed anisotropic operator.

The proof occupies the following subsections. In particular, (8) is not deduced from frequency positivity alone. The nonlinear continuation and actual-startup steps are essential. For $w = 19/20$, the theorem gives $r < 9128/2869$ for every bounded $Q \succeq I/19$, whereas the older three-tap general-coupling result fixes $Q = I/19$ and has its different closed cap.

Corollary 2.1 (Included curvature-independent sharp cap)**.** *For the literal map* (6)*, every* $0 < r \le 3$ *gives strong anchored geometric convergence for every bounded coercive self-adjoint* Q *and every proper lower-semicontinuous convex* f *on a real Hilbert space. This is the sharp class-wide cap: for every* $r > 3$ *some positive scalar* Q *and some such* f *have an actual nonconstant three-cycle. No convergence statement for* $Q = 0$*, or rate uniform as coercivity vanishes, is included.*

Proof. Choose $w = \max\{47/50, \|(I+Q)^{-1}\|\} < 1$. Spectral calculus gives $Q \succeq (w^{-1}-1)I$, and the exact identity

$$R_3(w) - 3 = \frac{(1-w)(19w+8)}{w(9w-1)} > 0 \tag{10}$$

places even $r = 3$ strictly inside the instance's certified interval. Conversely, $R_3(w) \to 3$ as $w \uparrow 1$. For any $r > 3$, choose $w < 1$ sufficiently close to 1 that $R_3(w) < r$ and apply the scalar hinge witness (37) with $q = w$. Its fixed-objective cycle exists at every product at or above $R_3(w)$ by Theorem 2. The included value 3 is therefore not a convergence assertion at the excluded endpoint $R_3(w)$. ∎

2.2 Why the graph weights must be common

The graph inequality for a normalized convex potential is an inequality in the original Hilbert inner product. Positivity need not survive an operator weight, even when that weight is positive definite. For example, $\Psi(p) = \frac{1}{2}(p_1+p_2)^2$, $p = (1,-2)$ and $v = \nabla\Psi(p) = (-1,-1)$ give $\langle p, v\rangle = 1$, while $\langle p, \mathrm{diag}(3,1)v\rangle = -1$. Thus scalar modewise certificates do not justify an operator-weighted nonlinear graph inequality. The same scalar lag weights must work for all modes, unless additional structure of f is assumed.

Write $z = e^{-i\omega}$ for the delay variable. Eliminating the scalar dual filter gives the residual symbol

$$V_{q,r}(z) = z - 1 - \frac{rqz(9-z)}{8(1-qz)}, \qquad P_{q,r}(t) = -(1+q^2-2qt)\,\mathrm{Re}\big(M(z)V_{q,r}(z)\big), \quad t = \mathrm{Re}\, z. \tag{11}$$

The product is MV, with no extra conjugation: the Fourier form of $\langle p, T_M v\rangle$ is $\overline{\widehat{p}}\, M\widehat{v}$, and $\widehat{v} = V\widehat{p}$.

A two-tap multiplier tuned to the critical worst mode can fail at other curvatures. At $w = 19/20$, put $a_0 = 14345/19397$. The symmetric attempted repair

$$1 - a_0 z - (1-a_0)z^{-2} + \varepsilon(2 - z^3 - z^{-3})$$

has nonnegative mass-one graph weights after division by $1+2\varepsilon$ for $\varepsilon \ge 0$. Nevertheless, at $r = 9128/2869$, positivity at $(q,t) = (1/100, 9/10)$ and $(19/20, 0)$ would respectively require

$$\varepsilon \ge \frac{23931461875}{177573269769} \quad \text{and} \quad \varepsilon \le \frac{1794}{19397}, \tag{12}$$

and the lower bound is strictly larger. More structurally, at $q = 0$ the base multiplier has a negative coefficient of $1 - t$ near $t = 1$ when $a_0 > 2/3$, whereas the symmetric correction begins at second order. This is a failure of that proof route, not an algorithmic counterexample.

An asymmetric third tap changes the first-order behavior. Define

$$\begin{aligned} L_w &= \frac{16w^3 + 21w^2 - 9w - 22}{-32w^3 + 39w^2 + 9w + 44}, & U_w &= \frac{4w^3 + 7w^2 - 3w - 6}{-12w^3 + 13w^2 + 3w + 16}, \\ c_3 &= \frac{L_w + U_w}{2}, & D_w &= (11 - 8w)(1 + w + w^2), \\ a &= \frac{w(9w - 1) + c_3(8w^3 - 12w^2 - 2w - 11)}{D_w}, & b &= 1 - a - c_3, \\ M(z) &= 1 - az - bz^{-2} - c_3 z^{-3}. \end{aligned} \tag{13}$$

The formula for a enforces the critical double zero at $q = w$, $r = R_3(w)$, $t = -1/2$. The admissible interval for the third tap has positive width

$$U_w - L_w = \frac{8(1 - w)(11 - 8w)(1 + w + w^2)^2}{(-12w^3 + 13w^2 + 3w + 16)(-32w^3 + 39w^2 + 9w + 44)} > 0. \tag{14}$$

For example, at $w = 19/20$ the midpoint gives

$$(a, b, c_3) = \left(\frac{1641387915}{2447923162}, \frac{1418804809}{4895846324}, \frac{194265685}{4895846324} \right).$$

These are scalar coefficients common to all spectral modes, rather than separate coefficients chosen as functions of the operator T.

2.3 Uniform frequency positivity and the zero-frequency repair

Lemma 1 (A common positive supply)**.** *For* (13) *and* $47/50 \le w < 1$*, the taps* a, b, c_3 *are strictly positive. At* $r = R_3(w)$*, the polynomial* $K(q, t) = P_{q,r}(t)/(1 - t)$ *satisfies* $K \ge 0$ *on* $0 \le q \le w$*,* $-1 \le t \le 1$*, with equality only at* $q = w$*,* $t = -1/2$*. For every strict* $0 < r < R_3(w)$ *and every* $q_- > 0$*, there exist* $\eta, \delta > 0$ *such that*

$$-\operatorname{Re}\big[(M(z) + \eta)V_{q,r}(z)\big] \ge \delta \quad (q \in [q_-, w],\ |z| = 1). \tag{15}$$

Proof. At the cap, K is quadratic in q. Put $K_0(t) = K(0, t)$, $K_w(t) = K(w, t)$, and let $K_2(t)$ be its q^2 coefficient. Exact expansion gives

$$K(q, t) = \left(1 - \frac{q}{w}\right) K_0(t) + \frac{q}{w} K_w(t) + q(w - q)\big[-K_2(t)\big]. \tag{16}$$

The three terms have a useful shape:

- K_0 is a decreasing, strictly positive convex quadratic, with $K_0(1) = 8(1 - w)(1 + w + w^2)/(-12w^3 + 13w^2 + 3w + 16)$;

- $K_w = (2t+1)^2 J_w(t)$, where J_w is decreasing affine and $J_w(1) = 8w(1-w)(1+w+w^2)/(-32w^3+39w^2+9w+44) > 0$;
- $-K_2$ is a strictly positive increasing convex quadratic.

The exact signs, including tap positivity, follow from the short polynomial certificate in Appendix G. Thus (16) proves the asserted nonnegativity and its only possible zero. No sampling of either parameter is involved.

At $r = 0$, the same multiplier satisfies

$$P_{q,0}(t) = (1 + q^2 - 2qt)(1-t)K_0(t). \tag{17}$$

The supply is affine in r. Interpolating (17) with the cap therefore makes $P_{q,r}(t)/(1-t)$ strictly positive, uniformly on the compact rectangle $[q_-, w]\times[-1,1]$, for a strict subcritical step. Only the mass-one zero at $t = 1$ remains. At that point

$$-\operatorname{Re} V_{q,r}(1) = \frac{rq}{1-q} > 0 \tag{18}$$

uniformly for $q \in [q_-, w]$. By continuity this sign persists on a common neighborhood of $t = 1$. On its compact complement the original supply has a positive minimum and $\operatorname{Re} V$ is bounded. A sufficiently small $\eta > 0$ consequently gives (15). Adding η to the constant coefficient preserves the graph inequality because it adds $\eta\langle p, v\rangle \geq 0$. The bound $q_- > 0$ is where boundedness of Q matters. At $r = R_3(w)$ the critical three-cycle zero is not removed by this argument. ∎

2.4 The soft graph inequality and Hilbert space continuation

The required trajectory argument uses a bilateral soft inequality, not a boundary-free finite-horizon inequality for anticausal taps. Let T_M now denote convolution with symbol $M + \eta$.

Lemma 2 (The common scalar graph inequality)**.** *Let Ψ be proper lower-semicontinuous convex with $\Psi(0) = 0$, $\Psi \geq 0$ and $0 \in \partial\Psi(0)$. If $p, v \in \ell_2(\mathbb{Z}; \mathcal{H})$ and $v_n \in \partial\Psi(p_n)$, then*

$$\langle p, T_M v\rangle_{\ell_2} \geq 0. \tag{19}$$

Moreover, every homotopy map $\alpha \operatorname{prox}_\Psi$, $0 \leq \alpha \leq 1$, has the same property, with graph variable $v = u - p$ when $p = \alpha \operatorname{prox}_\Psi(u)$.

Proof. Fenchel equality and normalization give $0 \leq \Psi(p_n), \Psi^*(v_n) \leq \langle p_n, v_n\rangle$. Both sequences of potentials are summable by Cauchy–Schwarz. Since $p_n \in \partial\Psi^*(v_n)$, convexity gives, for every integer j,

$$\langle p_n, v_n - v_{n-j}\rangle \geq \Psi^*(v_n) - \Psi^*(v_{n-j}).$$

The bilateral sum on the right is zero. Taking the nonnegative weighted sum with weights a, b, c_3, and adding $\eta \sum_n \langle p_n, v_n \rangle \geq 0$, proves (19). All sums are justified by absolute convergence.

For $0 < \alpha \leq 1$, the scaled proximal map is the proximal map of

$$\Psi_\alpha(p) = \alpha \Psi(p/\alpha) + \frac{1-\alpha}{2\alpha} \|p\|^2. \tag{20}$$

This follows directly from the subgradient optimality condition. It has the same normalization. At $\alpha = 0$, the zero map is the proximal map of the indicator of $\{0\}$. Thus the graph inequality holds throughout the homotopy. Each map is nonexpansive and fixes zero. ∎

Lemma 3 (Causal continuation from a soft inequality)**.** *Let G be a stable strictly causal bounded linear convolution operator on $\ell_2(\mathbb{N}; \mathcal{H})$ with norm g. Let Δ be memoryless, 1-Lipschitz and zero at zero. Suppose every $\alpha\Delta$, $0 \leq \alpha \leq 1$, satisfies* (19) *on square-summable input-output pairs, extended by zero to the negative indices, and suppose*

$$\mathrm{Re}\langle p, T_M(G - I)p \rangle \leq -\delta \|p\|^2 \qquad (p \in \ell_2). \tag{21}$$

Set $m = \|T_M\|$, $k = m/\delta$ and $C_0 = \max\{1 + kg, k\}$. For all $e, d \in \ell_2$, the causal feedback

$$p = e + \alpha\Delta(u), \qquad u = Gp + d \tag{22}$$

has a unique square-summable solution for every $\alpha \in [0, 1]$, with $\|p\| \leq C_0(\|e\| + \|d\|)$. For $e = 0$, the sharper bound is $\|p\| \leq k\|d\|$.

Proof. Strict causality gives unique finite-time solutions recursively, without assuming an infinite-horizon bound. First suppose a fixed homotopy parameter is already square-summably stable, and put $h = p - e$. The graph inequality and (21) give

$$0 \leq \mathrm{Re}\langle h, T_M[(G - I)h + Ge + d] \rangle \leq -\delta \|h\|^2 + m\|h\| \|Ge + d\|.$$

Thus $\|h\| \leq k\|Ge + d\|$ and the asserted gain follows, uniformly in this parameter. In particular, it gives the sharper $e = 0$ estimate.

Stability is immediate at $\alpha = 0$. Suppose it holds at α. The solution at $\alpha + \varepsilon \leq 1$ can be written as the stable α feedback with first disturbance $e + \varepsilon\Delta(u)$. For every finite prefix ending at N, truncate this disturbance and d to zero after N. They are square summable; causality makes the old stable solution agree with the new solution on that prefix. Hence

$$\begin{aligned} \|p\|_N &\leq C_0\big(\|e\|_N + \|d\|_N + \varepsilon \|\Delta(u)\|_N\big) \\ &\leq C_0\big(\|e\|_N + (1 + \varepsilon)\|d\|_N + \varepsilon g \|p\|_N\big). \end{aligned}$$

Here $\|Gp\|_N \leq g\|p\|_N$ follows by causally truncating p, and $\|\Delta(u)\|_N \leq \|u\|_N$ follows from its Lipschitz property. If $\varepsilon C_0 g < 1$, all prefix norms are uniformly bounded. Letting

$N \to \infty$ proves square summability at the new parameter; only now is its soft inequality used to restore the same constant C_0. A finite partition of $[0,1]$ with such increments completes the proof (the case $g = 0$ needs no restriction).

This uses Hilbert-space Cauchy–Schwarz, bounded convolution and causality. It neither invokes finite-dimensional compactness nor assumes an incremental graph inequality between two trajectories. In particular, no soft inequality has been applied circularly to an unproved square-summable orbit. ∎

2.5 Spectral lifting actual startup and strong convergence

Proof of Theorem 1 except the rate. The function $f(x) + \frac{1}{2}\langle Q^{-1}(x-c), x-c\rangle$ is proper lower-semicontinuous and uniformly strongly convex. It is coercive because the positive quadratic dominates an affine minorant of f. Weak lower semicontinuity and reflexivity give a minimizer, and strict convexity makes it unique. The subgradient sum rule with the continuous quadratic gives $-y_* \in \partial f(x_*)$ and $Qy_* = x_* - c$, proving unique existence of the KKT pair. The normalized F in (7) satisfies the hypotheses of Lemma 2.

For $n \geq 1$ the anchored recurrence is

$$v_n = p_{n-1} - p_n - rY_{n-1} \in \partial F(p_n), \qquad Y_n = T\left[Y_{n-1} + \frac{9p_n - p_{n-1}}{8}\right]. \tag{23}$$

The zero-state plant from p to $u = p + v$ has symbol

$$G_T(z) = zI - \frac{r}{8}Tz(9-z)(I - Tz)^{-1}. \tag{24}$$

Since $\|T\| \leq w < 1$, this is strictly causal and its impulse response is summable in operator norm. Spectral functional calculus and Hilbert-valued Parseval applied to (15) give (21) with $G = G_T$. Only this linear inequality is lifted modewise; the nonlinear graph uses the common scalar multiplier. The complex Fourier calculation represents the real Hilbert pairing by its real part and needs no finite-dimensional eigenspace decomposition.

Starting the one-sided signals at $n = 1$ with zero artificial past, the arbitrary physical states are represented exactly by

$$\begin{aligned} p &= \operatorname{prox}_F(G_T p + d), \qquad d_1 = p_0 - rY_0, \\ d_n &= -rT^{n-1}(Y_0 - p_0/8), \qquad n \geq 2. \end{aligned} \tag{25}$$

The first identity follows at $n = 1$ from the primal update and thereafter from the stable dual recurrence. The disturbance is in ℓ_2 because $\|T\| < 1$. Lemma 3, with $e = 0$, now gives $p \in \ell_2$. Stability of G_T gives $u \in \ell_2$, hence $v = u - p \in \ell_2$. The stable dual recurrence, including its exponentially decaying startup term, gives $Y \in \ell_2$. There was no evaluation of $F(p_0)$ and no artificial subgradient at index zero.

Finally, $0 \leq F(p_n) \leq \langle p_n, v_n\rangle$ gives summable gaps by Cauchy–Schwarz. Square summability yields strong convergence of both physical errors. This proves (8) for actual

trajectories. The failure statement follows from the existing scalar hinge witness (37), with $q = w$ and $\mu = (1 - w)/w < 8$. Its construction is inherited from the scalar phase and the established Seiler–Carrasco mechanism [4]; it is not inferred from a certificate zero. ∎

2.6 Uniform true restarts and anchored geometric convergence

The continuation proof also controls every restart in the ordinary physical pair norm. With $k = m/\delta$ and $g = \|G_T\|$ as above, set

$$D = 1 + r^2 + \frac{65r^2w^2}{64(1 - w^2)}. \tag{26}$$

For the disturbance (25),

$$\|d\|_{\ell_2}^2 \le D\|z_0\|^2,$$

since $\|p_0 - rY_0\|^2 \le (1 + r^2)\|z_0\|^2$ and $\|p_0/8 - Y_0\|^2 \le (65/64)\|z_0\|^2$. The sharper $e = 0$ estimate gives

$$\sum_{n\ge1} \|p_n\|^2 \le k^2 D\|z_0\|^2, \qquad \|u\|_{\ell_2} \le (gk + 1)\|d\|_{\ell_2}.$$

As $u_n = p_{n-1} - rY_{n-1}$,

$$\sum_{n\ge0} \|Y_n\|^2 \le \frac{2}{r^2}\left(\sum_{n\ge0} \|p_n\|^2 + \sum_{n\ge1} \|u_n\|^2\right).$$

Therefore the first bound in (9) holds with

$$C = 1 + \frac{2}{r^2} + D\left[\left(1 + \frac{2}{r^2}\right)k^2 + \frac{2}{r^2}(gk + 1)^2\right] > 1. \tag{27}$$

Every physical pair is an allowed initial pair, so the same constant holds after every true restart with the same KKT anchor.

For completeness, let S be the autonomous anchored update map and set

$$E(z) = \sum_{n\ge0} \|S^n z\|^2. \tag{28}$$

It is finite for every z, and $\|z\|^2 \le E(z) \le C\|z\|^2$. Autonomy gives the exact identity

$$E(Sz) = E(z) - \|z\|^2 \le (1 - C^{-1})E(z).$$

Iteration proves the second estimate in (9) and completes Theorem 1. This tail energy need not be quadratic, and the estimate is relative to one fixed KKT point. No contraction between arbitrary trajectories, positive margin at $R_3(w)$, or uniform rate as coercivity vanishes is asserted. Likewise, the proof supplies no extension to unbounded Q, $Q = 0$ or general L.

The multiplier, convex graph and homotopy ingredients belong to the established O'Shea–Zames–Falb and IQC methodology [3, 9, 11]. The additional conclusion here is the sharp anisotropic class theorem with a common scalar multiplier, a valid nonseparable Hilbert-space closure, and the included universal cap of Corollary 2.1. The polynomial data below support that argument; their number is not a separate mathematical contribution.

3 Complete sharp scalar curvature phase

3.1 The complete sharp scalar curvature phase

For the literal primal-first iteration with $\theta = 1/8$, $L = I$, $\sigma = 1$, $\tau = r$, and $Q = \mu I$, write $q = (1+\mu)^{-1}$. Thus

$$x_n = \operatorname{prox}_{rf}(x_{n-1} - r y_{n-1}), \qquad y_n = q\left(y_{n-1} + \frac{9x_n - x_{n-1}}{8} - c\right), \quad n \geq 1. \tag{29}$$

The following cap concerns the actual algorithm, uniformly over its convex primal objectives. It is distinct from the exact radius $\mathcal{R}(q) = R_{\mathrm{ans}}(q)$ of the specified three-history certificate. Set

$$\begin{gathered} q^{\dagger} = \frac{\sqrt{105} - 3}{8}, \qquad R_{\mathrm{lin}}(q) = \frac{8(1+q)}{5q}, \qquad R_3(q) = \frac{8(1+q+q^2)}{q(9q-1)}, \\ R_{\mathrm{cap}}(q) = \begin{cases} R_{\mathrm{lin}}(q), & 0 < q \leq q^{\dagger}, \\ R_3(q), & q^{\dagger} < q < 1. \end{cases} \end{gathered} \tag{30}$$

The expression for R_3 is used as a period-three cap only for $q > 1/9$. In particular, (30), rather than a literal minimum of its two rational expressions, is the definition when $q \leq 1/9$.

Theorem 2 (Complete sharp scalar curvature phase)**.** *Let $\mathcal{H}$ be any real Hilbert space, let $f : \mathcal{H} \to (-\infty, +\infty]$ be proper lower-semicontinuous convex, and let $c \in \mathcal{H}$. Fix $0 < q < 1$, put $\mu = (1-q)/q$, and take arbitrary $x_0, y_0 \in \mathcal{H}$. If $0 < r < R_{\mathrm{cap}}(q)$, every orbit of* (29) *converges strongly to the unique KKT pair*

$$x_* = \operatorname{prox}_{\mu f}(c), \qquad y_* = (x_* - c)/\mu. \tag{31}$$

For $p_n = x_n - x_*$, $w_n = y_n - y_*$, and

$$\begin{aligned} F(p) &= r\big[f(x_* + p) - f(x_*) + \langle y_*, p\rangle\big], \\ u_n &= (x_{n-1} - x_n)/r - y_{n-1} \in \partial f(x_n), \qquad v_n = r(u_n + y_*), \end{aligned} \tag{32}$$

one has

$$\sum_{n \geq 1} \big(\|p_n\|^2 + \|w_n\|^2 + \|v_n\|^2 + F(p_n)\big) < \infty. \tag{33}$$

No finite $f(x_0)$ is required. The ordinary physical pair also satisfies the uniform restart and geometric estimates of Theorem 9.

For every fixed $q \in (0,1)$ and every $r \geq R_{\mathrm{cap}}(q)$, there exist an admissible fixed scalar convex function and an actual initialization whose orbit does not converge. For $q \leq q^{\dagger}$, $f = 0, c = 0$ already has a nonzero two-cycle at the cap and an unbounded orbit above it. For $q > q^{\dagger}$, an actual three-cycle exists at every $r \geq R_3(q)$, including equality. These are existence-of-failure statements, not failure for every function or every initialization.

Proof. The proximal formula (31) gives $-y_* \in \partial f(x_*)$ and $x_* - c = \mu y_*$, and conversely follows from those KKT conditions. Thus the KKT pair exists uniquely on every real Hilbert space. The proof of convergence has three overlapping sources, with a disjoint selection available for the quantitative estimates:

$$\begin{array}{ll} \mathrm{F}: & q < 7/10 \text{ or } r < 3, \\ \mathrm{M}: & q \geq 7/10,\ r \geq 3,\ q \leq q^{\dagger}, \\ \mathrm{B}: & r \geq 3,\ q > q^{\dagger}. \end{array} \tag{34}$$

In F, Proposition 5.1 gives $\mathcal{R}(q) > 3$ for all $q \in (0,1)$ and $\mathcal{R}(q) = R_{\mathrm{lin}}(q)$ for $q \leq q_\star = (\sqrt{161} - 7)/8 > 7/10$. The identity specialization of Theorem 4, recorded in Corollary 3.2, therefore applies throughout this branch. In M, Theorem 3 and its actual graph-corrected energy supply exactly the required conclusion, including $q = q^{\dagger}$. In B, Proposition A.1 proves the remaining band by two distinct storages: independent squares for $q^{\dagger} < q \leq 907/1000$, and a cross-Gram form for $907/1000 \leq q < 1$. Both contain the common split. Their proof starts at the finite actual energy E_2, first descends at $n = 3$, establishes graph coercivity before telescoping, and removes the DC zero by actual graph monotonicity. The shared graph is (32); the branch-specific dissipations are not identified with one another. This covers $r = 3$ and the transition $q = q^{\dagger}$ without a gap. Appendix C.1 gives the uniform quantitative version.

For sharpness in the first branch, the actual $f = 0, c = 0$ matrix in coordinates $(x, y/q)$ is

$$C(r) = \begin{pmatrix} 1 & -rq \\ 1 & q - 9rq/8 \end{pmatrix}, \qquad C(R_{\mathrm{lin}}) \begin{pmatrix} 1 \\ 5/[4(1+q)] \end{pmatrix} = -\begin{pmatrix} 1 \\ 5/[4(1+q)] \end{pmatrix}. \tag{35}$$

Thus $x_0 = h$, $y_0 = 5qh/[4(1+q)]$, $h \neq 0$, gives the endpoint cycle. Above the cap, $\det(-I - C(r)) = 2(1+q) - 5rq/4 < 0$; its characteristic polynomial has a real root below -1, giving an unbounded actual orbit. For the second branch, Theorem 13 gives an actual fixed-objective three-cycle whenever

$$0 < \mu < 8, \qquad r \geq \frac{8(\mu^2 + 3\mu + 3)}{8 - \mu} = R_3(q).$$

Here $q > q^{\dagger} > 1/9$ implies $0 < \mu < 8$, so every parameter in the claimed failure range is covered. The explicit witness below also makes its initialization and fixed objective

visible. Finally,

$$R_3(q) - R_{\mathrm{lin}}(q) = -\frac{8(4q^2+3q-6)}{5q(9q-1)} \quad (q > 1/9). \tag{36}$$

Hence the caps agree at $q^\dagger$, and the safe piecewise upper bound $U(q)$ of (44) equals $R_{\mathrm{cap}}(q)$. ∎

An actual period-three boundary witness. For clarity, the period-three witness is a weak-dual-curvature example: $0 < \mu < 8$. Put $D = 8(\mu^2+3\mu+3)$ and, for any $r \geq D/(8-\mu)$,

$$\begin{aligned} y_0 &= \frac{(8-\mu)(\mu+1)}{\mu D} - \frac{c}{\mu}, \quad y_1 = \frac{8-\mu}{\mu D} - \frac{c}{\mu}, \\ y_2 &= \frac{9\mu^2+18\mu+8}{\mu D} - \frac{c}{\mu}, \quad \alpha = \frac{1}{r} - y_2, \qquad \beta = -\frac{1}{r} - y_1, \\ f(x) &= \max\{\alpha x, \beta x\}. \end{aligned} \tag{37}$$

The inclusions $\alpha, -y_0 \in \partial f(0)$ and $\beta \in \partial f(1)$ yield the actual orbit $(0, y_0) \mapsto (0, y_1) \mapsto (1, y_2) \mapsto (0, y_0)$. The inequalities establishing these inclusions and the KKT pair are proved in the period-three classification. This is the period-three unit-slope specialization of the Seiler–Carrasco construction [4, Proposition 1]; the sharpness assertion is not inferred from a failed frequency certificate.

Corollary 3.1 (The weak-curvature subband)**.** *For $19/20 \leq q < 1$, the exact class-universal interval is $0 < r < R_3(q)$. In particular, $q = 19/20$ gives $\mu = 1/19$ and $R_{\mathrm{cap}} = 9128/2869$.*

This is the restriction of the cross-storage branch, since $19/20 > 907/1000$. The single-interval certificate on $[19/20, 1]$ is retained as an alternative exact certificate in the source. The affine limit $q = 1$ has its separate conclusions and is excluded from Theorem 2.

3.2 The low q specialization

This low-q specialization uses the mixed graph energy proved in Appendix A.1. Throughout this subsection the literal iteration is

$$x_n = \mathrm{prox}_{rf}(x_{n-1} - r y_{n-1}), \qquad y_n = q\left(y_{n-1} + \frac{9x_n - x_{n-1}}{8} - c\right), \tag{38}$$

on a real Hilbert space $\mathcal{H}$, with f proper lower-semicontinuous convex, $c \in \mathcal{H}$, $0 < q < 1$, and arbitrary $x_0, y_0 \in \mathcal{H}$. Thus $L = I$, $\sigma = 1$, $\theta = 1/8$, and $\mu = q^{-1} - 1 > 0$. Set

$$q^\dagger = \frac{\sqrt{105}-3}{8}, \qquad r_{\mathrm{lin}}(q) = \frac{8(1+q)}{5q}. \tag{39}$$

The safe piecewise function U in (44) equals the full cap in (30); on the present low-q band it equals r_{lin}.

Theorem 3 (Sharp mixed identity band)**.** *For every $0 < q \le q^\dagger$ and $0 < r < r_{\mathrm{lin}}(q)$, every orbit of* (38) *converges strongly to the unique KKT pair*

$$x_* = \mathrm{prox}_{\mu f}(c), \qquad y_* = (x_* - c)/\mu.$$

With

$$\begin{aligned} p_n &= x_n - x_*, & R_n &= (y_n - y_*)/q, \\ F(p) &= r\big(f(x_* + p) - f(x_*) + \langle y_*, p\rangle\big), & v_n &= p_{n-1} - p_n - rqR_{n-1}, \end{aligned} \tag{40}$$

one has $\sum_{n\ge1}(\|p_n\|^2 + \|R_n\|^2 + \|v_n\|^2 + F(p_n)) < \infty$. At $r = r_{\mathrm{lin}}(q)$ an actual nonzero period-two orbit exists already for $f = 0, c = 0$ on $\mathbb{R}$; for every larger product there is an unbounded orbit for the same scalar problem. Thus the asserted universal all-initializations threshold is sharp. The curvature endpoint $q = q^\dagger$ is included, whereas the product endpoint is excluded. The separate full-cap rate is Theorem 9.

The actual endpoint obstruction is the linear example proved in Section 3.3. In the coordinates used here its matrix is $C(r) = \left(\begin{smallmatrix} 1 & -rq \\ 1 & q - 9rq/8 \end{smallmatrix}\right)$. At $r = r_{\mathrm{lin}}$, $(1, 5/[4(1+q)])^\top$ is a -1 eigenvector; thus the original initialization is $x_0 = h$, $y_0 = 5qh/[4(1+q)]$, $h \neq 0$. Above that product, $\det(-I - C(r)) = 2(1+q) - 5rq/4 < 0$, which gives a real eigenvalue below -1. This is the same actual linear counterexample, not an additional certificate obstruction. On the new local band,

$$r_3(q) - r_{\mathrm{lin}}(q) = \frac{8g(q)}{5q(9q-1)} \ge 0,$$

where $g = g_{\mathrm{low}} = 6 - 3q - 4q^2$, with equality at $q^\dagger$. The high-q bridge uses the opposite sign $g_{\mathrm{high}} = 4q^2 + 3q - 6 = -g_{\mathrm{low}}$ and a different storage. The period-three comparison therefore does not conflict with the convergence theorem. In particular, $q = 9/10$ gives the sharp interval $0 < r < 152/45$ directly; the alternate fixed-q storage in Appendix F.3 is not needed for this specialization.

Corollary 3.2 (Identity coupling at fixed scalar curvature)**.** *Let $X = Y = \mathcal{H}$, $L = I$, $Q = \mu I$, $\sigma = 1$, $\tau = r$, $\mu > 0$, and let f be any proper lower-semicontinuous convex function on the real Hilbert space $\mathcal{H}$. Set $q = (1+\mu)^{-1}$. If $0 < r < \mathcal{R}(q)$, the literal iteration converges strongly to its unique KKT pair. Both anchor errors, increments and ports are square summable, and its positive-output graph and original objective gaps are summable.*

Proof. The proximal formula (103) depends only on μ, f, c, not on the algorithmic product r, and gives the unique KKT pair for every $r > 0$. Use $\kappa_{\mathrm{cert}} = r$ in Theorem 4. Then $B = I$ and $u_n = p_n$, so the general L-image and dual conclusions give strong convergence and both anchor-error square sums, without separability of f. This corollary uses no rate assertion from the one-delay comparison. If one additionally applies the global growth corollary, its hypothesis is verified here by the objective's $1/\mu$ strong convexity; its constants remain certificate-dependent. ∎

3.3 Actual counterexamples and the meaning of sharpness

Proposition 3.1 (An actual endpoint two-cycle and divergent larger products)**.** Take $X = Y = \mathbb{R}$, $f = 0$, $Q = m > 0$, $L = 1$ and $c = 0$, with $q = (1 + \sigma m)^{-1}$ and $d = \tau\sigma$. The unique KKT pair is zero. At $d_* = 8(1 + q)/(5q) = (8/5)(2 + \sigma m)$ every nonzero a gives

$$(x_n, y_n) = (-1)^n (a, 2a/\tau). \tag{41}$$

At every $d > d_*$ there is a real eigenvector initialization for which the pair and its objective diverge exponentially. These are existence of failing orbits, not failure of every orbit.

Proof. The literal map and its trace and determinant are

$$M_d = \begin{pmatrix} 1 & -\tau \\ \sigma q & q(1 - 9d/8) \end{pmatrix}, \quad \operatorname{tr} M_d = 1 + q - 9qd/8, \quad \det M_d = q(1 - d/8). \tag{42}$$

The three real quadratic Schur quantities are

$$\begin{aligned} 1 - \operatorname{tr} M_d + \det M_d &= qd > 0, \\ 1 + \operatorname{tr} M_d + \det M_d &= 2 + 2q - 5qd/4, \\ 1 - \det M_d &= 1 - q + qd/8 > 0. \end{aligned}$$

Thus its stable interval is exactly $0 < d < d_*$. At $d = d_*$, direct multiplication gives $M_d(a, 2a/\tau) = -(a, 2a/\tau)$. The objective equals $a^2/(2m) > 0$ along the cycle. For $d > d_*$ the characteristic polynomial has $\chi_d(-1) = 2 + 2q - 5qd/4 < 0$, while $\chi_d(\lambda) \to +\infty$ as $\lambda \to -\infty$. Hence it has a real root $\lambda_- < -1$. The corresponding nonzero eigenvector satisfies $y = (1 - \lambda_-)x/\tau$, so $x \neq 0$ and both the pair and objective diverge. ∎

For contrast within the same class, take $Q = \operatorname{diag}(m, M)$, $M > m$, $X = \mathbb{R}$, $Y = \mathbb{R}^2$, and $Lx = (0, x)$, still with $f = 0, c = 0$. The first dual coordinate decays at $(1 + \sigma m)^{-1}$ and the coupled block has curvature M. Its stable cap is $(8/5)(2 + \sigma M)$, which exceeds the lower-curvature class bound. This demonstrates why class-wide sharpness does not imply fixed-pair necessity.

3.4 Combining the two known scalar counterexample families

The classification in Theorem 13 is independent of the linear example. In identity normalization $q = (1+\mu)^{-1}$, its positive-curvature range $0 < \mu < 8$ is exactly $1/9 < q < 1$. Define

$$r_3(q) = R(q^{-1} - 1) = \frac{8(1 + q + q^2)}{q(9q - 1)}, \qquad 1/9 < q < 1. \tag{43}$$

For every fixed q in that range and every $r \geq r_3(q)$, including onset, there is a fixed convex scalar objective with a KKT pair and a genuine least-period-three orbit. The objective may depend on the chosen parameters, but does not change along its orbit. For $0 < q \leq 1/9$, absence of such three-cycles is not a convergence theorem. The affine

point $q = 1$ and the excluded point $q = 0$ are not covered by this positive-curvature classification.

For physical scalar steps $Q = m > 0$, $L = 1$, write $\widetilde{y} = y/\sigma$, $\widetilde{f} = f/\sigma$ and $d = \tau\sigma$. The map becomes exactly

$$x^+ = \operatorname{prox}_{d\widetilde{f}}(x - d\widetilde{y}), \qquad \widetilde{y}^+ = q\left[\widetilde{y} + (9x^+ - x)/8 - c\right], \qquad q = (1 + \sigma m)^{-1}.$$

Thus $\widetilde{\mu} = \sigma m$ and the period-three region is $0 < \sigma m < 8$, $d \geq R(\sigma m)$. Rescaling the fixed witness by $f = \sigma\widetilde{f}$ also preserves the KKT correspondence. This is an admissible scalar subclass argument; it does not turn an eigenvalue of a noncommuting pair (Q, L) into an actual orbit for that pair.

To record the known actual failures, set

$$U(q) = \begin{cases} \min\{8(1+q)/(5q), r_3(q)\}, & 1/9 < q < 1, \\ 8(1+q)/(5q), & 0 < q \leq 1/9. \end{cases} \tag{44}$$

On $1/9 < q < 1$, direct subtraction gives

$$r_3(q) - r_{\mathrm{lin}}(q) = \frac{8(6 - 3q - 4q^2)}{5q(9q - 1)}.$$

The numerator is nonnegative exactly up to $q^\dagger$ in this interval. Together with the explicit first branch for $q \leq 1/9$, this proves $U(q) = R_{\mathrm{cap}}(q)$ as defined in (30). Thus for every fixed $0 < q < 1$ and every $r \geq U(q)$ there exists an actual fixed-convex-function counterexample of the preceding two types; Theorem 2 proves convergence for every smaller positive product. The definition never evaluates the rational period-three expression as a cap at $q \leq 1/9$. The endpoint $30337/10000$ at $q = 99/100$ remains included in the alternative causal certificate, with its exact distance to the full cap recorded in Section 5.3. At $q > q_\star$, the boundary $\mathcal{R}(q)$ alone is a strict-certificate boundary, not an actual-failure assertion. This complete scalar converse does not resolve the general two-function problem or a fixed noncommuting operator pair. The Wu–Ma theorem-level comparison remains open in the limited bibliographic sense explained in the comparison section.

4 General coupling and four certificate regimes

The common background is (3) on real Hilbert spaces, a bounded L, a proper lower-semicontinuous convex f, positive steps, and existence of a KKT pair. Write $d = \tau\sigma\|L\|^2$ and $K = [(1 + \sigma M)^{-1}, (1 + \sigma m)^{-1}]$ when $0 < mI \preceq Q \preceq MI$. The four regimes are as follows.

Certificate	Curvature and product condition	Quantitative scope
Three history	General coercive Q; $d < \min_{q\in K}\mathcal{R}(q)$	Original stability, restart and global single-anchor growth rates
Analytic mixed two tap	$\sigma m \geq (\sqrt{105}-9)/12$; $d < 2 + 2\sqrt{105}/15$; any finite M	Weak primal; strong dual and L-image; no new rate
Rational mixed two tap	$(1/9)I \preceq \sigma Q \preceq 9I$; $d \leq 337/100$	Same qualitative conclusions; closed product endpoint; no new rate
Three tap	Only $\sigma = 1$, $Q = I/19$; $d \leq 159/50$	Same qualitative conclusions; closed product endpoint; no new rate

The first three rows allow noncommuting Q and LL^*. The fourth allows every bounded L, including $L = 0$, with no injectivity, closed-range or compactness assumption. Finite-dimensional primal spaces upgrade the weak primal limits to norm limits. Each row has its own summability and energy proof. No universal inclusion among these four sufficient conditions is asserted. In particular, when $\sigma m \geq (\sqrt{161}-7)/14$, the first row gives $d < (8/5)(2+\sigma m)$ and grows with m.

4.1 Curvature dependent Hilbert convergence

This independent three-history certificate proves convergence for bounded coercive quadratic dual curvature and arbitrary bounded coupling on real Hilbert spaces. The energy is positive on attained subgradient histories, although its quadratic part need not be positive on arbitrary histories. Quadratic-plus-function-value energies and graph-restricted positivity are established tools, including the Hilbert-space framework of Upadhyaya–Banert–Taylor–Giselsson [6]. Our graph correction also has the causal/anticausal convex-potential decomposition used in Scherer's storage construction [9]; we give its precise decomposition below. The work specific to the present recurrence is the compact-band certificate, its operator lift, and the resulting convergence and rate conclusions.

4.1.1 The Hilbert-space convergence theorem

Throughout this section, X, Y are real Hilbert spaces, $L : X \to Y$ is bounded linear, $f : X \to (-\infty, +\infty]$ is proper, lower-semicontinuous and convex, and $Q : Y \to Y$ is bounded self-adjoint. Let $c \in Y$, $\tau, \sigma > 0$, and assume

$$0 < mI \preceq Q \preceq MI, \qquad 0 < m \leq M < \infty, \qquad d = \tau\sigma\|L\|^2 < \min_{q\in K}\mathcal{R}(q). \tag{45}$$

Here K is the compact resolvent band defined by

$$\begin{gathered} A = (I + \sigma Q)^{-1}, \qquad q_- = (1 + \sigma M)^{-1}, \qquad q_+ = (1 + \sigma m)^{-1}, \\ K = [q_-, q_+] \subset (0, 1), \end{gathered} \tag{46}$$

Choose a positive certificate product satisfying

$$d \le \kappa_{\mathrm{cert}} < \min_{q \in K} \mathcal{R}(q), \qquad \kappa_{\mathrm{cert}} > 0. \tag{47}$$

Such a choice is equivalent to the strict product hypothesis. In particular, if $L = 0$, choose a positive κ_{cert} below the minimum; no normalization divides by $\|L\|$. Continuity of $\mathcal{R}$ and compactness make the minimum attained and strictly greater than 3. Outside the proved closed-form band we retain this minimum and make no monotonicity substitution. For any fixed KKT anchor, put

$$\begin{gathered} p_n = x_n - x^*, \qquad w_n = y_n - y^*, \\ F(p) = \tau \left[f(x^* + p) - f(x^*) + \langle L^* y^*, p \rangle \right], \\ v_n = p_{n-1} - p_n - \tau L^* w_{n-1}, \qquad F_n = F(p_n) \quad (n \ge 1). \end{gathered} \tag{48}$$

Theorem 4 (Bounded coercive dual curvature on Hilbert spaces)**.** *Under* (45) *and the KKT condition* (2)*, with the data specified above, the orbit* (3) *is bounded, and there is a KKT pair* (x_∞, y_∞) *such that*

$$x_n \rightharpoonup x_\infty, \qquad y_n \to y_\infty, \qquad Lx_n \to Lx_\infty. \tag{49}$$

The last two convergences are in norm. All KKT pairs have the same dual coordinate and the same L-image. For every KKT anchor,

$$\begin{gathered} \sum_{n \ge 1} \|x_n - x_{n-1}\|^2 < \infty, \qquad \sum_{n \ge 0} \|L(x_n - x^*)\|^2 < \infty, \\ \sum_{n \ge 0} \|y_n - y^*\|^2 < \infty, \qquad \sum_{n \ge 1} \|v_n\|^2 < \infty, \qquad \sum_{n \ge 1} F_n < \infty, \\ \sum_{n \ge 1} \big[\Phi(x_n) - \Phi(x^*) \big] < \infty. \end{gathered} \tag{50}$$

All function gaps displayed here are nonnegative. If X is finite-dimensional, then $x_n \to x_\infty$ in norm, even when Y is infinite-dimensional.

No separability, eigenbasis for Q, compactness of L, commutation of Q with LL^*, primal strong convexity, or finite $f(x_0)$ is assumed. Products through 3*, singleton bands, and $L = 0$ are included. The boundary $d = \min_K \mathcal{R}$ is not included by this theorem. Without the additional growth assumption below, no general strong primal convergence in infinite dimensions or geometric rate is asserted.*

Corollary 4.1 (A sharp class-wide strong-curvature band)**.** *Retain the structural hypotheses of Theorem 4, and replace its product restriction by*

$$\sigma m \geq \frac{\sqrt{161}-7}{14}, \qquad \tau\sigma\|L\|^2 < \frac{8}{5}(2+\sigma m). \tag{51}$$

Then all its conclusions and its growth-free stability bound hold; its rate corollary also holds when the stated global single-anchor growth is assumed. For the class specified only by these curvature and norm-product bounds, the strict product cap cannot be enlarged or its endpoint included. This is not an optimality statement for each fixed pair (Q, L)*.*

Proof. The curvature assumption is equivalent to $q_+ = (1+\sigma m)^{-1} \leq q_\star = (\sqrt{161}-7)/8$. The closed formula in Proposition 5.1 gives

$$\min_{q\in K} \mathcal{R}(q) = \frac{8(1+q_+)}{5q_+} = \frac{8}{5}(2+\sigma m),$$

since the derivative of $8(1+q)/(5q)$ is $-8/(5q^2)$ on this band. The bounded upper curvature assumption remains essential to $q_- > 0$. Proposition 3.1 proves the class-wide converse. ∎

4.2 The two established mixed certificate realizations

The first three rows of the comparison above have the following conditions. Write $d = \tau\sigma\|L\|^2$ and $K = [(1+\sigma M)^{-1}, (1+\sigma m)^{-1}]$ for bounded coercive $0 < mI \preceq Q \preceq MI$, with $M < \infty$.

1. The three-history radius theorem, Theorem 4, requires $d < \min_{q\in K} \mathcal{R}(q)$. Its separate strong-curvature specialization, Corollary 4.1, gives $d < \frac{8}{5}(2+\sigma m)$ when $\sigma m \geq (\sqrt{161}-7)/14$. This cap grows with m; its energy and quantitative consequences retain their original hypotheses.

2. The analytic mixed certificate proved below requires

$$\sigma m \geq \mu_0 := \frac{\sqrt{105}-9}{12}, \qquad d < \kappa^\dagger := 2 + \frac{2\sqrt{105}}{15}. \tag{52}$$

   It permits any finite upper bound M and uses the fixed multiplier at $h = q^\dagger = (\sqrt{105}-3)/8$, even if the actual spectral maximum is smaller.

3. The rational mixed certificate requires

$$\frac{1}{9}I \preceq \sigma Q \preceq 9I, \qquad d \leq \frac{337}{100}. \tag{53}$$

   Its product endpoint is included, and its margin is explicitly computable.

The constant $\kappa^\dagger \simeq 3.3662601$ is strictly smaller than $337/100$. Thus neither mixed statement subsumes the other, and neither replaces the increasing cap in the first regime. The common consequence below is proved once for the two mixed certificates in Appendix B. It does not substitute a mixed storage into any of the three-history identities.

4.3 Two realizations of the mixed certificate

Proposition 4.1 (Analytic outer-band certificate)**.** Let $h = q^\dagger$, a, b, χ be as in (237), and set $\nu = hb$, $\lambda = a - \nu$. For every fixed $0 < q_{\min} \le h$ and $0 < \kappa < \kappa^\dagger = r_{\mathrm{lin}}(h)$, there exist a common $\eta > 0$, a continuous symmetric $P(q)$ on $[q_{\min}, h]$, and a common $\delta > 0$ satisfying (239).

Proposition 4.2 (Rational closed-product certificate)**.** For $h = 9/10$, $q \in [1/10, 9/10]$ and $0 < \kappa \le 337/100$, use

$$a = \frac{45}{71}, \quad b = \frac{26}{71}, \quad \chi = \frac{19}{71}, \quad \lambda = \frac{108}{355}, \quad \nu = \frac{117}{355}, \quad \eta = \frac{1}{1000}. \tag{54}$$

There is an explicitly specified continuous piecewise-affine certificate in q satisfying (239) with $\delta(\kappa) = \kappa/337000$.

4.4 The common Hilbert convergence consequence

Theorem 5 (Mixed graph convergence with arbitrary bounded coupling)**.** *Let X, Y be real Hilbert spaces, $L : X \to Y$ bounded linear, f proper lower-semicontinuous convex, $c \in Y$, and $0 < mI \preceq Q \preceq MI$ bounded self-adjoint with $M < \infty$. Let $\tau, \sigma > 0$. Suppose a KKT pair exists and either* (52) *or* (53) *holds. From arbitrary (x_0, y_0), the literal iteration* (3) *is bounded and admits a KKT pair (x_∞, y_∞) such that*

$$x_n \rightharpoonup x_\infty, \qquad y_n \to y_\infty, \qquad Lx_n \to Lx_\infty. \tag{55}$$

The last two limits are in norm. All KKT pairs have the same dual coordinate and L-image. For every KKT anchor, with (246) *and the objective* (1)*,*

$$\begin{aligned} &\sum_{n\ge1}\|x_n - x_{n-1}\|^2 < \infty, \quad \sum_{n\ge0}\|L(x_n - x^*)\|^2 < \infty, \quad \sum_{n\ge0}\|y_n - y^*\|^2 < \infty, \\ &\sum_{n\ge1}\|v_n\|^2 < \infty, \quad \sum_{n\ge1} F_n < \infty, \quad \sum_{n\ge1}\big[\Phi(x_n) - \Phi(x^*)\big] < \infty. \end{aligned} \tag{56}$$

Every displayed function gap is nonnegative, and the distance to every KKT primal anchor has a limit. If X is finite-dimensional, primal convergence is in norm. No general infinite-dimensional strong primal convergence or rate is asserted.

The analytic threshold is sharp as a universal norm-product threshold for the class $\sigma m \ge \mu_0$, finite M: the scalar instance $Q = \mu_0 I$, $L = I$, $\sigma = 1$, $f = 0$, $c = 0$ in Proposition 3.1 has an actual two-cycle at $\kappa^\dagger$ and an unbounded orbit above it. This is a class-wide statement, not failure for every fixed (Q, L) or every stronger actual lower-curvature bound. The rational endpoint is sufficient and is not claimed optimal. It strictly improves the old enclosing-band radius guarantee because the admissible witness $q = 9/10$, $t = -7/10$ gives $\mathcal{R}(9/10) \le 709784/214911 < 337/100$; no monotonicity of $\mathcal{R}$ is needed. The mixed theorem supplies no new rates and does not transfer any old growth-dependent rate to its larger parameter set.

4.5 A three tap certificate at one fixed weak curvature

The fourth operator regime uses an additional negative-delay tap. It applies at the fixed curvature $Q = I/19$ and fixed dual step $\sigma = 1$; its product endpoint is included. The proof uses one original convex graph and a separate residual storage, so it also covers a nontrivial kernel of L.

Theorem 6 (Fixed curvature bounded operator certificate)**.** *Let X, Y be real Hilbert spaces, let $L : X \to Y$ be bounded, and let $f : X \to (-\infty, +\infty]$ be proper lower-semicontinuous and convex. Fix $h \in Y$ and suppose a KKT pair exists:*

$$-L^*y^* \in \partial f(x^*), \qquad Lx^* = h + \frac{1}{19}y^*. \tag{57}$$

Set $q = 19/20$, $\sigma = 1$, $Q = I/19$ and $\theta = 1/8$. For arbitrary $x_0 \in X$, $y_0 \in Y$ and $\tau > 0$, consider the literal iteration

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^*y_{n-1}), \\ y_n &= q\left[y_{n-1} + \frac{1}{8}L(9x_n - x_{n-1}) - h\right], \qquad n \geq 1. \end{aligned} \tag{58}$$

If

$$0 \leq \tau\|L\|^2 \leq K := \frac{159}{50}, \tag{59}$$

then there is a KKT pair $(\bar{x}, \bar{y})$ such that $x_n \rightharpoonup \bar{x}$, $y_n \to \bar{y}$ and $Lx_n \to L\bar{x}$. The dual solution and the image of every primal solution are unique. For every KKT anchor (x^*, y^*), define

$$\begin{aligned} F(p) &= \tau\big[f(x^* + p) - f(x^*) + \langle L^*y^*, p\rangle\big], \\ p_n &= x_n - x^*, \qquad v_n = p_{n-1} - p_n - \tau L^*(y_{n-1} - y^*). \end{aligned} \tag{60}$$

Then

$$\begin{aligned} \sum_{n=1}^{\infty}\big(\|y_n - y^*\|^2 + \|L(x_n - x^*)\|^2 + \|x_n - x_{n-1}\|^2 + \|v_n\|^2 \\ + F(p_n) + F^*(v_n)\big) < \infty. \end{aligned} \tag{61}$$

For $\Phi(x) = f(x) + (19/2)\|Lx - h\|^2$, all attained objective gaps $\Phi(x_n) - \Phi(x^*)$, $n \geq 1$, are nonnegative and summable. If X is finite dimensional, $x_n \to \bar{x}$ in norm. No assumption $f(x_0) < \infty$ is required, and $L = 0$ is included.

The exact proof is in Appendix B.9. It uses the O'Shea–Zames–Falb multiplier and Fenchel graph-potential method discussed in [9], with explicitly verified rational matrices. The uncorrected quadratic storage is not required to be positive; positivity holds after adding one common original-graph correction. The extra tap requires the potential at index 3 and its first descent comparison at index 4.

This certificate strictly improves the three-history ansatz at $q = 19/20$. The admissible witness $t = -3/5$ in the formula for R_{ans} gives

$$R_{\mathrm{ans}}(19/20) \le \frac{38944}{12331} < \frac{159}{50} < \frac{9128}{2869} = R_3(19/20), \tag{62}$$

with the exact gaps

$$\frac{159}{50} - \frac{38944}{12331} = \frac{13429}{616550}, \qquad \frac{9128}{2869} - \frac{159}{50} = \frac{229}{143450}. \tag{63}$$

The common frequency interface of Section 5.4 covers every fixed strict budget below $R_3(19/20)$, whereas the actual operator theorem above is proved only through $159/50$. Neither a sharp operator threshold, a nonscalar-Q or arbitrary-σ extension, a convergence rate, nor strong primal convergence in a general Hilbert space is asserted here.

5 Exact certificate radii and multiplier families

5.1 The exact radius of the specified graph certificate

The scalar calculation in this section determines the exact strict feasibility region of one specified graph-storage ansatz. Its graph weights are $(3, 3, 1, 0)$, and its unrestricted-state supply is fixed below. Exactness refers to this ansatz: failure of its harmonic test need not be failure of an actual convex orbit. The closed-form strong-curvature band will be distinguished from this more general certificate boundary.

5.1.1 Literal ports and the certificate coordinates

Fix $0 < q < 1$ and $r > 0$, and put $\mu = q^{-1} - 1$. For identity coupling and $\sigma = 1$, the literal primal-first iteration is

$$\begin{aligned} x_n &= \mathrm{prox}_{rf}(x_{n-1} - r y_{n-1}), \\ y_n &= q\left[y_{n-1} + \frac{9x_n - x_{n-1}}{8} - c\right]. \end{aligned} \tag{64}$$

Here f is proper, lower-semicontinuous and convex on a real Hilbert space, and scalar curvature means the dual Hessian is μI; it does not impose scalar or separable f. Fix a KKT anchor satisfying $-y_* \in \partial f(x_*)$ and $x_* - c = \mu y_*$. Write $p_n = x_n - x_*$ and $w_n = y_n - y_*$. The dual coordinate used throughout this section is explicitly

$$\begin{aligned} R_n = w_n/q, \qquad R_n &= qR_{n-1} + \frac{9p_n - p_{n-1}}{8}, \\ F(p) &= r\big[f(x_* + p) - f(x_*) + \langle y_*, p\rangle\big], \\ v_n &= p_{n-1} - p_n - rqR_{n-1} \in \partial F(p_n). \end{aligned} \tag{65}$$

Thus $F \ge 0$, $F(0) = 0$, and $F(p_n) < \infty$ for $n \ge 1$. No graph point at index zero, and no assumption $f(x_0) < \infty$, is used. Solving the dual update in (65) for R_{n-1} gives the

second expression for the port,

$$v_n = \left(\frac{9r}{8} - 1\right) p_n + \left(1 - \frac{r}{8}\right) p_{n-1} - r R_n.$$

For unrestricted real state and input vectors $s = (s_0, s_1, s_2, R)^{\mathsf{T}}$ and $z = (s, h)^{\mathsf{T}}$, set

$$\begin{gathered} J = [I_4 \;\; 0], \qquad T_R(q) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1/8 & 0 & 0 & q & 9/8 \end{pmatrix}, \\ c_R(r) = \begin{pmatrix} 9r/8 - 1 \\ 1 - r/8 \\ 0 \\ -r \end{pmatrix}, \qquad a = \begin{pmatrix} 3 \\ 3 \\ 1 \\ 0 \end{pmatrix}, \\ v_R(q, r) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -rq \\ -1 \end{pmatrix}, \qquad w_\eta = \begin{pmatrix} 0 \\ -2 \\ -1 \\ 0 \\ 10 + \eta \end{pmatrix}, \qquad e_5 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \end{gathered} \tag{66}$$

For a square matrix M put $\operatorname{sym} M = (M + M^{\mathsf{T}})/2$. Define

$$\begin{aligned} G_R(q, r) &= \operatorname{sym}(c_R(r) a^{\mathsf{T}}), \\ S_R(q, r, \eta) &= \operatorname{sym}(w_\eta v_R(q, r)^{\mathsf{T}}) - 7 \operatorname{sym}\big(e_5 (J^{\mathsf{T}} c_R(r))^{\mathsf{T}}\big). \end{aligned} \tag{67}$$

The redundant argument q in G_R is convenient for parameter families. Direct multiplication gives $c_R(r)^{\mathsf{T}} T_R(q) = v_R(q, r)^{\mathsf{T}}$. On the actual before-step history $s = (p_{n-1}, p_{n-2}, p_{n-3}, R_{n-1})$ and input $h = p_n$, $c_R(r)^{\mathsf{T}} s = v_{n-1}$ and $v_R(q, r)^{\mathsf{T}} z = v_n$. These are identities of the literal recurrence, independent of any zero-state transfer calculation. The resulting causal supply is

$$\mathcal{Q}_n = \langle (10 + \eta) p_n - 2 p_{n-2} - p_{n-3}, v_n \rangle - 7 \langle p_n, v_{n-1} \rangle. \tag{68}$$

The ansatz asks for a real symmetric P_R and $\eta > 0$ such that

$$\begin{aligned} P_R + G_R &\succ 0, \\ J^{\mathsf{T}} P_R J - T_R^{\mathsf{T}} P_R T_R - S_R(q, r, \eta) &\succ 0. \end{aligned} \tag{69}$$

Both inequalities are on all real states, with sizes four and five, respectively. Positivity is required of the graph correction $P_R + G_R$, not of P_R alone. The scalar matrices also act on Hilbert-valued histories by replacing scalar products by Hilbert inner products.

5.1.2 The exact radius and a shared storage lemma

For $t \in [-1, 1]$ define

$$\begin{aligned} \Delta_q(t) &= 1 + q^2 - 2qt, \\ A_q(t) &= (9q + 7)t^2 + (14q - 34)t + 25q - 21, \\ \rho_q(t) &= \frac{8(1-t)^2\Delta_q(t)}{qA_q(t)} \quad \text{when } A_q(t) > 0, \\ \mathcal{R}(q) &= \min_{\substack{-1\le t\le 1 \\ A_q(t)>0}} \rho_q(t). \end{aligned} \tag{70}$$

This is an attained finite positive minimum. Indeed, $A_q(-1) = 20(q+1) > 0$ and $A_q(1) = 48(q-1) < 0$. The strictly convex quadratic A_q therefore has one root $\alpha_q \in (-1, 1)$ and its other root above one. Its positive set on $[-1, 1]$ is $[-1, \alpha_q)$, where explicitly

$$\alpha_q = \frac{17 - 7q - 2\sqrt{109 - 56q - 44q^2}}{9q + 7}.$$

The numerator of ρ_q is positive on this set and at its excluded endpoint. Hence $\rho_q(t) \to +\infty$ as $t \uparrow \alpha_q$, and the minimum is attained away from that endpoint.

Theorem 7 (Exact radius of the specified ansatz)**.** *For $0 < q < 1$ and $r > 0$, there exist $\eta > 0$ and a real symmetric P_R satisfying* (69) *if and only if*

$$0 < r < \mathcal{R}(q). \tag{71}$$

Let $\mathcal{C}$ be any nonempty compact subset of this feasible (q, r) region. There exists $\eta_* > 0$ such that, for every $0 < \eta \le \eta_*$, a continuous bounded symmetric matrix function $P_{R,\eta}(q, r)$ on $\mathcal{C}$ satisfies both inequalities in (69) *with a common positive lower margin $\delta_\eta I$. In particular, the same scalar η applies to all parameter pairs in $\mathcal{C}$.*

Theorem 7 is proved in Appendix B. The essential steps are a rank-two completed-square supply at zero multiplier, a three-state quotient whose full characteristic polynomial is retained, strict bounded-real storage, and a positive-multiplier repair of the two one-dimensional kernels. Necessity is a harmonic test on the unrestricted certificate and does not assert an attainable nonlinear orbit.

5.1.3 Continuity and the closed-form strong-curvature band

Proposition 5.1 (Radius bounds and continuity)**.** The function $\mathcal{R}$ is continuous on $(0, 1)$, and

$$3 < \mathcal{R}(q) \le r_{\text{lin}}(q) := \frac{8(1+q)}{5q} \qquad (0 < q < 1). \tag{72}$$

For

$$\begin{aligned} q_\star &= \frac{\sqrt{161} - 7}{8}, \\ \mathcal{R}(q) &= r_{\text{lin}}(q) \quad (0 < q \le q_\star), \\ \mathcal{R}(q) &< r_{\text{lin}}(q) \quad (q_\star < q < 1). \end{aligned} \tag{73}$$

The equality range is equivalently $\mu = q^{-1} - 1 \geq (\sqrt{161} - 7)/14$. In particular, $\mathcal{R}(1/2) = 24/5$. For every nonempty compact interval $K \subset (0,1)$, $\min_{q \in K} \mathcal{R}(q)$ exists and is strictly greater than three. No monotonicity outside the displayed closed-form band is asserted.

The strict estimate (205) supplies the low-product bridge used in the mixed theorem. The factorization (206) supplies its low-curvature-parameter bridge. The exact ansatz remains useful for every $q \in (0,1)$, including $q > q^{\dagger}$, and for its independently quantified operator consequences.

5.2 Static and delayed certificate comparisons

The exact radius $\mathcal{R}(q)$ concerns the specified three-history supply and its unrestricted-state inequalities. It is useful to compare that radius with simpler certificates, without identifying failure of a certificate with failure of the algorithm. Throughout the identity-coupled parts of this section, $\mu > 0$, $q = (1+\mu)^{-1}$, $\sigma = 1$, $\tau = r$, and $L = I$. Thus f remains an arbitrary proper lower-semicontinuous convex function on a real Hilbert space; scalar dual curvature does not impose separability of f.

Fix the KKT pair from (103) and put

$$\begin{aligned} p_n &= x_n - x^*, \qquad w_n = y_n - y^*, \qquad R_n = w_n/q, \\ F(p) &= r\big(f(x^* + p) - f(x^*) + \langle y^*, p\rangle\big), \\ v_n &= p_{n-1} - p_n - r w_{n-1} \in \partial F(p_n), \qquad n \geq 1. \end{aligned} \tag{74}$$

Here $F(0) = \min F = 0$. Every $F_n := F(p_n)$ with $n \geq 1$ is finite, but $F(p_0)$ need not be. The exact anchored recurrence is

$$R_n = qR_{n-1} + \frac{9p_n - p_{n-1}}{8}. \tag{75}$$

Use the forward spectral variable z and its delay $\zeta = z^{-1}$. With $t = \operatorname{Re}\zeta \in [-1,1]$ on the unit circle, set

$$\begin{aligned} V(\zeta) &= \zeta - 1 - \frac{rq\zeta(9-\zeta)}{8(1-q\zeta)}, \\ \Delta_q(t) &= 1 + q^2 - 2qt > 0, \qquad r_{\mathrm{lin}}(q) = \frac{8(1+q)}{5q}, \qquad q_\star = \frac{\sqrt{161} - 7}{8}. \end{aligned} \tag{76}$$

All frequency inequalities below refer to this orientation and port.

5.2.1 The variable-product static circle

Proposition 5.2 (Exact radius of the stated static supply)**.** For $0 < q < 1$, define

$$\begin{aligned} B_q(t) &= 2t^2 - (q+9)t + 9q - 1, \\ r_{\mathrm{static}}(q) &= \min_{\substack{-1 \leq t \leq 1 \\ B_q(t) > 0}} \frac{8(1-t)\Delta_q(t)}{qB_q(t)}. \end{aligned} \tag{77}$$

The minimum is finite and attained. A positive quadratic storage with strict unrestricted-state dissipation for the static supply $\langle p_n, v_n \rangle$ exists exactly for $0 < r < r_{\mathrm{static}}(q)$. In that range it gives strong convergence and square summability of the anchored identity-coupled state, as well as summability of the positive-output F_n, from every finite start. With $q_\circ = \sqrt{6} - 2$,

$$\begin{cases} r_{\mathrm{static}}(q) = \mathcal{R}(q) = r_{\mathrm{lin}}(q), & 0 < q \le q_\circ, \\ r_{\mathrm{static}}(q) < \mathcal{R}(q), & q_\circ < q < 1. \end{cases} \tag{78}$$

These are radii of the stated tests, not an optimality assertion over all static IQCs or nonlinear arguments.

5.2.2 The established OZF test and its precise scope

The one-delay construction below is a specialization of established O'Shea–Zames–Falb (OZF) machinery. We use the precise convention of Kharitenko–Scherer [11]: Section 2.3 and Theorem 1, equation (3), PDF page 2 of arXiv:2210.14992v1; Section 3.1, page 3, for Kronecker lifting; and Remark 9, page 6. These are pins to the inspected preprint, not an assertion that the journal version has the same numbering.

The exact negative-feedback plant for the anchored prox is

$$G(z) = \frac{(9rq/8 - 1)z + q(1 - r/8)}{z(z - q)}, \qquad V(\zeta) = -G(z) - 1, \qquad G_{\mathrm{source}} = -G = 1 + V. \tag{79}$$

It is strictly proper and Schur, with stable poles $0, q$ before any cancellation. In the source's positive-feedback, slope-sector $[0, 1]$ convention, its Hermitian multiplier and evaluated form are

$$\begin{gathered} \Pi = \begin{pmatrix} 0 & M^* \\ M & -(M + M^*) \end{pmatrix}, \\ \begin{pmatrix} G_{\mathrm{source}} \\ 1 \end{pmatrix}^* \Pi \begin{pmatrix} G_{\mathrm{source}} \\ 1 \end{pmatrix} = 2\,\mathrm{Re}\big[M(G_{\mathrm{source}} - 1)\big] = 2\,\mathrm{Re}(MV). \end{gathered} \tag{80}$$

In particular no delay reversal or change of the feedback sign is implicit in the test.

For $\phi = \mathrm{prox}_F$, one has $\phi(0) = 0$ and, as relations,

$$\phi \circ (I - \phi)^{-1} = (\partial F)^{-1} = \partial F^*. \tag{81}$$

Indeed $u = \mathrm{prox}_F(z)$ is equivalent to $z - u \in \partial F(u)$. Thus the transformed relation in the source is cyclically monotone, including for nonseparable multivariate f. The prox is total and single-valued; strict properness makes the causal feedback well posed. For the source's total finite-slope class the same well-posedness check uses that finite slope forces the pointwise relation to be single-valued. Finite initial states contribute a square-summable free response at the exogenous signal port. For

$$M(\zeta) = m - \gamma\zeta, \qquad m \ge \gamma \ge 0, \tag{82}$$

the diagonal Toeplitz coefficient is m, its sole off-diagonal coefficient is $-\gamma$, and row and column sums are nonnegative. It is therefore a doubly hyperdominant multiplier of the source class. Every strictly feasible multiplier used below has $m > \gamma$.

The printed source theorem gives finite-dimensional pointwise-signal, forced-ℓ^2 stability under the strict FDI. Its discussion of ℓ^2-valued nonlinearities in Remark 9 is not silently substituted for an arbitrary-real-Hilbert convergence theorem. That conclusion is proved directly in Proposition 5.4 below. Likewise, the source's Theorem 4 concerns uniform forced gain across dimensions. It does not turn failure of a chosen multiplier into a nonconvergent autonomous convex orbit.

5.2.3 Static or one delay covers the entire sharp-curvature band

Proposition 5.3 (The full open strong-curvature band)**.** For every

$$0 < q \le q_\star, \qquad 0 < r < r_{\mathrm{lin}}(q), \tag{83}$$

there is a multiplier $M = m - \gamma\zeta$ with $m > \gamma \ge 0$ such that $\mathrm{Re}(MV) < 0$ on the entire unit circle. The static choice suffices in the small-q and low-r regions used in the proof. At $q = 1/2$, the single fixed multiplier $M = 1 - \zeta/3$ works for every $0 < r < 24/5$. Consequently the identity-coupled conclusion in this sharp band does not require the particular three-history ansatz.

5.2.4 A direct Hilbert storage with first-output startup

The source attribution does not replace the following argument. Its different state and convex potential require only the first actual proximal output for initialization, rather than the three outputs used in the longer-history construction.

Proposition 5.4 (One-delay storage and arbitrary finite starts)**.** Suppose $0 < q < 1$, $r > 0$, and some $m > \gamma \ge 0$ satisfies $\mathrm{Re}[(m - \gamma\zeta)V(\zeta)] < 0$ on the unit circle. For every real Hilbert space and every proper lower-semicontinuous convex f, the identity-coupled iteration converges strongly to the unique KKT pair from every finite (x_0, y_0), allowing $f(x_0) = +\infty$. Relative to this pair, p_n, w_n, and v_n are square summable. The positive-output values F_n and the composite objective gaps are summable. The proof gives a finite energy V_1 and descent for $n \ge 2$. No uniform quadratic restart estimate or pointwise rate is asserted by this proposition.

5.2.5 A finite first-neighbor separation in weak curvature

Proposition 5.5 (An exact excluded simplex)**.** At $q = 999/1000$ and $r = 3$, no multiplier in the complete normalized nonnegative first-neighbor family

$$M(\zeta) = 1 - h_+\zeta - h_-\zeta^{-1}, \qquad h_+, h_- \ge 0, \qquad h_+ + h_- \le 1, \tag{84}$$

satisfies the strict negative FDI for the port (76). The specified three-history certificate is nevertheless strictly feasible, and the actual convex iteration converges at these same parameters.

5.3 Weak curvature and alternative causal certificates

The exact radius $\mathcal{R}(q)$ remains a certificate radius, including above $q^\dagger$. Its weak-curvature behavior distinguishes the limitation of that ansatz from actual periodic failure. Put $\varepsilon = 1 - q$.

Proposition 5.6 (Second-order gap and two finite-delay witnesses)**.** As $\varepsilon \downarrow 0$,

$$\begin{aligned} \mathcal{R}(1-\varepsilon) &= 3 + \frac{27}{8}\varepsilon - \frac{4597}{768}\varepsilon^2 + \frac{347875}{9216}\varepsilon^3 + O(\varepsilon^4), \\ r_3(1-\varepsilon) - \mathcal{R}(1-\varepsilon) &= \frac{8281}{768}\varepsilon^2 - \frac{288925}{9216}\varepsilon^3 + O(\varepsilon^4), \\ r_{\mathrm{lin}}(1-\varepsilon) - \mathcal{R}(1-\varepsilon) &= \frac{1}{5} - \frac{71}{40}\varepsilon + \frac{29129}{3840}\varepsilon^2 + O(\varepsilon^3). \end{aligned} \tag{85}$$

In addition, at $q = 99/100$ all conclusions of the convergence and summability part of Theorem 3 hold for

$$0 < r \le \frac{30337}{10000}, \qquad M(z) = \frac{1001}{1000} - \frac{14}{15}z - \frac{1}{15}z^6. \tag{86}$$

This is a two-tap multiplier with longest delay six and a strict frequency inequality at the included product endpoint. The simpler causal witness

$$q = \frac{9}{10}, \quad 0 < r \le \frac{10}{3}, \qquad M(z) = \frac{1001}{1000} - \frac{3}{4}z - \frac{1}{4}z^3 \tag{87}$$

has the same conclusions, but is now covered by the stronger mixed sharp interval $0 < r < 152/45$.

The $q = 99/100$ result is outside the band of Theorem 3 and strictly improves $\mathcal{R}(99/100)$: the exact comparisons are

$$\mathcal{R}(99/100) \le \frac{898874528}{296345709} < \frac{30337}{10000} < \frac{33944}{11187} = r_3(99/100) < \frac{1592}{495} = r_{\mathrm{lin}}(99/100). \tag{88}$$

The guaranteed improvement over the old radius is at least $1494493933/2963457090000$, and the certificate distance to the period-three threshold is $59981/111870000$. Theorem 2 now identifies that threshold as the actual cap at $q = 99/100$. The causal witness remains an alternative closed-endpoint certificate; no best FIR multiplier is identified by it. The global-minimum expansion and the complete finite-delay actual-startup argument are proved in Appendix F.2. The expansion is a limit from $q < 1$ and does not add a convergence theorem at $q = 1$.

5.4 The complete common family on three prescribed taps

At the fixed curvature $q = 19/20$, a negative third-delay tap allows a common residual and period-three frequency certificate. Throughout this subsection,

$$\begin{aligned} R = R_3(19/20) = \frac{9128}{2869}, \qquad V_r(z) = z - 1 - \frac{rqz(9-z)}{8(1-qz)}, \\ t = \operatorname{Re} z, \quad |z| = 1, \qquad \Delta(t) = 1 + q^2 - 2qt = \frac{761 - 760t}{400}. \end{aligned} \tag{89}$$

Here z represents one time delay. The sign convention pairs the positive tap z with $\langle p_n, v_{n-1}\rangle$ and the negative tap z^{-j} with $\langle p_{n-j}, v_n\rangle$.

Proposition 5.7 (Complete family for the prescribed support)**.** Among the nonnegative-tap multipliers $M(z) = 1 - \sum_{j\in\{-3,-2,1\}} h_j z^j$ with $\sum_j h_j \le 1$, the two full-circle inequalities $-\operatorname{Re}(MV_0) \ge 0$ and $-\Delta \operatorname{Re}(MV_R) \ge 0$ hold simultaneously if and only if

$$\begin{aligned} h_{-3} = x, \qquad h_{-2} = \frac{14345x + 5052}{19397}, \qquad h_1 = \frac{14345 - 33742x}{19397}, \\ \frac{4241}{120623} \le x \le \frac{897}{20294}. \end{aligned} \tag{90}$$

In particular, feasibility forces unit mass, and all three taps are strictly positive throughout this closed parameter interval.

The convenient interior choice $x = 1/25$ gives

$$M_0(z) = 1 - az - bz^{-2} - cz^{-3}, \qquad (a, b, c) = \frac{1}{484925}(324883, 140645, 19397). \tag{91}$$

For every fixed $0 < k < R$, these same taps admit one positive diagonal repair $M_\eta = M_0 + \eta$ for all effective products $0 \le r \le k$. Explicitly, with

$$h_0 = \frac{29196}{484925}, \qquad \gamma = \frac{(1 - k/R)h_0}{400}, \qquad 0 < \eta \le \frac{\gamma}{6}, \tag{92}$$

Appendix F.4 proves

$$-\Delta(t) \operatorname{Re}\big[M_\eta(z)V_r(z)\big] \ge \frac{\gamma}{2}(1 - t) + \frac{19\eta r}{400}, \qquad 0 \le r \le k. \tag{93}$$

Thus every positive effective product has a strict full-circle margin; the residual product $r = 0$ retains its necessary DC zero. There is no uniform positive margin as $r \downarrow 0$. The normalized symbol $M_\eta/(1 + \eta)$ has nonnegative taps of total mass $1/(1 + \eta) < 1$.

Proposition 5.7 is an exact result for the stated support and mass restriction, not an optimality statement over all multipliers. Its frequency inequality and its one-graph finite potential are not by themselves a general-L convergence theorem. The separate rational storage of Theorem 6 proves the fixed product budget $K = 159/50$. Its directly certified value $\eta = 807419/10^9$ is larger than the sufficient small-η bound in (92) at $k = K$; there is no conflict, because that bound is not necessary and the four storage inequalities are verified independently at the chosen η. The multiplier and convex-potential framework is the O'Shea–Zames–Falb and Fenchel method, with the attribution in [9].

5.4.1 A simpler static quadratic metric

The soft-IQC argument in Section 6 obtains a strict DC margin from composite growth. A different, static supply yields a genuine coercive metric when the scaled dual curvature is large enough. We retain the spaces, literal map and positive steps of Theorem 18, but replace its growth hypothesis by an existing KKT pair and the spectral restriction below. This branch reuses the same linear plant and KKT normalization; it does not deduce convergence merely from a non-strict soft IQC.

Theorem 8 (Quadratic dual with a bounded resolvent)**.** *Let $X = \mathbb{R}^d$, $Y = \mathbb{R}^m$, $Q \succ 0$, and let $L : X \to Y$ be arbitrary. Suppose f is proper lower-semicontinuous convex, a KKT pair exists, and*

$$\sigma Q \succeq \tfrac{1}{2} I, \qquad \tau\sigma\|L\|^2 \le 3.$$

Then the literal primal-first quadratic-dual iteration converges in paired-state norm to a KKT pair from every finite initialization. Its orbit is bounded. A fixed positive-definite quadratic metric, depending only on τ, σ, Q, L, makes the squared distance to every KKT pair nonincreasing. Relative to any KKT anchor, the sequences $L(x_n - x^*)$ and $y_n - y^*$ are square summable, as are the primal increments. Neither uniqueness of the primal minimizer nor a geometric rate for the full paired state is asserted.

6 Rates explicit constants and restricted robustness

6.1 Uniform restart estimates on the complete scalar cap

The exact cap in Theorem 2 also supports a geometric bound in the ordinary physical pair norm. There is no additional growth assumption on f. The identity-coupled composite itself is strongly convex, but the algorithmic rate below follows from quantitative startup, dissipation, and DC closure, rather than strong convexity alone.

Theorem 9 (Full-cap paired and attained-output rates)**.** *Under the hypotheses of Theorem 2, let $0 < r < R_{\mathrm{cap}}(q)$ and write*

$$e_n^2 = \|x_n - x_*\|^2 + \|y_n - y_*\|^2. \tag{94}$$

Appendix C.1 specifies a finite constant $C = C(q, r) > 1$ for which every actual restart $k \ge 0$ satisfies

$$\sum_{j \ge 0} e_{k+j}^2 \le C e_k^2. \tag{95}$$

The same chosen certificate and the same C work for every $f, c, \mathcal{H}$, initial pair, and restart index at these fixed q, r. With $\varrho = 1 - 1/C \in (0, 1)$, one has, for every $k, j \ge 0$,

$$e_{k+j}^2 \le C \varrho^j e_k^2, \qquad e_{k+j} \le \sqrt{C}\, \varrho^{j/2} e_k. \tag{96}$$

For the original composite and its one-output constant, put

$$\Phi(x) = f(x) + \frac{\|x - c\|^2}{2\mu}, \qquad D_\Phi = \frac{1 + r^2}{2}\left(\frac{1}{r} + \frac{1}{\mu}\right). \tag{97}$$

Every attained output obeys, for $k \geq 0$ and $j \geq 1$,

$$0 \leq \Phi(x_{k+j}) - \Phi(x_*) \leq D_\Phi C \varrho^{j-1} e_k^2, \tag{98}$$

$$\|u_{k+j} + y_*\|^2 \leq \frac{1 + r^2}{r^2} C \varrho^{j-1} e_k^2, \tag{99}$$

where u_n is the actual proximal subgradient in (32)*. No finite $\Phi(x_k)$ is needed to restart.*

The squared-error factor is ϱ, and the ordinary-norm factor is $\sqrt{\varrho}$. These estimates assert neither stepwise monotonicity of e_n nor one-step contraction between two moving trajectories. The constants are finite for each fixed admissible parameter and chosen certificate; they are not asserted uniform as $q \to 0$, $q \to 1$, $r \uparrow R_{\mathrm{cap}}(q)$, or across the algebraic branch switches. The cap is sharp by the actual counterexamples; the rate factors are not claimed optimal.

Relation to the earlier rates. Theorem 10 already gives a Hilbert-space paired geometric bound for $0 < r \leq 3$. Corollaries 3.2 and 6.1, together with Lemma 6, already give a certificate-dependent rate throughout $0 < r < \mathcal{R}(q)$, since Φ has global single-anchor quadratic growth coefficient $1/\mu$. The new range is the extension of that conclusion to the complete piecewise sharp cap. The previous explicit constants and their operator scalings are retained below. No new rate for general L, nonscalar Q, the affine endpoint, or the earlier restricted robustness regimes follows from this identity-coupled result. The full-cap scalar rate above extends the established $\mathcal{R}(q)$ rate region. The general-coupling mixed and three-tap theorems supply no new rate, and no old perturbation tolerance is extended by the scalar result. Each quantitative branch uses its own energy, normalization and growth hypothesis. The following index records the scope and the location of the actual constants.

1. Under $d < \min_K \mathcal{R}$, arbitrary three-output startup and fixed-anchor orbital stability are Lemma 6 and Corollary C.1. Global single-anchor quadratic growth adds Corollary 6.1. The product-three formulas retain K_0^2, $J = 20$, the coefficients $140, 60, 20$, the objective coefficient $3/2$, and the factor $20 + 15b$ in Appendix C.

2. Identity coupling with scalar $\mu > 0$ and $0 < r \leq 3$ has the explicit all-restart square-sum and geometric estimates of Theorem 10. The constants in (453) and (474) are different constructions. They need no additional growth assumption and are not uniform as $\mu \downarrow 0$.

3. Affine dual objectives have strong primal and weak dual convergence for $r < 3$ (Theorem 11). The scalar endpoint $r = 3$ has the restart constant 300 and the residual and cluster conclusions of Theorem 12. Exact affine-primal conjugacy holds for all $r > 0$ and real extrapolation parameters (Proposition C.1); its endpoint bound carries the condition number in (498).

4. The finite-dimensional soft-IQC bounds in Theorems 18 and 19 apply at product at most 3. General rectangular coupling assumes the stated global growth; the identity branch retains the sharper gain, rather than replacing it with the conservative rectangular estimate.

5. Theorem 20 gives Hilbert isotropic and finite-dimensional positive-definite perturbation tolerances. Theorem 21 requires global strong convexity of the original quadratic composite. Their actual perturbed anchor, tolerances and C_ε estimates, including the retained 1/2462 specialization, are proved in Appendix D.

The exact matrices, stable-filter estimates, low-product prerequisite certificate, restart arguments, and all physical scaling and output-index exponents are retained in Appendices C and D. Square summability alone is not used to assert a uniform geometric factor. The explicit scalar statements and the variable-radius growth corollary follow.

6.2 The map and the curvature transition

The following identity-coupled theorems retain their elementary proofs and fully explicit constants. They complement the full-cap estimate of Theorem 9 and the certificate-dependent range of Theorem 4; the exact product-three specialization is Theorem 16. All formulas in this branch retain $L = I$, $\tau = r$, $\sigma = 1$. No reordering, relaxation, correction, or changing stepsize is introduced. The paired norm here is the ordinary product norm. The affine and positive-curvature branches have distinct endpoint and growth conclusions.

Let $\mathcal{H}$ be a real Hilbert space, let $f : \mathcal{H} \to (-\infty, +\infty]$ be proper lower-semicontinuous convex, and set

$$h(y) = \frac{\mu}{2}\|y\|^2 + \langle c, y\rangle + d, \qquad c \in \mathcal{H}, \quad d \in \mathbb{R}, \quad \mu \geq 0. \tag{100}$$

For the first branch of the argument the curvature μ is a scalar. Section B.10 replaces it by a bounded coercive self-adjoint operator, and Section D.3 treats small smooth convex perturbations. For the present Hilbert branch the coupling is the identity, and the parameters are exactly $\theta = 1/8$, $\tau = r$, $\sigma = 1$. From an arbitrary $(x_0, y_0) \in \mathcal{H} \times \mathcal{H}$, the literal iteration is

$$\begin{aligned} x_{n+1} &= \operatorname{prox}_{rf}(x_n - r y_n), \\ y_{n+1} &= \frac{y_n + (9x_{n+1} - x_n)/8 - c}{1 + \mu}, \qquad n \geq 0. \end{aligned} \tag{101}$$

Here $\mathrm{prox}_{\eta f}(a)$ minimizes $\eta f(x) + \|x-a\|^2/2$. There is no relaxation, correction, or changing stepsize. The KKT set is

$$\mathcal{Z} = \{(x,y) : -y \in \partial f(x),\ x = \mu y + c\}. \tag{102}$$

The product norm is $\|(u,w)\|^2 = \|u\|^2 + \|w\|^2$.

Theorem 10 (Positive scalar curvature)**.** *Suppose $\mu > 0$ and $0 < r \le 3$. Then $\mathcal{Z}$ consists of the single pair*

$$x^* = \mathrm{prox}_{\mu f}(c), \qquad y^* = (x^* - c)/\mu. \tag{103}$$

For every initialization, (101) *converges strongly to this pair, and*

$$\sum_{n=0}^{\infty} \big(\|x_n - x^*\|^2 + \|y_n - y^*\|^2\big) < \infty. \tag{104}$$

More precisely, there are explicit constants $C_{r,\mu} > 1$ and $\rho_{r,\mu} = 1 - C_{r,\mu}^{-1} \in (0,1)$, given in Sections C.8 and C.9, such that

$$\|(x_n - x^*, y_n - y^*)\| \le \sqrt{C_{r,\mu}}\, \rho_{r,\mu}^{n/2} \|(x_0 - x^*, y_0 - y^*)\|, \qquad n \ge 0. \tag{105}$$

These constants depend only on r and μ, uniformly over f and the Hilbert space. Both increment sequences are square summable.

The bound is for each fixed positive curvature. No rate uniform as $\mu \downarrow 0$, no rate at $\mu = 0$, and this elementary theorem by itself does not assert an optimal stepsize threshold for fixed $\mu > 0$. The strong-curvature class-wide sharp range is given separately in Corollary 4.1. This theorem itself has isotropic quadratic scope. The later matrix and nonlinear extensions have separate hypotheses; geometric general-coupling bounds require the growth hypotheses in Corollary C.3 or Theorem 18, respectively.

6.3 The variable radius under global single anchor growth

Corollary 6.1 (Global single-anchor quadratic growth)**.** *In addition to the hypotheses of Theorem 4, suppose that for one KKT anchor and some $\mu_{\mathrm{QG}} > 0$,*

$$\Phi(x) - \Phi(x^*) \ge \frac{\mu_{\mathrm{QG}}}{2} \|x - x^*\|^2, \qquad x \in \mathrm{dom}\, f. \tag{106}$$

This is a global inequality to one anchor, rather than a bound by distance to a possibly non-singleton solution set or a local growth condition. It implies that x^* is the unique primal minimizer. With b from (235) *and the startup constants, set*

$$\begin{gathered} C_* = 1 + K_0^2 + K_0^4 + K_0^6 + C_0 \left\{ \frac{2}{\tau \mu_{\mathrm{QG}}} \left(\frac{1}{\eta} + \frac{\kappa_{\mathrm{cert}} b}{2\delta} \right) + \frac{1}{\delta} \right\} > 1, \\ \rho_{\mathrm{QG}} = 1 - C_*^{-1} \in (0,1). \end{gathered} \tag{107}$$

Then for every restart $k \geq 0$,

$$\sum_{j\geq 0} e_{k+j}^2 \leq C_* e_k^2, \qquad e_{k+j}^2 \leq C_* \rho_{\mathrm{QG}}^j e_k^2 \quad (j \geq 0). \tag{108}$$

Thus the primal and dual iterates converge geometrically in norm, with norm factor $\sqrt{\rho_{\mathrm{QG}}}$ and physical bounds

$$\|x_n - x^*\| \leq \sqrt{C_*}\, \rho_{\mathrm{QG}}^{n/2} e_0, \qquad \|y_n - y^*\| \leq \sqrt{\frac{\kappa_{\mathrm{cert}} \sigma C_*}{\tau}}\, \rho_{\mathrm{QG}}^{n/2} e_0. \tag{109}$$

The composite objective gaps obey

$$\begin{aligned} 0 \leq \Phi(x_{k+j}) - \Phi(x^*) &\leq \frac{a_{\mathrm{cert}}(2 + \kappa_{\mathrm{cert}} b/2) C_*}{\tau}\, \rho_{\mathrm{QG}}^{j-1} e_k^2, \qquad k \geq 0,\ j \geq 1, \\ \Phi(x_n) - \Phi(x^*) &\leq \frac{a_{\mathrm{cert}}(2 + \kappa_{\mathrm{cert}} b/2) C_*}{\tau}\, \rho_{\mathrm{QG}}^{n-1} \\ &\quad \times \left(\|x_0 - x^*\|^2 + \frac{\tau}{\kappa_{\mathrm{cert}} \sigma} \|y_0 - y^*\|^2 \right), \quad n \geq 1. \end{aligned} \tag{110}$$

The constants are explicit finite expressions given the selected certificate data. The arbitrary-band analytic existence argument does not itself supply numerical η, δ, M_P, and hence does not supply a certificate-free numerical rate in the problem parameters alone.

7 Affine boundary and actual periodic classifications

7.1 The affine boundary and its endpoint

The affine case $\mu = 0$, or $q = 1$, lies outside the positive-curvature phase. We use the literal map and KKT set in (101)–(102).

Theorem 11 (Identity-coupled Hilbert convergence)**.** *Suppose $\mu = 0$. Let $\mathcal{H}$ be a real Hilbert space, let $f : \mathcal{H} \to (-\infty, +\infty]$ be proper lower-semicontinuous convex, and let $h(y) = \langle c, y \rangle + d$ with $c \in \mathcal{H}$. Suppose*

$$\mathcal{Z} = \{(c, y) : -y \in \partial f(c)\} \neq \varnothing.$$

For $0 < r < 3$, run the actual map (101) on $\mathcal{H} \times \mathcal{H}$, with identity coupling, from any initialization. Then for some $(c, y_\infty) \in \mathcal{Z}$,

$$x_n \longrightarrow c \text{ strongly}, \qquad y_n \rightharpoonup y_\infty \text{ weakly}.$$

The paired orbit is bounded and

$$\sum_{n=0}^{\infty} \|x_n - c\|^2 < +\infty, \tag{111}$$

$$\sum_{n=0}^{\infty}\big(\|x_{n+1}-x_n\|^2+\|y_{n+1}-y_n\|^2\big)<+\infty. \tag{112}$$

In finite dimensions, paired convergence is in norm. For $3/4<r<3$, every KKT anchor p and restart $m\geq 0$ satisfy

$$\sup_{n\geq m}\|(x_n,y_n)-p\|_{\mathcal{H}\times\mathcal{H}}\leq B_r^{\mathcal{H}}\|(x_m,y_m)-p\|_{\mathcal{H}\times\mathcal{H}}, \tag{113}$$

with a finite constant depending only on r, independent of the Hilbert space and objective. No strong dual convergence in a general infinite-dimensional space and no Hilbert endpoint conclusion at $r=3$ are asserted.

Theorem 12 (Endpoint dynamics)**.** *Assume $\mu=0$, $\mathcal{H}=\mathbb{R}$, and $\mathcal{Z}\neq\varnothing$. Set $r=3$ and let T be the actual map* (101)*. Every orbit is bounded, and every KKT pair p and every restart $m\geq 0$ satisfy*

$$\sup_{n\geq m}\|(x_n,y_n)-p\|_2\leq 300\|(x_m,y_m)-p\|_2. \tag{114}$$

Moreover,

$$\sum_{n=2}^{\infty}|x_n+x_{n-1}+x_{n-2}-3c|^2<+\infty, \tag{115}$$

$$\sum_{n=0}^{\infty}\|(x_{n+3},y_{n+3})-(x_n,y_n)\|_2^2<+\infty. \tag{116}$$

Every cluster point belongs to $\mathrm{Fix}(T^3)$ and hence lies on an actual orbit of least period one or three for the same fixed objectives.

Theorem 12 asserts neither convergence to a single cycle nor convergence of the three phase subsequences. Its cluster points need not be KKT. For the general class of two arbitrary convex scalar objectives, the interval $(r_*,3)$ remains an open convergence question. A cycle exclusion alone would not resolve that question. No Hilbert endpoint conclusion or publication-priority claim is made here.

7.2 Exact scalar quadratic period-three boundary

Consider the actual primal-first map

$$x_{n+1}=\mathrm{prox}_{rf}(x_n-ry_n),\qquad (1+\mu)y_{n+1}=y_n+\frac{9}{8}x_{n+1}-\frac{1}{8}x_n-c, \tag{117}$$

where $r>0$, $\mu>0$, $c\in\mathbb{R}$, and f is any fixed proper lower-semicontinuous convex function on $\mathbb{R}$. Thus the dual objective is $h(y)=\mu y^2/2+cy$.

Theorem 13 (Scalar quadratic period three)**.** *For each fixed $\mu > 0$ and $c \in \mathbb{R}$, the set of $r > 0$ for which some such f admits a genuine period-three actual orbit is*

$$\begin{cases} [R(\mu), \infty), & 0 < \mu < 8, \\ \varnothing, & \mu \geq 8, \end{cases} \qquad R(\mu) = \frac{8(\mu^2 + 3\mu + 3)}{8 - \mu}.$$

The same classification holds if existence of a KKT pair is additionally required. At $r = R(\mu)$ every nonconstant three-periodic primal triple has exactly two equal values. This theorem makes no claim that exclusion of period three implies global convergence.

7.3 Actual four- and five-period obstructions

The following results concern existence of actual orbits of one fixed convex objective. They use (38) with $c = 0$, $L = I$, $\sigma = 1$, $\theta = 1/8$, and $\mu = q^{-1} - 1$. They are distinct from the failure of a storage ansatz.

Theorem 14 (Four-period classification in the stated band)**.** *For every $q_\star < q < 1$, the set of $r > 0$ admitting a least-period-four orbit for some proper lower-semicontinuous convex f on some real Hilbert space is exactly $(r_{\mathrm{lin}}(q), \infty)$. At $r = r_{\mathrm{lin}}(q)$ every four-periodic paired orbit has period dividing two. For $0 < r < r_{\mathrm{lin}}(q)$ every four-periodic paired orbit is constant. These exclusions hold on every real Hilbert space and require no KKT hypothesis. The least-period-four examples use a convex-hull indicator in $\mathbb{R}^3$ and have KKT pair $(0, 0)$.*

Theorem 15 (Two exact five-period onsets)**.** *At $q = 9/10$ and $q = 19/20$, respectively, the set admitting a least-period-five orbit over the same Hilbert-space class is exactly $[R_5(q), \infty)$, where*

$$R_5(9/10) = \frac{70162 + 2\sqrt{183969001}}{28755}, \qquad R_5(19/20) = \frac{1237104 + 8\sqrt{3690736609}}{530765}. \quad (118)$$

Necessity covers arbitrary real Hilbert spaces and every one fixed convex f, without a KKT assumption. Attainment uses a fixed convex-hull indicator in $\mathbb{R}^4$ with KKT pair $(0, 0)$; the same indicator and paired state list work at every larger product.

Both five-period onsets exceed the already known actual-failure envelope: $R_5(9/10) > 152/45 = r_{\mathrm{lin}}(9/10)$ and $R_5(19/20) > 9128/2869 = r_3(19/20)$. Likewise the four-period onset cannot lower that envelope because $U(q) \leq r_{\mathrm{lin}}(q)$. These onsets do not lower the completed scalar sharp cap. The classifications retain their distinct dynamical content and do not by themselves classify all longer-period or aperiodic failures. No scalar period-five classification or all-q period-five formula is asserted. Appendix E.3 proves the Hilbert-space necessity, finite-dimensional converse, endpoint signs, and original normal-cone inclusions, and includes an entirely rational least-period-five witness.

A Shared tools and the complete scalar sharp band

We use the following strict storage lemma, including for nonminimal realizations. Its proof is shared by all the storage constructions below.

Lemma 4 (Strict bounded-real storage)**.** *Let A_0 be a finite real Schur matrix, and let B_0, C_0 be real matrices of compatible sizes. If*

$$\sup_{|\lambda|=1} \|C_0(\lambda I - A_0)^{-1}B_0\| < 1,$$

then some $Z = Z^{\mathsf{T}} \succ 0$ and $\varepsilon > 0$ satisfy

$$\begin{aligned}(A_0x + B_0e)^{\mathsf{T}}Z(A_0x + B_0e) - x^{\mathsf{T}}Zx + \|C_0x\|^2 - \|e\|^2 \\ \leq -\varepsilon(\|x\|^2 + \|e\|^2) \qquad \textit{for all } x, e.\end{aligned} \tag{119}$$

No controllability or observability assumption is needed.

Proof. All sequence spaces in this proof are one-sided real ℓ_2 spaces. Schur stability makes the maps $\mathcal{O}, \mathcal{X}$ from an initial state to its unforced output and state sequences bounded. It also makes the zero-state convolution maps $\mathcal{T}, \mathcal{U}$ from inputs to output and state sequences bounded. The state sequences include the state at time zero. Parseval's identity and the transfer bound imply $\|\mathcal{T}\| < 1$. Choose $\varepsilon > 0$ so small that

$$\|\mathcal{T}\|^2 + \varepsilon\|\mathcal{U}\|^2 < 1 - \varepsilon.$$

For an initial state x define

$$W(x) = \sup_{\mathbf{e}\in\ell_2} \left\{ \|\mathcal{O}x + \mathcal{T}\mathbf{e}\|^2 + \varepsilon\|\mathcal{X}x + \mathcal{U}\mathbf{e}\|^2 - (1-\varepsilon)\|\mathbf{e}\|^2 \right\}.$$

The quadratic coefficient in $\mathbf{e}$ is bounded above by a negative multiple of the identity. Completing the square in ℓ_2 therefore gives a finite quadratic form $W(x) = x^{\mathsf{T}}Zx$ with Z real and symmetric. Taking $\mathbf{e} = 0$ gives $W(x) \geq \varepsilon\|x\|^2$, so $Z \succ 0$. For any first input e, splitting the remaining input into its ℓ_2 tail gives

$$W(x) \geq \|C_0x\|^2 + \varepsilon\|x\|^2 - (1-\varepsilon)\|e\|^2 + W(A_0x + B_0e).$$

Rearrangement proves (119). This argument uses only ℓ_2 inputs and requires no stability property of an optimizing feedback. ∎

A.1 Mixed supply and an unrestricted-state identity

Proof of the convergence assertion. We first prove the new part

$$7/10 \leq q \leq q^{\dagger}, \qquad 3 \leq r < r_{\mathrm{lin}}(q). \tag{120}$$

The proximal formula in the theorem exists uniquely and gives $-y_* \in \partial f(x_*)$ and $x_* - c = \mu y_*$. Conversely these KKT conditions imply that formula. Subtracting its fixed-point equations yields

$$R_n = qR_{n-1} + \frac{9p_n - p_{n-1}}{8}, \qquad v_n \in \partial F(p_n). \tag{121}$$

Here $F(0) = 0$, $F \geq 0$, and $0 \in \partial F(0)$. Every $n \geq 1$ is an attained proximal output, so

$$F_n = F(p_n) < \infty, \qquad B_n = F^*(v_n) = \langle p_n, v_n \rangle - F_n \geq 0, \qquad p_n \in \partial F^*(v_n). \tag{122}$$

In particular, no finite value of $f(x_0)$ is required. Eliminating the preceding dual state also gives the actual after-step port

$$v_n = (-1 + 9r/8)p_n - rR_n + (1 - r/8)p_{n-1}. \tag{123}$$

Put

$$a = \frac{5q}{9q-1}, \quad b = \frac{4q-1}{9q-1}, \quad g = 6 - 3q - 4q^2, \quad \rho = \frac{r-3}{r_{\mathrm{lin}}(q) - 3}. \tag{124}$$

On (120), $a, b > 0$, $a + b = 1$, $g \geq 0$, and $0 \leq \rho < 1$. For $n \geq 3$ define

$$\mathcal{Q}_n = \langle p_n, v_n \rangle - a\langle p_n, v_{n-1} \rangle - b\langle p_{n-2}, v_n \rangle, \qquad \mathcal{A}_n = aB_n + b(F_n + F_{n-1}) \quad (n \geq 2). \tag{125}$$

Convexity of F^* and F, respectively, makes the following two gaps nonnegative:

$$\mathcal{C}_n = B_{n-1} - B_n + \langle p_n, v_n - v_{n-1} \rangle, \qquad \mathcal{G}_n = F_{n-2} - F_n + \langle p_n - p_{n-2}, v_n \rangle.$$

Their exact cancellation is

$$\mathcal{Q}_n - (\mathcal{A}_n - \mathcal{A}_{n-1}) = a\mathcal{C}_n + b\mathcal{G}_n \geq 0. \tag{126}$$

This realizes the mixed multiplier by actual past quantities. It is not an application of a causal-FIR positive-storage theorem to an anticausal tap.

For the before-step state $s = (p, R, p_2) = (p_{n-1}, R_{n-1}, p_{n-2})$ and the new output $h = p_n$, write

$$s^+ = As + dh, \qquad A = \begin{pmatrix} 0 & 0 & 0 \\ -1/8 & q & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad d = \begin{pmatrix} 1 \\ 9/8 \\ 0 \end{pmatrix}, \quad v = p - h - rqR, \tag{127}$$

and $v^- = (1 - r/8)p_2 + (-1 + 9r/8)p - rR$. Thus $\mathcal{Q}_r = hv - ahv^- - bp_2v$ on scalar coordinates. The following rows are defined on every state and input:

$$\begin{aligned} w_1 &= h + \frac{1-q}{9q-1}p - \frac{8q(1-q)}{9q-1}R, \\ w_2 &= h - \frac{q+7}{9q-1}p + \frac{8(1-q)^2}{9q-1}R, \\ w_3 &= h - \frac{q^2+24q-9}{q(9q-1)}p + \frac{p_2}{q} + \frac{8(q-1)^3}{q(9q-1)}R, \\ w_e &= h + \frac{9-q}{9q-1}p + \frac{8(q^2-1)}{9q-1}R, \\ W_1 &= 4w_1 - 4w_2 + w_3, \qquad W_2 = 2w_2 - w_3, \qquad W_3 = w_3. \end{aligned} \tag{128}$$

Define

$$\begin{aligned} c_1 &= \frac{-237q^3+179q^2+50q-16}{128(9q-1)}, \quad & c_2 &= \frac{-136q^3+120q^2+13q-8}{64(9q-1)}, \\ c_3 &= \frac{5q(q+1)(8-7q)}{128(9q-1)}, & c_e &= g/10, \\ D_3 &= c_1W_1^2+c_2W_2^2+c_3W_3^2, & D_\ell &= c_e w_e^2. \end{aligned} \tag{129}$$

All c_j for $1 \le j \le 3$ are strictly positive on the stated band; $c_e \ge 0$ may vanish. A compact rational recipe specifies every storage entry. For $(j, D_j) = (3, D_3)$ or $(r_{\mathrm{lin}}, D_\ell)$ set

$$S_j = \tfrac{1}{2}\nabla^2_{(p,R,p_2)}(\mathcal{Q}_j + D_j), \qquad P_j = S_j + A^\top S_j A + \frac{(A^2)^\top S_j A^2}{1-q^2}, \qquad P = (1-\rho)P_3 + \rho P_\ell. \tag{130}$$

The Hessian is constant in h. Since $A^3 = qA^2$, this is the exact finite Lyapunov sum for its state-state coefficients. The remaining input-state and input-square coefficients cancel as well; their explicit polynomial verification is described in Appendix A.2. Thus, for unrestricted s, h,

$$\langle s^+, Ps^+\rangle - \langle s, Ps\rangle + \mathcal{Q}_r = -(1-\rho)D_3 - \rho D_\ell. \tag{131}$$

Here and below scalar row squares mean squared Hilbert norms after lifting. Bilinearity lifts the identity without assuming a finite-dimensional or coordinatewise convex objective. The matrix P itself need not be positive.

Graph coercivity before telescoping

$$\begin{gathered} \text{Let } s_n = (p_n, R_n, p_{n-1}) \text{ and } E_n = \langle s_n, Ps_n\rangle + \mathcal{A}_n. \text{ Set} \\ c_r = (-1+9r/8, -r, 1-r/8)^\top, \quad H = \tfrac{1}{2}(e_1c_r^\top + c_re_1^\top), \quad J = \tfrac{1}{2}(e_3c_r^\top + c_re_3^\top), \\ \lambda = 2P_{12}/r, \qquad \nu = 2P_{23}/r, \qquad T = P + \lambda H + \nu J. \end{gathered} \tag{132}$$

Fenchel–Young at p_{n-1}, v_n gives the exact decomposition

$$\begin{aligned} \mathcal{A}_n - \lambda(F_n + B_n) - \nu\langle p_{n-1}, v_n\rangle = (a-\lambda-\nu)B_n + (b-\lambda)F_n + (b-\nu)F_{n-1} \\ + \nu\big(F_{n-1} + B_n - \langle p_{n-1}, v_n\rangle\big). \end{aligned} \tag{133}$$

The four required budgets are exactly $\nu \ge 0$, $b-\lambda \ge 0$, $b-\nu \ge 0$, and $a-\lambda-\nu \ge 0$. There is no condition $\lambda \ge 0$: indeed at $(q, r) = (7/10, 3)$, $\lambda = -12609993/145899460$. Appendix A.2 proves these budgets on the whole band and proves $T_{12} = T_{23} = 0$ together with

$$rT = (1-\rho)^2K_0 + 2\rho(1-\rho)K_1 + \rho^2K_2, \qquad K_0, K_1 \succ 0, \quad K_2 \succeq 0. \tag{134}$$

At $q = q^\dagger$, K_2 has rank one, but the strictly positive $(1-\rho)^2K_0$ term remains. Hence $T \succ 0$ for every $r < r_{\mathrm{lin}}$, and (133) yields, for some $\kappa(q, r) > 0$,

$$E_n \ge \langle s_n, Ts_n\rangle \ge \kappa(q,r)(\|p_n\|^2 + \|R_n\|^2 + \|p_{n-1}\|^2) \ge 0. \tag{135}$$

A real orthogonal diagonalization on the three-coordinate index space justifies the same bound for Hilbert-valued states.

Actual startup and removal of the DC zero

The first energy is E_2: its only function values are B_2, F_2, F_1, and all its state coordinates are actual finite outputs. The first descent step is $n = 3$. Combining (126), (131), and (135) gives

$$E_n - E_{n-1} \le -(1-\rho)\sum_{j=1}^{3} c_j \|W_{j,n}\|^2 - \rho c_e \|w_{e,n}\|^2, \quad n \ge 3. \tag{136}$$

Telescoping against $E_N \ge 0$ proves $W_1, W_2, W_3 \in \ell_2$. Consequently $w_1 = (W_1 + 2W_2 + W_3)/4 \in \ell_2$. Neither a finite F_0 nor an artificial v_0 nor a zero-state initialization has entered.

It remains to control the constant mode, which cannot be inferred from a difference output alone. Substitution of the actual adjacent dual update into (128) gives, with $k = rq/(1-q) > 0$,

$$\begin{aligned} &w_{1,n} - q w_{1,n-1} = p_n - p_{n-1}, \\ &e_n := v_n + k p_n = (k-1) w_{1,n} + q(1 - r/8) w_{1,n-1}, \quad n \ge 3. \end{aligned} \tag{137}$$

The finite $w_{1,2}$ is harmless, so $e \in \ell_2$. Graph monotonicity yields $0 \le \langle p_n, v_n \rangle = -k\|p_n\|^2 + \langle p_n, e_n \rangle$, and hence $k\|p_n\| \le \|e_n\|$. Therefore $p \in \ell_2$, and $v = e - kp \in \ell_2$. The actual row identity

$$R_{n-1} = \frac{9q-1}{8q(1-q)} \left(p_n + \frac{1-q}{9q-1} p_{n-1} - w_{1,n} \right) \tag{138}$$

gives $R \in \ell_2$. Finally $0 \le F_n \le \langle p_n, v_n \rangle \le \|p_n\|\|v_n\|$, so Cauchy–Schwarz for series gives $\sum F_n < \infty$. The omitted prefixes contain only finitely many finite attained values. Square summability gives the asserted strong convergence. The port is the anchored residual $v_n = r(u_n + y_*)$, where $u_n = (x_{n-1} - x_n)/r - y_{n-1} \in \partial f(x_n)$; no summability of the raw unanchored u_n is asserted.

The lower-product and lower-curvature bridge

The new construction proves precisely (120). Proposition 5.1 supplies $\mathcal{R}(q) > 3$ for every $0 < q < 1$ and $\mathcal{R}(q) = r_{\mathrm{lin}}(q)$ for $0 < q \le q_\star = (\sqrt{161} - 7)/8$. Theorem 4 supplies the same summability conclusions in the identity normalization (40) whenever $0 < r < \mathcal{R}(q)$; its KKT hypothesis is automatic by the proximal formula above. Since $7/10 < q_\star < q^\dagger$, the old theorem covers the full required range for $q \le q_\star$, and covers $0 < r < 3$ for $q_\star < q \le q^\dagger$. The local construction covers the remaining $3 \le r < r_{\mathrm{lin}}$. This explicit overlap proves the full band, including its summability claims, without extending the local construction outside its proved scope. ∎

A.2 Exact mixed identity calculations

This appendix supplies the continuum signs and unrestricted-state algebra used in Section A.1. All definitions refer to (124)–(132).

A.2.1 Frequency factorization and the actual-state recipe

With z the unit-circle delay variable and $t = \Re z$, elimination of the literal dual recurrence gives

$$V_r(z) = z - 1 - \frac{rqz(9-z)}{8(1-qz)}, \quad \Delta_q(t) = 1 + q^2 - 2qt, \quad M_0(z) = 1 - az - bz^{-2}. \tag{139}$$

Direct reciprocal symmetrization gives, with $B_r(t) = -\Delta_q(t)\Re(M_0(z)V_r(z))$,

$$\begin{aligned} B_{r_{\mathrm{lin}}}(t) &= \frac{2g(q)}{5}(1-t^2), \\ B_3(t) &= 8(1-t)\{c_1(1+t)^2 + c_2(1-t)(1+t) + c_3(1-t)^2\}, \\ B_r &= (1-\rho)B_3 + \rho B_{r_{\mathrm{lin}}}. \end{aligned} \tag{140}$$

The rows w_j have transfer numerators $(1-z)^j$, $j = 1, 2, 3$, over $1 - qz$; w_e has numerator $1 - z^2$. The W_j numerators are $(1-z)(1+z)^2$, $(1-z)^2(1+z)$, and $(1-z)^3$. For example, the numerator of $h + \alpha p + \beta R + \gamma p_2$ is $(1 + \alpha z + \gamma z^2)(1 - qz) + \beta z(9-z)/8$. Matching these polynomials gives exactly (128). These transfer identities motivate the rows; the proof retains their free initial states through the following unrestricted polynomial calculation.

Write $\mathcal{Q}_j + D_j = s^\top S_j s + 2h\ell_j^\top s + \chi_j h^2$ using the displayed rows and coefficients. The three coefficient identities are

$$A^\top P_j A - P_j + S_j = 0, \qquad A^\top P_j d + \ell_j = 0, \qquad d^\top P_j d + \chi_j = 0. \tag{141}$$

The first follows by substituting (130) and $A^3 = qA^2$. Substituting the row numerators above and c_1, c_2, c_3, c_e into the last two identities gives zero component by component at $j = 3, r_{\mathrm{lin}}$. These are four explicit rational coefficient identities in q, with denominators products of q, $1 - q^2$, and $9q - 1$. Thus the recipe and (141) specify and verify all ten coefficients of the quadratic identity, including its free-input coefficients. Affinity in r then proves (131) for unrestricted states.

A conventional strictly positive interior frequency inequality is available as well, although it is not used in place of graph coercivity. Write $C_r(t) = -\Delta_q \Re V_r = At^2 + Bt + C$ and $L = 2|A| + |B|$. Then $C_r(1) = rq(1-q) > 0$ and $C_r(t) \geq rq(1-q) - L(1-t)$. If $m_c = \min(c_1, c_2, c_3) > 0$, (140) gives $B_3 \geq 24m_c(1-t)$. Hence

$$\eta = \frac{12(1-\rho)m_c}{1+L} > 0 \quad \Longrightarrow \quad B_r + \eta C_r \geq 12(1-\rho)m_c(1-t) + \eta rq(1-q) > 0.$$

After normalization by $1 + \eta$, the nonnegative delayed weights have total mass less than one. This observation does not supply an anticausal storage or justify dropping any boundary term.

A.2.2 The graph budgets and matrix coefficients

For each budget $\nu, b-\lambda, b-\nu, a-\lambda-\nu$, its product with r is affine in r. It therefore suffices to check its two endpoint values. Define $Z(\rho) = rT$, $r = 3 + \rho(r_{\mathrm{lin}} - 3)$, and

$$K_0 = Z(0), \qquad K_1 = Z(0) + \tfrac{1}{2}Z'(0), \qquad K_2 = Z(1). \tag{142}$$

The R cross terms of each coefficient vanish. Expansion of the rational recipes gives the degree-two identity (134). The endpoint coefficient has

$$\begin{aligned} (K_2)_{11} &= -\frac{8(q+1)(16q^3 - 132q^2 - 357q + 241)}{25q(9q-1)^2}, \\ (K_2)_{22} &= \frac{256(1-q)(q+1)^2 g(q)}{25q(9q-1)^2}, \\ (K_2)_{11}(K_2)_{33} - (K_2)_{13}^2 &= \frac{256(q+1)^2(4q-1)^2 g(q)(5q^2+4q-5)}{125q^2(9q-1)^4}. \end{aligned} \tag{143}$$

In particular, the only nonstrict scalar factors needed at this endpoint are the displayed $g(q)$ factors. The fourth endpoint budget is

$$(a-\lambda-\nu)|_{r=r_{\mathrm{lin}}} = \frac{8qg(q)}{(9q-1)^2}.$$

All remaining signs are strictly positive on the rational enclosing interval

$$I = [l, u] = [7/10, 90587/100000]. \tag{144}$$

Indeed $g(l) = 97/50$, $g(u) = -29569/2500000000$, and $g' = -3 - 8q < 0$; thus $[7/10, q^\dagger] \subset I$ and $g \geq 0$ on the target band.

Here is finite, exact rational arithmetic establishing every required strict sign. For a polynomial N of degree m, expand $N(l + (u-l)x) = \sum_{k=0}^{m} d_k x^k$ and form

$$\beta_i(N) = \sum_{k=0}^{i} d_k \frac{\binom{i}{k}}{\binom{m}{k}}, \qquad N(l + (u-l)x) = \sum_{i=0}^{m} \beta_i(N) \binom{m}{i} x^i (1-x)^{m-i}. \tag{145}$$

For each rational expression listed below, numerator and denominator are taken exactly as printed, before cancellation. Substitution in (145) gives

$$\min_i \beta_i(N) > 1/1000, \qquad \min_i \beta_i(D) > 1. \tag{146}$$

These deliberately coarse rational bounds permit a short certificate; they are not floating-point estimates. They can be checked solely by binomial arithmetic on the polynomials printed next. Positivity of the Bernstein basis and its partition of unity imply $N/D > 0$ on the entire closed I. The full rational coefficient arrays and two different reconstruction checkers are also supplied in the source bundle.

For $c_1 = N_1/D_1$,

$$N_1 = -237q^3 + 179q^2 + 50q - 16$$
$$D_1 = 1152q - 128.$$

For $c_2 = N_2/D_2$,

$$N_2 = -136q^3 + 120q^2 + 13q - 8$$
$$D_2 = 576q - 64.$$

For $c_3 = N_3/D_3$,

$$N_3 = -5q\,(q+1)\,(7q-8)$$
$$D_3 = 1152q - 128.$$

For $\nu|_{r=3} = N_4/D_4$,

$$N_4 = 101q^5 + 4q^4 + 150q^3 - 90q^2 + 8 - 29q$$
$$D_4 = 6q^2\,(9q-1)^2\,.$$

For $\nu|_{r=r_{\text{lin}}} = N_5/D_5$,

$$N_5 = q\,(4q-1)$$
$$D_5 = 9q - 1.$$

For $(b-\lambda)|_{r=3} = N_6/D_6$,

$$N_6 = -179q^6 - 231q^5 + 96q^4 - 338q^3 + 455q^2 + 269q - 72$$
$$D_6 = 6q^2\,(9q-1)^3\,.$$

For $(b-\lambda)|_{r=r_{\text{lin}}} = N_7/D_7$,

$$N_7 = (q-1)\left(4q^2 - 42q - 1\right)$$
$$D_7 = (9q-1)^2\,.$$

For $(b-\nu)|_{r=3} = N_8/D_8$,

$$N_8 = -101q^5 + 212q^4 - 228q^3 + 96q^2 + 29q - 8$$
$$D_8 = 6q^2\,(9q-1)^2\,.$$

For $(b-\nu)|_{r=r_{\text{lin}}} = N_9/D_9$,

$$N_9 = (1-q)\,(4q-1)$$
$$D_9 = 9q - 1.$$

For $(a-\lambda-\nu)|_{r=3} = N_{10}/D_{10}$,

$$N_{10} = -544q^6 + 160q^5 - 436q^4 + 260q^3 + 316q^2 + 84q - 32$$
$$D_{10} = 3q^2\,(9q-1)^3\,.$$

For $(a-\lambda-\nu)|_{r=r_{\text{lin}}}/g = N_{11}/D_{11}$,

$$\begin{aligned} N_{11} &= 8q \\ D_{11} &= (9q-1)^2 . \end{aligned}$$

For $(K_0)_{11} = N_{12}/D_{12}$,

$$\begin{aligned} N_{12} &= 6802q^6 - 1938q^5 + 38370q^4 - 2045q^3 \\ &\quad - 6790q^2 - 2743q + 792 \\ D_{12} &= 32q^2 (9q-1)^3 . \end{aligned}$$

For $(K_0)_{22} = N_{13}/D_{13}$,

$$\begin{aligned} N_{13} &= 816q^7 - 1056q^6 + 894q^5 - 1044q^4 \\ &\quad - 84q^3 + 348q^2 + 174q - 48 \\ D_{13} &= q^2 (9q-1)^3 . \end{aligned}$$

For $\det(K_0)_{\{1,3\}} = N_{14}/D_{14}$,

$$\begin{aligned} N_{14} &= -16760836q^{12} - 28365622q^{11} + 76040616q^{10} - 98886276q^9 \\ &\quad + 93906897q^8 + 6432029q^7 - 28508871q^6 - 5364543q^5 \\ &\quad + 8296414q^4 - 772528q^3 - 730880q^2 + 208896q \\ &\quad - 16384 \\ D_{14} &= 64q^4 (9q-1)^6 . \end{aligned}$$

For $(K_1)_{11} = N_{15}/D_{15}$,

$$\begin{aligned} N_{15} &= -1304q^7 + 28802q^6 + 25800q^5 - 6117q^4 \\ &\quad + 2877q^3 - 4381q^2 - 2421q + 648 \\ D_{15} &= 40q^3 (9q-1)^3 . \end{aligned}$$

For $(K_1)_{22} = N_{16}/D_{16}$,

$$\begin{aligned} N_{16} &= 2816q^8 + 784q^7 - 4680q^6 - 1016q^5 \\ &\quad + 1232q^4 + 64q^3 + 696q^2 + 168q \\ &\quad - 64 \\ D_{16} &= 5q^3 (9q-1)^3 . \end{aligned}$$

For $\det(K_1)_{\{1,3\}} = N_{17}/D_{17}$,

$$\begin{aligned} N_{17} &= -8867282q^{12} - 6543884q^{11} + 3798052q^{10} + 9639176q^9 \\ &\quad + 4423015q^8 + 1347605q^7 - 7946q^6 - 1004474q^5 \\ &\quad - 519879q^4 - 99255q^3 + 36232q^2 + 19568q \\ &\quad - 3456 \\ D_{17} &= 50q^4 (9q-1)^6 . \end{aligned}$$

For $(K_2)_{11} = N_{18}/D_{18}$,

$$N_{18} = (-8q - 8)\left(16q^3 - 132q^2 - 357q + 241\right)$$
$$D_{18} = 25q\,(9q - 1)^2\,.$$

For $(K_2)_{22}/g = N_{19}/D_{19}$,

$$N_{19} = (256 - 256q)\,(q + 1)^2$$
$$D_{19} = 25q\,(9q - 1)^2\,.$$

For $\det(K_2)_{\{1,3\}}/g = N_{20}/D_{20}$,

$$N_{20} = 20480q^6 + 47104q^5 + 5376q^4 - 37376q^3 - 6144q^2 + 8704q - 1280$$
$$D_{20} = 125q^2\,(9q - 1)^4\,.$$

The determinant notation in this list means the determinant of the principal $(1, 3)$ block, not of the full matrix. For K_0, K_1, positivity of entry 11, entry 22, and that block determinant proves positive definiteness. For K_2, the same list and (143) prove positive semidefiniteness on the target band. At $q = q^\dagger$ the separate entry 22 and block determinant vanish, while entry 11 stays positive, so its rank is one. The positive coefficient of K_0 in (134) gives $T \succeq (1 - \rho)^2 K_0/r \succ 0$ for $\rho < 1$. The endpoint checks for r times each budget give the four nonnegative budgets throughout $[3, r_{\mathrm{lin}}]$; the fourth is strictly positive for $r < r_{\mathrm{lin}}$, even at $q = q^\dagger$. This completes the sign proof used before the energy telescope.

A.3 Two actual storages reaching the period-three cap

We prove the part of Theorem 2 with $q > q^\dagger$ and $r \geq 3$. All scalar quadratic identities below are identities on unrestricted finite states. Replacing products by Hilbert inner products therefore lifts them to the original space, without separability or finite-dimensional assumptions on f.

Proposition A.1 (The period-three bridge)**.** For $q^\dagger < q < 1$ and $3 \leq r < R_3(q)$, every actual orbit of (29) converges strongly to (31) and satisfies (33).

The common actual convex graph. Use F, p_n, v_n from (32) and set $R_n = (y_n - y_*)/q$. Subtracting the anchor equations gives

$$R_n = qR_{n-1} + \frac{9p_n - p_{n-1}}{8}, \qquad v_n = p_{n-1} - p_n - rqR_{n-1} \in \partial F(p_n). \tag{147}$$

The KKT condition gives $F(0) = 0$, $F \geq 0$, and $0 \in \partial F(0)$. Every $n \geq 1$ is an attained proximal output. Hence

$$F_n = F(p_n) < \infty, \qquad B_n = F^*(v_n) = \langle p_n, v_n \rangle - F_n \geq 0, \qquad p_n \in \partial F^*(v_n). \tag{148}$$

There is no F_0 or virtual v_0. The predecessor elimination in (147) also yields

$$v_n = (9r/8 - 1)p_n - rR_n + (1 - r/8)p_{n-1}. \tag{149}$$

These are exactly the graph, Fenchel, and after-port identities (121)–(123); only the multiplier and energy will change.

Endpoint taps and their limited uniqueness claim. For a unit-circle delay z, write $t = \Re z$ and

$$\begin{aligned} V_r(z) &= z - 1 - \frac{rqz(9-z)}{8(1-qz)}, & \Delta_q(t) &= 1 + q^2 - 2qt, \\ D(q) &= (11 - 8q)(1 + q + q^2), & g_{\text{high}}(q) &= 4q^2 + 3q - 6, \\ a &= \frac{q(9q-1)}{D(q)}, & b &= 1 - a, \\ c_e &= \frac{q(q+1)g_{\text{high}}(q)}{D(q)}, & M(z) &= 1 - az - bz^{-2}. \end{aligned} \tag{150}$$

Both taps are positive. Indeed $a > 0$ and the numerator of b is $11 + 4q - 6q^2 - 8q^3$, whose derivative is negative on $[9/10, 1]$ and whose value at 1 is 1. Also $g_{\text{high}} > 0$ on $q > q^\dagger$. The low-q mixed proof instead uses $g_{\text{low}} = 6 - 3q - 4q^2 = -g_{\text{high}}$; its signs are not substituted here. Direct reciprocal symmetrization gives, for $\mathcal{B}_r(t) = -\Delta_q(t)\Re(M(z)V_r(z))$,

$$\mathcal{B}_{R_3}(t) = 2c_e(1 - t)(2t + 1)^2, \qquad |1 - z^3|^2 = 2(1 - t)(2t + 1)^2. \tag{151}$$

Within the mass-one two-tap family $M_A = 1 - Az - (1 - A)z^{-2}$, $\mathcal{B}_{R_3,A}(-1/2) = 0$ for every A, and

$$\mathcal{B}'_{R_3,A}(-1/2) = -\frac{3\{A(8q - 11)(1 + q + q^2) + q(9q - 1)\}}{9q - 1}. \tag{152}$$

Nonnegativity at this interior zero forces the derivative to vanish, uniquely giving $A = a$. Equation (151) proves sufficiency within that family. This is not optimality among all multipliers, and a frequency factor alone is not an actual convergence proof.

Two lower-anchor representations and their domains. Define

$$\begin{aligned} c_1 &= \frac{304q^5 + 283q^4 + 334q^3 - 739q^2 - 214q + 176}{128D(q)}, \\ c_2 &= \frac{152q^5 - 146q^4 - 197q^3 - 186q^2 + 241q + 88}{64D(q)}, \\ c_3 &= \frac{q(q+1)(8 - 7q)(9q - 1)}{128D(q)}. \end{aligned} \tag{153}$$

Then

$$\mathcal{B}_3(t) = 8(1 - t)\{c_1(1 + t)^2 + c_2(1 - t)(1 + t) + c_3(1 - t)^2\}. \tag{154}$$

Set

$$q_0 = \frac{907}{1000}, \qquad \ell = \frac{226467}{250000}, \qquad \ell < q^\dagger < q_0. \tag{155}$$

The independent-square storage is used on $q^\dagger < q \le q_0$; the cross-Gram storage is used on $q_0 \le q < 1$. Both include q_0. The rational enclosure $[\ell, q_0]$ is used only after removing explicit g_{high} factors: $g_{\text{high}}(\ell) < 0$, $g_{\text{high}}(q_0) > 0$, and $g'_{\text{high}} > 0$ on $q > 0$. Thus no positivity of g_{high} is claimed on the unused part $[\ell, q^\dagger]$. The need for two representations is algebraic. With

$$G_c = \begin{pmatrix} c_1 & -c_2/2 \\ -c_2/2 & c_3 \end{pmatrix}, \qquad \det G_c = \frac{(1-q)(19q+8)H_8(q)}{16384 D(q)^2}, \tag{156}$$

one has

$$\begin{aligned} H_8(q) &= 1216q^8 - 624q^7 - 1197q^6 - 1171q^5 + 3914q^4 \\ &\quad + 2118q^3 + 2011q^2 - 4147q - 968, \\ H_8(q^\dagger) &= \frac{455281295 - 44431765\sqrt{105}}{8192} < 0. \end{aligned} \tag{157}$$

The last sign follows by squaring the positive integers in its numerator. Therefore the same cross-Gram matrix cannot be continued down to $q^\dagger$. The independent-square form supplies that interval.

Finite rows and unrestricted storage identities. For the new output $h = p_n$, use the before-step state

$$s = (p, R, p_2) = (p_{n-1}, R_{n-1}, p_{n-2}).$$

Set

$$\begin{gathered} s^+ = As + dh, \qquad A = \begin{pmatrix} 0 & 0 & 0 \\ -1/8 & q & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad d = \begin{pmatrix} 1 \\ 9/8 \\ 0 \end{pmatrix}, \\ v = p - h - rqR, \qquad v^- = (1 - r/8)p_2 + (9r/8 - 1)p - rR, \\ \mathcal{Q}_r = hv - ahv^- - bp_2 v. \end{gathered} \tag{158}$$

On an actual history, $v^- = v_{n-1}$ by (149). The finite output rows are

$$\begin{aligned} w_1 &= h + \frac{1-q}{9q-1}p - \frac{8q(1-q)}{9q-1}R, \\ w_2 &= h - \frac{q+7}{9q-1}p + \frac{8(1-q)^2}{9q-1}R, \\ w_3 &= h - \frac{q^2+24q-9}{q(9q-1)}p + \frac{p_2}{q} + \frac{8(q-1)^3}{q(9q-1)}R, \\ W_1 &= 4w_1 - 4w_2 + w_3, \qquad W_2 = 2w_2 - w_3, \qquad W_3 = w_3, \\ w_c &= 3w_1 - 3w_2 + w_3. \end{aligned} \tag{159}$$

The numerator over $1 - qz$ of $\alpha_0 h + \alpha_1 p + \alpha_2 R + \alpha_3 p_2$ is

$$(\alpha_0 + \alpha_1 z + \alpha_3 z^2)(1 - qz) + \alpha_2 z(9 - z)/8.$$

Matching coefficients gives numerators $(1-z)^j$ for w_j, $(1-z)(1+z)^2$ for W_1, $(1-z)^2(1+z)$ for W_2, $(1-z)^3$ for W_3, and $1-z^3$ for w_c. For the two respective branches put

$$\begin{aligned} D_3^{\mathrm{sq}} &= c_1 W_1^2 + c_2 W_2^2 + c_3 W_3^2, \\ D_3^{\mathrm{cr}} &= c_1 W_1^2 - c_2 W_1 W_3 + c_3 W_3^2, \qquad D_U = c_e w_c^2. \end{aligned} \tag{160}$$

Both realize (154), because

$$\Re\big[(1-z)(1+z)^2\overline{(1-z)^3}\big] = -8(1-t)^2(1+t).$$

They are different finite-time dissipations, so their lower-anchor storage matrices must be constructed separately. For either selected D_3 and for $(j, D_j) = (3, D_3)$ or (R_3, D_U), define

$$S_j = \tfrac{1}{2}\nabla^2_{(p,R,p_2)}(\mathcal{Q}_j + D_j), \qquad P_j = S_j + A^\top S_j A + \frac{(A^2)^\top S_j A^2}{1-q^2}. \tag{161}$$

The Hessian is independent of h. Write $\mathcal{Q}_j + D_j = s^\top S_j s + 2h\ell_j^\top s + \chi_j h^2$. Direct substitution of the displayed rows and coefficients gives

$$A^\top P_j A - P_j + S_j = 0, \qquad A^\top P_j d + \ell_j = 0, \qquad d^\top P_j d + \chi_j = 0. \tag{162}$$

The first identity also follows from $A^3 = qA^2$: the last term in (161) is the sum of all Lyapunov terms with powers at least two. The last two identities verify the three free input-state coefficients and the free input-square coefficient. Thus all ten scalar coefficients, including the initial-state terms, vanish in

$$(s^+)^\top P_j s^+ - s^\top P_j s + \mathcal{Q}_j = -D_j. \tag{163}$$

In particular, this is more than a zero-state frequency identity. Since

$$R_3 - 3 = \frac{(1-q)(19q+8)}{q(9q-1)} > 0, \qquad \theta_m = \frac{r-3}{R_3-3}, \qquad P = (1-\theta_m)P_3 + \theta_m P_U, \tag{164}$$

affinity of $\mathcal{Q}_r$ in r yields

$$\langle s^+, Ps^+\rangle - \langle s, Ps\rangle + \mathcal{Q}_r = -(1-\theta_m)D_3 - \theta_m D_U. \tag{165}$$

The Hilbert cross term is $-c_2\langle W_1, W_3\rangle$; the scalar squares are squared Hilbert norms.

Actual-past supply and four graph budgets. For $n \geq 3$, convexity gives the genuine nonnegative gaps

$$\begin{aligned} \mathcal{C}_n &= B_{n-1} - B_n + \langle p_n, v_n - v_{n-1}\rangle \geq 0, \\ \mathcal{G}_n &= F_{n-2} - F_n + \langle p_n - p_{n-2}, v_n\rangle \geq 0. \end{aligned} \tag{166}$$

Set $\mathcal{A}_n = aB_n + b(F_n + F_{n-1})$ for $n \geq 2$. Using $a + b = 1$ and Fenchel equality gives the exact cancellation

$$\mathcal{Q}_n - (\mathcal{A}_n - \mathcal{A}_{n-1}) = a\mathcal{C}_n + b\mathcal{G}_n \geq 0. \tag{167}$$

Thus the negative delay in M uses the actual past point p_{n-2}, with no anticausal boundary, discarded prefix, or invented graph value. For $s_n = (p_n, R_n, p_{n-1})$ define

$$E_n = \langle s_n, P s_n \rangle + \mathcal{A}_n. \tag{168}$$

There is no requirement that P itself be positive. Let

$$\begin{gathered} c_r = (9r/8 - 1, -r, 1 - r/8)^\top, \qquad H = \operatorname{sym}(e_1 c_r^\top), \qquad J = \operatorname{sym}(e_3 c_r^\top), \\ \lambda = 2P_{12}/r, \qquad \nu = 2P_{23}/r, \qquad T = P + \lambda H + \nu J, \end{gathered} \tag{169}$$

where $\operatorname{sym} K = (K + K^\top)/2$. By (149), $\langle s_n, H s_n \rangle = F_n + B_n$ and $\langle s_n, J s_n \rangle = \langle p_{n-1}, v_n \rangle$. The exact budget identity is

$$\begin{aligned} \mathcal{A}_n - \lambda(F_n + B_n) - \nu \langle p_{n-1}, v_n \rangle = (a - \lambda - \nu) B_n + (b - \lambda) F_n + (b - \nu) F_{n-1} \\ + \nu \big[F_{n-1} + B_n - \langle p_{n-1}, v_n \rangle \big]. \end{aligned} \tag{170}$$

The bracket is nonnegative by Fenchel–Young. The required conditions are precisely

$$\nu \geq 0, \quad b - \lambda \geq 0, \quad b - \nu \geq 0, \quad a - \lambda - \nu \geq 0, \qquad T \succ 0. \tag{171}$$

No sign restriction is imposed on λ alone.

Finite continuum positivity proof. Here is a reproducible rational proof of all the signs, for each of the two distinct P_3 in (161). Set $r = 3 + \theta_m(R_3 - 3)$, $Z(\theta_m) = rT(r)$ and

$$K_0 = Z(0), \qquad K_1 = Z(0) + \tfrac{1}{2} Z'(0), \qquad K_2 = Z(1). \tag{172}$$

Substitution first gives $T_{12} = T_{23} = 0$ and then the exact identity

$$Z(\theta_m) = (1 - \theta_m)^2 K_0 + 2\theta_m (1 - \theta_m) K_1 + \theta_m^2 K_2. \tag{173}$$

For each of the four budgets in (171), multiplication by r gives an affine function of r. It suffices to check both endpoints $3, R_3$. For the matrices it suffices to check, for $i = 0, 1, 2$,

$$(K_i)_{11} > 0, \quad (K_i)_{22} > 0, \quad (K_i)_{11}(K_i)_{33} - (K_i)_{13}^2 > 0. \tag{174}$$

Their 12 and 23 entries vanish, so this is the exact two-by-two block criterion. The complete list consists of 23 rational functions per branch: a, b, c_e, c_1, c_3; c_2 in the square branch or $\det G_c$ in the cross branch; the eight endpoint budgets; and the nine quantities in (174).

For each test, cancel to the reduced coprime integer numerator and denominator. Remove from the numerator every exact factor $g_{\text{high}}(q)$ and $1 - q$, recording the identity before removal. Orient the remaining denominator to be positive at the left endpoint.

For the squares use $[L, H] = [\ell, q_0]$; for the cross form use $[L, H] = [q_0, 1]$. If a residual polynomial satisfies $N(L + (H - L)x) = \sum_{k=0}^{m} d_k x^k$, its exact Bernstein coefficients are

$$\begin{aligned} \beta_i(N) &= \sum_{k=0}^{i} d_k \frac{\binom{i}{k}}{\binom{m}{k}}, \\ N(L + (H - L)x) &= \sum_{i=0}^{m} \beta_i(N) \binom{m}{i} x^i (1 - x)^{m-i}. \end{aligned} \tag{175}$$

Apply the same rule to the residual denominator. Substituting the finite recipes above gives the following exact bounds for every coefficient of all 46 residual polynomials on each branch:

$$\begin{array}{ll} \text{squares:} & \beta_i(N) > 1/200, \quad \beta_i(D_0) > 1/200, \\ \text{cross:} & \beta_i(N) \geq 1, \quad \beta_i(D_0) \geq 1. \end{array} \tag{176}$$

These are deliberately coarse rational inequalities, obtained by the binomial formula, not numerical eigenvalue bounds. For example, over all square-branch numerator and denominator coefficients the minimum is

$$\frac{10617343450610166117922 7}{15258789062500000000000000} > \frac{1}{200};$$

the cross-branch minimum is exactly 1. The exact coefficient arrays are supplied in `new_scalar/certificates/bridge.json`; their residual degrees, which also specify the finite size of the calculation, are recorded in Table 1. The Bernstein basis is nonnegative and sums to one on $[0, 1]$. Thus every residual numerator and denominator is positive on its whole closed enclosure. Restoring g_{high} and $1 - q$ only on the actual range $q^\dagger < q < 1$ proves the required original strict signs.

The scalar tests show $c_1, c_2, c_3 > 0$ on the square branch and $G_c \succ 0$ on the cross branch. In particular, in either case

$$D_3 \geq \gamma(q)(\|W_1\|^2 + \|W_3\|^2), \qquad \gamma(q) = \begin{cases} \min(c_1, c_3), & \text{square branch}, \\ \det(G_c)/\operatorname{tr}(G_c), & \text{cross branch}. \end{cases} \tag{177}$$

In the second case $\lambda_{\max}(G_c) \leq \operatorname{tr}(G_c)$, so the displayed positive rational number is a valid lower bound for $\lambda_{\min}(G_c)$. The endpoint-budget tests and their r-affinity prove all four budgets throughout $3 \leq r \leq R_3$. The nine minor tests and (173) give $T \succ 0$ on that interval. Consequently (170) gives, for fixed q, r,

$$E_n \geq \langle s_n, T s_n \rangle \geq \kappa(q, r)(\|p_n\|^2 + \|R_n\|^2 + \|p_{n-1}\|^2), \qquad \kappa(q, r) > 0. \tag{178}$$

Orthogonal diagonalization of the scalar three-by-three matrix proves the same Hilbert-valued bound. No uniform positive margin at $q = q^\dagger$ or $q = 1$ is asserted; the former is supplied by the separate low-q mixed proof.

Table 1: Degrees of the residual numerator and denominator in the exact positivity calculation. The two entries in each column are $(\deg N, \deg D_0)$.

Test	Squares	Cross
a	$(2,3)$	$(2,3)$
b	$(3,3)$	$(3,3)$
c_e	$(2,3)$	$(2,3)$
c_1	$(5,3)$	$(5,3)$
c_3	$(4,3)$	$(4,3)$
c_2 or $\det G_c$	$(5,3)$	$(9,6)$
$\nu\|_3$	$(7,6)$	$(6,5)$
$\nu\|_{R_3}$	$(2,3)$	$(2,3)$
$(b-\lambda)\|_3$	$(7,7)$	$(5,5)$
$(b-\lambda)\|_{R_3}$	$(3,4)$	$(3,4)$
$(b-\nu)\|_3$	$(6,6)$	$(5,5)$
$(b-\nu)\|_{R_3}$	$(2,3)$	$(2,3)$
$(a-\lambda-\nu)\|_3$	$(8,7)$	$(0,2)$
$(a-\lambda-\nu)\|_{R_3}$	$(0,2)$	$(0,2)$
$(K_0)_{11}$	$(8,7)$	$(6,5)$
$(K_0)_{22}$	$(8,7)$	$(0,2)$
K_0 block determinant	$(16,14)$	$(11,9)$
$(K_1)_{11}$	$(9,9)$	$(6,6)$
$(K_1)_{22}$	$(8,7)$	$(2,4)$
K_1 block determinant	$(18,16)$	$(12,10)$
$(K_2)_{11}$	$(5,6)$	$(5,6)$
$(K_2)_{22}$	$(4,6)$	$(4,6)$
K_2 block determinant	$(8,10)$	$(8,10)$

The single-interval weak-curvature certificate. On $19/20 \le q < 1$, the cross branch has exactly the same taps, finite rows, D_3, D_U, P_3, P_U, graph budgets and DC identities as the single-interval construction. Thus its actual proof is a specialization, not a separate energy being discarded. The alternative 23 rational sign certificates on $[19/20, 1]$ are retained in `new_scalar/certificates/weak_band.json`. In that representation, remove $1-q$ from $\det G_c$, from the endpoint tests for $b-\lambda$ and $b-\nu$, and from $(K_i)_{22}$; all remaining numerators and denominators have Bernstein coefficients at least $17043/20000 > 4/5$. This is a second certificate for the identical rational objects. At $q = 19/20$, the exact taps are $a = 14345/19397$, $b = 5052/19397$ and $R_3 = 9128/2869$. Neither the wider proof nor its storage construction depends on this narrower certificate.

Completion of Proposition A.1. The first energy E_2 contains the actual finite state (p_2, R_2, p_1) and only B_2, F_2, F_1. The first step using both convex gaps is $n = 3$. Combining (165), (167), and (178) gives

$$E_n - E_{n-1} \le -(1-\theta_m) D_{3,n} - \theta_m c_e \|w_{c,n}\|^2, \qquad n \ge 3. \tag{179}$$

Since $r < R_3$, $1-\theta_m > 0$. Telescoping against $E_N \ge 0$ and using (177) proves $W_1, W_3 \in \ell_2$ from index three. This conclusion is not asserted at $r = R_3$, where the lower-anchor coefficient vanishes.

To remove the DC zero, substitute the genuine predecessor relation

$$R_{n-2} = \big[R_{n-1} - (9p_{n-1} - p_{n-2})/8\big]/q$$

into the finite rows. Direct cancellation gives

$$W_{1,n} - W_{3,n} = 4w_{1,n-1}, \qquad w_{1,n} - q w_{1,n-1} = p_n - p_{n-1}, \quad n \ge 3. \tag{180}$$

Thus $w_1 \in \ell_2$ from index two; $w_{1,2}$ uses only p_2, p_1, R_1. Set $\kappa_{\mathrm{DC}} = rq/(1-q) > 0$. A second actual identity is

$$z_n := v_n + \kappa_{\mathrm{DC}} p_n = (\kappa_{\mathrm{DC}} - 1) w_{1,n} + q(1 - r/8) w_{1,n-1}, \qquad n \ge 3. \tag{181}$$

Hence $z \in \ell_2$. Graph monotonicity against $(0,0)$ gives $0 \le \langle p_n, v_n\rangle = -\kappa_{\mathrm{DC}}\|p_n\|^2 + \langle p_n, z_n\rangle$, so $\|p_n\| \le \|z_n\|/\kappa_{\mathrm{DC}}$, also when $p_n = 0$. Therefore $p \in \ell_2$ and then $v = z - \kappa_{\mathrm{DC}} p \in \ell_2$. The actual row rearranges to

$$R_{n-1} = \frac{9q-1}{8q(1-q)} \left[p_n + \frac{1-q}{9q-1} p_{n-1} - w_{1,n} \right], \tag{182}$$

which gives $R \in \ell_2$. All omitted prefixes are finite actual values. Thus both physical errors tend strongly to zero. Finally, $0 \le F_n \le \langle p_n, v_n\rangle \le \|p_n\|\|v_n\|$; Cauchy–Schwarz for series proves $\sum_{n\ge1} F_n < \infty$. Since $v_n = r(u_n + y_*)$, this is square summability of the anchored proximal residual, not of the generally nonzero raw subgradient u_n. ∎

The multipliers and primal/conjugate potentials above apply the O'Shea–Zames–Falb and Fenchel framework already used for the mixed band. The two-storage split and its exact graph signs extend its actual scalar range; no general methodological or priority claim is required.

B Exact radii and general coupling storage

B.1 Proof of the exact three history radius

Proof of Theorem 7.

A rank-two supply at zero multiplier. In this proof the arguments (q, r) of the matrices are suppressed. Introduce the corrected supply and storage

$$\begin{gathered}\widehat{S}_\eta = S_R(q,r,\eta) + J^{\mathsf{T}} G_R J - T_R^{\mathsf{T}} G_R T_R, \qquad H = P_R + G_R, \\ H \succ 0, \qquad J^{\mathsf{T}} H J - T_R^{\mathsf{T}} H T_R - \widehat{S}_\eta \succ 0.\end{gathered} \tag{183}$$

The second line is equivalent to (69). At $\eta = 0$ set

$$\begin{gathered}b = \begin{pmatrix} 17/2 - 63r/16 \\ 7r/16 - 2 \\ 1/2 \\ 7r(1-q)/2 \end{pmatrix}, \qquad k = \begin{pmatrix} 63r - 88 \\ 80 - 7r \\ 8 \\ 56r(q-1) \end{pmatrix}, \\ e = h - \frac{b^{\mathsf{T}} s}{7}, \qquad y = \frac{k^{\mathsf{T}} s}{112}.\end{gathered} \tag{184}$$

Expanding (66)–(67) gives

$$\begin{gathered}(\widehat{S}_0)_{hh} = -7, \qquad (\widehat{S}_0)_{sh} = b, \qquad (\widehat{S}_0)_{ss} + \frac{bb^{\mathsf{T}}}{7} = \frac{kk^{\mathsf{T}}}{1792}, \\ z^{\mathsf{T}} \widehat{S}_0 z = 7(y^2 - e^2).\end{gathered} \tag{185}$$

The third entry of k is nonzero, and e has a nonzero h coefficient. Thus the h-Schur complement has rank one and $\widehat{S}_0$ has inertia $(1, 1, 3)$, in the order positive, negative, zero. This remains true at every product r; the reduction below uses this zero-multiplier identity only.

The actual quotient realization. Write $T_R = [F_0 \;\; B_0]$, so that $B_0 = (1, 0, 0, 9/8)^{\mathsf{T}}$, and let

$$K_q = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & q-1 \end{pmatrix}, \qquad d_q = \begin{pmatrix} 1 \\ 1 \\ 1 \\ (1-q)^{-1} \end{pmatrix}. \tag{186}$$

The first three columns of K_q have determinant one, so K_q has rank three and kernel $\operatorname{span}\{d_q\}$. Also $b^{\mathsf{T}} d_q = 7$ and $k^{\mathsf{T}} d_q = 0$. With $X = K_q s$ and $h = e + b^{\mathsf{T}} s/7$, direct multiplication gives

$$\begin{gathered}X^+ = \overline{A} X + \overline{B} e, \qquad y = \overline{C} X, \\ \overline{A} = \begin{pmatrix} 3/14 - r/16 & -1/14 & -r/2 \\ 1 & 0 & 0 \\ (9q-1)(24-7r)/896 & (1-9q)/112 & q + r(1-9q)/16 \end{pmatrix}, \\ \overline{B} = \begin{pmatrix} 1 \\ 0 \\ (9q-1)/8 \end{pmatrix}, \qquad \overline{C} = \begin{pmatrix} r/16 - 11/14 & -1/14 & r/2 \end{pmatrix}.\end{gathered} \tag{187}$$

The identities being used are

$$K_q(F_0 + B_0 b^{\mathsf{T}}/7) = \overline{A} K_q, \qquad K_q B_0 = \overline{B}, \qquad \overline{C} K_q = k^{\mathsf{T}}/112.$$

The actual characteristic polynomial, before any pole-zero cancellation, is $D_{q,r}/112$, and the transfer is $N_{q,r}/D_{q,r}$, where

$$\begin{aligned} D_{q,r}(\lambda) &= 112\lambda^3 + (63qr - 112q - 24)\lambda^2 \\ &\quad + (24q + 8 - 7qr)\lambda - 8q, \\ N_{q,r}(\lambda) &= (63qr - 88)\lambda^2 + (88q - 7qr - 8)\lambda + 8q. \end{aligned} \tag{188}$$

Consequently the argument covers $q = 1/9$ and other nonminimal realizations, including any cancelled state modes. For $|\lambda| = 1$ and $t = \operatorname{Re}\lambda$, Laurent expansion yields the exact affine-in-r gap

$$\begin{aligned} |D_{q,r}(\lambda)|^2 - |N_{q,r}(\lambda)|^2 &= W_{q,r}(t), \\ W_{q,r}(t) &= 448\big[8(1-t)^2\Delta_q(t) - rqA_q(t)\big]. \end{aligned} \tag{189}$$

Schur stability and reduced storage below the radius. Suppose $0 < r < \mathcal{R}(q)$. On the positive-A_q domain, (70) gives $W_{q,r}(t) > 0$. Where $A_q(t) \le 0$, the two terms in the bracket in (189) are nonnegative; the first can vanish only at $t = 1$, where $A_q(1) < 0$. Hence the gap is strictly positive on the entire frequency interval. The same argument applies at every product between zero and r. At product zero the actual characteristic polynomial factors as

$$D_{q,0}(\lambda) = (\lambda - q)(112\lambda^2 - 24\lambda + 8). \tag{190}$$

Its roots are q and a conjugate pair of modulus $1/\sqrt{14}$, so the actual matrix is Schur there. At every intervening positive product, the strict gap precludes a unit-circle zero of $D_{q,r}$. Continuity of polynomial roots, with leading coefficient fixed at 112, therefore prevents a root from crossing the unit circle or escaping to infinity. Thus $\overline{A}$ is Schur throughout the interval. This does not require a strict gain gap at product zero, where stability was established directly. Compactness of the unit circle now gives

$$\sup_{|\lambda|=1} |N_{q,r}(\lambda)/D_{q,r}(\lambda)| < 1.$$

Lemma 4 supplies $Z \succ 0$ and $\varepsilon > 0$ for the quotient realization.

The two exact kernels and their repair. Define

$$\begin{aligned} H_0 = 7K_q^{\mathsf{T}} Z K_q, \qquad \mathcal{W}_{q,r} &= \begin{pmatrix} K_q & 0 \\ -b^{\mathsf{T}}/7 & 1 \end{pmatrix}, \\ D_0 &= J^{\mathsf{T}} H_0 J - T_R^{\mathsf{T}} H_0 T_R - \widehat{S}_0. \end{aligned} \tag{191}$$

The square identity, the quotient identities, and (119) imply

$$D_0 \succeq 7\varepsilon \mathcal{W}_{q,r}^{\mathsf{T}} \mathcal{W}_{q,r}. \tag{192}$$

Since $Z \succ 0$, H_0 has exact kernel $\operatorname{span}\{d_q\}$. Since $b^{\mathsf{T}} d_q = 7$, $\mathcal{W}_{q,r}$ has rank four and exact kernel $\operatorname{span}\{z_0\}$, where $z_0 = (d_q, 1)^{\mathsf{T}}$. Furthermore

$$T_R z_0 = J z_0 = d_q, \qquad \widehat{S}_0 z_0 = 0.$$

Indeed, both square forms in (185) vanish on z_0, so their symmetric rank-one matrix difference annihilates it. Since $H_0 d_q = 0$, we obtain $D_0 z_0 = 0$; combined with (192), this identifies the exact kernel of D_0. For $e_0 = z_0/\|z_0\|$ there is therefore $\gamma > 0$ with $w^{\mathsf{T}} D_0 w \geq \gamma \|w\|^2$ whenever $w \perp e_0$.

Now put

$$\begin{aligned} H_\eta = H_0 + \eta I_4, \qquad P_{R,\eta} = H_\eta - G_R, \\ D_1 = J^{\mathsf{T}} J - T_R^{\mathsf{T}} T_R - \operatorname{sym}(e_5 v_R^{\mathsf{T}}). \end{aligned} \tag{193}$$

As $\widehat{S}_\eta = \widehat{S}_0 + \eta \operatorname{sym}(e_5 v_R^{\mathsf{T}})$, the new dissipation matrix is $D_\eta = D_0 + \eta D_1$. The storage difference vanishes on z_0, and $z_0^{\mathsf{T}} \operatorname{sym}(e_5 v_R^{\mathsf{T}}) z_0 = -rq/(1-q)$. It follows that

$$m := e_0^{\mathsf{T}} D_1 e_0 = \frac{rq(1-q)}{4(1-q)^2 + 1} > 0. \tag{194}$$

Take $L_1 \geq \max\{1, \|D_1\|\}$ and

$$0 < \eta \leq \min\left\{ \frac{\gamma}{2L_1}, \frac{\gamma m}{8L_1^2} \right\}.$$

For $z = \alpha e_0 + w$ with $w \perp e_0$, Young's inequality gives

$$\begin{aligned} z^{\mathsf{T}} D_\eta z &\geq (\gamma - \eta L_1)\|w\|^2 + \eta m \alpha^2 - 2\eta L_1 |\alpha| \|w\| \\ &\geq \frac{\gamma}{4}\|w\|^2 + \frac{\eta m}{2}\alpha^2. \end{aligned}$$

Here the mixed term is bounded above by $(\eta m/2)\alpha^2 + (2\eta L_1^2/m)\|w\|^2$. Since $H_\eta \succeq \eta I_4$, both inequalities in (69) are strict. This proves sufficiency.

Harmonic necessity for every positive multiplier. Let $|\lambda| = 1$ and $\zeta = \lambda^{-1}$. The harmonic state of unit new-primal amplitude is

$$z_\lambda = \left(\zeta, \zeta^2, \zeta^3, \frac{\zeta(9-\zeta)}{8(1-q\zeta)}, 1 \right)^{\mathsf{T}}, \qquad T_R z_\lambda = \lambda J z_\lambda. \tag{195}$$

Thus its Hermitian storage difference vanishes. A strict real symmetric dissipation inequality is strictly positive on every nonzero complex vector, by applying real positivity

to its real and imaginary parts. Consequently (69) requires $z_\lambda^* S_R(q,r,\eta) z_\lambda < 0$. The port and supply multiplier are

$$\begin{aligned} V(\zeta) &= \zeta - 1 - \frac{rq\zeta(9-\zeta)}{8(1-q\zeta)}, \\ M_\eta(\zeta) &= 10 + \eta - 7\zeta - 2\zeta^{-2} - \zeta^{-3}. \end{aligned} \tag{196}$$

With $t = \operatorname{Re}\zeta$, direct expansion of the Hermitian supply gives

$$\begin{aligned} \Delta_q(t)\operatorname{Re}[M_\eta(\zeta)V(\zeta)] &= \frac{(t-1)W_{q,r}(t)}{896} + \eta E_{q,r}(t), \\ E_{q,r}(t) &= (t-1)\Delta_q(t) + \frac{rq}{8}\big[2t^2 - (q+9)t + 9q - 1\big]. \end{aligned} \tag{197}$$

This expression must be strictly negative, since $\Delta_q(t) > 0$. Let t_* attain the minimum in (70); necessarily $t_* < 1$. At a boundary product $r = \rho_q(t)$ with $A_q(t) > 0$,

$$E_{q,\rho_q(t)}(t) = \frac{2(1-t)(t+1)(t+2)(5-4q-t)\Delta_q(t)}{A_q(t)}. \tag{198}$$

For $-1 < t < 1$, every factor on the right is positive, including $5-4q-t > 0$. In contrast, $E_{q,0}(t) = -(1-t)\Delta_q(t) < 0$. Since $E_{q,r}(t)$ is affine in r, its slope is positive and its value remains positive for every $r \geq \rho_q(t)$. At the remaining possible minimizing frequency $t = -1$, the full expression is

$$\Delta_q(-1)\operatorname{Re}[M_\eta(-1)V(-1)] = \frac{(\eta+16)(q+1)(5qr-8q-8)}{4}. \tag{199}$$

It is nonnegative at and above $\rho_q(-1) = 8(1+q)/(5q)$.

If $r \geq \mathcal{R}(q)$, use the minimizing t_*. Then $W_{q,r}(t_*) \leq 0$, and $t_* - 1 < 0$, so the first term on the right of (197) is nonnegative. For an interior minimizer its η term is positive for every $\eta > 0$. For $t_* = -1$, (199) is nonnegative. Strict negativity is impossible in either case, for every P_R and every $\eta > 0$. This proves necessity. The harmonic state is a necessary test on the unrestricted-state certificate; no assertion that it is an attainable orbit of a convex f has been made.

One multiplier on every compact feasible set. The feasible region is open: by (189) and the attained minimum, it is exactly the region where $W_{q,r}$ is strictly positive on the compact frequency interval. For each $(q_0, r_0) \in \mathcal{C}$, select a strict reduced storage Z_0 from Lemma 4. Keeping Z_0 fixed, its positivity and strict reduced dissipation persist in a relative open neighborhood, because (187) depends continuously on (q,r). A finite cover and a continuous nonnegative partition of unity subordinate to it give a continuous $Z(q,r)$ by convex combinations of these fixed matrices. At each parameter pair the reduced dissipation matrix is affine in Z, so both it and Z remain positive. Compactness yields a common $\varepsilon_{\mathcal{C}} > 0$ for (119).

The resulting $H_0, D_0, \mathcal{W}_{q,r}$ are continuous with the exact kernels already established. The minimum of $w^{\mathsf{T}} D_0(q,r) w$ on the compact set

$$\{(q,r,w) : (q,r) \in \mathcal{C},\ \|w\| = 1,\ w \perp z_0(q)\}$$

is positive; call it $\gamma_{\mathcal{C}}$. Also define

$$\begin{aligned} m_{\mathcal{C}} &= \min_{(q,r)\in\mathcal{C}} \frac{rq(1-q)}{4(1-q)^2+1} > 0, \\ L_{\mathcal{C}} &= \max\{1, \max_{(q,r)\in\mathcal{C}} \|D_1(q,r)\|\}, \\ \eta_* &= \min\left\{\frac{\gamma_{\mathcal{C}}}{2L_{\mathcal{C}}}, \frac{\gamma_{\mathcal{C}} m_{\mathcal{C}}}{8L_{\mathcal{C}}^2}\right\} > 0. \end{aligned} \tag{200}$$

The preceding perturbation estimate now holds simultaneously on $\mathcal{C}$. For every fixed $0 < \eta \le \eta_*$, $P_{R,\eta} = 7K_q^{\mathsf{T}} Z(q,r) K_q + \eta I_4 - G_R(q,r)$ is continuous and bounded, with the common margin

$$\delta_\eta = \min\{\eta, \gamma_{\mathcal{C}}/4, \eta m_{\mathcal{C}}/2\} > 0. \tag{201}$$

This is one scalar multiplier, not a mode-dependent choice, and completes the proof. ∎

Remark (Positive multipliers change the supply rank). The rank-one Schur complement used above is specific to $\eta = 0$. For completeness, put

$$\begin{aligned} t_0 = (1,0,0,-rq)^{\mathsf{T}}, \qquad \ell = b - \frac{7}{2} t_0 &= \begin{pmatrix} 5 - 63r/16 \\ 7r/16 - 2 \\ 1/2 \\ 7r/2 \end{pmatrix}, \\ e_\eta = h - \frac{(b + (\eta/2)t_0)^{\mathsf{T}} s}{7+\eta}&, \\ z^{\mathsf{T}} \widehat{S}_\eta z = \frac{(k^{\mathsf{T}} s)^2}{1792} + \frac{\eta}{4}(t_0^{\mathsf{T}} s)^2& \\ - \frac{\eta(\ell^{\mathsf{T}} s)^2}{7(7+\eta)} - (7+\eta) e_\eta^2&. \end{aligned} \tag{202}$$

Thus the h-Schur complement M_η has rank at most three. For $\eta > 0$, $r > 0$, and $0 < q < 1$, its principal minors include

$$\begin{aligned} \det M_\eta[\{1,2,3\}] &= -\frac{\eta^2 (r-8)^2}{1024(\eta+7)}, \\ \det M_\eta[\{1,2,4\}] &= -\frac{9\eta^2 r^2 (5qr - 8q - 4)^2}{256(\eta+7)}, \\ \det M_\eta[\{3,4\}] &= \frac{\eta r^2 (q-1)}{4(\eta+7)} < 0, \qquad (M_\eta)_{33} = \frac{1}{4(\eta+7)} > 0. \end{aligned} \tag{203}$$

The two negative three-by-three minors can vanish simultaneously only at $(r,q) = (8, 1/8)$. Away from that point at least one is negative, so M_η has rank three; its signed-square

representation allows at most two positive and one negative eigenvalues, and therefore fixes its inertia as $(2,1,1)$. At $(r,q)=(8,1/8)$, direct substitution gives $k=16\ell+840t_0$, so the rank is at most two. The displayed negative two-by-two minor gives rank two and inertia $(1,1,2)$. Adding the negative h pivot, the inertia of $\widehat{S}_\eta$ is $(2,2,1)$ generically and $(1,2,2)$ at the exceptional point. In particular, no one-output bounded-real assertion at positive η is being assumed. The proof first uses the zero-multiplier quotient and then repairs its two kernels.

B.1.1 Proof of the radius bounds and continuity

Proof of Proposition 5.1.

The strict gap at product three. Expanding (189) at $r=3$ gives

$$\begin{aligned} W_{q,3}(t) &= 3584(1-q)^2(t-1)^2 \\ &\quad - 448q(1-q)(16t^3-27t^2-54t-79) \\ &\quad + 1792q^2(1-t)(2t+1)^2. \end{aligned} \tag{204}$$

For $-1\le t\le 1$, the cubic in parentheses is at most $16+54-79=-9$. Therefore

$$W_{q,3}(t)\ge 4032q(1-q)>0. \tag{205}$$

For fixed q, compactness of the frequency interval and continuous dependence on r give strict positivity for some $r>3$ as well. Equivalently, the attained positive-A_q minimum is greater than three. The upper bound in (72) follows by using $t=-1$ in (70).

The closed-form band. At $r=r_{\mathrm{lin}}(q)$ the gap factors exactly as

$$\begin{aligned} W_{q,r_{\mathrm{lin}}(q)}(t) &= -\frac{7168}{5}(t+1)B_q(t), \\ B_q(t) &= 5qt^2+(2q^2-7q+1)t+10q^2+2q-13, \\ B_q(1) &= 12(q^2-1)<0, \qquad B_q(-1)=8q^2+14q-14. \end{aligned} \tag{206}$$

The quadratic B_q is strictly convex. It is nonpositive throughout $[-1,1]$ exactly when its two endpoint values are nonpositive, which holds precisely for $q\le q_\star$. In this case $W_{q,r_{\mathrm{lin}}(q)}\ge 0$ and its only zero is $t=-1$. On the other hand,

$$W_{q,0}(t)=3584(1-t)^2\Delta_q(t)$$

is nonnegative with its only zero at $t=1$. Since $W_{q,r}$ is affine in r, every strict interpolation $0<r<r_{\mathrm{lin}}(q)$ is strictly positive at every frequency. Thus $\mathcal{R}(q)\ge r_{\mathrm{lin}}(q)$, and the opposite inequality was already established.

If $q > q_\star$, then $B_q(-1) > 0$, and the cap gap is negative for t just above -1. Since A_q is positive near -1, (189) implies $\rho_q(t) < r_{\mathrm{lin}}(q)$ at such a frequency. Therefore $\mathcal{R}(q) < r_{\mathrm{lin}}(q)$. Equivalently, the exact derivative at the endpoint is

$$\rho_q'(-1) = -\frac{8(4q^2+7q-7)}{25q(q+1)} < 0 \qquad (q > q_\star). \tag{207}$$

At $q = 1/2$, the cap factor becomes particularly simple:

$$W_{1/2,24/5}(t) = \frac{3584}{5}(t+1)(19+4t-5t^2). \tag{208}$$

The last factor is at least ten on $[-1,1]$, confirming the full open certificate interval $0 < r < 24/5$ analytically.

Continuity of the attained minimum. Fix $q_0 \in (0,1)$ and a minimizing frequency t_0 for $\mathcal{R}(q_0)$. Since $A_{q_0}(t_0) > 0$, it remains positive near q_0. Evaluation at this same frequency gives

$$\limsup_{q\to q_0} \mathcal{R}(q) \le \lim_{q\to q_0} \rho_q(t_0) = \mathcal{R}(q_0).$$

For the lower limit, take any sequence $q_j \to q_0$, pass to a subsequence realizing its lower limit, and choose minimizing t_j. After a further subsequence, $t_j \to t \in [-1,1]$. The values $\mathcal{R}(q_j)$ are bounded above by the continuous endpoint values $8(1+q_j)/(5q_j)$. Since $A_{q_j}(t_j) > 0$, its limit cannot be negative. If $A_{q_0}(t) = 0$, then $t \neq 1$ because $A_{q_0}(1) < 0$. The numerator $8(1-t)^2\Delta_{q_0}(t)$ is then strictly positive, while the positive denominators tend to zero. This would force $\rho_{q_j}(t_j) \to +\infty$, a contradiction. Hence $A_{q_0}(t) > 0$, and continuity of the ratio gives

$$\liminf_{j\to\infty} \mathcal{R}(q_j) = \rho_{q_0}(t) \ge \mathcal{R}(q_0).$$

Together with the upper limit this proves continuity. Its minimum on compact K is attained and is greater than three by (72). ∎

Remark (Scope of the boundary). At $q > q_\star$, infeasibility at and above $\mathcal{R}(q)$ excludes all P_R and all positive η for the exact supply and weights in (67)–(69). It does not exclude different multipliers or storages, nor does it construct a convex nonlinear orbit. Actual convergence follows from the separate convex-graph argument, and actual linear and period-three obstructions must be compared with their own parameter ranges. The compact-set assertion gives no uniform strict margin as q approaches zero or one, as r approaches zero, or as r reaches the ansatz boundary.

B.2 Operator lift and the attained-history energy

Proof of Theorem 4.

Continuous functional calculus. The spectrum of A lies in K. Select the common scalar $\eta > 0$, margin $\delta > 0$, and continuous symmetric family $P_R(q, \kappa_{\mathrm{cert}})$ from Theorem 7 on $K \times \{\kappa_{\mathrm{cert}}\}$. The scalar certificate uses dual coordinate $R = t/q$. To use the physical normalized dual coordinate t, set

$$\begin{aligned} D_4(q) = \mathrm{diag}(1,1,1,q^{-1}), \quad D_5(q) = \mathrm{diag}(1,1,1,q^{-1},1), \\ P(q) = D_4(q)^{\mathsf{T}} P_R(q,\kappa_{\mathrm{cert}}) D_4(q), \quad T_R D_5 = D_4 T, \quad J D_5 = D_4 J. \end{aligned} \tag{209}$$

In the new coordinates define

$$\begin{aligned} T(q) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -q/8 & 0 & 0 & q & 9q/8 \end{pmatrix}, \quad c(q) = \begin{pmatrix} 9\kappa_{\mathrm{cert}}/8 - 1 \\ 1 - \kappa_{\mathrm{cert}}/8 \\ 0 \\ -\kappa_{\mathrm{cert}}/q \end{pmatrix}, \\ v = (1,0,0,-\kappa_{\mathrm{cert}},-1)^{\mathsf{T}}, \qquad G(q) = \mathrm{sym}(c(q)a^{\mathsf{T}}), \\ S(q,\eta) = \mathrm{sym}(w_\eta v^{\mathsf{T}}) - 7\,\mathrm{sym}\big(e_5 (J^{\mathsf{T}} c(q))^{\mathsf{T}}\big). \end{aligned} \tag{210}$$

Here J, a, w_η, e_5 have the scalar module's meanings. Directly, $G = D_4^{\mathsf{T}} G_R D_4$, $S = D_5^{\mathsf{T}} S_R D_5$ and $c^{\mathsf{T}} T = v^{\mathsf{T}}$. Since $D_i^{\mathsf{T}} D_i \succeq I$, both scalar inequalities retain the same positive margin:

$$\begin{aligned} P(q) + G(q) \succeq \delta I_4, \\ J^{\mathsf{T}} P(q) J - T(q)^{\mathsf{T}} P(q) T(q) - S(q,\eta) \succeq \delta I_5, \qquad q \in K. \end{aligned} \tag{211}$$

The transformed P is continuous and bounded since $q_- > 0$; all bounds M_P below refer to this P, not to P_R. For a continuous real matrix function $N(q)$ on K, write

$$N[A] = (N_{ij}(A))_{ij}$$

for its block-operator lift by continuous functional calculus. This operation preserves sums, compatible products and adjoints. If $N(q)$ is symmetric and nonnegative for all $q \in K$, its finite-matrix square root $R(q) = N(q)^{1/2}$ is continuous; for example, approximate the square root uniformly by polynomials on a common compact interval containing all the spectra. Consequently

$$N[A] = R[A]^* R[A] \succeq 0. \tag{212}$$

Thus both inequalities in (211) lift to block operators on Y^4 and Y^5, with the same δ. Real continuous functional calculus suffices; alternatively one can complexify and restrict the real-valued calculus to the real subspace. No eigenvector expansion is used. Define the finite certificate bound

$$M_P = \sup_{q \in K} \max_i \sum_j |P(q)_{ij}|. \tag{213}$$

The pointwise inequalities $-M_P I \preceq P(q) \preceq M_P I$ and (212) imply $\|P[A]\| \le M_P$.

One convex graph and its actual histories. Fix a KKT anchor. The function F in (48) is proper lower-semicontinuous convex, with $F \geq 0$ and $F(0) = 0$. The proximal optimality condition gives

$$v_n \in \partial F(p_n), \qquad F_n < \infty \quad (n \geq 1). \tag{214}$$

In particular $0 \leq F_n \leq \langle p_n, v_n \rangle$. Introduce

$$\begin{gathered} B = \sqrt{\tau\sigma/\kappa_{\mathrm{cert}}}\, L, \qquad C = (I - B^*B)^{1/2}, \qquad B^*B + C^*C = I, \\ u_n = Bp_n, \qquad r_n = Cp_n, \qquad t_n = \sqrt{\tau/(\kappa_{\mathrm{cert}}\sigma)}\, w_n. \end{gathered} \tag{215}$$

Both B and C are contractions, and the dual update becomes

$$t_n = A\left(t_{n-1} + \frac{9u_n - u_{n-1}}{8}\right). \tag{216}$$

The physical factor is exactly

$$\kappa_{\mathrm{cert}} B^* t_{n-1} = \tau L^* w_{n-1}. \tag{217}$$

With $z_n = r_{n-1} - r_n$ and $g_n = u_{n-1} - u_n - \kappa_{\mathrm{cert}} t_{n-1}$, the identity $B^*B + C^*C = I$ gives

$$v_n = C^* z_n + B^* g_n, \qquad \langle p_i, v_j \rangle = \langle r_i, z_j \rangle + \langle u_i, g_j \rangle. \tag{218}$$

Solving (216) for t_{n-1} yields

$$\begin{gathered} s_n = (u_n, u_{n-1}, u_{n-2}, t_n), \\ g_n = (9\kappa_{\mathrm{cert}}/8 - 1)u_n + (1 - \kappa_{\mathrm{cert}}/8)u_{n-1} - \kappa_{\mathrm{cert}} A^{-1} t_n = c[A]^* s_n, \\ s_n = T[A](s_{n-1}, u_n). \end{gathered} \tag{219}$$

Here $c(q)$ and $T(q)$ are the original-coordinate column and transition matrix in (210). These component ports are not separate subgradients: every graph inequality below uses only the single relation (214). When $L = 0$, these definitions give $B = 0$, $C = I$, $u_n = 0$, $t_n = At_{n-1}$ and $v_n = p_{n-1} - p_n$ literally. The chosen κ_{cert} remains positive, and all arguments continue without division by $\|L\|$. No relation requiring A to commute with BB^* is used.

Quadratic dissipation. For $n \geq 3$ define

$$\begin{gathered} P_{\mathrm{res}} = \begin{pmatrix} (19+\eta)/2 & -7 & 1 \\ -7 & 5 & -3/2 \\ 1 & -3/2 & 1/2 \end{pmatrix}, \qquad R_n = (r_n, r_{n-1}, r_{n-2}), \\ E_n = \langle R_n, (P_{\mathrm{res}} \otimes I) R_n \rangle + \langle s_n, P[A] s_n \rangle. \end{gathered} \tag{220}$$

For $n \geq 4$ the full supply is

$$\mathcal{Q}_n = \langle (10+\eta)p_n - 2p_{n-2} - p_{n-3}, v_n \rangle - 7\langle p_n, v_{n-1} \rangle. \tag{221}$$

Its residual part, obtained from (218), obeys the exact inner-product identity

$$\begin{aligned}&\langle R_n, (P_{\mathrm{res}} \otimes I) R_n\rangle - \langle R_{n-1}, (P_{\mathrm{res}} \otimes I) R_{n-1}\rangle \\ &\quad + \langle (10+\eta) r_n - 2r_{n-2} - r_{n-3}, z_n\rangle - 7\langle r_n, z_{n-1}\rangle \\ &= -\frac{1}{2}\|r_n - 3r_{n-1} + 3r_{n-2} - r_{n-3}\|^2 - \frac{\eta}{2}\|r_n - r_{n-1}\|^2.\end{aligned} \tag{222}$$

It follows by expanding both sides and is valid in every real Hilbert space. Applying the lifted dissipation certificate to $(s_{n-1}, u_n) \in Y^5$, and using (219), supplies the coupled part. Together they prove

$$E_n - E_{n-1} + \mathcal{Q}_n \le -D_n, \qquad n \ge 4, \tag{223}$$

where

$$\begin{aligned}D_n = &\frac{1}{2}\|r_n - 3r_{n-1} + 3r_{n-2} - r_{n-3}\|^2 + \frac{\eta}{2}\|r_n - r_{n-1}\|^2 \\ &+ \delta\big(\|s_{n-1}\|^2 + \|u_n\|^2\big).\end{aligned} \tag{224}$$

The convex-potential correction. Set, using only attained outputs,

$$G_n = 7\langle p_n, v_n\rangle - 4F_n + 3F_{n-1} + F_{n-2}, \qquad V_n = E_n + G_n, \qquad n \ge 3. \tag{225}$$

The exact decomposition

$$\begin{aligned}&\mathcal{Q}_n - (G_n - G_{n-1}) - \eta\langle p_n, v_n\rangle \\ &= 7\big[F_n - F_{n-1} - \langle v_{n-1}, p_n - p_{n-1}\rangle\big] \\ &\quad + 2\big[F_{n-2} - F_n - \langle v_n, p_{n-2} - p_n\rangle\big] \\ &\quad + \big[F_{n-3} - F_n - \langle v_n, p_{n-3} - p_n\rangle\big] \ge 0 \qquad (n \ge 4)\end{aligned} \tag{226}$$

is a sum of subgradient inequalities. Since $\langle p_n, v_n\rangle \ge F_n$, it gives $\mathcal{Q}_n \ge G_n - G_{n-1} + \eta F_n$. For identification with convex-potential storage, Fenchel equality at the attained pair gives

$$\begin{aligned}F^*(v_n) &= \langle p_n, v_n\rangle - F_n, \\ G_n &= 7F^*(v_n) + 3F_n + 3F_{n-1} + F_{n-2} \\ &= 7F^*(v_n) + 2(F_n + F_{n-1}) + (F_n + F_{n-1} + F_{n-2}).\end{aligned} \tag{227}$$

Thus the correction combines delay-two and delay-three primal potentials with a delay-one conjugate potential. The primal potential increments satisfy $F_n - F_{n-k} \le \langle p_n - p_{n-k}, v_n\rangle$, while $F^*(v_n) - F^*(v_{n-1}) \le \langle p_n, v_n - v_{n-1}\rangle$. Their weights $2, 1, 7$ give the same graph descent, in the causal/anticausal template of [9]. Fenchel equality makes every displayed conjugate value finite; no function value at the initial state is introduced.

Coercivity on attained graph histories. Two more subgradient inequalities give

$$\begin{aligned}&G_n - \langle v_n, 3p_n + 3p_{n-1} + p_{n-2}\rangle \\ &= 3\big[F_{n-1} - F_n - \langle v_n, p_{n-1} - p_n\rangle\big] \\ &\quad + \big[F_{n-2} - F_n - \langle v_n, p_{n-2} - p_n\rangle\big] \ge 0.\end{aligned} \tag{228}$$

Equivalently this is weighted Fenchel–Young at the three outputs. The residual identity for this lower bound is

$$\langle R_n, (P_{\mathrm{res}} \otimes I)R_n\rangle + \langle z_n, 3r_n + 3r_{n-1} + r_{n-2}\rangle = \frac{\eta}{2}\|r_n\|^2 + \langle R_n, (H_{\mathrm{res}} \otimes I)R_n\rangle,$$
$$H_{\mathrm{res}} = \begin{pmatrix} 13/2 & -7 & 1/2 \\ -7 & 8 & -1 \\ 1/2 & -1 & 1/2 \end{pmatrix} \succeq 0. \tag{229}$$

The eigenvalues of this fixed scalar matrix are 0 and $(15 \pm 3\sqrt{21})/2$, and its kernel is the constant-history line. For the coupled part, (219) and the lifted coercivity certificate give

$$\langle s_n, P[A]s_n\rangle + \langle g_n, 3u_n + 3u_{n-1} + u_{n-2}\rangle \geq \delta\|s_n\|^2.$$

Combining these facts proves

$$V_n \geq \frac{\eta}{2}\|r_n\|^2 + \delta\|s_n\|^2 \geq \beta\big(\|p_n\|^2 + \|t_n\|^2\big), \qquad \beta = \min\{\eta/2, \delta\} > 0. \tag{230}$$

Combining (223) and (226) proves the actual-history descent

$$V_n - V_{n-1} \leq -D_n - \eta F_n, \qquad n \geq 4. \tag{231}$$

Separate positivity of $P[A]$ on arbitrary histories is neither needed nor asserted. This is a graph-corrected energy, not an ordinary squared-distance Fejér inequality.

Startup, boundedness, and summable gaps and residuals. The quantity V_3 is finite: its function values and selected subgradients have indices $1, 2, 3$, all actual proximal outputs. In particular $f(x_0)$ can be infinite. Telescoping (231) and using (230) gives

$$\sup_{n\geq 3}(\|p_n\|^2 + \|t_n\|^2) \leq V_3/\beta, \qquad \sum_{n\geq 4} D_n \leq V_3, \qquad \sum_{n\geq 4} F_n \leq V_3/\eta. \tag{232}$$

The finite prefix causes no difficulty. By (224), (u_n) and (t_n) are square summable and converge in norm to zero; also $F_n \to 0$. The increments $r_n - r_{n-1}$ are square summable. Since

$$\|p_n - p_{n-1}\|^2 = \|r_n - r_{n-1}\|^2 + \|u_n - u_{n-1}\|^2 \tag{233}$$

and first differences of a square-summable sequence are square summable, the full primal increments are square summable and converge in norm to zero. The identity $v_n = p_{n-1} - p_n - \kappa_{\mathrm{cert}} B^* t_{n-1}$ then gives square summability of (v_n). This proves the first five series in (50).

Expanding the quadratic term at the KKT anchor yields

$$\tau[\Phi(x_n) - \Phi(x^*)] = F_n + \frac{\kappa_{\mathrm{cert}}}{2}\langle(\sigma Q)^{-1}u_n, u_n\rangle, \qquad n \geq 1. \tag{234}$$

The linear cross term cancels by $Qy^* = Lx^* - c$. As

$$0 \preceq (\sigma Q)^{-1} \preceq bI, \qquad b = \frac{q_+}{1 - q_+} = \frac{1}{\sigma m}, \tag{235}$$

the composite gaps are nonnegative and summable. This does not assert square summability of $\|p_n\|$ without additional growth.

Distance limits for every KKT anchor. For the fixed anchor, V_n is nonnegative and decreasing for $n \geq 3$, so it has a finite limit. We claim

$$V_n - \frac{\eta}{2}\|p_n\|^2 \longrightarrow 0. \tag{236}$$

Boundedness and strong asymptotic regularity imply $R_n - (r_n, r_n, r_n) \to 0$ in X^3. The bounded residual quadratic form therefore differs by $o(1)$ from its value on this constant history. The sum of the entries of P_{res} is $\eta/2$, so that value is $(\eta/2)\|r_n\|^2$. Meanwhile $s_n \to 0$ in Y^4 and $\|P[A]\| \leq M_P$, whence $\langle s_n, P[A]s_n\rangle \to 0$. Also $G_n \to 0$ because (p_n) is bounded, $v_n \to 0$, and $F_n, F_{n-1}, F_{n-2} \to 0$. Finally $\|p_n\|^2 = \|r_n\|^2 + \|u_n\|^2$ and $u_n \to 0$. These facts prove (236) and hence the existence of $\lim_n \|x_n - x^*\|$. The same argument applies separately to every KKT anchor; it needs no uniform bound on the set of anchors.

Weak cluster points and whole-sequence convergence. A bounded sequence in a Hilbert space has a weakly convergent subsequence, including in the nonseparable case: restrict to the separable closed linear span of the sequence and use weak sequential compactness of its closed ball; its orthogonal complement does not affect weak convergence. Choose $p_{n_k} \rightharpoonup \bar{p}$. Bounded linearity of B and $Bp_{n_k} \to 0$ imply $B\bar{p} = 0$, hence $L\bar{p} = 0$. Moreover F is weakly lower-semicontinuous, because its convex norm-closed sublevel sets are weakly closed. Thus

$$0 \leq F(\bar{p}) \leq \liminf_k F_{n_k} = 0.$$

It follows that $0 \in \partial F(\bar{p})$, so $(x^* + \bar{p}, y^*)$ is KKT. Alternatively, weak–strong graph closure follows directly: for every $(z, a) \in \operatorname{gra} \partial F$, $\langle p_{n_k} - z, v_{n_k} - a\rangle \geq 0$. Since $v_{n_k} \to 0$ and (p_{n_k}) is bounded, passage to the limit gives $\langle \bar{p} - z, -a\rangle \geq 0$; maximal monotonicity of ∂F puts $(\bar{p}, 0)$ in its graph. Neither argument uses continuity of f at a domain boundary or strong compactness of the orbit.

For two KKT pairs (x^1, y^1) and (x^2, y^2), monotonicity gives

$$0 \leq \langle x^1 - x^2, -L^*(y^1 - y^2)\rangle = -\langle Q(y^1 - y^2), y^1 - y^2\rangle.$$

Coercivity of Q implies $y^1 = y^2$ and hence $Lx^1 = Lx^2$. Let a, b be two weak cluster points of (x_n). Both are KKT primal anchors. The distance limits already proved show that

$$\|x_n - a\|^2 - \|x_n - b\|^2 = 2\langle x_n, b - a\rangle + \|a\|^2 - \|b\|^2$$

has a limit. Along subsequences weakly converging to a and b this limit is respectively $-\|a - b\|^2$ and $\|a - b\|^2$, so $a = b$. If the whole sequence failed to converge weakly to this unique cluster point, a bounded linear functional would separate a subsequence from its proposed limit. That subsequence has a further weakly convergent subsequence, contradicting uniqueness. Therefore $x_n \rightharpoonup x_\infty$. Already $t_n \to 0$ and $u_n \to 0$ give the asserted norm limits for y_n and Lx_n. In finite-dimensional X, weak convergence is norm convergence, regardless of the dimension of Y. ∎

B.3 The mixed operator storage and graph interface

We first specify the common finite-dimensional interface. For $7/10 \le h \le q^{\dagger}$, set

$$a = \frac{5h}{9h-1}, \qquad b = \frac{4h-1}{9h-1}, \qquad \chi = a - b = \frac{h+1}{9h-1}, \qquad r_{\mathrm{lin}}(h) = \frac{8(1+h)}{5h}. \tag{237}$$

Here $a + b = 1$, $\chi > 0$ and $2b - a = (1 - 3\chi)/2 > 0$. For a positive certificate product κ, let

$$\begin{gathered} T(q) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ -1/8 & q & 0 & 9/8 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} I_3 & 0 \end{pmatrix}, \qquad e = (0,0,0,1)^T, \\ c_\kappa = (9\kappa/8 - 1, -\kappa, 1 - \kappa/8)^T, \qquad v_{q,\kappa} = (1, -\kappa q, 0, -1)^T, \\ w_\eta = (0, 0, -b, 1+\eta)^T, \qquad S_{q,\kappa,\eta} = \mathrm{sym}(w_\eta v_{q,\kappa}^T) - a\,\mathrm{sym}\big(e(J^T c_\kappa)^T\big), \\ G_\kappa = \mathrm{sym}\big(c_\kappa(\lambda, 0, \nu)\big), \qquad \mathrm{sym}(D) = (D + D^T)/2. \end{gathered} \tag{238}$$

The sufficient certificate is one choice of the scalar numbers $\eta > 0$, $\nu \in [\chi, b]$, $\lambda = a - \nu$, one continuous symmetric $P(q)$ on $[q_{\min}, h] \subset (0,1)$, and $\delta > 0$ such that

$$\begin{aligned} P(q) + G_\kappa &\succeq \delta I_3, \\ D(q) := J^T P(q) J - T(q)^T P(q) T(q) - S_{q,\kappa,\eta} &\succeq \delta I_4. \end{aligned} \tag{239}$$

These inequalities concern unrestricted actual-history coordinates, including $q = 1/9$; no state is cancelled or divided by $9q - 1$. The quadratic matrix P itself need not be positive. The same η, λ, ν must satisfy both inequalities and the original convex graph bounds, rather than being selected separately for each spectral value.

B.4 One convex graph and two linear ports

Let F be a single proper lower-semicontinuous convex function on the original primal Hilbert space, with $F(0) = \min F = 0$, and suppose $v_n \in \partial F(p_n)$ for the actual outputs $n \ge 1$. Put

$$F_n = F(p_n), \qquad H_n = F^*(v_n) = \langle p_n, v_n \rangle - F_n, \qquad \mathcal{A}_n = aH_n + b(F_n + F_{n-1}) \quad (n \ge 2). \tag{240}$$

All these attained function values are finite and nonnegative. For the mixed supply $\mathcal{Q}_n = \langle p_n, v_n \rangle - a\langle p_n, v_{n-1} \rangle - b\langle p_{n-2}, v_n \rangle$, exact cancellation gives, for $n \ge 3$,

$$\begin{aligned} \mathcal{Q}_n - (\mathcal{A}_n - \mathcal{A}_{n-1}) &= a\big[F_n - F_{n-1} - \langle v_{n-1}, p_n - p_{n-1} \rangle\big] \\ &\quad + b\big[F_{n-2} - F_n - \langle v_n, p_{n-2} - p_n \rangle\big] \ge 0. \end{aligned} \tag{241}$$

Adding $\eta\langle p_n, v_n \rangle$ bounds the last expression below by ηF_n. The graph lower bound follows from the exact Fenchel identity

$$\begin{aligned} \mathcal{A}_n - \langle v_n, \lambda p_n + \nu p_{n-1} \rangle &= (a - \lambda - \nu) H_n + (b - \lambda) F_n + (b - \nu) F_{n-1} \\ &\quad + \nu\big[H_n + F_{n-1} - \langle v_n, p_{n-1} \rangle\big] \ge 0, \end{aligned} \tag{242}$$

provided the four budgets $\nu, b-\lambda, b-\nu, a-\lambda-\nu$ are nonnegative. There is no separate sign requirement on λ. In (239), $\lambda = a-\nu$, $\chi \le \nu \le b$ ensures these budgets. Neither F nor F^* is acted on by spatial functional calculus.

For an arbitrary residual sequence r_n, let $z_n = r_{n-1} - r_n$ and

$$E_n^{\mathrm{res}} = \frac{\eta}{2}\|r_n\|^2 - a\langle r_n, z_n\rangle + \frac{\chi}{2}\|z_n\|^2. \tag{243}$$

Direct inner-product expansion on unrestricted histories yields

$$\begin{aligned} E_n^{\mathrm{res}} - E_{n-1}^{\mathrm{res}} + \langle r_n, z_n\rangle - a\langle r_n, z_{n-1}\rangle - b\langle r_{n-2}, z_n\rangle + \eta\langle r_n, z_n\rangle \\ = -\big(2b - a + \eta/2\big)\|z_n\|^2 - \frac{\chi}{2}\|z_n - z_{n-1}\|^2, \\ E_n^{\mathrm{res}} + \langle z_n, \lambda r_n + \nu r_{n-1}\rangle = \frac{\eta}{2}\|r_n\|^2 + (\chi/2 + \nu)\|z_n\|^2. \end{aligned} \tag{244}$$

For example, substitute $r_{n-1} = r_n + z_n$ and $r_{n-2} = r_n + z_n + z_{n-1}$ and collect coefficients. Its two-history matrix is

$$P_{\mathrm{res}}^{\eta} = \begin{pmatrix} 1/2 + \chi + \eta/2 & -(1+3\chi)/4 \\ -(1+3\chi)/4 & \chi/2 \end{pmatrix}. \tag{245}$$

Thus the residual correction uses exactly the same scalar graph weights.

For the literal iteration (3), fix a KKT anchor (2), choose $\kappa > 0$ with $d \le \kappa$, and define

$$\begin{aligned} A = (I + \sigma Q)^{-1}, \quad p_n = x_n - x^*, \quad w_n = y_n - y^*, \\ F(p) = \tau\big[f(x^* + p) - f(x^*) + \langle L^* y^*, p\rangle\big], \quad v_n = p_{n-1} - p_n - \tau L^* w_{n-1}, \\ B = \sqrt{\tau\sigma/\kappa}\, L, \quad C = (I - B^*B)^{1/2}, \quad u_n = Bp_n, \quad r_n = Cp_n, \\ t_n = \sqrt{\tau/(\kappa\sigma)}\, w_n, \quad z_n = r_{n-1} - r_n, \quad g_n = u_{n-1} - u_n - \kappa t_{n-1}. \end{aligned} \tag{246}$$

The proximal inclusion gives $v_n \in \partial F(p_n)$. Since $B^*B + C^*C = I$, subtraction of the literal recurrence gives

$$\begin{aligned} t_n = A\big[t_{n-1} + (9u_n - u_{n-1})/8\big], \qquad v_n = C^* z_n + B^* g_n, \\ \langle p_i, v_j\rangle = \langle r_i, z_j\rangle + \langle u_i, g_j\rangle \quad \text{for every } i, j, \qquad \|p_n\|^2 = \|r_n\|^2 + \|u_n\|^2. \end{aligned} \tag{247}$$

There is a single genuine convex graph. The two ports separately need not be subgradient graphs. For the coupled after-state $s_n = (u_n, A^{-1}t_n, u_{n-1})$, the scalar transition is precisely $T(q)$, and the current port is $c_\kappa^T s_n$. This proves the coordinate and supply interpretation of (238).

For completeness, the frequency lift uses congruence, not commutation. For unit-circle delay ζ, put

$$M_\eta(\zeta) = 1 + \eta - a\zeta - b\zeta^{-2}, \qquad V_{q,\kappa}(\zeta) = \zeta - 1 - \frac{\kappa q \zeta(9-\zeta)}{8(1-q\zeta)}. \tag{248}$$

The vector $\xi = (\zeta, \zeta(9-\zeta)/(8(1-q\zeta)), \zeta^2, 1)^T$ satisfies $T\xi = \zeta^{-1} J\xi$. Consequently the second LMI implies

$$\mathrm{Re}(M_\eta V_{q,\kappa}) = \xi^* S_{q,\kappa,\eta}\xi = -\xi^* D(q)\xi \le -\delta. \tag{249}$$

This uses the very same η as graph coercivity. Entrywise continuous functional calculus on A lifts continuous positive matrix functions: if $N(q) \succeq 0$, its continuous matrix square root satisfies $N[A] = N^{1/2}[A]^* N^{1/2}[A] \succeq 0$. On the complexification the full primal plant is $V_L = C^*(\zeta - 1)C + B^* V_{A,\kappa} B$, so

$$\operatorname{Re}(M_\eta V_L) \preceq -(1 - 3\chi + \eta)(1 - \operatorname{Re}\zeta) C^*C - \delta B^*B. \tag{250}$$

Indeed $\operatorname{Re}[M_0(\zeta)(\zeta-1)] = -(1-t)[1-\chi(2t+1)]$ for $t = \operatorname{Re}\zeta$. No relation $ABB^* = BB^*A$ or eigenbasis is used. At DC, $V_L(1) = -\tau L^* Q^{-1} L$ can vanish on $\ker L$. The residual identity (244), rather than strict primal frequency dissipation, supplies control of that direction.

B.5 Proofs of the mixed certificate realizations

The first argument proves Proposition 4.1.

Proof. The full calculation is given in Appendix B.7. Its construction is important: in DC-null coordinates the endpoint $\kappa = r_{\mathrm{lin}}(h)$ has an explicit continuous semidefinite storage whose corrected graph form is positive definite for $q < h$ and rank one at $q = h$. A zero-product storage has the complementary one-dimensional kernel and strictly positive dissipation off DC. For the fixed strict κ, their convex interpolation makes the graph form positive definite and leaves exactly the DC dissipation kernel. One uniformly small positive η repairs that kernel by a Schur complement, without changing the corrected state matrix. Compactness supplies a common positive margin after the coordinate congruence. The appendix establishes the root branch, signs and common η explicitly; frequency feasibility alone is not used to infer graph positivity. ∎

The following argument proves Proposition 4.2.

Proof. At $\kappa_* = 337/100$, the public rational data give 33 matrices at $q_i = 1/10 + i/40$, with common denominator 10^9. On the 32 closed intervals interpolate these matrices affinely. The graph inequalities are certified at the 33 nodes; the dissipation polynomial on each interval is cubic and all four of its Bernstein coefficient matrices are certified. Exact Sylvester tests on all 128 Bernstein matrices give the common margin $\delta_* = 1/100000$. Appendix B.8 prints the reconstruction recipe and the zero-product identities. Interpolating the endpoint storage with the zero-product storage, with weight κ/κ_* and the same η, λ, ν, gives both LMIs with margin $(\kappa/\kappa_*)\delta_* = \kappa/337000$. The four graph budgets are $117/355, 22/355, 13/355, 0$. This proves the entire continuum, including $q = 1/9$ and the closed product endpoint, by exact arithmetic rather than sampling. ∎

B.6 Proof of mixed Hilbert convergence

Proof. *Certificate selection and lift.* Under (52), choose $0 < \kappa < \kappa^\dagger$ with $d \le \kappa$, and use $q_{\min} = (1 + \sigma M)^{-1} > 0$ and $h = q^\dagger$ in Proposition 4.1. The lower curvature condition

is exactly $(1+\sigma m)^{-1} \le h$. Under (53), use $\kappa = 337/100$ and Proposition 4.2, for which $\operatorname{spec}(A) \subset [1/10, 9/10]$. In particular κ is positive even when $L = 0$. Define the ports by (246); A^{-1} and $P[A]$ are bounded. The continuous square-root argument above lifts both LMIs to Y^3 and Y^4.

Coercivity before telescoping and actual startup. For $s_n = (u_n, A^{-1}t_n, u_{n-1})$, set

$$\mathcal{E}_n = E_n^{\mathrm{res}} + \langle s_n, P[A]s_n\rangle + \mathcal{A}_n, \qquad n \ge 2. \tag{251}$$

Apply the original-space graph bound (242) and split it by the all-pairs identity (247). The residual correction is (244), and the coupled correction is the first LMI. Hence

$$\mathcal{E}_n \ge \frac{\eta}{2}\|r_n\|^2 + \delta\|s_n\|^2 \ge \min\{\eta/2, \delta\}\big(\|p_n\|^2 + \|t_n\|^2\big). \tag{252}$$

Here $\|A^{-1}t_n\| \ge \|t_n\|$. The second LMI, the residual identity and (241) give, for $n \ge 3$,

$$\begin{aligned}\mathcal{E}_n - \mathcal{E}_{n-1} \le &-(2b - a + \eta/2)\|z_n\|^2 - \frac{\chi}{2}\|z_n - z_{n-1}\|^2 \\ &- \delta\big(\|s_{n-1}\|^2 + \|u_n\|^2\big) - \eta F_n.\end{aligned} \tag{253}$$

The energy $\mathcal{E}_2$ is finite: its graph terms use only actual proximal outputs 1 and 2. The first descent uses graph indices $1, 2, 3$. Neither $f(x_0) < \infty$, an initial subgradient nor artificial prehistory is required. Coercivity (252) has been proved without assuming any infinite sum finite, and now justifies summing (253).

Summability and objective values. Telescoping gives boundedness of p_n, t_n, square summability of u, t, s, z, and $\sum_{n\ge1} F_n < \infty$, after restoring finite prefixes. The fixed finite-delay expression for g_n gives $g \in \ell^2$; then $v_n = C^*z_n + B^*g_n$ gives $v \in \ell^2$. The isometry also gives

$$\|p_n - p_{n-1}\|^2 = \|r_n - r_{n-1}\|^2 + \|u_n - u_{n-1}\|^2. \tag{254}$$

Thus primal increments are square summable. The scalings in (246) give the dual and image-error sums in (56), and their strong limits. The exact KKT expansion of the objective is

$$\Phi(x_n) - \Phi(x^*) = \frac{F_n}{\tau} + \frac{1}{2}\|Q^{-1/2}Lp_n\|^2 \ge 0. \tag{255}$$

Boundedness of Q^{-1} proves summability of these objective gaps. The graph and objective sums start at $n = 1$, so no possibly infinite initial function value is included.

Distance limits in the residual direction. Fix any KKT anchor. Because $F_n \to 0$, $v_n \to 0$ and p_n is bounded, $0 \le H_n = \langle p_n, v_n\rangle - F_n \to 0$, and hence $\mathcal{A}_n \to 0$. Also $s_n \to 0$, so the bounded coupled quadratic storage tends to zero. From (243), bounded r_n and $z_n \to 0$ give

$$\mathcal{E}_n - \frac{\eta}{2}\|x_n - x^*\|^2 \longrightarrow 0, \tag{256}$$

since $\|p_n\|^2 = \|r_n\|^2 + \|u_n\|^2$ and $u_n \to 0$. The nonnegative decreasing energy has a limit. Therefore the distance $\|x_n - x^*\|$ has a limit for every KKT primal anchor, including kernel directions invisible to L.

KKT cluster points and the whole sequence. For two KKT pairs, monotonicity of ∂f gives

$$0 \le \langle x^* - x^{**}, -L^*(y^* - y^{**})\rangle = -\langle Q(y^* - y^{**}), y^* - y^{**}\rangle.$$

Coercivity of Q forces $y^* = y^{**}$ and then $Lx^* = Lx^{**}$. The bounded primal sequence has a weakly convergent subsequence. If $x_{n_j} \rightharpoonup \widehat{x}$, the strong image limit implies $L\widehat{x} = Lx^*$. Weak lower semicontinuity of F and $F_{n_j} \to 0$ give $F(\widehat{x} - x^*) = 0$. This is a minimum of F, so $-L^*y^* \in \partial f(\widehat{x})$ and $(\widehat{x}, y^*)$ is KKT. If $\widehat{x}, \widetilde{x}$ were two weak cluster points, the limit of

$$\|x_n - \widehat{x}\|^2 - \|x_n - \widetilde{x}\|^2 = 2\langle x_n, \widetilde{x} - \widehat{x}\rangle + \|\widehat{x}\|^2 - \|\widetilde{x}\|^2$$

exists by the distance-limit result. Evaluating on subsequences converging weakly to the two cluster points makes the difference between those limits equal to $2\|\widehat{x} - \widetilde{x}\|^2$. They coincide. Finally, any subsequence outside a weak neighborhood of this unique cluster point would have a further weakly convergent subsequence, by weak sequential compactness of bounded Hilbert sequences, contradicting that exclusion. Hence the entire primal sequence converges weakly. The already established strong dual and image limits finish the proof. No separability or compactness of L has been used. ∎

B.7 Analytic mixed storage and the DC repair

We prove Proposition 4.1 by explicit unrestricted-history calculations. Throughout this appendix, $h = q^\dagger$, $U_\dagger = r_{\mathrm{lin}}(h) = \kappa^\dagger$, and a, b, χ are given by (237). Set $\nu = hb$, $\lambda = a - \nu$ once and for all. The algebraic relation $4h^2 + 3h - 6 = 0$ gives

$$a = bh(h+1), \qquad U_\dagger a(1-h) = 2(2-3a), \qquad a - \frac{1}{2} - \frac{aU_\dagger}{16} = 0. \tag{257}$$

In particular $2 - 3a > 0$, $\chi > 0$ and $b - \chi = 2b - a > 0$. Moreover $\nu < b$ and $\nu - \chi = 5(1-h)/(9h-1) > 0$, so the common graph budgets are admissible. This choice $\nu = hb$ is being used only at the algebraic endpoint h; it is not asserted admissible throughout $[7/10, q^\dagger]$.

B.7.1 An exact DC-null identity

Write a before-state as (p, R, p_2) and the new output as u. Use

$$\omega = R - \frac{p}{1-q}, \quad \ell = p_2 - p, \quad e = u - p, \quad K_q = \frac{\kappa q}{1-q}, \quad m_q = \frac{9q-1}{8(1-q)}, \quad L_\kappa = 1 - \frac{\kappa}{8}. \tag{258}$$

The transformed state $\zeta = (p, \omega, \ell)$ updates according to

$$\zeta^+ = (p + e, q\omega - m_q e, -e), \qquad v = -K_q p - \kappa q\omega - e, \qquad v^- = -K_q p - \kappa\omega + L_\kappa \ell. \tag{259}$$

Thus the unregularized supply and the graph lower form are respectively

$$\mathcal{Q} = (ap + e - b\ell)v - a(p+e)v^-, \qquad \mathcal{G} = v^-(ap + \nu\ell).$$

For free entries x, z, w, define a symmetric matrix $\widetilde{P}$ by

$$\begin{aligned} \widetilde{P}_{11} &= (2-a)K_q/2, \quad \widetilde{P}_{12} = a\kappa/2, \\ \widetilde{P}_{13} &= \frac{9a\kappa q - a\kappa - 8aq + 8a - 8\kappa q}{16(q-1)}, \quad \widetilde{P}_{22} = x, \quad \widetilde{P}_{23} = z, \quad \widetilde{P}_{33} = w. \end{aligned} \tag{260}$$

Expanding the unrestricted polynomial $\zeta^T \widetilde{P}\zeta - (\zeta^+)^T \widetilde{P}\zeta^+ - \mathcal{Q}$ cancels its entire p row and column and gives $(\omega, \ell, e)^T \widehat{D}(\omega, \ell, e)$, where

$$\begin{aligned} \widehat{D}_{11} &= (1-q^2)x, & \widehat{D}_{12} &= z - b\kappa q/2, \\ \widehat{D}_{13} &= qm_q x + qz + \kappa[q - a(1+q)]/2, & \widehat{D}_{22} &= w, \\ \widehat{D}_{23} &= a - 1/2 - a\kappa/16, & \widehat{D}_{33} &= b + aK_q/2 - m_q^2 x - 2m_q z - w. \end{aligned} \tag{261}$$

Independently, the graph-corrected state matrix is

$$\widetilde{W} = \widetilde{P} + \mathcal{G} = \begin{pmatrix} (2-3a)K_q/2 & 0 & K_q(b-\nu)/2 \\ 0 & x & z - \kappa\nu/2 \\ K_q(b-\nu)/2 & z - \kappa\nu/2 & w + \nu L_\kappa \end{pmatrix}. \tag{262}$$

Here addition of $\mathcal{G}$ means addition of its symmetric quadratic matrix. Return to the original state by

$$P(q) = H(q)^T \widetilde{P}(q) H(q), \qquad H(q) = \begin{pmatrix} 1 & 0 & 0 \\ -1/(1-q) & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}. \tag{263}$$

All these coordinates are continuous and invertible for $q < 1$, including $q = 1/9$. The zero m_q there causes no pole.

B.7.2 Continuous semidefinite storage at the endpoint

At $\kappa = U_\dagger$ set $z = bU_\dagger q/2$ and $w = 0$. To avoid confusing the following scalar coefficients with the coupling operators, write

$$C_q = \frac{U_\dagger}{2}[bq(q+1) - a], \quad F_q = b + \frac{aK_q}{2} - m_q b U_\dagger q, \quad E_q = (1-q^2)F_q - 2qm_q C_q. \tag{264}$$

The ℓ row of (261) vanishes by (257). Its remaining block is

$$D_2 = \begin{pmatrix} (1-q^2)x & qm_q x + C_q \\ qm_q x + C_q & F_q - m_q^2 x \end{pmatrix}, \qquad f_q(x) := \det D_2 = E_q x - m_q^2 x^2 - C_q^2. \tag{265}$$

Define the following continuous coefficients:

$$\begin{aligned} \overline{C}_q &= -\frac{U_\dagger b(q+h+1)}{2}, \qquad \overline{F}_q = \frac{9U_\dagger b}{8}\frac{q + (4h-1)/9}{1-q}, \\ \overline{H}_+ &= b(h+1), \qquad \overline{H}_- = \frac{U_\dagger b[(18+5h)q + 3h - 2]}{8(1-q)}, \\ \overline{E}_q &= (1-q^2)\overline{F}_q - 2qm_q\overline{C}_q, \qquad \overline{\Delta}_q = (1-q^2)\overline{H}_-\overline{H}_+, \\ x_\star(q) &= (h-q)\frac{2\overline{C}_q^2}{\overline{E}_q + \sqrt{\overline{\Delta}_q}}. \end{aligned} \tag{266}$$

The square root is the positive real one. Direct substitution of $4h^2+3h-6=0$ gives

$$\begin{aligned} C_q &= (h-q)\overline{C}_q, \quad F_q = (h-q)\overline{F}_q, \quad E_q = (h-q)\overline{E}_q, \\ (1+q)\overline{F}_q + 2m_q\overline{C}_q &= \overline{H}_+, \qquad (1-q)\overline{F}_q - 2m_q\overline{C}_q = \overline{H}_-, \\ \overline{E}_q &= \frac{(1+q)\overline{H}_- + (1-q)\overline{H}_+}{2}, \qquad \overline{E}_q^2 - 4m_q^2\overline{C}_q^2 = \overline{\Delta}_q. \end{aligned} \tag{267}$$

Both $\overline{H}_\pm$ are strictly positive for $0 \le q \le h$. Thus $\overline{E}_q, \overline{\Delta}_q > 0$, the denominator in (266) never vanishes, and $x_\star$ is continuous on $[0,h]$, positive for $q<h$ and zero at $q=h$. For $q<h$ it is the smaller positive root of f_q: this is the rationalized quadratic formula when $m_q \neq 0$, and the expression C_q^2/E_q when $m_q = 0$. Hence $D_2 \succeq 0$, since its first diagonal entry is positive and its determinant vanishes. At $q=h$ all its entries are zero. The full endpoint dissipation is therefore semidefinite.

Graph positivity requires a separate sign, not just the preceding determinant zero. For $0 < q \le h$, eliminate the positive first entry $(2-3a)K_q/2$ from (262). The Schur complement is

$$\begin{pmatrix} x_\star & U_\dagger b(q-h)/2 \\ U_\dagger b(q-h)/2 & (b^2/a)(h-q)/(1-q) \end{pmatrix}. \tag{268}$$

For $q<h$, it is positive definite exactly when $x_\star > x_{\min} := aU_\dagger^2(h-q)(1-q)/4$. The exact evaluations

$$\begin{aligned} f_q(x_{\min}) &= -\frac{4(h+1)^2(h^2-78h+121)(h-q)^2(1-q)^2}{25h^2(9h-1)^2} < 0, \\ f_q'(x_{\min}) &= \frac{(h-q)[(41h+90)q+31h-10]}{8h(9h-1)} > 0 \end{aligned} \tag{269}$$

place $x_{\min}$ strictly to the left of the smaller root of the concave quadratic f_q. The first sign uses $h^2-78h+121 > 43$ for $0<h<1$, and the second uses $h > 9/10$. At $m_q = 0$, f_q is increasing affine and the same argument applies. Consequently $\widetilde{W}_\star \succ 0$ for $0<q<h$. At $q=h$ the Schur complement vanishes, so $\widetilde{W}_\star \succeq 0$ has rank one and is strictly positive on every nonzero vector $(p,0,0)$.

B.7.3 The complementary zero-product kernel and interpolation

At $\kappa = 0$ choose

$$z_0 = 0, \qquad w_0 = \chi/2, \qquad x_0 = \frac{(b-\chi)(1-q^2)}{2(1+m_q^2)} > 0. \tag{270}$$

Then the corrected state form is $\widetilde{W}_0 = \operatorname{diag}(0, x_0, \chi/2+\nu)$, whose kernel is exactly the p axis. Its non-DC dissipation is

$$\widehat{D}_0 = \begin{pmatrix} (1-q^2)x_0 & 0 & qm_qx_0 \\ 0 & \chi/2 & \chi/2 \\ qm_qx_0 & \chi/2 & b - m_q^2x_0 - \chi/2 \end{pmatrix}. \tag{271}$$

Eliminating the first two positive diagonal entries leaves

$$b-\chi-\frac{m_q^2 x_0}{1-q^2}=(b-\chi)\left(1-\frac{m_q^2}{2(1+m_q^2)}\right)\geq(b-\chi)/2>0.$$

Thus $\widehat{D}_0\succ 0$; the full four-coordinate form has precisely the DC kernel $\omega=\ell=e=0$. For the target fixed $0<\kappa<U_\dagger$, set $\rho=\kappa/U_\dagger$ and

$$\widetilde{P}_\kappa=(1-\rho)\widetilde{P}_0+\rho\widetilde{P}_\star. \tag{272}$$

The transition is independent of κ, while the supply and scalar graph correction are affine in it. Therefore $\widetilde{W}_\kappa=(1-\rho)\widetilde{W}_0+\rho\widetilde{W}_\star\succ 0$: any vector in the kernel of $\widetilde{W}_0$ is strictly controlled by $\widetilde{W}_\star$. Likewise $\widehat{D}_\kappa=(1-\rho)\widehat{D}_0+\rho\widehat{D}_\star\succ 0$. This proves state positivity and dissipation off DC for every fixed strict product, with the same λ,ν for all q.

B.7.4 One uniformly small positive regularization

Keep (272) unchanged and add ηuv to the supply. The full dissipation becomes

$$(\omega,\ell,e)^T\widehat{D}_\kappa(\omega,\ell,e)+\eta(p+e)(K_q p+\kappa q\omega+e). \tag{273}$$

For the fixed compact positive band $[q_{\min},h]$, let

$$\begin{aligned}\mu&=\min_{q\in[q_{\min},h]}\lambda_{\min}(\widehat{D}_\kappa(q))>0,\quad K_{\min}=\frac{\kappa q_{\min}}{1-q_{\min}},\quad K_{\max}=\frac{\kappa h}{1-h},\\ R_{\max}^2&=\frac{(\kappa h)^2+(1+K_{\max})^2}{4},\qquad \eta_{\mathrm{DC}}=\frac{\mu}{2(1+\kappa h/2+R_{\max}^2/K_{\min})}.\end{aligned} \tag{274}$$

In the split p versus (ω,ℓ,e), the perturbation has first entry K_q, cross vector $(\kappa q/2,0,(1+K_q)/2)$ and lower block bounded below by $-(\kappa h/2)I$. Its Schur complement for $0<\eta\leq\eta_{\mathrm{DC}}$ is bounded below by $\mu-\eta(\kappa h/2+R_{\max}^2/K_{\min})\geq\mu/2$. Thus (273) is uniformly positive definite.

One may also make compatibility with the direct scalar frequency certificate explicit. With $t=\operatorname{Re}\zeta$ and $\Delta_q=1+q^2-2qt$, put $B_\kappa=-\Delta_q\operatorname{Re}(M_0V_{q,\kappa})$. The endpoint factorization $B_\star=(1-t)K_\star$ has $K_\star$ affine in t and

$$\begin{aligned}K_\star(h,q,-1)&=\frac{10(h-q)(q+1)}{9h-1},\\ K_\star(h,q,1)&=\beta_0(1-q/h)^2+2\beta_1(q/h)(1-q/h)+\beta_2(q/h)^2,\\ \beta_0&=\frac{2(3h-2)}{9h-1},\quad \beta_1=\frac{33h^2+111h-22}{5(9h-1)},\quad \beta_2=\frac{4(6-3h-4h^2)}{5}=0.\end{aligned} \tag{275}$$

Hence $B_\star\geq 0$. At zero product, $B_0=\Delta_q(1-t)[1-\chi(2t+1)]$, so affinity gives $B_\kappa\geq\gamma(1-t)$ with $\gamma=(1-\kappa/U_\dagger)(1-h)^2(1-3\chi)>0$. For $C_\kappa=-\Delta_q\operatorname{Re}V_{q,\kappa}$, direct expansion gives

$$C_\kappa=(1-t)(1+q^2-2qt)+\frac{\kappa q}{8}[-2t^2+(9+q)t+1-9q].$$

In particular $C_\kappa(1) = \kappa q(1-q)$. With $L_0 = 4h + \kappa h/2 + (1+h)^2 + \kappa h(9+h)/8$, differentiation bounds $C_\kappa(t) \geq \kappa q_{\min}(1-h) - L_0(1-t)$. Choose

$$\eta = \min\left\{\eta_{\mathrm{DC}}, \frac{\gamma}{2(1+L_0)}\right\} > 0. \tag{276}$$

Then $B_\kappa + \eta C_\kappa \geq \gamma(1-t)/2 + \eta\kappa q_{\min}(1-h) > 0$. The same η therefore works in the Schur repair, residual storage and scalar FDI. Adding it did not alter $\widetilde{W}_\kappa$. Both final LMI matrices are continuous and positive definite after (263) and its input-coordinate congruence. Their smallest eigenvalues attain a positive common minimum on the compact band, giving δ in (239). Singleton bands are covered. There is no margin uniform as $q_{\min} \downarrow 0$ or $\kappa \uparrow U_\dagger$, and no strict endpoint certificate is asserted.

B.8 The rational Bernstein certificate and its product extension

Use the weights (54), $\kappa_* = 337/100$, $\delta_* = 1/100000$, and $q_i = 1/10 + i/40$, $0 \leq i \leq 32$. The public file `mixed_operator/certificates/rational_certificate.json` contains 33 integer six-tuples in the order $(11, 12, 13, 22, 23, 33)$, each divided by 10^9 to obtain a symmetric matrix P_i. For $q \in [q_i, q_{i+1}]$ put $\xi = 40(q - q_i)$ and

$$P_*(q) = (1-\xi)P_i + \xi P_{i+1}. \tag{277}$$

This defines one continuous matrix on the full closed band. At each node, all three leading principal determinants of $P_i + G_{\kappa_*} - \delta_* I_3$ are positive. On each segment put $P_0 = P_i$, $\dot{P} = P_{i+1} - P_i$, $T_0 = T(q_i)$, $\dot{T} = T(q_{i+1}) - T_0$, $S_0 = S_{q_i,\kappa_*,\eta}$ and $\dot{S} = S_{q_{i+1},\kappa_*,\eta} - S_0$. The dissipation polynomial is $D_i(\xi) = A_0 + \xi A_1 + \xi^2 A_2 + \xi^3 A_3$, with

$$\begin{aligned}
A_0 &= J^T P_0 J - T_0^T P_0 T_0 - S_0,\\
A_1 &= J^T \dot{P} J - T_0^T \dot{P} T_0 - \dot{T}^T P_0 T_0 - T_0^T P_0 \dot{T} - \dot{S},\\
A_2 &= -\dot{T}^T \dot{P} T_0 - T_0^T \dot{P} \dot{T} - \dot{T}^T P_0 \dot{T},\\
A_3 &= -\dot{T}^T \dot{P} \dot{T}.
\end{aligned} \tag{278}$$

Its four cubic Bernstein matrices are

$$B_0 = A_0, \quad B_1 = A_0 + A_1/3, \quad B_2 = A_0 + 2A_1/3 + A_2/3, \quad B_3 = A_0 + A_1 + A_2 + A_3. \tag{279}$$

All four leading principal determinants of each $B_j - \delta_* I_4$ are strictly positive for all 32 segments. Sylvester's criterion therefore proves these matrices positive definite. Since $D_i(\xi) = \sum_{j=0}^{3} \binom{3}{j} \xi^j (1-\xi)^{3-j} B_j$ is a convex combination for every $\xi \in [0,1]$, the second LMI follows throughout the interval. The first LMI is the affine interpolation of its positive endpoint matrices. There are $33 \cdot 3 + 128 \cdot 4 = 611$ strict rational leading-minor tests. The two interior Bernstein matrices on each interval are essential; checking only endpoint values of D_i would not establish continuum positivity. The public author checker

reconstructs (278); the independent checker instead reconstructs the polynomial by exact Bernstein-basis interpolation and checks positive rational LDL pivots against Leibniz determinant sums. The integer data and these finite checks complement the preceding mathematical positivity argument.

For the product extension keep the same η, λ, ν and embed (245) as

$$P_0^\eta = \begin{pmatrix} 1/2+\chi+\eta/2 & 0 & -(1+3\chi)/4 \\ 0 & 0 & 0 \\ -(1+3\chi)/4 & 0 & \chi/2 \end{pmatrix}. \tag{280}$$

For an arbitrary state $s = (p, R, p_2)^T$ and new output u, put $\widehat{s} = (p, R, p_2, u)^T$. Expansion gives

$$\begin{aligned} s^T(P_0^\eta + G_0)s &= \frac{\eta}{2}p^2 + (\chi/2+\nu)(p_2-p)^2, \\ \widehat{s}^T\big[J^T P_0^\eta J - T^T P_0^\eta T - S_{q,0,\eta}\big]\widehat{s} &= (2b-a+\eta/2)(p-u)^2 + \frac{\chi}{2}(2p-u-p_2)^2. \end{aligned} \tag{281}$$

Both forms are semidefinite for every q; their zero R direction is legitimate zero-product degeneracy. Define for $0 < \kappa \le \kappa_*$

$$P_\kappa(q) = \left(1 - \frac{\kappa}{\kappa_*}\right) P_0^\eta + \frac{\kappa}{\kappa_*} P_*(q). \tag{282}$$

The transition has no κ dependence and, with fixed η, both S and G are affine in κ. Thus each LMI is exactly the convex combination of its zero-product and endpoint versions, with margin $\delta(\kappa) = \kappa\delta_*/\kappa_* = \kappa/337000$. Varying η with κ here would introduce a bilinear supply term; no such interpolation is used. This completes the proof of the rational certificate and its compatibility with the common graph and residual storage.

B.9 Rational three tap storage for bounded coupling

We prove Theorem 6 directly from the literal iteration. Throughout this subsection,

$$q = \frac{19}{20}, \quad K = \frac{159}{50}, \quad (a,b,c) = \frac{1}{484925}(324883, 140645, 19397), \quad a+b+c = 1. \tag{283}$$

The scalar repair, common graph coefficients and strict matrix margin are

$$\eta = \frac{807419}{10^9}, \qquad (\lambda_0, \lambda_1, \lambda_2) = \frac{1}{10^9}(305936985, 318753189, 39983507), \qquad \delta = \frac{1}{10000}. \tag{284}$$

The single repaired multiplier is $M(z) = 1 + \eta - az - bz^{-2} - cz^{-3}$. Its normalized off-diagonal mass is $1/(1+\eta) < 1$.

B.9.1 Explicit rational reconstruction of the four inequalities

In the state order $(u, R, u_{\mathrm{prev}}, u_{\mathrm{prev2}})$, take

$$P = \frac{1}{10^9}\begin{pmatrix} -78793629 & 466733895 & -138486398 & -15593973 \\ 466733895 & 3595908 & 493416316 & 59783196 \\ -138486398 & 493416316 & 33479885 & 3850230 \\ -15593973 & 59783196 & 3850230 & 23901422 \end{pmatrix}. \tag{285}$$

For the residual state $(r, r_{\mathrm{prev}}, r_{\mathrm{prev2}})$, set

$$
\begin{aligned}
(x_{12}, x_{13}, x_{23}) &= \frac{1}{10^9}(-578921373, 27945310, -42117043),\\
P_{\mathrm{res}} &= \begin{pmatrix} a/2+\eta/2-x_{12}-x_{13} & x_{12} & x_{13} \\ x_{12} & -a/2-x_{12}-x_{23} & x_{23} \\ x_{13} & x_{23} & -x_{13}-x_{23} \end{pmatrix}.
\end{aligned}
\tag{286}
$$

Neither uncorrected matrix is assumed positive. For $\mathrm{sym}(N) = (N+N^T)/2$, the complete coupled reconstruction is

$$
\begin{aligned}
T &= \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ -1/8 & q & 0 & 0 & 9/8 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}, \qquad J = (I_4 \;\; 0),\\
e &= (0,0,0,0,1)^T, \quad w = (0,0,-b,-c,1+\eta)^T, \quad \ell = (\lambda_0, 0, \lambda_1, \lambda_2)^T,\\
d(\kappa) &= (9\kappa/8 - 1, -\kappa, 1-\kappa/8, 0)^T, \qquad v(\kappa) = (1, -\kappa q, 0, 0, -1)^T,\\
S(\kappa) &= \mathrm{sym}(wv(\kappa)^T) - a\,\mathrm{sym}\big(e(J^T d(\kappa))^T\big),\\
G(\kappa) &= \mathrm{sym}(d(\kappa)\ell^T), \qquad D(\widetilde{P}, \kappa) = J^T\widetilde{P}J - T^T\widetilde{P}T - S(\kappa).
\end{aligned}
\tag{287}
$$

For the residual reconstruction use

$$
\begin{aligned}
T_{\mathrm{res}} &= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad J_{\mathrm{res}} = (I_3 \;\; 0), \quad e_{\mathrm{res}} = (0,0,0,1)^T,\\
w_{\mathrm{res}} &= (0, -b, -c, 1+\eta)^T, \quad v_{\mathrm{res}} = (1,0,0,-1)^T, \quad v_{\mathrm{prev}} = (-1,1,0,0)^T,\\
d_{\mathrm{res}} &= (-1,1,0)^T, \qquad \ell_{\mathrm{res}} = (\lambda_0, \lambda_1, \lambda_2)^T,\\
S_{\mathrm{res}} &= \mathrm{sym}(w_{\mathrm{res}} v_{\mathrm{res}}^T) - a\,\mathrm{sym}(e_{\mathrm{res}} v_{\mathrm{prev}}^T), \qquad G_{\mathrm{res}} = \mathrm{sym}(d_{\mathrm{res}}\ell_{\mathrm{res}}^T),\\
D_{\mathrm{res}} &= J_{\mathrm{res}}^T P_{\mathrm{res}} J_{\mathrm{res}} - T_{\mathrm{res}}^T P_{\mathrm{res}} T_{\mathrm{res}} - S_{\mathrm{res}}.
\end{aligned}
\tag{288}
$$

Let H be the leading 3×3 block of D_{res} and put $\mathcal{R} = (I_3 \;\; -\mathbf{1}_3)$. Exact expansion yields

$$
D_{\mathrm{res}}\mathbf{1}_4 = 0, \qquad D_{\mathrm{res}} = \mathcal{R}^T H \mathcal{R}, \qquad \mathbf{1}_3^T P_{\mathrm{res}} \mathbf{1}_3 = \frac{\eta}{2}. \tag{289}
$$

The four strict rational inequalities are

$$
\begin{aligned}
A_{\mathrm{cg}} &:= P + G(K) - \delta I_4 \succ 0, \quad A_{\mathrm{cd}} := D(P,K) - \delta I_5 \succ 0,\\
A_{\mathrm{rg}} &:= P_{\mathrm{res}} + G_{\mathrm{res}} - \delta I_3 \succ 0, \qquad A_{\mathrm{rd}} := H - \delta I_3 \succ 0.
\end{aligned}
\tag{290}
$$

Here is a compact exact-arithmetic certificate for every leading principal minor. If $\Delta_j(A) = \det A_{1:j,1:j}$, each displayed exponent e asserts the rational strict inequality $\Delta_j(A) > 10^{-e} > 0$; a dash means that the matrix has smaller order.

A	$j=1$	$j=2$	$j=3$	$j=4$	$j=5$
A_{cg}	1	3	5	6	–
A_{cd}	2	7	12	13	18
A_{rg}	1	3	7	–	–
A_{rd}	1	3	5	–	–

(291)

All fifteen comparisons follow by integer cross-multiplication from (283)–(288); the complete unrounded fractions are supplied in `exact_determinants.json`. For example,

$$\Delta_1(A_{\mathrm{cg}}) = \frac{56772715987}{80000000000} > 10^{-1}.$$

The accompanying exact checker reconstructs these matrices from the rational parameters, evaluates all fifteen determinants, and verifies both their full fractions and (291). Its entry point is `verify_operator.py`. The printed recipes determine every entry without numerical data or an optimization result. Sylvester's criterion now proves

$$\begin{aligned} P + G(K) &\succeq \delta I_4, \qquad & D(P,K) &\succeq \delta I_5, \\ P_{\mathrm{res}} + G_{\mathrm{res}} &\succeq \delta I_3, \qquad & D_{\mathrm{res}} &\succeq \delta \mathcal{R}^T \mathcal{R}. \end{aligned} \tag{292}$$

These are inequalities on unrestricted tuples. Diagonalizing a finite real symmetric matrix lifts each inequality to Hilbert direct sums by replacing scalar squares by Hilbert norms. No zero initial condition or reachable-state restriction is involved.

B.9.2 Zero product and fixed parameter interpolation

Define the fixed embeddings

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad P_0 = E P_{\mathrm{res}} E^T. \tag{293}$$

The zero-product identities are

$$P_0 + G(0) = E(P_{\mathrm{res}} + G_{\mathrm{res}})E^T \succeq 0, \qquad D(P_0, 0) = Z^T D_{\mathrm{res}} Z \succeq 0. \tag{294}$$

For $0 \le s \le 1$ let $\kappa = sK$ and $P_\kappa = (1-s)P_0 + sP$. The transition T is independent of κ, and all scalar parameters, including η and the three λ_j, remain fixed. Thus direct expansion gives the exact affine identities

$$\begin{aligned} P_\kappa + G(\kappa) &= (1-s)[P_0 + G(0)] + s[P + G(K)], \\ D(P_\kappa, \kappa) &= (1-s)D(P_0, 0) + sD(P, K). \end{aligned} \tag{295}$$

Both are bounded below by $s\delta$ times the relevant identity for $0 < s \le 1$. At $s = 0$, the R coordinate is intentionally semidefinite. This interpolation is an independent algebraic property; the actual bounded-L proof below always uses the positive normalization K, even when the actual product is zero.

The intended frequency sign follows from the storage, rather than being used to postulate its existence. On $|z| = 1$ put

$$\xi = \left(z, \frac{z(9-z)}{8(1-qz)}, z^2, z^3, 1 \right)^T, \qquad \xi_{\mathrm{res}} = (z, z^2, z^3, 1)^T. \tag{296}$$

The exact identities

$$\begin{aligned} T\xi = z^{-1}J\xi, \qquad T_{\mathrm{res}}\xi_{\mathrm{res}} = z^{-1}J_{\mathrm{res}}\xi_{\mathrm{res}}, \\ \xi^* S(\kappa)\xi = \mathrm{Re}[M(z)V_\kappa(z)], \qquad \xi_{\mathrm{res}}^* S_{\mathrm{res}}\xi_{\mathrm{res}} = \mathrm{Re}[M(z)(z-1)] \end{aligned} \tag{297}$$

cancel the storage differences. Therefore $-\mathrm{Re}(MV_K) \geq \delta\|\xi\|^2$, and for $0 < \kappa = sK \leq K$ the lower bound is $s\delta\|\xi\|^2$. The residual inequality is

$$-\mathrm{Re}[M(z)(z-1)] \geq \delta \sum_{j=1}^{3} |z^j - 1|^2. \tag{298}$$

Its DC zero is necessary and is not removed by this argument.

B.9.3 The original graph potential and its full boundary terms

The following elementary statement is shared with the frequency interface. It records explicitly the O'Shea–Zames–Falb convex-potential and Fenchel calculation used here; see [9] for that framework.

Lemma 5 (One graph with three taps)**.** *Let F be proper lower-semicontinuous convex on a real Hilbert space, with $F(0) = \min F = 0$. Suppose $v_n \in \partial F(p_n)$ at actual outputs $n \geq 1$. Let $a, b, c \geq 0$, $a + b + c = 1$, and $\eta \geq 0$. Set*

$$F_n = F(p_n), \qquad B_n = F^*(v_n) = \langle p_n, v_n\rangle - F_n \quad (n \geq 1), \tag{299}$$

and define

$$\mathcal{A}_n = aB_n + (b+c)(F_n + F_{n-1}) + cF_{n-2}, \qquad n \geq 3. \tag{300}$$

All these attained values are finite and nonnegative. For $n \geq 4$, put

$$\mathcal{Q}_n = (1+\eta)\langle p_n, v_n\rangle - a\langle p_n, v_{n-1}\rangle - b\langle p_{n-2}, v_n\rangle - c\langle p_{n-3}, v_n\rangle. \tag{301}$$

Then the exact three-gap identity is

$$\begin{aligned} \mathcal{Q}_n - (\mathcal{A}_n - \mathcal{A}_{n-1}) = a\big[F_n - F_{n-1} - \langle v_{n-1}, p_n - p_{n-1}\rangle\big] \\ + b\big[F_{n-2} - F_n - \langle v_n, p_{n-2} - p_n\rangle\big] \\ + c\big[F_{n-3} - F_n - \langle v_n, p_{n-3} - p_n\rangle\big] \\ + \eta(F_n + B_n) \;\geq\; \eta(F_n + B_n). \end{aligned} \tag{302}$$

For any real $\lambda_0, \lambda_1, \lambda_2$, define $\ell_n = \langle v_n, \lambda_0 p_n + \lambda_1 p_{n-1} + \lambda_2 p_{n-2}\rangle$. The exact lower decomposition is

$$\begin{aligned} \mathcal{A}_n - \ell_n = (a - \lambda_0 - \lambda_1 - \lambda_2)B_n + (b + c - \lambda_0)F_n \\ + (b + c - \lambda_1)F_{n-1} + (c - \lambda_2)F_{n-2} \\ + \lambda_1\big[B_n + F_{n-1} - \langle v_n, p_{n-1}\rangle\big] \\ + \lambda_2\big[B_n + F_{n-2} - \langle v_n, p_{n-2}\rangle\big]. \end{aligned} \tag{303}$$

In particular $\mathcal{A}_n \geq \ell_n$ under (689)*; no separate sign assumption on λ_0 is needed.*

Proof. Fenchel equality gives (299); nonnegativity of F and F^* follows from $0 \in \partial F(0)$ and $F(0) = 0$. Substitute $B_j = \langle p_j, v_j \rangle - F_j$ into $\mathcal{Q}_n - \mathcal{A}_n + \mathcal{A}_{n-1}$, and use $a+b+c = 1$. The coefficients of every function value and pairing are precisely those in (302). Each bracket is a subgradient inequality for the same F. Expanding the two Fenchel brackets in (303) gives $\mathcal{A}_n - \ell_n$ term by term; Fenchel's inequality makes those brackets nonnegative. This proves the claims using only actual graph outputs. The term cF_{n-2} in (300) supplies the third boundary contribution and cannot be removed. The first finite potential is $\mathcal{A}_3$ and the first supply comparison is at $n = 4$; no F_0 or v_0 is invoked. ∎

For the fixed coefficients (284), $\lambda_1, \lambda_2 > 0$ and the remaining exact budgets are

$$\begin{aligned} b + c - \lambda_0 &= \frac{93484060391}{3879400000000} > 0, \\ b + c - \lambda_1 &= \frac{218824392967}{19397000000000} > 0, \\ c - \lambda_2 &= \frac{16493}{1000000000} > 0, \\ a - \lambda_0 - \lambda_1 - \lambda_2 &= \frac{102644609643}{19397000000000} > 0. \end{aligned} \tag{304}$$

Thus one triple of scalar coefficients is admissible for the original nonlinear graph and both linear ports.

B.9.4 Actual ports and coercivity before telescoping

Fix any KKT anchor in (57). Set $p_n = x_n - x^*$, $w_n = y_n - y^*$ and use F, v_n from (60). Primal proximal optimality and the KKT condition give

$$v_n \in \partial F(p_n) \quad (n \geq 1), \qquad F(0) = \min F = 0. \tag{305}$$

Every positive-index proximal output has finite f value. In particular Lemma 5 applies even if $f(x_0) = +\infty$. For every actual product satisfying (59), keep the same positive constant K and define

$$\begin{gathered} \mathsf{B} = \sqrt{\tau/K}\, L, \qquad \mathsf{C} = (I - \mathsf{B}^*\mathsf{B})^{1/2}, \qquad u_n = \mathsf{B}p_n, \quad r_n = \mathsf{C}p_n, \\ d_n = \sqrt{\tau/K}\, w_n, \quad R_n = d_n/q, \qquad g_n = u_{n-1} - u_n - Kd_{n-1}, \quad z_n = r_{n-1} - r_n. \end{gathered} \tag{306}$$

The bounded positive square root exists because $\|\mathsf{B}\| \leq 1$. The literal iteration gives, for $n \geq 1$,

$$\begin{gathered} d_n = q[d_{n-1} + (9u_n - u_{n-1})/8], \qquad R_n = -u_{n-1}/8 + qR_{n-1} + 9u_n/8, \\ v_n = \mathsf{C}^* z_n + \mathsf{B}^* g_n, \qquad \langle p_i, v_j \rangle = \langle r_i, z_j \rangle + \langle u_i, g_j \rangle, \\ \|p_n\|^2 = \|r_n\|^2 + \|u_n\|^2. \end{gathered} \tag{307}$$

Indeed $\mathsf{C}^*\mathsf{C} + \mathsf{B}^*\mathsf{B} = I$ and $K\mathsf{B}^* d_{n-1} = \tau L^* w_{n-1}$ prove the primal identities. This is a linear decomposition of pairings in one original graph, not a claim that either port is separately a subgradient graph.

For $n \geq 2$ let

$$s_n = (u_n, R_n, u_{n-1}, u_{n-2}), \qquad \mathfrak{h}_n = (r_n, r_{n-1}, r_{n-2}). \tag{308}$$

The current coupled port has the after-state form $g_n = d(K)^T s_n$. At each descent step $n \geq 4$ the unrestricted before/input tuple is

$$(s_{n-1}, u_n) = (u_{n-1}, R_{n-1}, u_{n-2}, u_{n-3}, u_n),$$

and its after-state is exactly $T(s_{n-1}, u_n) = s_n$. Likewise $T_{\mathrm{res}}(\mathfrak{h}_{n-1}, r_n) = \mathfrak{h}_n$. On these tuples $S(K)$ and S_{res} represent respectively

$$\begin{aligned} &[(1+\eta)u_n - bu_{n-2} - cu_{n-3}] \cdot g_n - a\, u_n \cdot g_{n-1}, \\ &[(1+\eta)r_n - br_{n-2} - cr_{n-3}] \cdot z_n - a\, r_n \cdot z_{n-1}, \end{aligned} \tag{309}$$

where the dots denote the appropriate Hilbert inner products. Their sum is $\mathcal{Q}_n$. In the same way, the two corrected quadratic forms $G(K)$ and G_{res} sum to ℓ_n. All coefficients, including the three graph coefficients, are shared.

Define the actual attained-history energy

$$\mathcal{V}_n = \langle \mathfrak{h}_n, P_{\mathrm{res}} \mathfrak{h}_n \rangle + \langle s_n, P s_n \rangle + \mathcal{A}_n, \qquad n \geq 3. \tag{310}$$

Matrices act as scalar multiples of the identity on each Hilbert coordinate. Before summing any dissipation inequality, the single graph bound $\mathcal{A}_n \geq \ell_n$, the pairing split, and (292) give

$$\mathcal{V}_n \geq \delta(\|\mathfrak{h}_n\|^2 + \|s_n\|^2) \geq \delta(\|p_n\|^2 + \|R_n\|^2) \geq 0. \tag{311}$$

The two actual-history dissipation inequalities and (302) then give, for every $n \geq 4$,

$$\begin{aligned} \mathcal{V}_n - \mathcal{V}_{n-1} \leq\ & -\delta \sum_{j=1}^{3} \|r_{n-j} - r_n\|^2 - \delta(\|s_{n-1}\|^2 + \|u_n\|^2) \\ & - \eta(F_n + B_n). \end{aligned} \tag{312}$$

The initial energy $\mathcal{V}_3$ is finite. Its only graph values are at indices $1, 2, 3$, and its histories are finite actual iterates. Thus the first comparison at $n = 4$ is legitimate, with no fabricated history, zero-state premise, initial subgradient, or assumed convergence.

B.9.5 Summability and the residual DC direction

Summing (312) using (311), then restoring finitely many finite initial terms, gives

$$u, R, d \in \ell^2(\{0, 1, \ldots\}), \qquad z \in \ell^2(\{1, 2, \ldots\}), \qquad \sum_{n=1}^{\infty} (F_n + B_n) < \infty. \tag{313}$$

For clarity, the state dissipation starts with s_3 and the supply dissipation with $n = 4$; all omitted u, R, d, z, F, B terms have actual finite indices. Also p_n is bounded by coercivity.

The formula for g_n is a fixed finite combination of u, d and shifts, so $g \in \ell^2$. Boundedness of B, C gives $v \in \ell^2$. Finally,

$$\|p_n - p_{n-1}\|^2 = \|r_n - r_{n-1}\|^2 + \|u_n - u_{n-1}\|^2 \tag{314}$$

shows that primal increments are square summable. Restoring the fixed positive scalings in (306) proves (61). In particular, $y_n \to y^*$ and $Lx_n \to Lx^*$ strongly. The objective identity is

$$\Phi(x_n) - \Phi(x^*) = \frac{F_n}{\tau} + \frac{19}{2}\|Lp_n\|^2 \geq 0, \qquad n \geq 1, \tag{315}$$

so these gaps are summable as well.

Any two KKT pairs have the same dual component and the same primal image: monotonicity gives

$$0 \leq \langle x^* - x^{**}, -L^*(y^* - y^{**})\rangle = -\frac{1}{19}\|y^* - y^{**}\|^2. \tag{316}$$

It remains to control primal directions in $\ker L$. Fix any KKT primal anchor. The nonnegative decreasing energy $\mathcal{V}_n$ has a finite limit. Equation (313) gives $F_n, B_n \to 0$ and $\mathcal{A}_n \to 0$, while $s_n \to 0$. Since $z_n \to 0$, the three bounded residual histories synchronize:

$$\mathfrak{h}_n - (r_n, r_n, r_n) \longrightarrow 0.$$

For bounded tuples a_n, b_n, the estimate

$$|\langle a_n, P_{\mathrm{res}} a_n\rangle - \langle b_n, P_{\mathrm{res}} b_n\rangle| \leq \|P_{\mathrm{res}}\|\|a_n - b_n\|(\|a_n\| + \|b_n\|)$$

and the exact DC identity (289) imply

$$\mathcal{V}_n - \frac{\eta}{2}\|r_n\|^2 \longrightarrow 0, \qquad \mathcal{V}_n - \frac{\eta}{2}\|x_n - x^*\|^2 \longrightarrow 0. \tag{317}$$

The second limit uses $u_n \to 0$ and the norm split in (307). The coefficient $\eta/2 = 807419/2000000000$ is strictly positive. Hence $\|x_n - x^*\|$ has a limit for every KKT primal anchor. This is the actual energy argument that closes the residual DC direction.

B.9.6 Weak cluster points and whole sequence convergence

The bounded primal sequence has weakly convergent subsequences. If $x_{n_j} \rightharpoonup \widehat{x}$, bounded linearity of L and the already established strong image limit imply $L\widehat{x} = Lx^*$. Convex weak lower semicontinuity and $F_{n_j} \to 0$ give $F(\widehat{x} - x^*) = 0$. Since zero is the minimum of F, $0 \in \partial F(\widehat{x} - x^*)$, or equivalently $-L^*y^* \in \partial f(\widehat{x})$. Therefore every weak cluster point is a KKT primal point with the same y^*.

To see uniqueness of the weak cluster point without a compactness upgrade, let $\widehat{x}, \widetilde{x}$ be two such points. The distance limits in (317) show that

$$\|x_n - \widehat{x}\|^2 - \|x_n - \widetilde{x}\|^2 = 2\langle x_n, \widetilde{x} - \widehat{x}\rangle + \|\widehat{x}\|^2 - \|\widetilde{x}\|^2$$

has a limit. Along subsequences converging weakly to the two points its limits differ by $2\|\widehat{x}-\widetilde{x}\|^2$, forcing equality of the points. If the whole bounded sequence failed to converge weakly to that point, a subsequence outside one weak neighborhood would have a further weakly convergent subsequence, contradicting uniqueness. Hence $x_n \rightharpoonup \bar{x}$ along the whole sequence. In finite-dimensional X, boundedness and uniqueness of the cluster point give norm convergence. This completes the proof of Theorem 6.

The argument includes $L = 0$: it never divides by $\tau\|L\|^2$, keeping $K = 159/50 > 0$ in the normalization. The residual port and stable dual recurrence are therefore both covered in this degenerate case.

B.9.7 Exact comparison and the scope of the repair

For the existing three-history ansatz,

$$R_{\mathrm{ans}}(q) = \min_{\substack{-1\le t\le 1\\ A_q(t)>0}} \rho_q(t), \qquad \rho_q(t) = \frac{8(1-t)^2(1+q^2-2qt)}{qA_q(t)}, \tag{318}$$
$$A_q(t) = (9q+7)t^2 + (14q-34)t + 25q - 21.$$

At the admissible witness $q = 19/20$, $t = -3/5$,

$$A_q(-3/5) = \frac{2596}{125} > 0, \qquad \rho_q(-3/5) = \frac{38944}{12331}. \tag{319}$$

Taking a minimum proves the upper bound on R_{ans} in (62); direct subtraction proves both exact gaps in (63). This argument uses a valid witness, not a numerical minimizer or an unproved monotonicity assertion.

At $k = K$, the sufficient repair bound of (92) is strictly smaller than the present η:

$$0 < \frac{(1 - K/R_3(19/20))h_0}{2400} < \frac{807419}{10^9}. \tag{320}$$

That sufficient frequency estimate is not used in (290); the latter verifies the larger fixed η directly, with the same triple of graph coefficients at both ports. No conclusion is inferred for $K < \tau\|L\|^2 < R_3(19/20)$, other curvatures, nonscalar Q, arbitrary σ, rates, or optimality in a larger multiplier or storage class.

B.10 The product three specialization and exact constants

The product-three statements and displays in this section fix $\kappa_{\mathrm{cert}} = 3$. Their matrices, ports, energies and constants retain precisely that normalization. The separately marked variable-product rational paragraph B.10.3 is the sole exception and uses its stated R-coordinate certificate. Since $\mathcal{R}(q) > 3$ and is continuous, every compact band admits this choice. The general proof in Section 4.1 therefore proves all conclusions here. The formulas below record its exact product-three specialization and the earlier explicit certificate data, rather than introducing a second convergence argument.

B.10.1 The Hilbert-space convergence theorem

Throughout the graph-storage argument, X, Y are real Hilbert spaces, $L : X \to Y$ is bounded linear, $f : X \to (-\infty, +\infty]$ is proper, lower-semicontinuous and convex, and $Q : Y \to Y$ is bounded self-adjoint. Let $c \in Y$, $\tau, \sigma > 0$, and assume

$$0 < mI \preceq Q \preceq MI, \qquad 0 < m \le M < \infty, \qquad \tau\sigma\|L\|^2 \le 3. \tag{321}$$

Use the notation

$$\begin{gathered} A = (I + \sigma Q)^{-1}, \qquad q_- = (1 + \sigma M)^{-1}, \qquad q_+ = (1 + \sigma m)^{-1}, \\ K = [q_-, q_+] \subset (0, 1), \end{gathered} \tag{322}$$

and use the composite objective (1). Suppose a KKT pair (x^*, y^*) satisfying (2) exists. Starting from arbitrary $(x_0, y_0) \in X \times Y$, consider the literal primal-first iteration (3). For any fixed KKT anchor, put

$$\begin{aligned} p_n &= x_n - x^*, \qquad w_n = y_n - y^*, \\ F(p) &= \tau\left[f(x^* + p) - f(x^*) + \langle L^* y^*, p\rangle\right], \\ v_n &= p_{n-1} - p_n - \tau L^* w_{n-1}, \qquad F_n = F(p_n) \quad (n \ge 1). \end{aligned} \tag{323}$$

Theorem 16 (Bounded coercive dual curvature on Hilbert spaces)**.** *Under* (321) *and the KKT condition* (2)*, with the data specified above, the orbit* (3) *is bounded, and there is a KKT pair* (x_∞, y_∞) *such that*

$$x_n \rightharpoonup x_\infty, \qquad y_n \to y_\infty, \qquad Lx_n \to Lx_\infty. \tag{324}$$

The last two convergences are in norm. All KKT pairs have the same dual coordinate and the same L-image. For every KKT anchor,

$$\begin{aligned} &\sum_{n\ge1}\|x_n - x_{n-1}\|^2 < \infty, \qquad \sum_{n\ge0}\|L(x_n - x^*)\|^2 < \infty, \\ &\sum_{n\ge0}\|y_n - y^*\|^2 < \infty, \qquad \sum_{n\ge1}\|v_n\|^2 < \infty, \qquad \sum_{n\ge1} F_n < \infty, \\ &\sum_{n\ge1}\big[\Phi(x_n) - \Phi(x^*)\big] < \infty. \end{aligned} \tag{325}$$

All function gaps displayed here are nonnegative. If X is finite-dimensional, then $x_n \to x_\infty$ in norm, even when Y is infinite-dimensional.

No separability, eigenbasis for Q, compactness of L, commutation of Q with LL^*, primal strong convexity, or finite $f(x_0)$ is assumed. The endpoint $\tau\sigma\|L\|^2 = 3$, singleton bands, and $L = 0$ are included. Without the additional growth assumption below, no general strong primal convergence in infinite dimensions or geometric rate is asserted.

B.10.2 A compact-band graph certificate

We first establish the scalar matrix inequalities used in the Hilbert-space argument. For a square matrix M write $\operatorname{sym} M = (M + M^{\mathsf{T}})/2$. For $0 < q < 1$ set

$$J = [I_4 \;\; 0], \qquad T(q) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -q/8 & 0 & 0 & q & 9q/8 \end{pmatrix},$$
$$c(q) = \begin{pmatrix} 19/8 \\ 5/8 \\ 0 \\ -3/q \end{pmatrix}, \qquad a = \begin{pmatrix} 3 \\ 3 \\ 1 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -3 \\ -1 \end{pmatrix}, \qquad w_\eta = \begin{pmatrix} 0 \\ -2 \\ -1 \\ 0 \\ 10 + \eta \end{pmatrix}. \tag{326}$$

With $e_5 = (0, 0, 0, 0, 1)^{\mathsf{T}}$, define

$$S(q, \eta) = \operatorname{sym}(w_\eta v^{\mathsf{T}}) - 7 \operatorname{sym}\big(e_5 (J^{\mathsf{T}} c(q))^{\mathsf{T}}\big). \tag{327}$$

Thus $c(q)^{\mathsf{T}} T(q) = v^{\mathsf{T}}$, and for $z = (s_0, s_1, s_2, t, h)^{\mathsf{T}}$,

$$z^{\mathsf{T}} S(q, \eta) z = [(10 + \eta)h - 2s_1 - s_2](s_0 - h - 3t) - 7h \left(\frac{19}{8} s_0 + \frac{5}{8} s_1 - \frac{3}{q} t \right).$$

In particular, these matrices retain the original causal state and input coordinates.

Theorem 17 (Compact-band graph certificate)**.** *For every nonempty compact interval $K = [q_-, q_+] \subset (0, 1)$ there is $\eta_* > 0$ such that, for every $0 < \eta \le \eta_*$, there are a continuous symmetric matrix function $P_\eta : K \to \mathbb{R}^{4\times 4}$ and $\delta_\eta > 0$ satisfying, for every $q \in K$,*

$$\begin{aligned} P_\eta(q) + \operatorname{sym}\big(c(q) a^{\mathsf{T}}\big) &\succeq \delta_\eta I_4, \\ J^{\mathsf{T}} P_\eta(q) J - T(q)^{\mathsf{T}} P_\eta(q) T(q) - S(q, \eta) &\succeq \delta_\eta I_5. \end{aligned} \tag{328}$$

The same scalar η and margin δ_η apply throughout K. The function P_η is bounded on K; positivity is required of its graph correction in the first inequality, rather than of P_η itself.

Exact product-three specialization. Theorem 7 and $\mathcal{R}(q) > 3$ supply the certificate. The exact identities below specialize its construction at $r = 3$. Here the legacy scaled dual coordinate is denoted by r in the state (s_0, s_1, s_2, r) only; it satisfies $t = qr$ and is not the stepsize parameter of the preceding section. Explicitly, $s = (s_0, s_1, s_2, r)^{\mathsf{T}}$ and $\widetilde{z} = (s, h)$. Write $G = \operatorname{sym}(\widetilde{c} a^{\mathsf{T}})$, $R_\eta = \widetilde{S}_\eta + J^{\mathsf{T}} G J - \widetilde{T}^{\mathsf{T}} G \widetilde{T}$, $H = \widetilde{P} + G$, and $Q_1 = \operatorname{sym}(e_5 \widetilde{v}^{\mathsf{T}})$. Also $\widetilde{T} = [F_0 \;\; B_0]$, $B_0 = (1, 0, 0, 9/8)^{\mathsf{T}}$.

Scaling and a completed square.

$$\widetilde{T} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1/8 & 0 & 0 & q & 9/8 \end{pmatrix}, \qquad \widetilde{c} = \begin{pmatrix} 19/8 \\ 5/8 \\ 0 \\ -3 \end{pmatrix}, \qquad \widetilde{v} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -3q \\ -1 \end{pmatrix}, \tag{329}$$

$$\widetilde{S}_\eta = \operatorname{sym}(w_\eta \widetilde{v}^{\mathsf{T}}) - 7\operatorname{sym}\big(e_5 (J^{\mathsf{T}}\widetilde{c})^{\mathsf{T}}\big), \qquad \widetilde{c}^{\mathsf{T}}\widetilde{T} = \widetilde{v}^{\mathsf{T}}.$$

$$G = \operatorname{sym}(\widetilde{c}a^{\mathsf{T}}), \qquad R_\eta = \widetilde{S}_\eta + J^{\mathsf{T}}GJ - \widetilde{T}^{\mathsf{T}}G\widetilde{T}.$$

$$H \succ 0, \qquad J^{\mathsf{T}}HJ - \widetilde{T}^{\mathsf{T}}H\widetilde{T} - R_\eta \succ 0. \tag{330}$$

$$b = \begin{pmatrix} -53/16 \\ -11/16 \\ 1/2 \\ 21(1-q)/2 \end{pmatrix}, \qquad k = \begin{pmatrix} 101 \\ 59 \\ 8 \\ 168(q-1) \end{pmatrix}, \tag{331}$$

$$e = h - \frac{b^{\mathsf{T}}s}{7}, \qquad y = \frac{k^{\mathsf{T}}s}{112}.$$

$$\widetilde{z}^{\mathsf{T}} R_0 \widetilde{z} = 7(y^2 - e^2). \tag{332}$$

Removing the constant-state direction. Set $x = K_q s$ and $h = e + b^{\mathsf{T}}s/7$; here x denotes the three-dimensional quotient state, not the primal iterate.

$$K_q = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & q-1 \end{pmatrix}, \qquad d(q) = \begin{pmatrix} 1 \\ 1 \\ 1 \\ (1-q)^{-1} \end{pmatrix}. \tag{333}$$

$$x^+ = \overline{A}(q)x + \overline{B}(q)e, \qquad y = \overline{C}x,$$

$$\overline{A}(q) = \begin{pmatrix} 3/112 & -1/14 & -3/2 \\ 1 & 0 & 0 \\ (27q-3)/896 & (1-9q)/112 & (3-11q)/16 \end{pmatrix}, \tag{334}$$

$$\overline{B}(q) = \begin{pmatrix} 1 \\ 0 \\ (9q-1)/8 \end{pmatrix}, \qquad \overline{C} = \begin{pmatrix} -67/112 & -1/14 & 3/2 \end{pmatrix}.$$

Schur stability and the strict frequency gap. The actual characteristic polynomial of $\overline{A}(q)$ is $D_q/112$, and the transfer from e to y is N_q/D_q, before any cancellation. In the following frequency formulas $|\lambda| = 1$ and $\xi = \operatorname{Re}\lambda \in [-1, 1]$.

$$\begin{aligned} D_q(\lambda) &= 112\lambda^3 + (77q-24)\lambda^2 + (3q+8)\lambda - 8q, \\ N_q(\lambda) &= (189q-88)\lambda^2 + (67q-8)\lambda + 8q. \end{aligned} \tag{335}$$

$$\begin{aligned} |D_q(\lambda)|^2 - |N_q(\lambda)|^2 = {} & 3584(1-q)^2(\xi-1)^2 \\ & - 448q(1-q)(16\xi^3 - 27\xi^2 - 54\xi - 79) \\ & + 1792q^2(1-\xi)(2\xi+1)^2. \end{aligned} \tag{336}$$

$$|D_q(\lambda)|^2 - |N_q(\lambda)|^2 \geq 4032q(1-q) > 0 \qquad (0 < q < 1). \tag{337}$$
$$\sup_{|\lambda|=1} |N_q(\lambda)/D_q(\lambda)| < 1.$$

Continuous selection on the band. Choose positive reduced storages locally by Lemma 4. The finite partition-of-unity construction in the shared proof gives a continuous positive $Z(q)$ and a common strict margin $\varepsilon_K > 0$.

$$\begin{aligned} x^{\mathsf{T}} Z(q) x - (\overline{A}(q)x + \overline{B}(q)e)^{\mathsf{T}} Z(q)(\overline{A}(q)x + \overline{B}(q)e) \\ - |\overline{C}x|^2 + |e|^2 \geq \varepsilon_K(\|x\|^2 + |e|^2) \qquad (q \in K). \end{aligned} \tag{338}$$

The two exact kernels. Write $z_0(q) = (d(q), 1)^{\mathsf{T}}$. The following exact matrices have kernels $\operatorname{span}\{d(q)\}$ and $\operatorname{span}\{z_0(q)\}$, respectively, as in the shared proof.

$$H_0(q) = 7K_q^{\mathsf{T}} Z(q) K_q, \qquad W_q = \begin{pmatrix} K_q & 0 \\ -b^{\mathsf{T}}/7 & 1 \end{pmatrix}, \tag{339}$$
$$D_0(q) = J^{\mathsf{T}} H_0(q) J - \widetilde{T}^{\mathsf{T}} H_0(q) \widetilde{T} - R_0.$$

$$D_0(q) \succeq 7\varepsilon_K W_q^{\mathsf{T}} W_q. \tag{340}$$

$$\widetilde{T} z_0(q) = J z_0(q) = d(q), \qquad R_0 z_0(q) = 0.$$

$$w^{\mathsf{T}} D_0(q) w \geq \gamma_K \|w\|^2 \quad \text{whenever } w \perp z_0(q), \quad q \in K. \tag{341}$$

One common positive multiplier. Use $H_\eta = H_0 + \eta I_4$, $\widetilde{P}_\eta = H_\eta - G$, $D_1 = J^{\mathsf{T}} J - \widetilde{T}^{\mathsf{T}} \widetilde{T} - Q_1$, $D_\eta = D_0 + \eta D_1$, and $e_0 = z_0/\|z_0\|$. For the estimate below decompose $z = \alpha e_0 + w$, $w \perp e_0$.

$$H_\eta(q) = H_0(q) + \eta I_4, \qquad \widetilde{P}_\eta(q) = H_\eta(q) - G, \qquad D_1(q) = J^{\mathsf{T}} J - \widetilde{T}^{\mathsf{T}} \widetilde{T} - Q_1.$$

$$e_0(q)^{\mathsf{T}} D_1(q) e_0(q) = \frac{3q(1-q)}{4(1-q)^2 + 1} > 0. \tag{342}$$

$$\begin{aligned} m_K = \min_{q \in K} \frac{3q(1-q)}{4(1-q)^2+1} > 0, \qquad L_K = \max\left\{1, \max_{q \in K} \|D_1(q)\|\right\}, \\ \eta_* = \min\left\{\frac{\gamma_K}{2L_K}, \frac{\gamma_K m_K}{8L_K^2}\right\} > 0. \end{aligned} \tag{343}$$

$$\begin{aligned} z^{\mathsf{T}} D_\eta(q) z &\geq \gamma_K \|w\|^2 + \eta m_K \alpha^2 - 2\eta L_K |\alpha| \|w\| - \eta L_K \|w\|^2 \\ &\geq \frac{\gamma_K}{4} \|w\|^2 + \frac{\eta m_K}{2} \alpha^2. \end{aligned}$$

$$\delta_\eta = \min\{\eta, \gamma_K/4, \eta m_K/2\} > 0. \tag{344}$$

Returning to the original coordinates. Let $U_q = \operatorname{diag}(1,1,1,1/q)$ and $\widehat{U}_q = \operatorname{diag}(U_q, 1)$.

$$P_\eta(q) = U_q^{\mathsf{T}} \widetilde{P}_\eta(q) U_q.$$

$$\begin{aligned} P_\eta(q) + \operatorname{sym}(c(q)a^{\mathsf{T}}) &= U_q^{\mathsf{T}} H_\eta(q) U_q, \\ J^{\mathsf{T}} P_\eta(q) J - T(q)^{\mathsf{T}} P_\eta(q) T(q) - S(q,\eta) &= \widehat{U}_q^{\mathsf{T}} D_\eta(q) \widehat{U}_q. \end{aligned}$$

The cubic frequency term is at most $16 + 54 - 79 = -9$ on $[-1,1]$. At $q = 0$ the actual characteristic polynomial is $8\lambda(14\lambda^2 - 3\lambda + 1)$, with root moduli $0, 1/\sqrt{14}$. The displayed strict gap prevents unit-circle crossings for $0 < q < 1$, including the nonminimal point $q = 1/9$. The common bounded-real lemma gives local positive Z; the finite partition-of-unity construction in the general proof gives continuous $Z(q)$ and the displayed ε_K. The exact kernels are $\ker H_0 = \operatorname{span}\{d(q)\}$ and $\ker D_0 = \operatorname{span}\{z_0(q)\}$, with $z_0 = (d(q), 1)$, $b^{\mathsf{T}} d = 7$ and $R_0 z_0 = 0$. In the perturbation formulas $H_\eta = H_0 + \eta I_4$, $\widetilde{P}_\eta = H_\eta - G$, $D_1 = J^{\mathsf{T}} J - \widetilde{T}^{\mathsf{T}} \widetilde{T} - Q_1$, $D_\eta = D_0 + \eta D_1$, and $e_0 = z_0/\|z_0\|$. Decompose $z = \alpha e_0 + w$, $w \perp e_0$. The displayed Young estimate proves the retained exact η_* and δ_η for every $0 < \eta \le \eta_*$, not just for one value. Finally $U_q = \operatorname{diag}(1,1,1,1/q)$ and $\widehat{U}_q = \operatorname{diag}(U_q, 1)$ have $U_q^{\mathsf{T}} U_q \succeq I$, $\widehat{U}_q^{\mathsf{T}} \widehat{U}_q \succeq I$; the congruences preserve the margin. Since $q_- > 0$, the original-coordinate P_η remains continuous and bounded. These are precisely the specializations of the general construction, with all product-three coefficients displayed explicitly. ∎

Remark (Dependence on the spectral band). The frequency lower bound (337) and the kernel slope (342) degenerate at the endpoints. The theorem asserts no uniform positive margin up to $q = 0$ or $q = 1$. In fact one fixed $\eta > 0$ cannot satisfy this supply certificate for all q arbitrarily close to one. To see this directly in the original coordinates, let $|\lambda| = 1$ and set

$$\zeta = \left(\lambda^{-1}, \lambda^{-2}, \lambda^{-3}, \frac{q(9\lambda - 1)}{8\lambda(\lambda - q)}, 1\right)^{\mathsf{T}}.$$

Then $T(q)\zeta = \lambda J \zeta$. Applying the real symmetric inequality (328) to real and imaginary parts therefore requires $\zeta^* S(q,\eta)\zeta < 0$. At $\operatorname{Re}\lambda = -1/2$, direct substitution gives

$$\zeta^* S(q,\eta)\zeta = \frac{3}{16(1 + q + q^2)}\big(11\eta q^2 - 8\eta + 171q^2 - 99q - 72\big).$$

For $q > \sqrt{8/11}$, necessity of a negative sign implies

$$\eta < \frac{9(1-q)(19q+8)}{11q^2 - 8}, \tag{345}$$

whose right-hand side tends to zero as $q \uparrow 1$. This is compatible with a common positive η on each fixed compact band.

B.10.3 Operator lift and the attained-history energy

Specialization of the shared Hilbert proof. Apply Theorem 4 with $\kappa_{\mathrm{cert}} = 3$ and the certificate of Theorem 17. In the following record $P = P_\eta$ is in the normalized dual coordinate t; for a continuous matrix function N, $N[A]$ denotes its entrywise continuous functional calculus and $R(q) = N(q)^{1/2}$ when $N \succeq 0$. Thus the operator positivity and row-sum bound are exactly those proved there. Write $z_n = r_{n-1} - r_n$ and $g_n = u_{n-1} - u_n - 3t_{n-1}$. The graph function F is always the single tilted function in (323). Every displayed identity and estimate below is the literal $\kappa_{\mathrm{cert}} = 3$ substitution in that shared proof.

Continuous functional calculus.

$$N[A] = (N_{ij}(A))_{ij}$$

$$N[A] = R[A]^* R[A] \succeq 0. \tag{346}$$

$$M_P = \sup_{q \in K} \max_i \sum_j |P(q)_{ij}|. \tag{347}$$

One convex graph and its actual histories.

$$v_n \in \partial F(p_n), \qquad F_n < \infty \quad (n \geq 1). \tag{348}$$

$$\begin{gathered} B = \sqrt{\tau\sigma/3}\, L, \qquad C = (I - B^*B)^{1/2}, \qquad B^*B + C^*C = I, \\ u_n = Bp_n, \qquad r_n = Cp_n, \qquad t_n = \sqrt{\tau/(3\sigma)}\, w_n. \end{gathered} \tag{349}$$

$$t_n = A\left(t_{n-1} + \frac{9u_n - u_{n-1}}{8}\right). \tag{350}$$

$$v_n = C^* z_n + B^* g_n, \qquad \langle p_i, v_j \rangle = \langle r_i, z_j \rangle + \langle u_i, g_j \rangle. \tag{351}$$

$$\begin{gathered} s_n = (u_n, u_{n-1}, u_{n-2}, t_n), \\ g_n = \frac{19}{8} u_n + \frac{5}{8} u_{n-1} - 3A^{-1} t_n = c[A]^* s_n, \qquad s_n = T[A](s_{n-1}, u_n). \end{gathered} \tag{352}$$

Quadratic dissipation.

$$\begin{gathered} P_{\mathrm{res}} = \begin{pmatrix} (19+\eta)/2 & -7 & 1 \\ -7 & 5 & -3/2 \\ 1 & -3/2 & 1/2 \end{pmatrix}, \qquad R_n = (r_n, r_{n-1}, r_{n-2}), \\ E_n = \langle R_n, (P_{\mathrm{res}} \otimes I) R_n \rangle + \langle s_n, P[A] s_n \rangle. \end{gathered} \tag{353}$$

$$\mathcal{Q}_n = \langle (10+\eta) p_n - 2p_{n-2} - p_{n-3}, v_n \rangle - 7 \langle p_n, v_{n-1} \rangle. \tag{354}$$

$$\begin{aligned} &\langle R_n, (P_{\mathrm{res}} \otimes I) R_n \rangle - \langle R_{n-1}, (P_{\mathrm{res}} \otimes I) R_{n-1} \rangle \\ &\quad + \langle (10+\eta) r_n - 2r_{n-2} - r_{n-3}, z_n \rangle - 7 \langle r_n, z_{n-1} \rangle \\ &= -\frac{1}{2} \| r_n - 3r_{n-1} + 3r_{n-2} - r_{n-3} \|^2 - \frac{\eta}{2} \| r_n - r_{n-1} \|^2. \end{aligned} \tag{355}$$

$$E_n - E_{n-1} + \mathcal{Q}_n \le -D_n, \qquad n \ge 4, \tag{356}$$

$$\begin{aligned} D_n = \frac{1}{2}\|r_n - 3r_{n-1} + 3r_{n-2} - r_{n-3}\|^2 + \frac{\eta}{2}\|r_n - r_{n-1}\|^2 \\ + \delta\big(\|s_{n-1}\|^2 + \|u_n\|^2\big). \end{aligned} \tag{357}$$

The convex-potential correction.

$$G_n = 7\langle p_n, v_n\rangle - 4F_n + 3F_{n-1} + F_{n-2}, \qquad V_n = E_n + G_n, \qquad n \ge 3. \tag{358}$$

$$\begin{aligned} &\mathcal{Q}_n - (G_n - G_{n-1}) - \eta\langle p_n, v_n\rangle \\ &= 7\big[F_n - F_{n-1} - \langle v_{n-1}, p_n - p_{n-1}\rangle\big] \\ &\quad + 2\big[F_{n-2} - F_n - \langle v_n, p_{n-2} - p_n\rangle\big] \\ &\quad + \big[F_{n-3} - F_n - \langle v_n, p_{n-3} - p_n\rangle\big] \ge 0 \qquad (n \ge 4) \end{aligned} \tag{359}$$

$$\begin{aligned} F^*(v_n) &= \langle p_n, v_n\rangle - F_n, \\ G_n &= 7F^*(v_n) + 3F_n + 3F_{n-1} + F_{n-2} \\ &= 7F^*(v_n) + 2(F_n + F_{n-1}) + (F_n + F_{n-1} + F_{n-2}). \end{aligned} \tag{360}$$

Coercivity on attained graph histories.

$$\begin{aligned} &G_n - \langle v_n, 3p_n + 3p_{n-1} + p_{n-2}\rangle \\ &= 3\big[F_{n-1} - F_n - \langle v_n, p_{n-1} - p_n\rangle\big] \\ &\quad + \big[F_{n-2} - F_n - \langle v_n, p_{n-2} - p_n\rangle\big] \ge 0. \end{aligned} \tag{361}$$

$$\begin{aligned} \langle R_n, (P_{\mathrm{res}} \otimes I)R_n\rangle + \langle z_n, 3r_n + 3r_{n-1} + r_{n-2}\rangle = \frac{\eta}{2}\|r_n\|^2 + \langle R_n, (H_{\mathrm{res}} \otimes I)R_n\rangle, \\ H_{\mathrm{res}} = \begin{pmatrix} 13/2 & -7 & 1/2 \\ -7 & 8 & -1 \\ 1/2 & -1 & 1/2 \end{pmatrix} \succeq 0. \end{aligned} \tag{362}$$

$$\langle s_n, P[A]s_n\rangle + \langle g_n, 3u_n + 3u_{n-1} + u_{n-2}\rangle \ge \delta\|s_n\|^2.$$

$$V_n \ge \frac{\eta}{2}\|r_n\|^2 + \delta\|s_n\|^2 \ge \beta\big(\|p_n\|^2 + \|t_n\|^2\big), \qquad \beta = \min\{\eta/2, \delta\} > 0. \tag{363}$$

$$V_n - V_{n-1} \le -D_n - \eta F_n, \qquad n \ge 4. \tag{364}$$

Startup, boundedness, and summable gaps and residuals.

$$\sup_{n\ge 3}(\|p_n\|^2 + \|t_n\|^2) \le V_3/\beta, \qquad \sum_{n\ge 4} D_n \le V_3, \qquad \sum_{n\ge 4} F_n \le V_3/\eta. \tag{365}$$

$$\|p_n - p_{n-1}\|^2 = \|r_n - r_{n-1}\|^2 + \|u_n - u_{n-1}\|^2 \tag{366}$$

$$\tau[\Phi(x_n) - \Phi(x^*)] = F_n + \frac{3}{2}\langle(\sigma Q)^{-1}u_n, u_n\rangle, \qquad n \ge 1. \tag{367}$$

$$0 \preceq (\sigma Q)^{-1} \preceq bI, \qquad b = \frac{q_+}{1 - q_+} = \frac{1}{\sigma m}, \tag{368}$$

Distance limits for every KKT anchor.

$$V_n - \frac{\eta}{2}\|p_n\|^2 \longrightarrow 0. \tag{369}$$

Weak cluster points and whole-sequence convergence.

$$0 \le F(\bar{p}) \le \liminf_k F_{n_k} = 0.$$

$$0 \le \langle x^1 - x^2, -L^*(y^1 - y^2)\rangle = -\langle Q(y^1 - y^2), y^1 - y^2\rangle.$$

$$\|x_n - a\|^2 - \|x_n - b\|^2 = 2\langle x_n, b - a\rangle + \|a\|^2 - \|b\|^2$$

The residual lower matrix has eigenvalues $0, (15\pm 3\sqrt{21})/2$ and constant-history kernel. Its storage entries sum to $\eta/2$. The functional-calculus, single-graph, coercivity, telescoping, every-anchor distance-limit and weak-cluster arguments of the shared proof apply with these exact objects. In particular V_3 uses only outputs $1, 2, 3$ and descent starts at 4; no F_0 or artificial graph point is introduced. The dual and L-image are the same at every KKT pair. This gives every series and convergence in the statement, including singleton bands and $L = 0$, without primal growth. ∎

An exactly evaluable finite-band specialization. If $\frac{1}{19}I \preceq \sigma Q \preceq I$, then $\operatorname{spec}(A) \subset [1/2, 19/20]$. On this band one can take $\eta = 1/10$ and $\delta = 1/200$: the companion rational certificate `certificates/graph_interval_18_piece.json` contains nineteen knot matrices defining eighteen continuous affine pieces. Its exact Bernstein check proves (328) on every piece, using 144 positive coefficient matrices. The same Hilbert proof applies without change. This explicit special case neither replaces the analytic arbitrary-band argument nor requires a second convergence proof. Since P is affine on each piece, a verified M_P is the maximum absolute row sum of the knot matrices, namely 1442367203/50000000.

An exact variable-product interval at fixed curvature. A separate rational certificate has $q = 1/2$ and $3 \le r \le 479/100$, with $\eta = 1/10$ and $\delta = 1/100$ in the R coordinate of (69). Its two endpoint matrices are stored in `certificates/curvature_q_half_interval.json`. Interpolate P_R affinely in r. The transition T_R is independent of r, while G_R and S_R are affine in r; hence both certificate matrices are affine in r. Exact rational positive-pivot or principal-minor tests after subtracting δI at the two endpoints prove the entire closed interval. The coordinate congruence (209) preserves the lower margin. This dataset is independent of the preceding 18-piece q-band certificate, whose different margin and original matrices are unchanged. The analytic proof, rather than extrapolation of these data, establishes the larger full open interval $0 < r < 24/5$.

C Quantitative estimates and affine dynamics

C.1 Complete scalar-cap constants and actual restarts

This appendix proves Theorem 9. We retain the physical error $w_n = y_n - y_* = qR_n$ and $e_n^2 = \|p_n\|^2 + \|w_n\|^2$ throughout. In particular a normalized state containing R_n is not silently identified with the physical pair. Use the disjoint branch rule (34); within B choose the square form for $q \le q_0$ and the cross form for $q > q_0$. Both are valid at q_0, but this convention chooses one constant.

A homogeneous actual-output bound. The translated proximal input is $a_n = p_{n-1} - rw_{n-1}$, and $p_n = \mathrm{prox}_F(a_n)$, $a_n = p_n + v_n$. Since $\langle p_n, v_n\rangle \ge 0$ by the graph anchor,

$$\|p_n\|^2 + \|v_n\|^2 \le \|a_n\|^2 \le (1+r^2)e_{n-1}^2, \qquad n \ge 1. \tag{370}$$

Fenchel equality then gives, with $H = (1+r^2)/2$,

$$0 \le F_n + B_n = \langle p_n, v_n\rangle \le He_{n-1}^2. \tag{371}$$

The actual dual equation and the proximal nonexpansiveness bound imply componentwise

$$\begin{pmatrix} \|p_n\| \\ \|w_n\| \end{pmatrix} \le \begin{pmatrix} 1 & r \\ 5q/4 & q(1+9r/8) \end{pmatrix} \begin{pmatrix} \|p_{n-1}\| \\ \|w_{n-1}\| \end{pmatrix}.$$

Its squared Frobenius norm yields the convenient nonoptimal constant

$$M = 1 + r^2 + q^2\left\{\frac{25}{16} + (1+9r/8)^2\right\} > 1, \qquad e_n^2 \le Me_{n-1}^2 \le M^n e_0^2. \tag{372}$$

All three inequalities use only the preceding pair and the next actual output. Thus they are uniform over $f, c, \mathcal{H}$, initialization and restart. They never evaluate F_0.

Branch F and the strict four-state certificate. For $q < 7/10$ or $r < 3$, the coverage argument in Theorem 2 ensures $r < \mathcal{R}(q)$. The strict certificate of Theorem 7 is as follows:

$$\begin{gathered} T_F = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1/8 & 0 & 0 & q & 9/8 \end{pmatrix}, \qquad J_F = [I_4 \;\; 0], \\ c_F = (9r/8 - 1, 1 - r/8, 0, -r)^\top, \qquad a_g = (3,3,1,0)^\top, \\ v_F = (1,0,0,-rq,-1)^\top, \qquad e_5 = (0,0,0,0,1)^\top, \\ \ell_\eta = (0,-2,-1,0,10+\eta)^\top, \qquad G_F = \mathrm{sym}(c_F a_g^\top), \\ Q_\eta = \mathrm{sym}(\ell_\eta v_F^\top) - 7\,\mathrm{sym}\big(e_5(J_F^\top c_F)^\top\big). \end{gathered} \tag{373}$$

Choose any $\eta > 0$ and symmetric P_F with

$$P_F + G_F \succ 0, \qquad D_F = J_F^\top P_F J_F - T_F^\top P_F T_F - Q_\eta \succ 0. \tag{374}$$

Set $\delta = \lambda_{\min}(D_F) > 0$ and $m_F = \|P_F\|_{\mathrm{F}}$, the finite scalar Frobenius norm. A certified positive lower bound for D_F or upper bound for $\|P_F\|$ may replace these values. They depend on the selected certificate, not on the function or trajectory. Analytic certificate existence does not by itself specify a certificate-free numerical rate.

For $s_n^F = (p_n, p_{n-1}, p_{n-2}, R_n)$, the actual energy is

$$V_n = \langle s_n^F, P_F s_n^F \rangle + \mathcal{A}_n^F, \qquad \mathcal{A}_n^F = 7B_n + 3F_n + 3F_{n-1} + F_{n-2}, \quad n \geq 3. \tag{375}$$

Its graph lower bound is explicit:

$$\begin{aligned} \mathcal{A}_n^F - \langle s_n^F, G_F s_n^F \rangle = {} & 3\big(F_{n-1} + B_n - \langle p_{n-1}, v_n \rangle\big) \\ & + \big(F_{n-2} + B_n - \langle p_{n-2}, v_n \rangle\big) \geq 0. \end{aligned}$$

Hence $V_n \geq 0$ by (374). For completeness the actual supply in Q_η is

$$\big\langle (10+\eta)p_n - 2p_{n-2} - p_{n-3}, v_n \big\rangle - 7\langle p_n, v_{n-1} \rangle.$$

Subtract $\mathcal{A}_n^F - \mathcal{A}_{n-1}^F$ from it. The result is $7\mathcal{C}_n + 2\mathcal{G}_{n,2} + \mathcal{G}_{n,3} + \eta\langle p_n, v_n \rangle$, where

$$\mathcal{G}_{n,j} = F_{n-j} - F_n + \langle p_n - p_{n-j}, v_n \rangle \geq 0 \quad (j = 2, 3)$$

and $\mathcal{C}_n$ is the actual conjugate gap in (166). These are all attained values when $n \geq 4$. Since $\langle p_n, v_n \rangle = F_n + B_n \geq F_n$, the certificate yields

$$V_n - V_{n-1} \leq -\delta\big(\|s_{n-1}^F\|^2 + \|p_n\|^2\big) - \eta F_n, \qquad n \geq 4. \tag{376}$$

This is the identity specialization of the three-history graph proof, with its positive-output startup made quantitative next.

By (372) and $w_3 = qR_3$,

$$\|s_3^F\|^2 \leq q^{-2}e_3^2 + e_2^2 + e_1^2 \leq (q^{-2}M^3 + M^2 + M)e_0^2. \tag{377}$$

Moreover $7B_3 + 3F_3 \leq 7(B_3 + F_3)$, so

$$\mathcal{A}_3^F \leq H(7M^2 + 3M + 1)e_0^2.$$

Define

$$\begin{aligned} K_F &= m_F(q^{-2}M^3 + M^2 + M) + H(7M^2 + 3M + 1), \\ C_F &= 1 + M + M^2 + K_F/\delta > 1. \end{aligned} \tag{378}$$

Thus $0 \leq V_3 \leq K_F e_0^2$ without any initial function value. Telescoping (376) from $n = 4$ gives

$$\sum_{j \geq 3} e_j^2 \leq \sum_{j \geq 3} \|s_j^F\|^2 \leq V_3/\delta.$$

The first inequality uses $q < 1$ and the physical scaling $w_j = qR_j$. Adding the genuine pair prefixes $0, 1, 2$ from (372) proves $\sum_{j \geq 0} e_j^2 \leq C_F e_0^2$.

Branches M and B and their specified matrices. Both use the common A, d, v, v^- in (158) and the finite rows (159); their denominators are nonzero because $q \geq 7/10$. They use different taps and dissipations. For M, namely $7/10 \leq q \leq q^\dagger$ and $3 \leq r < R_{\mathrm{lin}}(q)$, set

$$\begin{aligned} R_{\mathrm{top}} &= R_{\mathrm{lin}}(q), & a &= \frac{5q}{9q-1}, & b &= \frac{4q-1}{9q-1}, \\ c_1 &= \frac{-237q^3+179q^2+50q-16}{128(9q-1)}, \\ c_2 &= \frac{-136q^3+120q^2+13q-8}{64(9q-1)}, \\ c_3 &= \frac{5q(q+1)(8-7q)}{128(9q-1)}, & c_e &= \frac{6-3q-4q^2}{10}, \\ w_e &= h + \frac{9-q}{9q-1}p + \frac{8(q^2-1)}{9q-1}R, \\ D_3 &= c_1W_1^2 + c_2W_2^2 + c_3W_3^2, & D_{\mathrm{top}} &= c_e w_e^2. \end{aligned} \tag{379}$$

Here $c_1, c_2, c_3 > 0$, while $c_e \geq 0$ and equals zero at $q = q^\dagger$. The graph budgets, matrix positivity and actual supply are precisely those of (133)–(136), with all their endpoint signs proved in Appendix A.2. For B, set $R_{\mathrm{top}} = R_3(q)$ and use the taps (150), coefficients (153), $D_{\mathrm{top}} = c_e w_c^2$, and the square or cross D_3 from (160) according to the stated split. No branch uses the other's sign of g or its storage matrix.

The following formulas specify the matrices in all three mixed choices:

$$\begin{aligned} \mathcal{Q}_r &= hv - ahv^- - bp_2 v, \\ S_j &= \tfrac{1}{2}\nabla^2_{(p,R,p_2)}(\mathcal{Q}_j + D_j), & j &= 3, R_{\mathrm{top}}, \\ P_j &= S_j + A^\top S_j A + (A^2)^\top S_j A^2/(1-q^2), \\ \theta_m &= (r-3)/(R_{\mathrm{top}}-3), \qquad P = (1-\theta_m)P_3 + \theta_m P_{\mathrm{top}}, \qquad m_P = \|P\|_{\mathrm{F}}. \end{aligned} \tag{380}$$

Use the explicitly positive constants

$$\gamma = \begin{cases} \min(c_1, c_3), & \text{M or square B}, \\ \det(G_c)/\operatorname{tr}(G_c), & \text{cross B}, \end{cases} \qquad d_{\mathrm{mix}} = (1-\theta_m)\gamma > 0. \tag{381}$$

For cross B, the eigenvalue argument following (177) justifies the rational bound. Thus $D_3 \geq \gamma(\|W_1\|^2 + \|W_3\|^2)$ in every case. The respective actual graph proofs give

$$\begin{aligned} E_n &= \langle s_n, Ps_n\rangle + aB_n + b(F_n + F_{n-1}) \geq 0, \quad s_n = (p_n, R_n, p_{n-1}), \quad n \geq 2, \\ E_n - E_{n-1} &\leq -(1-\theta_m)D_{3,n} - \theta_m D_{\mathrm{top},n}, \qquad n \geq 3. \end{aligned} \tag{382}$$

The lower bound comes from graph correction and remains valid if P is indefinite. At $q = q^\dagger$, M retains $d_{\mathrm{mix}} > 0$ for every strict product, despite its zero endpoint coefficient c_e.

Uniform E_2 startup. The physical norm scaling and the prefix estimates give

$$\|s_2\|^2 \le (q^{-2}M^2 + M)e_0^2, \qquad aB_2 + bF_2 + bF_1 \le H\{\max(a,b)M + b\}e_0^2.$$

Define

$$K_E = m_P(q^{-2}M^2 + M) + H\{\max(a,b)M + b\}. \tag{383}$$

Then $0 \le E_2 \le K_E e_0^2$. Its values are B_2, F_2, F_1, not independent starting graph data. Telescoping (382) yields

$$\sum_{n\ge 3}(\|W_{1,n}\|^2 + \|W_{3,n}\|^2) \le \frac{K_E}{d_{\mathrm{mix}}}e_0^2. \tag{384}$$

Quantitative DC closure and all prefixes. The actual lag identity (180) applies to M and B because their rows coincide. Its index $n = 3$ includes the actual $w_{1,2}$, and the two-term norm inequality gives

$$K_w = \frac{K_E}{8d_{\mathrm{mix}}}, \qquad \sum_{n\ge 2}\|w_{1,n}\|^2 \le K_w e_0^2. \tag{385}$$

Indeed $16\|w_{1,n-1}\|^2 \le 2(\|W_{1,n}\|^2 + \|W_{3,n}\|^2)$. Here the filter output $w_{1,n}$ is distinguished from the physical dual error w_n.

Put $\kappa_{\mathrm{DC}} = rq/(1-q) > 0$. The actual identity (181) and monotonicity at the anchor give $\|p_n\| \le \|z_n\|/\kappa_{\mathrm{DC}}$ for $n \ge 3$. Consequently, defining

$$\beta_{\mathrm{DC}} = \frac{2\{(\kappa_{\mathrm{DC}} - 1)^2 + q^2(1 - r/8)^2\}}{\kappa_{\mathrm{DC}}^2}, \qquad K_p = \beta_{\mathrm{DC}}K_w, \tag{386}$$

one obtains

$$\sum_{n\ge 3}\|p_n\|^2 \le K_p e_0^2. \tag{387}$$

No convergence rate has been assumed to derive this DC estimate.

Multiplying (182) by q gives the physical rather than normalized dual reconstruction

$$\begin{aligned} w_{n-1} &= A_y\big[p_n + b_y p_{n-1} - w_{1,n}\big], \qquad n \ge 3, \\ A_y &= \frac{9q-1}{8(1-q)}, \qquad b_y = \frac{1-q}{9q-1}. \end{aligned} \tag{388}$$

Using the three-term norm inequality and the actual prefix $\|p_2\|^2 \le M^2 e_0^2$, define

$$K_y = 3A_y^2\big[(1 + b_y^2)K_p + b_y^2 M^2 + K_w\big]. \tag{389}$$

Summing (388) for $n \ge 3$ proves

$$\sum_{n\ge 2}\|w_n\|^2 \le K_y e_0^2. \tag{390}$$

In particular w_2 has not been dropped. Combining the pair prefixes $n = 0, 1$, the single primal prefix p_2, and the last two tails gives

$$C_{MB} = 1 + M + M^2 + K_p + K_y > 1, \qquad \sum_{n\geq 0} e_n^2 \leq C_{MB} e_0^2. \tag{391}$$

This accounts separately for the lower starting indices of the filter, primal and dual sums. All constants are finite at each fixed admissible q, r, with the particular finite storage already specified.

True restart and geometric tail contraction. Choose $C = C_F$ or C_{MB} using (34). The preceding bounds are proved for an arbitrary initial pair, using its next two or three actual outputs. Since (29) is autonomous and first order in the physical pair, starting it from (x_k, y_k) produces exactly (x_{k+j}, y_{k+j}). The same F, anchor, certificate and constants apply. No predecessor history is fabricated, and no startup bound evaluates $F(p_k)$. Therefore the same proof yields (95) for every actual $k \geq 0$.

Apply the finite-tail implication already proved in (475)–(476), with the explicit substitutions $a_m = e_m^2$, $V_m^{\text{tail}} = S_m = \sum_{n\geq m} e_n^2$, and $C = C_F$ or C_{MB} selected above. Its premises, finiteness and the same restarted inequality $S_m \leq C e_m^2$, have just been established from the branch-specific actual energies. Thus

$$S_{m+1} \leq \varrho S_m, \qquad e_{k+j}^2 \leq C \varrho^j e_k^2, \qquad \varrho = 1 - 1/C. \tag{392}$$

The ordinary-norm estimate in (96) follows by taking square roots. This application uses the proved finite-tail bound; it does not assume a tail Lyapunov property to obtain that bound.

Original objective and anchored proximal residual. The KKT relation $x_* - c = \mu y_*$ gives the exact attained-output identity

$$\Phi(x_n) - \Phi(x_*) = F_n/r + \|p_n\|^2/(2\mu), \qquad n \geq 1. \tag{393}$$

In particular Φ has global quadratic growth coefficient $1/\mu$ about its unique minimizer. Equations (370) and (371) imply

$$0 \leq \Phi(x_n) - \Phi(x_*) \leq H(1/r + 1/\mu) e_{n-1}^2 = D_\Phi e_{n-1}^2.$$

Apply the restarted pair rate at $k + j - 1$ to obtain (98). The exponent is $j - 1$, with $j \geq 1$; an arbitrary restart point can lie outside $\operatorname{dom} f$. Finally $v_n = r(u_n + y_*)$ and (370) give (99) with the necessary factor $1/r^2$. The decaying residual is anchored at $-y_*$; the raw u_n need not tend to zero. This completes the proof of Theorem 9.

Preservation of the earlier quantitative scope. The earlier identity specialization of Corollary 6.1 already has the constant

$$C_{\mathrm{old}} = 1 + K_0^2 + K_0^4 + K_0^6 + C_0 \left(\frac{2\mu}{r\eta} + \frac{2}{\delta} \right) > 1, \tag{394}$$

where K_0, C_0, η, δ are its own startup and certificate constants. Branch F is an alternative physical-norm startup estimate, not a new low-radius rate. Branches M and B extend the uniform tail estimate to the full sharp interval. All earlier direct, low-product, product-three, general-coupling growth and robustness constants keep their stated hypotheses and their original norm scalings. In particular this new scalar theorem does not imply a no-growth rate for general L or extend any perturbation regime.

C.2 Uniform startup and anchored orbital stability

The next estimate uses no growth assumption. It supplies both a uniform stability bound and the startup needed for a geometric rate. Fix any KKT anchor and the selected compact-band certificate, and set

$$\begin{gathered} e_n^2 = \|p_n\|^2 + \|t_n\|^2, \qquad M_0 = \frac{35+\eta}{2}, \qquad a_{\mathrm{cert}} = 1 + \kappa_{\mathrm{cert}}^2, \qquad J_{\mathrm{cert}} = 2a_{\mathrm{cert}}, \\ K_0^2 = 1 + \kappa_{\mathrm{cert}}^2 + q_+^2 \left[\frac{25}{16} + (1 + 9\kappa_{\mathrm{cert}}/8)^2 \right], \\ C_0 = (M_0 + M_P)(K_0^6 + K_0^4 + K_0^2) + J_{\mathrm{cert}}(7K_0^4 + 3K_0^2 + 1). \end{gathered} \tag{395}$$

Lemma 6 (Homogeneous three-output startup)**.** *Under the hypotheses of Theorem 4,*

$$e_j \le K_0 e_{j-1}, \qquad F_j \le J_{\mathrm{cert}} e_{j-1}^2 \quad (j \ge 1), \qquad 0 \le V_3 \le C_0 e_0^2. \tag{396}$$

The same constants apply to every orbit restart and every KKT anchor.

Proof. The translated proximal map is prox_F and fixes zero. By its nonexpansiveness and (216), for $j \ge 1$,

$$\begin{aligned} \|p_j\| &\le \|p_{j-1}\| + \kappa_{\mathrm{cert}} \|t_{j-1}\| \le \sqrt{a_{\mathrm{cert}}}\, e_{j-1}, \\ \|t_j\| &\le q_+ \left(\frac{5}{4} \|p_{j-1}\| + (1 + 9\kappa_{\mathrm{cert}}/8) \|t_{j-1}\| \right). \end{aligned} \tag{397}$$

The Frobenius norm of the nonnegative comparison matrix with rows $(1, \kappa_{\mathrm{cert}})$ and $(5q_+/4, q_+(1 + 9\kappa_{\mathrm{cert}}/8))$ is K_0. Consequently $e_j \le K_0 e_{j-1} \le K_0^j e_0$. Moreover

$$\begin{aligned} \|v_j\| &\le 2\|p_{j-1}\| + 2\kappa_{\mathrm{cert}} \|t_{j-1}\| \le 2\sqrt{a_{\mathrm{cert}}}\, e_{j-1}, \\ 0 \le F_j &\le \langle p_j, v_j \rangle \le J_{\mathrm{cert}} e_{j-1}^2 \le J_{\mathrm{cert}} K_0^{2j-2} e_0^2. \end{aligned} \tag{398}$$

The absolute row sums of P_{res} are $(35+\eta)/2$, $27/2$, and 3, hence $\|P_{\mathrm{res}}\| \le M_0$. Since

$$\|R_3\|^2 \le e_3^2 + e_2^2 + e_1^2, \qquad \|s_3\|^2 \le e_3^2 + e_2^2 + e_1^2,$$

we obtain

$$\begin{aligned} E_3 &\le (M_0 + M_P)(K_0^6 + K_0^4 + K_0^2)e_0^2, \\ G_3 &\le J_{\mathrm{cert}}(7K_0^4 + 3K_0^2 + 1)e_0^2, \\ 0 \le V_3 &\le C_0 e_0^2. \end{aligned} \tag{399}$$

For the second line, discard $-4F_3$ and use (398) on $7\langle p_3, v_3\rangle$, $3F_2$ and F_1. Only actual proximal outputs are used, so $f(x_0)$ may be infinite.

The recurrence is autonomous and first order in (p, t). Starting it at (p_k, t_k) regenerates the same future orbit, and the displayed proof uses only the next three outputs $k+1, k+2, k+3$. It gives $V_{k+3} \le C_0 e_k^2$ with the actual history already present in that energy. There is no artificial prehistory, no reset of old entries, and no evaluation of $F(p_k)$. Finally, the proximal map fixes zero for every KKT anchor and all comparison constants are unchanged. ∎

Corollary C.1 (Uniform anchored orbital stability)**.** *Under the hypotheses of Theorem 4, without any growth assumption, define*

$$\begin{aligned} &M_{\mathrm{stab}} = \max\{1, K_0^2, K_0^4, C_0/\beta\}, \\ &\sup_{j\ge 0} e_{k+j}^2 \le M_{\mathrm{stab}} e_k^2 \qquad (k \ge 0). \end{aligned} \tag{400}$$

The constant is independent of the initialization, restart index, and KKT anchor. This is stability relative to a fixed KKT equilibrium in the normalized product norm; it does not assert contraction between two evolving trajectories, monotonicity of e_n, or a convergence rate.

Proof. For $j = 0, 1, 2$, Lemma 6 bounds e_{k+j}^2 by $1, K_0^2, K_0^4$ times e_k^2, respectively. For $j \ge 3$, coercivity, descent, and the restarted startup bound give

$$\beta e_{k+j}^2 \le V_{k+j} \le V_{k+3} \le C_0 e_k^2.$$

At $j = 3$ the middle inequality is equality; for $j \ge 4$ descent starts at global index $k+4$. No additional K_0^6 prefix term is needed. ∎

C.3 Proof of the variable radius growth rate

Proof. Equations (232) and (399) give

$$\sum_{n\ge 4} F_n \le \frac{C_0}{\eta} e_0^2, \qquad \sum_{n\ge 4} \|u_n\|^2 \le \frac{C_0}{\delta} e_0^2, \qquad \sum_{n\ge 4} \|t_n\|^2 \le \frac{C_0}{\delta} e_0^2. \tag{401}$$

The last estimate follows because the dissipation actually controls t_n for $n \ge 3$. Combining (106), (234) and (235) yields

$$\sum_{n\ge 4} \|p_n\|^2 \le \frac{2C_0}{\tau \mu_{\mathrm{QG}}} \left(\frac{1}{\eta} + \frac{\kappa_{\mathrm{cert}} b}{2\delta}\right) e_0^2.$$

Add the first four state terms, bounded by $(1 + K_0^2 + K_0^4 + K_0^6)e_0^2$, and the dual tail to obtain $\sum_{n\ge 0} e_n^2 \le C_* e_0^2$. Apply the proved startup estimate to the autonomous orbit starting at (p_k, t_k), as in Lemma 6. This gives the same square-sum estimate at every restart, without evaluating $F(p_k)$ or changing a history.

Set $\mathcal{S}_k = \sum_{n\ge k} e_n^2$. Then $\mathcal{S}_k \le C_* e_k^2$, so

$$\mathcal{S}_{k+1} = \mathcal{S}_k - e_k^2 \le (1 - C_*^{-1})\mathcal{S}_k = \rho_{\mathrm{QG}}\mathcal{S}_k.$$

Induction gives $e_{k+j}^2 \le \mathcal{S}_{k+j} \le \rho_{\mathrm{QG}}^j \mathcal{S}_k \le C_* \rho_{\mathrm{QG}}^j e_k^2$, proving (108) and (109).

At every actual output $n \ge 1$, the startup estimates give $F_n \le J_{\mathrm{cert}} e_{n-1}^2$ and $\|u_n\|^2 \le \|p_n\|^2 \le a_{\mathrm{cert}} e_{n-1}^2$. Hence

$$0 \le \tau[\Phi(x_n) - \Phi(x^*)] \le a_{\mathrm{cert}}(2 + \kappa_{\mathrm{cert}} b/2) e_{n-1}^2. \tag{402}$$

Insert the geometric state estimate at the preceding state, and repeat from a restart k, to obtain (110). The exponent is $j - 1$ at output $k + j$, and $n - 1$ at output n from the initial state. There is no $\Phi(x_0)$ or F_0 term. The gap factor is ρ_{QG}, while the norm-error factor is $\sqrt{\rho_{\mathrm{QG}}}$. ∎

The analytic compact-band selection establishes existence of the certificate-dependent constants. The rational specializations recorded in Appendix B additionally supply concrete storage data with which they can be evaluated. Neither conclusion is endpoint-uniform in $q \in (0, 1)$, and no optimized rate is claimed. All statements in this graph-storage argument concern the stated quadratic dual objective; they make no extension to nonlinear dual perturbations.

C.4 Product three startup and rate constants

C.4.1 Uniform startup and anchored orbital stability

The next estimate uses no growth assumption. It supplies both a uniform stability bound and the startup needed for a geometric rate. Fix any KKT anchor and the selected compact-band certificate, and set

$$\begin{gathered} e_n^2 = \|p_n\|^2 + \|t_n\|^2, \qquad M_0 = \frac{35 + \eta}{2}, \qquad K_0^2 = 10 + \frac{1325}{64} q_+^2, \\ C_0 = (M_0 + M_P)(K_0^6 + K_0^4 + K_0^2) + 140 K_0^4 + 60 K_0^2 + 20. \end{gathered} \tag{403}$$

Lemma 7 (Homogeneous three-output startup)**.** *Under the hypotheses of Theorem 16,*

$$e_j \le K_0 e_{j-1}, \qquad F_j \le 20 e_{j-1}^2 \quad (j \ge 1), \qquad 0 \le V_3 \le C_0 e_0^2. \tag{404}$$

The same constants apply to every orbit restart and every KKT anchor.

Proof. Lemma 6 specializes with $a_{\mathrm{cert}} = 10$, $J_{\mathrm{cert}} = 20$. Its exact one-step and startup calculations become

$$\begin{aligned} \|p_j\| &\le \|p_{j-1}\| + 3\|t_{j-1}\| \le \sqrt{10}\, e_{j-1}, \\ \|t_j\| &\le q_+ \left(\frac{5}{4}\|p_{j-1}\| + \frac{35}{8}\|t_{j-1}\| \right). \end{aligned} \tag{405}$$

$$\begin{aligned} \|v_j\| &\le 2\|p_{j-1}\| + 6\|t_{j-1}\| \le \sqrt{40}\, e_{j-1}, \\ 0 \le F_j &\le \langle p_j, v_j \rangle \le 20 e_{j-1}^2 \le 20 K_0^{2j-2} e_0^2. \end{aligned} \tag{406}$$

$$\begin{aligned} \|R_3\|^2 \le e_3^2 + e_2^2 + e_1^2, \qquad \|s_3\|^2 \le e_3^2 + e_2^2 + e_1^2, \\ E_3 \le (M_0 + M_P)(K_0^6 + K_0^4 + K_0^2) e_0^2, \\ G_3 \le (140 K_0^4 + 60 K_0^2 + 20) e_0^2, \\ 0 \le V_3 \le C_0 e_0^2. \end{aligned} \tag{407}$$

The comparison matrix rows are $(1, 3)$ and $(5q_+/4, 35q_+/8)$; its Frobenius norm is K_0. The absolute residual row sums are $(35+\eta)/2, 27/2, 3$. Dropping only $-4F_3$ produces the displayed $140, 60, 20$ coefficients. The shared proof uses only the next three outputs after each actual restart and never evaluates $F(p_k)$. ∎

Corollary C.2 (Uniform anchored orbital stability)**.** *Under the hypotheses of Theorem 16, without any growth assumption, define*

$$\begin{aligned} M_{\mathrm{stab}} &= \max\{1, K_0^2, K_0^4, C_0/\beta\}, \\ \sup_{j \ge 0} e_{k+j}^2 &\le M_{\mathrm{stab}} e_k^2 \qquad (k \ge 0). \end{aligned} \tag{408}$$

The constant is independent of the initialization, restart index, and KKT anchor. This is stability relative to a fixed KKT equilibrium in the normalized product norm; it does not assert contraction between two evolving trajectories, monotonicity of e_n, or a convergence rate.

Proof. For $j = 0, 1, 2$, Lemma 7 bounds e_{k+j}^2 by $1, K_0^2, K_0^4$ times e_k^2, respectively. For $j \ge 3$, coercivity, descent, and the restarted startup bound give

$$\beta e_{k+j}^2 \le V_{k+j} \le V_{k+3} \le C_0 e_k^2.$$

At $j = 3$ the middle inequality is equality; for $j \ge 4$ descent starts at global index $k + 4$. No additional K_0^6 prefix term is needed. ∎

C.4.2 Global single-anchor growth: geometric iterates and gaps

Corollary C.3 (Global single-anchor quadratic growth)**.** *In addition to the hypotheses of Theorem 16, suppose that for one KKT anchor and some $\mu > 0$,*

$$\Phi(x) - \Phi(x^*) \ge \frac{\mu}{2}\|x - x^*\|^2, \qquad x \in \operatorname{dom} f. \tag{409}$$

This is a global inequality to one anchor, rather than a bound by distance to a possibly non-singleton solution set or a local growth condition. It implies that x^* is the unique primal minimizer. With b from (368) *and the startup constants, set*

$$C_* = 1 + K_0^2 + K_0^4 + K_0^6 + C_0 \left\{ \frac{2}{\tau\mu} \left(\frac{1}{\eta} + \frac{3b}{2\delta} \right) + \frac{1}{\delta} \right\} > 1, \qquad \gamma = 1 - C_*^{-1} \in (0,1). \tag{410}$$

Then for every restart $k \geq 0$,

$$\sum_{j \geq 0} e_{k+j}^2 \leq C_* e_k^2, \qquad e_{k+j}^2 \leq C_* \gamma^j e_k^2 \quad (j \geq 0). \tag{411}$$

Thus the primal and dual iterates converge geometrically in norm, with norm factor $\sqrt{\gamma}$ and physical bounds

$$\|x_n - x^*\| \leq \sqrt{C_*}\, \gamma^{n/2} e_0, \qquad \|y_n - y^*\| \leq \sqrt{\frac{3\sigma C_*}{\tau}}\, \gamma^{n/2} e_0. \tag{412}$$

The composite objective gaps obey

$$\begin{aligned} 0 \leq \Phi(x_{k+j}) - \Phi(x^*) &\leq \frac{(20+15b)C_*}{\tau}\, \gamma^{j-1} e_k^2, \qquad k \geq 0,\ j \geq 1, \\ \Phi(x_n) - \Phi(x^*) &\leq \frac{(20+15b)C_*}{\tau}\, \gamma^{n-1} \left(\|x_0 - x^*\|^2 + \frac{\tau}{3\sigma} \|y_0 - y^*\|^2 \right), \quad n \geq 1. \end{aligned} \tag{413}$$

The constants are explicit finite expressions given the selected certificate data. The arbitrary-band analytic existence argument does not itself supply numerical η, δ, M_P, and hence does not supply a certificate-free numerical rate in the problem parameters alone.

Proof. Corollary 6.1, with $\kappa_{\mathrm{cert}} = 3$ and $\mu_{\mathrm{QG}} = \mu$, gives these claims. For completeness the exact tail and one-output specializations used in its proof are

$$\sum_{n \geq 4} F_n \leq \frac{C_0}{\eta} e_0^2, \qquad \sum_{n \geq 4} \|u_n\|^2 \leq \frac{C_0}{\delta} e_0^2, \qquad \sum_{n \geq 4} \|t_n\|^2 \leq \frac{C_0}{\delta} e_0^2. \tag{414}$$

$$\sum_{n \geq 4} \|p_n\|^2 \leq \frac{2C_0}{\tau\mu} \left(\frac{1}{\eta} + \frac{3b}{2\delta} \right) e_0^2.$$

$$\mathcal{S}_{k+1} = \mathcal{S}_k - e_k^2 \leq (1 - C_*^{-1}) \mathcal{S}_k = \gamma \mathcal{S}_k.$$

$$0 \leq \tau[\Phi(x_n) - \Phi(x^*)] \leq (20 + 15b) e_{n-1}^2. \tag{415}$$

Here $\mathcal{S}_k = \sum_{n \geq k} e_n^2$. The first four state terms contribute $1 + K_0^2 + K_0^4 + K_0^6$, and the displayed tails give C_*. The common restart argument yields $\mathcal{S}_k \leq C_* e_k^2$, and induction gives the geometric estimate. The one-output bounds are $F_n \leq 20 e_{n-1}^2$, $\|u_n\|^2 \leq 10 e_{n-1}^2$. Thus the gap exponent is $n-1$, or $j-1$ from restart k, whereas the pair norm factor is $\sqrt{\gamma}$. No initial function value is used. ∎

The analytic compact-band selection establishes existence of the certificate-dependent constants. The rational finite-band specialization additionally supplies concrete storage data with which they can be evaluated. Neither conclusion is endpoint-uniform in $q \in (0,1)$, and no optimized rate is claimed. All statements in this graph-storage argument concern the stated quadratic dual objective; they make no extension to nonlinear dual perturbations.

C.5 Normalization and finite Fenchel slacks

The proximal optimality condition for (103) is $(c - x^*)/\mu \in \partial f(x^*)$. Thus (103) is always a KKT pair when $\mu > 0$. If another pair (x', y') satisfies (102), monotonicity gives

$$0 \le \langle x' - x^*, -y' + y^* \rangle = -\mu \|y' - y^*\|^2,$$

so the pair is unique. When $\mu = 0$, fix any existing pair (c, y^*). In either case put

$$\begin{aligned} u_n = x_n - x^*, \quad w_n = y_n - y^*, \quad q = (1+\mu)^{-1}, \\ F(u) = r\big(f(x^* + u) - f(x^*) + \langle y^*, u \rangle\big). \end{aligned} \tag{416}$$

Then F is proper lower-semicontinuous convex, $F \ge 0$, $F(0) = 0$, and $0 \in \partial F(0)$. The shifted map, with positive output index, is

$$u_n = \operatorname{prox}_F(u_{n-1} - r w_{n-1}), \qquad w_n = q\big(w_{n-1} + (9u_n - u_{n-1})/8\big), \quad n \ge 1. \tag{417}$$

For $n \ge 1$ define actual graph pairs and their values by

$$\begin{aligned} v_n = u_{n-1} - u_n - r w_{n-1} \in \partial F(u_n), \\ f_n = F(u_n), \quad A_n = F^*(v_n), \quad B_n = \langle u_n, v_n \rangle = f_n + A_n. \end{aligned} \tag{418}$$

Every f_n and A_n here is finite by the proximal graph inclusion and Fenchel equality. They are nonnegative: $F \ge 0$ and $F^*(v) \ge 0$ follows by testing $u = 0$. Thus $B_n \ge 0$. Neither $F(u_0)$ nor $F^*(v_0)$ is used. For any positive i, j the finite Fenchel slack is

$$H_{ij} = f_i + A_j - \langle u_i, v_j \rangle \ge 0, \qquad H_{ii} = 0. \tag{419}$$

All pairs belong to the same fixed graph, including at any restart.

Put

$$\begin{aligned} b = 1 - r/8, \qquad c_r = 9r/8 - 2, \\ s_n = u_n + c_r u_{n-1} + b u_{n-2}, \\ \ell_n = u_n + (9qr/8 - 1 - q) u_{n-1} + qb u_{n-2}. \end{aligned} \tag{420}$$

Eliminating the actual w variables from (417)–(418) gives, for $n \ge 2$,

$$v_n - q v_{n-1} = -\ell_n, \qquad \ell_n = q s_n + (1-q)(u_n - u_{n-1}). \tag{421}$$

This scalar constant q is where the quadratic-dual structure enters.

For $n \geq 3$, define

$$\begin{aligned} Q_n &= 8\langle u_n, \ell_n\rangle - \langle u_{n-2}, \ell_n + \ell_{n-1}\rangle, \\ T_n &= 7A_n - qA_{n-1} + 8(1-q)f_n + 2(1-q)f_{n-1}. \end{aligned} \tag{422}$$

The formula for T_n also defines it at $n = 2$. First expand before bounding:

$$\begin{aligned} Q_n = \; &-8B_n + 8q\langle u_n, v_{n-1}\rangle + \langle u_{n-2}, v_n\rangle \\ &+ (1-q)\langle u_{n-2}, v_{n-1}\rangle - qB_{n-2}. \end{aligned} \tag{423}$$

Substitution of (419) yields the exact equality

$$\begin{aligned} T_n - T_{n-1} + Q_n + 6(1-q)B_{n-1} = \; &-8qH_{n,n-1} - H_{n-2,n} \\ &- (1-q)H_{n-2,n-1} \leq 0. \end{aligned} \tag{424}$$

All three slack weights are nonnegative since $0 < q \leq 1$. The signed term $-qA_{n-1}$ is retained; positivity of T_n is not assumed.

C.6 A coercive storage for the direct interval

For $3/4 < r \leq 3$, define

$$\lambda = (7-r)/2, \qquad d_r = \frac{4(r-2)}{7-r}, \qquad \delta = \frac{2(3-r)(4r-3)}{7-r}. \tag{425}$$

Here $\lambda > 0$, and $\delta > 0$ for $r < 3$, whereas $\delta = 0$ at $r = 3$. For three Hilbert-vector slots set

$$\begin{aligned} P_1(s,t,z) &= \frac{r+9}{2}\|s\|^2 + (5r-8)\langle s,t\rangle + \frac{r+5}{2}\|t\|^2 + b\langle t,z\rangle, \\ P_0(s,t,z) &= \frac{7}{2}\|s\|^2 - \frac{1}{2}\|t\|^2, \\ P_q &= qP_1 + (1-q)P_0. \end{aligned} \tag{426}$$

The same symbol denotes a quadratic form; its symmetric coefficient matrix will be denoted $\mathbf{P}_q$. Explicitly,

$$\mathbf{P}_q = \begin{pmatrix} [q(r+9)+7(1-q)]/2 & q(5r-8)/2 & 0 \\ q(5r-8)/2 & [q(r+5)-(1-q)]/2 & qb/2 \\ 0 & qb/2 & 0 \end{pmatrix}. \tag{427}$$

It is not assumed positive.

C.6.1 The exact quadratic identities

Direct expansion gives the affine identity

$$\begin{aligned} &8\langle u_n, s_n\rangle - \langle u_{n-2}, s_n + s_{n-1}\rangle \\ &\quad = P_1(u_n, u_{n-1}, u_{n-2}) - P_1(u_{n-1}, u_{n-2}, u_{n-3}) \\ &\qquad + \lambda\|u_n + d_r u_{n-1} + u_{n-2}\|^2 + \delta\|u_{n-1}\|^2. \end{aligned} \tag{428}$$

For $\ell_n^{(0)} = u_n - u_{n-1}$ a second direct expansion gives

$$\begin{aligned} 8\langle u_n, \ell_n^{(0)}\rangle - \langle u_{n-2}, \ell_n^{(0)} + \ell_{n-1}^{(0)}\rangle \\ = P_0(u_n, u_{n-1}, u_{n-2}) - P_0(u_{n-1}, u_{n-2}, u_{n-3}) \\ + 4\|u_n - u_{n-1}\|^2 + \tfrac{1}{2}\|u_n - u_{n-2}\|^2. \end{aligned} \tag{429}$$

Their q and $1-q$ combination, using (421), proves

$$Q_n = P_q(u_n, u_{n-1}, u_{n-2}) - P_q(u_{n-1}, u_{n-2}, u_{n-3}) + D_n, \tag{430}$$

where

$$\begin{aligned} D_n = q\lambda\|u_n + d_r u_{n-1} + u_{n-2}\|^2 + q\delta\|u_{n-1}\|^2 \\ + (1-q)\big(4\|u_n - u_{n-1}\|^2 + \tfrac{1}{2}\|u_n - u_{n-2}\|^2\big). \end{aligned} \tag{431}$$

Consequently the finite storage

$$E_n = T_n + P_q(u_n, u_{n-1}, u_{n-2}), \qquad n \geq 2, \tag{432}$$

satisfies, for $n \geq 3$,

$$\begin{aligned} E_n - E_{n-1} + D_n + 6(1-q)B_{n-1} = & -8qH_{n,n-1} - H_{n-2,n} \\ & - (1-q)H_{n-2,n-1} \leq 0. \end{aligned} \tag{433}$$

C.6.2 Coercivity on actual histories

The graph relation $qv_{n-1} = v_n + \ell_n$ implies

$$qA_{n-1} = \langle u_{n-1}, v_n + \ell_n\rangle - qf_{n-1}.$$

Define the two-slot forms

$$\begin{aligned} R_1(s,t) &= \frac{r+9}{2}\|s\|^2 + (5r-9)\langle s,t\rangle + \left(\frac{9}{2} - \frac{5r}{8}\right)\|t\|^2, \\ R_0(s,t) &= \frac{7}{2}\|s\|^2 - \langle s,t\rangle + \frac{1}{2}\|t\|^2, \\ R_q &= qR_1 + (1-q)R_0. \end{aligned} \tag{434}$$

The oldest slot cancels in $P_q(s,t,z) - \langle t, \ell(s,t,z)\rangle = R_q(s,t)$. Using $H_{n-1,n}$ now gives an exact coercivity decomposition:

$$\begin{aligned} E_n = 6A_n + 8(1-q)f_n + (1-q)f_{n-1} \\ + R_q(u_n, u_{n-1}) + H_{n-1,n}. \end{aligned} \tag{435}$$

The symmetric coefficient matrices of R_1, R_0 obey

$$\begin{aligned} (\mathbf{R}_1)_{11} = (r+9)/2 > 0, \qquad \det \mathbf{R}_1 = \frac{3r(117-35r)}{16} > 0, \\ (\mathbf{R}_0)_{11} = 7/2 > 0, \qquad \det \mathbf{R}_0 = 3/2 > 0. \end{aligned} \tag{436}$$

Thus R_q is positive definite for $3/4 < r \le 3$ and $0 < q \le 1$. If $\kappa_{r,q} > 0$ is the least eigenvalue of its real 2×2 matrix,

$$E_n \ge R_q(u_n, u_{n-1}) \ge \kappa_{r,q}(\|u_n\|^2 + \|u_{n-1}\|^2) \ge 0. \tag{437}$$

Finite scalar matrix inequalities lift to Hilbert-vector slots by orthogonal diagonalization, or by expanding the finitely many vectors in an orthonormal basis of their span. In particular, no finite-dimensional assumption on $\mathcal{H}$ has been made. The whole storage is coercive on actual histories; a free quadratic form or a frequency inequality alone would not establish (437).

C.6.3 Strict dissipation and the endpoint Gram form

For $3/4 < r < 3$, telescoping (433) and using (437) gives

$$q\delta \sum_{n=2}^{\infty} \|u_n\|^2 \le E_2. \tag{438}$$

At $r = 3$, $\lambda = 2$, $d_r = 1$, and the dissipation instead is

$$D_n = 2q\|u_n + u_{n-1} + u_{n-2}\|^2 + (1-q)\left(4\|u_n - u_{n-1}\|^2 + \tfrac{1}{2}\|u_n - u_{n-2}\|^2\right). \tag{439}$$

For $0 < q < 1$ this is positive definite in the three slots: the linear forms $s+t+z$, $s-t$, $s-z$ have determinant 3, and all square weights are positive. Its Gram determinant is $36q(1-q)^2 > 0$. For an explicit bound let $A = s+t+z$, $B = s-t$, $C = s-z$. Expansion gives

$$\frac{\|A\|^2}{3} + \|B\|^2 + \|C\|^2 - \left(\|s\|^2 + \|t\|^2 + \|z\|^2\right) = \frac{1}{3}\|B + C\|^2 \ge 0. \tag{440}$$

It follows that the endpoint dissipation is at least

$$\alpha_{3,q}(\|s\|^2 + \|t\|^2 + \|z\|^2), \qquad \alpha_{3,q} = \min\{6q, (1-q)/2\} > 0.$$

Its middle-slot bound replaces $q\delta$ in (438). For $q = 1$ this endpoint argument fails, in agreement with Theorem 12 and the cycles of Section C.13.

Lemma 8 (The affine specialization)**.** *At $q = 1$, the formulas above become*

$$\ell_n = s_n, \quad v_n - v_{n-1} = -s_n, \quad E_n = 7A_n - A_{n-1} + P_1(u_n, u_{n-1}, u_{n-2}).$$

For $3/4 < r \le 3$ and $n \ge 3$,

$$E_n - E_{n-1} \le -\lambda\|u_n + d_r u_{n-1} + u_{n-2}\|^2 - \delta\|u_{n-1}\|^2,$$

and for $n \ge 2$,

$$E_n \ge 6A_n + R_1(u_n, u_{n-1}).$$

Thus the affine storage, its signed conjugate coefficients, square dissipation, and actual-history coercivity are all contained in the same parameterized calculation.

Proof. Set $q = 1$ in (421), (433), and (435). The remaining slack is nonnegative. ∎

C.7 Control from arbitrary initial states

For $3/4 < r \le 3$ and $0 < q \le 1$, the common storage has a homogeneous initial bound. To obtain a uniform restart constant, put $D^2 = \|u_0\|^2 + \|w_0\|^2$ and define

$$K_0 = \sqrt{1+r^2}, \qquad L_q = q\left(1 + \frac{9K_0+1}{8}\right), \qquad M_0 = K_0 + rL_q. \tag{441}$$

Since prox_F fixes zero and is nonexpansive,

$$\begin{aligned} \|u_1\| \le K_0 D, \quad \|w_1\| \le L_q D, \quad \|u_2\| \le M_0 D, \\ \|v_1\| \le 2K_0 D, \qquad \|v_2\| \le 2M_0 D, \\ B_1 \le 2K_0^2 D^2, \qquad B_2 \le 2M_0^2 D^2. \end{aligned} \tag{442}$$

Use the ordinary operator norm of the real matrix (427) and set

$$C_{\mathrm{init}} = 16M_0^2 + 4(1-q)K_0^2 + \|\mathbf{P}_q\|(M_0^2 + K_0^2 + 1). \tag{443}$$

Indeed $7A_2 + 8(1-q)f_2 \le 8B_2$, $-qA_1 \le 0$, and $2(1-q)f_1 \le 2(1-q)B_1$. Thus the finite storage satisfies

$$0 \le E_2 \le C_{\mathrm{init}} D^2. \tag{444}$$

No initial function value is required, and no positivity of $\mathbf{P}_q$ is used in this upper bound.

For the affine branch $q = 1$, a slightly sharper bound retains the original constants. Write

$$K = K_0 = \sqrt{1+r^2}, \qquad L = L_1 = 1 + (9K+1)/8, \qquad M = M_0 = K + rL. \tag{445}$$

Let $P_r = \mathbf{P}_1$, let $R_r = \mathbf{R}_1$, and let $\kappa_r = \lambda_{\min}(R_r) > 0$. Since $A_2 \le B_2 \le 2M^2D^2$, $A_1 \ge 0$, and $E_2 = 7A_2 - A_1 + P_1(u_2, u_1, u_0)$,

$$0 \le E_2 \le C_r D^2, \qquad C_r = 14M^2 + \|P_r\|_{\mathrm{op}}(M^2 + K^2 + 1). \tag{446}$$

Consequently, for $3/4 < r \le 3$,

$$\kappa_r(\|u_n\|^2 + \|u_{n-1}\|^2) \le E_n \le E_2, \qquad n \ge 2. \tag{447}$$

In particular,

$$U_r = \max\{1, \sqrt{C_r/\kappa_r}\}, \qquad \|u_n\| \le U_r D \quad (n \ge 0). \tag{448}$$

The value $n = 1$ is covered by the two-slot bound at $n = 2$. On $3/4 < r < 3$, summing the affine descent gives

$$\sum_{n=0}^{\infty} \|u_n\|^2 \le S_r D^2, \qquad S_r = 1 + K^2 + C_r/\delta. \tag{449}$$

All constants are independent of the objective, anchor, initialization, and Hilbert space. Only positive-output function and graph values have been used. The same facts therefore hold at every autonomous restart.

C.8 The actual dual filter and homogeneous bounds

Assume $\mu > 0$, so $0 < q < 1$. Solving the literal second equation in (417), with $g_j = (9u_j - u_{j-1})/8$, yields

$$w_n = q^n w_0 + \sum_{j=1}^{n} q^{n-j+1} g_j. \tag{450}$$

If $U^2 = \sum_{n\geq 0} \|u_n\|^2$, then the triangle inequality on $\ell^2(\mathcal{H})$ gives $\|g\|_{\ell^2} \leq 5U/4$. The kernel $(q, q^2, \ldots)$ has ℓ^1 norm $q/(1-q)$. Thus

$$\left(\sum_{n\geq 0} \|w_n\|^2\right)^{1/2} \leq \frac{\|w_0\|}{\sqrt{1-q^2}} + \frac{5q}{4(1-q)} U. \tag{451}$$

Together with Section C.6, this already proves paired error square summability and strong convergence on $3/4 < r \leq 3$. The factor $q/(1-q) = 1/\mu$ explains why the estimate is not uniform at zero curvature.

For $3/4 < r \leq 3$ define the positive constant

$$\alpha_{r,q} = \begin{cases} q\dfrac{2(3-r)(4r-3)}{7-r}, & 3/4 < r < 3, \\ \min\{6q, (1-q)/2\}, & r = 3. \end{cases} \tag{452}$$

Set

$$\begin{aligned} S &= 1 + K_0^2 + C_{\text{init}}/\alpha_{r,q}, \\ W &= (1-q^2)^{-1/2} + \frac{5q}{4(1-q)}\sqrt{S}, \\ C_{r,\mu} &= S + W^2 > 1. \end{aligned} \tag{453}$$

The primal dissipation, the filter bound, and (444), with the endpoint replacement when appropriate, prove the homogeneous bound

$$\sum_{n\geq 0} (\|u_n\|^2 + \|w_n\|^2) \leq C_{r,\mu}(\|u_0\|^2 + \|w_0\|^2). \tag{454}$$

The two finite primal prefix terms $n = 0, 1$ are covered by $1 + K_0^2$ in S. The constants in (453) are deliberately nonoptimal.

C.9 The low product interval

For $0 < r \leq 3/4$ we use one fixed certificate from [1], together with a new residual estimate. The precise dependency is its four-pair storage, equations (29)–(33), and the exact coefficient data in its Appendix A. The unchanged paper and its portable exact verifiers accompany this manuscript. This step uses a quantitative storage estimate, not an inference from weak convergence.

C.9.1 The affine low-product conclusion

At $\mu = 0$, the Hilbert theorem of [1] applies to the same literal map for $0 < r \le 3/4 < r_*$. It supplies weak KKT convergence and square-summable primal and dual increments from arbitrary initialization. The affine dual equation gives

$$u_n = (w_n - w_{n-1}) - \tfrac{1}{8}(u_n - u_{n-1}), \qquad n \ge 1.$$

Hence the primal errors are square summable, and $u_n \to 0$ strongly. Adding the finite initial term proves all remaining conclusions of Theorem 11. No homogeneous low-product affine restart constant is imported or claimed.

For $\mu > 0$, weak convergence alone would not prove the asserted uniform geometric estimate. The following quantitative use of the same exact certificate supplies the needed bound.

C.9.2 A fixed certificate and exact scaling

Let

$$\varepsilon = 10^{-5}, \qquad r_* = \frac{88 + 16\sqrt{91}}{81}, \qquad R = r_* - \varepsilon > 3/4. \tag{455}$$

At this fixed parameter, write $\mathbf{P}_c = P(\varepsilon)$, $\mathbf{c}_c = c(\varepsilon)$, and $\mathbf{D}_8 = D_8(\varepsilon)$ for the 8×8 storage matrix, eight value coefficients, and leading 8×8 dissipation block of [1]. That certificate proves $\mathbf{D}_8 > 0$ by exact arithmetic. Fix

$$\beta = \lambda_{\min}(\mathbf{D}_8) > 0. \tag{456}$$

This is the unscaled dissipation block D_8, rather than RD_8.

Use the unique KKT anchor and define

$$U_n = \sqrt{R/r}\, u_n, \qquad V_n = w_n, \qquad K = \sqrt{r/R}\, I, \qquad Z_n = (U_n, V_n). \tag{457}$$

Then $\|K\| \le 1$, and the exact normalized equations are

$$\begin{aligned} U_n &= \operatorname{prox}_{R\phi}(U_{n-1} - RK^*V_{n-1}), \\ V_n &= \operatorname{prox}_{\psi}\big(V_{n-1} + K(9U_n - U_{n-1})/8\big), \end{aligned} \tag{458}$$

where

$$\begin{aligned} \phi(U) &= f(x^* + \sqrt{r/R}\, U) - f(x^*) + \langle y^*, \sqrt{r/R}\, U\rangle, \\ \psi(V) &= \mu\|V\|^2/2. \end{aligned} \tag{459}$$

Both functions are nonnegative, vanish at zero, and have zero as a subgradient there. The normalization does not change the actual stepsize r in (101). For positive indices the actual graph values are

$$\begin{aligned} a_n &= (U_{n-1} - U_n)/R - K^*V_{n-1} \in \partial\phi(U_n), \\ b_n &= V_{n-1} - V_n + K(9U_n - U_{n-1})/8 \in \partial\psi(V_n). \end{aligned} \tag{460}$$

In particular, the finite positive-output values satisfy

$$0 \le \phi(U_n) \le \langle a_n, U_n \rangle, \qquad 0 \le \psi(V_n) \le \langle b_n, V_n \rangle. \tag{461}$$

For clarity, the contraction embedding in [1] uses isometries

$$\mathsf{J}U = (KU, (I - K^*K)^{1/2}U), \qquad \mathsf{I}V = (V, 0)$$

into $\mathcal{H} \oplus \mathcal{H}$. Define the four-pair history and value vector

$$h_k = (\mathsf{J}U_k, \mathsf{I}V_k, \ldots, \mathsf{J}U_{k+3}, \mathsf{I}V_{k+3}), \quad \xi_k = (\phi(U_k), \psi(V_k), \ldots, \phi(U_{k+3}), \psi(V_{k+3})).$$

The storage supplied by the cited certificate is

$$\mathcal{V}_k = \langle h_k, \mathbf{P}_c h_k \rangle + \mathbf{c}_c^T \xi_k \ge 0, \qquad k \ge 1, \tag{462}$$

and it satisfies

$$\mathcal{V}_k - \mathcal{V}_{k+1} \ge \beta \|Z_{k+4} - Z_{k+3}\|^2. \tag{463}$$

These are exactly the nonnegative-storage and increment-dissipation consequences of [1, equations (31) and (33)]. Its finite nonnegative convexity slacks are evaluated at actual positive-index outputs. Neither positivity of $\mathbf{P}_c$ nor a sign restriction on $\mathbf{c}_c$ is assumed.

C.9.3 A homogeneous first storage bound

Define constants depending only on R:

$$\begin{gathered} A = \sqrt{1 + R^2}, \qquad B = 1 + (9A + 1)/8, \qquad \Gamma = \sqrt{A^2 + B^2} > 1, \\ G_a = (1 + \Gamma)/R + 1, \qquad G_b = 1 + \Gamma + (9\Gamma + 1)/8, \qquad G = \max\{G_a, G_b\}. \end{gathered} \tag{464}$$

Both proximal maps in (458) fix zero and are nonexpansive. Cauchy–Schwarz and $\|K\| \le 1$ give

$$\|U_n\| \le A\|Z_{n-1}\|, \qquad \|V_n\| \le B\|Z_{n-1}\|, \qquad \|Z_n\| \le \Gamma\|Z_{n-1}\|. \tag{465}$$

With $D_0 = \|Z_0\|$, induction and (460) yield, for $1 \le i \le 4$,

$$\|Z_i\| \le \Gamma^i D_0, \qquad \|a_i\| \le G_a \Gamma^{i-1} D_0, \qquad \|b_i\| \le G_b \Gamma^{i-1} D_0. \tag{466}$$

Each function value in ξ_1 is therefore at most $G\Gamma^{2i-1}D_0^2 \le G\Gamma^7 D_0^2$. The separate-slot isometries imply $\|h_1\|^2 = \sum_{i=1}^4 \|Z_i\|^2$. Consequently

$$\begin{gathered} 0 \le \mathcal{V}_1 \le C_V D_0^2, \\ C_V = \|\mathbf{P}_c\| \sum_{i=1}^{4} \Gamma^{2i} + \|\mathbf{c}_c\|_1 G\Gamma^7. \end{gathered} \tag{467}$$

The absolute values in $\|\mathbf{c}_c\|_1$ are essential because the value coefficients may have either sign.

For $1 \le i \le 4$, (465) also implies $\|Z_i - Z_{i-1}\| \le (\Gamma+1)\Gamma^{i-1}D_0$. Telescoping (463) controls the remaining increments, starting at index 5. Hence

$$\begin{aligned} \sum_{n=1}^{\infty} \|Z_n - Z_{n-1}\|^2 &\le \mathcal{J} D_0^2, \\ \mathcal{J} &= (\Gamma+1)^2 \sum_{i=0}^{3} \Gamma^{2i} + C_V/\beta. \end{aligned} \tag{468}$$

Returning to the original errors gives

$$\sum_{n\ge 1} (\|\Delta u_n\|^2 + \|\Delta w_n\|^2) \le \mathcal{J} \frac{R}{r} (\|u_0\|^2 + \|w_0\|^2), \tag{469}$$

where $\Delta u_n = u_n - u_{n-1}$ and $\Delta w_n = w_n - w_{n-1}$. Indeed $\|Z_0\|^2 \le (R/r)(\|u_0\|^2 + \|w_0\|^2)$, whereas $\|\Delta Z_n\|^2 \ge \|\Delta u_n\|^2 + \|\Delta w_n\|^2$.

C.9.4 Monotonicity converts increments into errors

From the literal map and actual primal graph value, define

$$\begin{aligned} e_n &= u_n - \mu w_n = \Delta w_n - \tfrac{1}{8}\Delta u_n, \\ t_n &= v_n + r w_n = -\Delta u_n + r \Delta w_n. \end{aligned} \tag{470}$$

Monotonicity relative to $(0,0) \in \operatorname{graph}(\partial F)$ gives $0 \le \langle u_n, v_n \rangle = \langle u_n, -r w_n + t_n \rangle$. Substitute $w_n = (u_n - e_n)/\mu$ and multiply by μ/r:

$$\|u_n\|^2 \le \langle u_n, e_n \rangle + (\mu/r)\langle u_n, t_n \rangle.$$

Division by $\|u_n\|$ when nonzero proves

$$\|u_n\| \le \|e_n\| + (\mu/r)\|t_n\|. \tag{471}$$

It also holds when $u_n = 0$. The identities in (470) and $w_n = (u_n - e_n)/\mu$ now give

$$\begin{aligned} \|u_n\| &\le (1/8 + \mu/r)\|\Delta u_n\| + (1+\mu)\|\Delta w_n\|, \\ \|w_n\| &\le (1/r + 1/(4\mu))\|\Delta u_n\| + (1 + 2/\mu)\|\Delta w_n\|. \end{aligned} \tag{472}$$

The second line is a valid nonoptimal bound. Put

$$\begin{aligned} M_{r,\mu} = {} & (1/8 + \mu/r)^2 + (1+\mu)^2 \\ & + (1/r + 1/(4\mu))^2 + (1 + 2/\mu)^2. \end{aligned} \tag{473}$$

Cauchy–Schwarz in the two scalar increment norms gives

$$\|u_n\|^2 + \|w_n\|^2 \le M_{r,\mu}(\|\Delta u_n\|^2 + \|\Delta w_n\|^2).$$

Combining this with (469) and adding the initial error proves (454) on $0 < r \le 3/4$ with the explicit constant

$$C_{r,\mu} = 1 + M_{r,\mu}\mathcal{J}(R/r) > 1. \tag{474}$$

Thus paired error square summability follows from the exact residual estimate, without promoting weak convergence to strong convergence.

C.10 Restarting the error bound gives a geometric rate

Both (453) and (474) depend only on r and μ. The iteration is autonomous, the two functions are fixed, and the KKT pair is unique. The same proof therefore applies from every restart m, with exactly the same constant $C = C_{r,\mu}$. Function and graph values after a restart are again needed only at positive output indices. Set

$$a_m = \|u_m\|^2 + \|w_m\|^2, \qquad V_m^{\mathrm{tail}} = \sum_{n=m}^{\infty} a_n \le C a_m. \tag{475}$$

Since these sums are finite,

$$V_{m+1}^{\mathrm{tail}} = V_m^{\mathrm{tail}} - a_m \le (1 - C^{-1}) V_m^{\mathrm{tail}}. \tag{476}$$

With $\rho = 1 - C^{-1} \in (0, 1)$, induction gives

$$a_n \le V_n^{\mathrm{tail}} \le \rho^n V_0^{\mathrm{tail}} \le C \rho^n a_0.$$

Taking square roots proves (105), including its prefactor. Finally, for any square-summable error sequence, $\|z_n - z_{n-1}\|^2 \le 2\|z_n\|^2 + 2\|z_{n-1}\|^2$. Both increment sequences are therefore square summable, completing the proof of Theorem 10.

C.11 The affine strict branch

Assume $\mu = 0$ and $3/4 < r < 3$. The common storage and (449) already give $u_n \to 0$ strongly. Put $\theta = 1/8$. The dual bound and its weak limit require a separate argument.

C.11.1 A shifted-dual energy with summable positive variation

Put $z_n = w_n - \theta u_n$. The affine dual equation gives $z_n - z_{n-1} = u_n$. The graph value can also be written

$$v_n = -u_n + (1 - r\theta) u_{n-1} - r z_{n-1}.$$

Expanding the squared norm increment gives the exact identity

$$\begin{aligned} \|z_n\|^2 - \|z_{n-1}\|^2 = {} & \left(1 - \frac{2}{r}\right) \|u_n\|^2 + \left(\frac{2}{r} - 2\theta\right) \langle u_{n-1}, u_n \rangle \\ & - \frac{2}{r} \langle u_n, v_n \rangle. \end{aligned} \tag{477}$$

Monotonicity against $(0, 0)$ gives $\langle u_n, v_n \rangle \ge 0$. Define

$$\alpha = 1 - 2/r, \qquad \beta = 2/r - 1/4 > 0,$$

and the nonnegative sequence

$$e_n = \max\{\alpha, 0\} \|u_n\|^2 + \frac{\beta}{2} (\|u_{n-1}\|^2 + \|u_n\|^2).$$

It is summable by (449), and (477) implies

$$\|z_n\|^2 - \|z_{n-1}\|^2 \le e_n.$$

Indeed $\|z_n\|^2 + \sum_{k=n+1}^{\infty} e_k$ is nonnegative and nonincreasing, so it has a finite limit. The tail tends to zero. Therefore $\|z_n\|^2$ has a finite limit, and z_n is bounded. Since $u_n \to 0$, also $w_n = z_n + \theta u_n$ is bounded and $\|w_n\|^2$ has a limit.

The argument holds at every KKT anchor. Thus $\|y_n - y'\|$ has a limit for each KKT dual value y'. This proof uses only inner-product monotonicity.

The same estimate is quantitative. Since $\|z_0\|^2 \le (1+\theta^2)D^2$, define

$$Z_r = \left[1 + \theta^2 + (\max\{\alpha, 0\} + \beta)S_r\right]^{1/2}. \tag{478}$$

Then $\|z_n\| \le Z_r D$ and

$$\|w_n\| \le (Z_r + \theta\sqrt{S_r})D.$$

A valid paired stability constant is

$$B_r^{\mathcal{H}} = \left[S_r + (Z_r + \theta\sqrt{S_r})^2\right]^{1/2}. \tag{479}$$

Re-anchoring and restarting prove the claimed homogeneous estimate at every time. If $D = 0$, the anchor is fixed and all the bounds hold trivially.

C.11.2 Weak cluster points and the full limit

The orbit is bounded and $u_n \to 0$ strongly. Choose any weakly convergent subsequence $w_{n_j} \rightharpoonup \bar{w}$, which exists by weak sequential compactness of bounded sets in a Hilbert space. The dual update and strong convergence of u_n give

$$w_n - w_{n-1} \to 0, \qquad u_n - u_{n-1} \to 0 \quad \text{strongly.}$$

Therefore $v_{n_j} \rightharpoonup -r\bar{w}$. The subgradient graph is closed in this strong-times-weak limit; in the present setting this can be checked without an additional closure theorem. The subgradient inequality against zero yields

$$0 \le F(u_{n_j}) \le \langle u_{n_j}, v_{n_j} \rangle \longrightarrow 0,$$

because the weakly convergent v_{n_j} are bounded. For any $t \in \mathcal{H}$ the subgradient inequality at u_{n_j} then passes to the limit as

$$F(t) \ge \langle -r\bar{w}, t \rangle.$$

This is $-r\bar{w} \in \partial F(0)$, or equivalently

$$-(y^* + \bar{w}) \in \partial f(c).$$

Thus every weak paired cluster point is KKT.

For every KKT dual point y', the distance $\|y_n - y'\|$ has a limit. If y' and y'' were two distinct weak cluster points, the expression

$$\|y_n - y'\|^2 - \|y_n - y''\|^2 = 2\langle y_n, y'' - y'\rangle + \|y'\|^2 - \|y''\|^2$$

would have a limit, but its limits along the two weakly convergent subsequences would be $-\|y' - y''\|^2$ and $+\|y' - y''\|^2$. They must coincide, so $y' = y''$. There is a unique weak cluster point. Boundedness then implies weak convergence of the whole sequence: a subsequence separated from the claimed weak limit by a continuous linear functional would have a weakly convergent further subsequence, a contradiction.

Equation (449) proves (111). The bound

$$\|u_n - u_{n-1}\|^2 \le 2\|u_n\|^2 + 2\|u_{n-1}\|^2$$

and the identity $w_n - w_{n-1} = u_n + \theta(u_n - u_{n-1})$ prove (112). In finite dimensions weak convergence is norm convergence. This proves the direct-range part of Theorem 11. The low-product interval is completed in Section C.9.

C.12 The endpoint at step product three

Assume $\mu = 0$ and $\mathcal{H} = \mathbb{R}$. At $r = 3$, write $P = P_1$ and $R_3 = \mathbf{R}_1|_{r=3}$. The storage formulas specialize to

$$P(s,t,z) = 6s^2 + 7st + 4t^2 + \tfrac{5}{8}tz, \tag{480}$$

$$R_3 = \begin{pmatrix} 6 & 3 \\ 3 & 21/8 \end{pmatrix}, \tag{481}$$

$$E_n - E_{n-1} \le -2(u_n + u_{n-1} + u_{n-2})^2, \quad n \ge 3. \tag{482}$$

The determinant of R_3 is $27/4$. More quantitatively,

$$R_3 - \tfrac{3}{4}I \quad \text{has leading entry } \tfrac{21}{4} \quad \text{and determinant } \tfrac{27}{32},$$

so

$$E_n \ge \tfrac{3}{4}(u_n^2 + u_{n-1}^2) \ge 0. \tag{483}$$

C.12.1 The scalar order barrier

Lemma 9 (Scalar dual barrier)**.** *Suppose an orbit of* (417) *with* $q = 1$*, anchored at a KKT pair, satisfies* $|u_n| \le UD$ *for all* $n \ge 0$*, with* $U \ge 1$*. Then*

$$|w_n| \le VD \quad (n \ge 0), \qquad V = \max\left\{\frac{9}{8}, \left(\frac{2}{r} + \frac{11}{8}\right)U\right\} + \frac{U}{8}. \tag{484}$$

Proof. If $D = 0$, the initial point is the chosen KKT point, which is fixed; the result is immediate. Suppose $D > 0$ and set

$$z_n = w_n - u_n/8.$$

The actual dual update gives the exact identity

$$z_n - z_{n-1} = u_n. \tag{485}$$

The uniform primal bound implies

$$|u_n - u_{n-1}| \le 2UD, \qquad |w_n - w_{n-1}| \le \frac{5}{4}UD.$$

Put $H = (2/r + 5/4)UD$. If $w_n > H$ at a positive index, then $w_{n-1} > 2UD/r$, and thus

$$v_n = -(u_n - u_{n-1}) - rw_{n-1} < 0.$$

Since both (u_n, v_n) and $(0, 0)$ belong to the same monotone scalar graph, $u_n v_n \ge 0$. Hence $u_n \le 0$, and (485) gives $z_n \le z_{n-1}$. Reversing signs proves that $w_n < -H$ implies $z_n \ge z_{n-1}$.

Let $J = (2/r + 11/8)UD$. If $z_n > J$, then

$$w_n = z_n + u_n/8 > J - UD/8 = H,$$

so $z_n \le z_{n-1}$. This prevents a first crossing above $\max\{|z_0|, J\}$. The lower-barrier argument is identical. Therefore

$$|z_n| \le \max\{|z_0|, J\}.$$

Using $|z_0| \le |w_0| + |u_0|/8 \le 9D/8$ and $|w_n| \le |z_n| + UD/8$ proves (484). ∎

For the strict scalar interval, taking $U = U_r$ in this lemma gives

$$B_r = \sqrt{U_r^2 + V_r^2}, \qquad V_r = \max\{9/8, (2/r + 11/8)U_r\} + U_r/8. \tag{486}$$

This also supplies a scalar proof of paired convergence: primal error summability gives asymptotic regularity, boundedness supplies a KKT cluster point by continuity of the actual map, and restarting the bound around that point forces the full tail to converge. The conclusion is already contained in Theorem 11; here the order barrier is needed for the endpoint where primal error summability is unavailable.

C.12.2 An explicit constant for every restart

Let $D = \|(u_0, w_0)\|_2$. From the first two literal updates,

$$|u_1| \le \sqrt{10}D \le 4D, \qquad |w_1| \le D + (9|u_1| + |u_0|)/8 \le 6D,$$
$$|u_2| \le |u_1| + 3|w_1| \le 22D, \qquad |v_2| \le |u_1| + |u_2| + 3|w_1| \le 44D.$$

Thus $A_2 \le u_2 v_2 \le 968 D^2$. The symmetric matrix of (480) is

$$P_3 = \begin{pmatrix} 6 & 7/2 & 0 \\ 7/2 & 4 & 5/16 \\ 0 & 5/16 & 0 \end{pmatrix}.$$

For a symmetric matrix its Euclidean operator norm is at most the largest absolute row sum, which here is $19/2$. Since $A_1 \ge 0$,

$$\begin{aligned} E_2 &\le 7(968)D^2 + \tfrac{19}{2}(22^2 + 4^2 + 1)D^2 \\ &= \tfrac{23071}{2} D^2. \end{aligned} \tag{487}$$

Equations (482) and (483) imply

$$u_n^2 + u_{n-1}^2 \le \tfrac{4}{3} E_2 \le \tfrac{46142}{3} D^2 < 125^2 D^2 \quad (n \ge 2).$$

Including initialization gives $|u_n| \le 125D$ for all $n \ge 0$. Apply Lemma 9 at $r = 3$, $U = 125$. Its maximum is the second term, giving

$$|w_n| \le \left(\tfrac{2}{3} + \tfrac{11}{8} + \tfrac{1}{8}\right) 125D = \tfrac{1625}{6} D.$$

Consequently

$$\|(u_n, w_n)\|_2^2 \le \left(125^2 + \left(\tfrac{1625}{6}\right)^2\right) D^2 = \tfrac{3203125}{36} D^2 < 300^2 D^2. \tag{488}$$

For $D = 0$ the orbit is fixed. The constants are independent of the objective and anchor. Re-anchoring and restarting prove (114).

C.12.3 Three-step residuals and cluster dynamics

Define $\chi_n = u_n + u_{n-1} + u_{n-2}$ for $n \ge 2$. Summing (482) gives

$$2 \sum_{n=3}^{N} \chi_n^2 \le E_2 - E_N \le E_2.$$

Adding the single finite term χ_2^2 proves (115), and in particular $\chi_n \to 0$. For all $n \ge 3$, the following identities are exact:

$$u_n - u_{n-3} = \chi_n - \chi_{n-1}, \qquad w_n - w_{n-3} = \frac{9\chi_n - \chi_{n-1}}{8}. \tag{489}$$

The second follows by summing three actual dual increments. Since a linear combination of a square-summable sequence and its shift is square summable, (489) proves (116). In particular,

$$\|T^3(x_n, y_n) - (x_n, y_n)\|_2 \longrightarrow 0.$$

Let z be any cluster point and choose $(x_{n_j}, y_{n_j}) \to z$. Continuity of T^3 gives

$$(x_{n_j+3}, y_{n_j+3}) = T^3(x_{n_j}, y_{n_j}) \longrightarrow T^3 z.$$

The three-step residual tends to zero, so the same sequence tends to z. Hence $T^3 z = z$. A point fixed by the third iterate has least period one or three. This proves Theorem 12 and uses neither invertibility nor a classification of the entire original orbit.

C.13 Same-class three-cycles and sharpness

The following explicit construction is the affine-dual specialization of the scalar onset construction in the scalar onset classification in [2, Theorem 5]. Its complete verification is short enough to give here, so that no external cycle construction is a proof dependency.

For every $r \geq 3$, put

$$\alpha = \frac{1}{r} - \frac{11}{24}, \qquad \beta = \frac{1}{3} - \frac{1}{r}, \qquad f(x) = \max\{\alpha x, \beta x\}, \qquad h(y) = y/3. \tag{490}$$

Here $\alpha \leq -1/8 < 0 \leq \beta$ and $\alpha \leq \beta$, so the two functions are fixed proper closed convex functions. Their subgradients are

$$\partial f(0) = [\alpha, \beta], \qquad \partial f(x) = \{\beta\} \ (x > 0), \qquad \partial h(y) = \{1/3\}.$$

The pair

$$(x_*, y_*) = (1/3, 1/r - 1/3) \tag{491}$$

is KKT because $-y_* = \beta \in \partial f(1/3)$.

Recall the actual proximal characterization

$$x' = \mathrm{prox}_{rf}(x - ry) \quad \Longleftrightarrow \quad (x - x')/r - y \in \partial f(x').$$

Starting at $(0, 0)$, the choice $x' = 0$ has residual $0 \in \partial f(0)$ and produces $y' = -1/3$. Starting at $(0, -1/3)$, the choice $x' = 1$ has residual $1/3 - 1/r = \beta \in \partial f(1)$ and produces

$$y' = -1/3 + 9/8 - 1/3 = 11/24.$$

Starting at $(1, 11/24)$, the choice $x' = 0$ has residual $1/r - 11/24 = \alpha \in \partial f(0)$ and produces

$$y' = 11/24 - 1/8 - 1/3 = 0.$$

Uniqueness of each proximal output proves the genuine three-cycle

$$(0, 0) \longmapsto (0, -1/3) \longmapsto (1, 11/24) \longmapsto (0, 0). \tag{492}$$

At $r = 3$, $f(x) = \max\{-x/8, 0\}$, and the cycle still consists of three distinct pairs. Relative to the KKT primal coordinate $c = 1/3$, the three primal coordinates are $-1/3, -1/3, 2/3$, whose sum is zero. Thus the endpoint dissipation vanishes along this cycle, consistently with Theorem 12.

C.14 An explicit stabilization example

At $r = 3$ on $\mathbb{R}$, consider the fixed primal function and initialization

$$f(x) = \max\{0, -x/8\}, \qquad h_\mu(y) = \mu y^2/2 + y/3, \qquad (x_0, y_0) = (0, 0). \tag{493}$$

For every $\mu \geq 0$, the unique KKT pair is $(1/3, 0)$. Indeed $x = 1/3 > 0$ makes $\partial f(x) = \{0\}$, and no $x \leq 0$ can satisfy both $x = \mu y + 1/3$ and $-y \in \partial f(x)$. At $\mu = 0$, the exact orbit is the cycle (492) at $r = 3$. To verify it, the primal proximal map at step 3 is

$$\operatorname{prox}_{3f}(a) = \begin{cases} a + 3/8, & a < -3/8, \\ 0, & -3/8 \leq a \leq 0, \\ a, & a > 0. \end{cases}$$

Substituting the three pairs in the literal recurrence proves (492). For every fixed $\mu > 0$, however, Theorem 10 gives geometric convergence to $(1/3, 0)$.

This family also rules out a simultaneous uniform prefactor and rate as $\mu \downarrow 0$. Each finite iterate is continuous in μ, since the primal proximal map is continuous and the dual update divides by $1 + \mu$. If fixed $M < \infty$ and $\eta \in (0, 1)$ satisfied

$$\|(x_n^\mu - 1/3, y_n^\mu)\| \leq M\eta^n \|(-1/3, 0)\|$$

for every n and all sufficiently small $\mu > 0$, taking $\mu \downarrow 0$ at each fixed $n = 3k$ would give $1 \leq M\eta^{3k}$ for every k, a contradiction. The primal function and initialization are fixed here; the dual function varies with μ. This argument concerns a common prefactor and factor together and does not assert how any one chosen rate factor must behave.

C.15 An exact conjugacy when the primal objective is affine

The one-affine-objective conclusions also apply with the affine objective on the primal side. The update order requires a change of state variables rather than simply exchanging the names of the two coordinates.

Proposition C.1 (Affine-primal conjugacy)**.** Let $\mathcal{H}$ be a real Hilbert space, $r > 0$, $\theta \in \mathbb{R}$, and $f(x) = \langle a, x \rangle + d$. Let h be proper lower-semicontinuous convex. The actual map

$$x' = x - r(y + a), \qquad y' = \operatorname{prox}_h\big(y + (1 + \theta)x' - \theta x\big) \tag{494}$$

is conjugate, through the affine homeomorphism

$$U = y + a, \qquad Z = -x/r + (1 + \theta)U, \tag{495}$$

to the zero-affine-dual map

$$U' = \operatorname{prox}_{rG}(U - rZ), \qquad Z' = Z + (1 + \theta)U' - \theta U, \qquad G(U) = h(U - a)/r. \tag{496}$$

The inverse is $y = U - a$, $x = r[(1 + \theta)U - Z]$. The original KKT pair $(x_*, -a)$, with $x_* \in \partial h(-a)$, maps exactly to the transformed KKT pair $(0, -x_*/r)$.

Proof. Translation and positive scaling preserve the objective class. Changing variables in the proximal minimization gives

$$\operatorname{prox}_{rG}(v) = a + \operatorname{prox}_h(v - a).$$

Using $x' = x - rU$, the transformed first coordinate is

$$\begin{aligned} U' &= \operatorname{prox}_{rG}\big(U + (1+\theta)x' - \theta x\big) \\ &= \operatorname{prox}_{rG}\big(U + x - r(1+\theta)U\big) = \operatorname{prox}_{rG}(U - rZ). \end{aligned}$$

The other coordinate satisfies

$$Z' = -x'/r + (1+\theta)U' = -x/r + U + (1+\theta)U' = Z + (1+\theta)U' - \theta U.$$

These identities hold for every state and every initialization. The displayed inverse proves the homeomorphism. Original KKT conditions are $y_* = -a$ and $x_* \in \partial h(-a)$. Since $\partial G(0) = r^{-1}\partial h(-a)$, they are equivalent to $U_* = 0$ and $-Z_* \in \partial G(0)$. No KKT-existence assumption is needed for the conjugacy itself. ∎

Corollary C.4 (Transported strict and scalar endpoint conclusions)**.** *Assume the KKT set for* (494) *is nonempty and* $\theta = 1/8$*. For every* $0 < r < 3$*, there is* $x_\infty \in \partial h(-a)$ *such that*

$$y_n \to -a \text{ strongly}, \qquad x_n \rightharpoonup x_\infty \text{ weakly}.$$

The paired orbit is bounded, $\sum_{n\geq 0} \|y_n + a\|^2 < +\infty$*, and both increment sequences are square summable. The paired limit is strong in finite dimensions. For* $3/4 < r < 3$*, the homogeneous KKT restart constant can be taken as* $B_r^{\mathcal{H}}\kappa(M_r)$*, where*

$$M_r = \begin{pmatrix} 0 & 1 \\ -1/r & 9/8 \end{pmatrix}, \qquad \kappa(M_r) = \|M_r\|_2\|M_r^{-1}\|_2. \tag{497}$$

In the scalar space at $r = 3$*, the orbit is bounded with homogeneous KKT restart constant* $300\kappa(M_3)$*. The three-term dual residual*

$$y_n + y_{n-1} + y_{n-2} + 3a \quad (n \geq 2)$$

and the paired three-step residual are square summable, and every cluster point is fixed by the third iterate of the original map. No unique-cycle or phase-subsequence convergence is asserted.

Proof. Apply Theorem 11 to G and the zero dual objective. It gives $U_n \to 0$ strongly, Z_n weakly convergent, square summability of U_n, and square summability of the two transformed increment sequences. The affine inverse gives the claimed convergence topologies and KKT identification. Its bounded linear part preserves boundedness and square summability of paired increments. The transformed stability bound transports through

$$\begin{pmatrix} \Delta U \\ \Delta Z \end{pmatrix} = M_r \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}, \qquad M_r^{-1} = \begin{pmatrix} 9r/8 & -r \\ 1 & 0 \end{pmatrix},$$

so its factor is multiplied by $\kappa(M_r)$. These real matrices act on Hilbert-vector slots with the same norm inequalities.

In the scalar endpoint case, apply Theorem 12. The transformed primal residual is exactly the displayed three-term dual residual. The bounded inverse preserves square summability of the paired three-step residual. If Ψ denotes (495), the exact identity $\Psi T = S\Psi$ gives $\Psi T^3 = S^3\Psi$. A cluster point z has $S^3\Psi(z) = \Psi(z)$, and injectivity yields $T^3 z = z$.

For clarity, the change of variables is not isometric. Its squared singular values are $(q \pm \sqrt{q^2 - 4/r^2})/2$, where $q = 1 + r^{-2} + (9/8)^2$. Thus

$$\kappa(M_3) = \frac{1369 + \sqrt{1726705}}{384}, \tag{498}$$

and the transported scalar endpoint constant is 300 times this quantity, rather than 300. ∎

No strong primal convergence for affine f in a general infinite-dimensional Hilbert space follows from this transport. At least one of the two objectives remains affine throughout these conclusions. In particular, this conjugacy does not assert a corresponding theorem for a quadratic primal objective.

The affine-primal boundary also admits an explicit genuine cycle. At $r = 3$ take

$$f(x) = 0, \qquad h(y) = \max\{0, -(3y + 1)/8\}.$$

Both functions are finite continuous convex, and $(0, 0)$ is KKT because h is locally constant at zero. The actual map has the cycle

$$(-9/8, -1/3) \longmapsto (-1/8, -1/3) \longmapsto (7/8, 2/3) \longmapsto (-9/8, -1/3). \tag{499}$$

Indeed the primal update is $x' = x - 3y$, and the dual proximal inputs in that order are $-1/3$, $2/3$, and $-17/24$. The proximal map is

$$\operatorname{prox}_h(t) = \begin{cases} t + 3/8, & t < -17/24, \\ -1/3, & -17/24 \le t \le -1/3, \\ t, & t > -1/3. \end{cases}$$

It sends those inputs to $-1/3$, $2/3$, and $-1/3$, respectively, including both boundary inputs exactly. The three pairs are distinct. Thus the strict affine-primal convergence conclusion cannot be extended universally to $r = 3$, consistently with the transported endpoint theorem.

C.16 Why the same storage does not settle the general scalar gap

Consider now two arbitrary fixed proper closed convex scalar objectives, shifted by a KKT pair so that both have value zero and a zero subgradient at zero. Write $F = rf$ for

the shifted primal objective and H for the shifted dual objective. Use s_n from (420) and the affine trial storage $E_n = 7F^*(v_n) - F^*(v_{n-1}) + P_1(u_n, u_{n-1}, u_{n-2})$. The actual graph values become

$$\begin{aligned} v_n &= u_{n-1} - u_n - rw_{n-1} \in \partial F(u_n), \\ a_n &= w_{n-1} - w_n + (9u_n - u_{n-1})/8 \in \partial H(w_n). \end{aligned}$$

For affine dual objectives after centering, $a_n = 0$. In the general case the same elimination gives

$$v_n - v_{n-1} = -s_n + ra_{n-1}. \tag{500}$$

Repeating the common Fenchel calculation at $q = 1$, as specialized in Lemma 8 produces

$$\begin{aligned} E_n - E_{n-1} \le & -\lambda(u_n + d_r u_{n-1} + u_{n-2})^2 - \delta u_{n-1}^2 \\ & + r\big[8u_n a_{n-1} - u_{n-2}(a_{n-1} + a_{n-2})\big]. \end{aligned} \tag{501}$$

The corresponding lower bound is only

$$E_n \ge 6F^*(v_n) + \begin{pmatrix} u_n & u_{n-1} \end{pmatrix} \mathbf{R}_1 \begin{pmatrix} u_n \\ u_{n-1} \end{pmatrix} + ru_{n-1}a_{n-1}. \tag{502}$$

The extra terms are not signed by scalar monotonicity, which relates a_n to w_n, rather than to u_n.

There is an exact example in which this specific storage increases even though the actual orbit converges geometrically. Set

$$r = \frac{299}{100}, \qquad f(x) = \frac{x^2}{198r}, \qquad h = \iota_{\{0\}}, \qquad (x_0, y_0) = (0, 1), \tag{503}$$

where $\iota_{\{0\}}$ is zero at zero and $+\infty$ elsewhere. The unique KKT pair is $(0, 0)$: the dual variable must be zero and $f'(x) = x/(99r)$ then forces $x = 0$. The actual map is

$$x_n = \frac{99}{100}(x_{n-1} - ry_{n-1}), \qquad y_n = 0.$$

It follows that

$$x_1 = -\frac{29601}{10000}, \quad x_2 = -\frac{2930499}{1000000}, \quad x_3 = -\frac{290119401}{100000000},$$

and thereafter $x_n = (99/100)^{n-1}x_1 \to 0$. For $F(x) = x^2/198$, the actual graph values satisfy $v_n = x_n/99$ and

$$F^*(v_n) = x_n^2/198.$$

Substitution into the affine specialization of (432), with $P = P_1$, yields

$$E_3 - E_2 = \frac{5012843039176242699}{20000000000000000000} > 0. \tag{504}$$

The rational comparison $(81r - 88)^2 > 16^2 \cdot 91$, with $81r - 88 > 0$, shows $r > r_*$; also $r < 3$. Thus (504) is a failure of this storage in the open general scalar gap. It is not a nonconvergence example and implies no error in an algorithm or external theorem. A valid extension to two nonlinear objectives would need an additional argument controlling both new terms in (501)–(502).

D Fixed metrics IQC gains and restricted robustness

D.1 Proof of the fixed positive metric theorem

Proof of Theorem 8. *The shared normalization and a static supply.* Fix any KKT pair and write $p_n = x_n - x^*$, $w_n = y_n - y^*$ and

$$F(p) = \tau\big(f(x^* + p) - f(x^*) + \langle L^* y^*, p\rangle\big).$$

As before, $F(0) = 0 = \min F$, and from the very first output,

$$v_n = p_{n-1} - p_n - \tau L^* w_{n-1} \in \partial F(p_n), \qquad \langle p_n, v_n\rangle \geq 0. \tag{505}$$

There is no initial-domain assumption. Define

$$A = (I + \sigma Q)^{-1}, \quad B = \sqrt{\tau\sigma/3}\, L, \quad S = (I - B^*B)^{1/2}, \quad u_n = Bp_n, \quad t_n = \sqrt{\tau/(3\sigma)}\, w_n. \tag{506}$$

Here $B^*B \preceq I$, so S is well defined. The recurrence and its static supply decompose exactly as

$$t_n = A\left(t_{n-1} + \frac{9u_n - u_{n-1}}{8}\right), \qquad v_n = p_{n-1} - p_n - 3B^* t_{n-1}, \tag{507}$$

$$\langle p_n, v_n\rangle = \langle Sp_n, S(p_{n-1} - p_n)\rangle + \langle u_n, u_{n-1} - u_n - 3t_{n-1}\rangle. \tag{508}$$

This is a linear supply decomposition, with no weighted or coordinatewise convexity assertion about f.

The product-three scalar plant. In an orthogonal eigenbasis of A, every eigenvalue satisfies $0 < q \leq 2/3$. For one coordinate, let (a, b) be the previous state and h the current input. The linear update and supply output are

$$(a', b') = (h, q[b + (9h - a)/8]), \qquad v = a - h - 3b.$$

Its state matrix and input column are

$$A_0 = \begin{pmatrix} 0 & 0 \\ -q/8 & q \end{pmatrix}, \qquad B_0 = \begin{pmatrix} 1 \\ 9q/8 \end{pmatrix}.$$

Thus A_0 is Schur. The scalar transfer, a product-three specialization of the shared plant, is

$$V_q(z) = (z - 1) - \frac{3zq(9 - z)}{8(1 - qz)},$$

where z is a delay. Put $a_f = \cos\vartheta$ and $D = 1 - 2qa_f + q^2 > 0$. The exact numerator is

$$D\,\mathrm{Re}\, V_q(z) = W(q, a_f) := \frac{(19 + 5a_f)q^2 - (3 + 11a_f + 10a_f^2)q + 8(a_f - 1)}{8}. \tag{509}$$

For $a_f \in [-1,1]$, this is strictly convex in q. Its endpoint values are $W(0,a_f) = a_f - 1 \le 0$ and

$$W(2/3,a_f) = -\frac{30a_f^2 - 13a_f + 7}{36} = -\frac{5}{6}(a_f - 13/60)^2 - \frac{671}{4320} < 0.$$

Convex interpolation gives

$$W(q,a_f) \le (1-3q/2)(a_f-1) - \frac{3q}{2}\frac{671}{4320} < 0. \tag{510}$$

For each fixed q the real part is uniformly strictly negative on the entire unit circle, including DC.

Strict KYP and the separate positivity argument. Use the strict discrete-time KYP lemma in [3, Lemma 12, pp. 6–7]. Its assumptions are a real state matrix without unit-circle eigenvalues and a Hermitian supply matrix; no controllability hypothesis is needed. Here the supply in state/input order (a,b,h) is the real symmetric matrix

$$\begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & -3/2 \\ 1/2 & -3/2 & -1 \end{pmatrix},$$

representing $h(a-h-3b)$, and (510) supplies the strict frequency condition. The source uses the advance variable $\lambda = 1/z$. Its nonnegative supply multiplier must be positive: if it were zero, evaluating the strict matrix inequality on $((\lambda I - A_0)^{-1}B_0, 1)$ for $|\lambda| = 1$ would cancel the storage difference and contradict strict negativity. Divide by this positive multiplier. We obtain a symmetric P_q and $\epsilon_q > 0$ such that, for all real a, b, h,

$$E_q(a',b') - E_q(a,b) + h(a-h-3b) \le -\epsilon_q(a^2+b^2+h^2), \qquad E_q(a,b) = \begin{pmatrix} a \\ b \end{pmatrix}^{\mathsf{T}} P_q \begin{pmatrix} a \\ b \end{pmatrix}. \tag{511}$$

Symmetry alone does not imply positivity. To prove it here, set the free plant input h to zero at every subsequent step. Then the supply vanishes and $\xi_k = A_0^k \xi_0 \to 0$. Telescoping (511) gives

$$E_q(\xi_0) \ge \epsilon_q \sum_{k \ge 0} \|A_0^k \xi_0\|^2 \ge \epsilon_q \|\xi_0\|^2. \tag{512}$$

Thus $P_q \succ 0$. This zero-input argument concerns the linear plant only; it assumes no boundedness or summability of the nonlinear orbit.

For completeness, $\det[B_0, A_0 B_0] = q(9q-1)/8$, whose only zero in this range is $q = 1/9$. Although the cited lemma includes this mode, there is also a direct full-state certificate. Put $\zeta = b - a/8$; then $\zeta' = \zeta/9$, $v = 5a/8 - h - 3\zeta$, and set $E(a,\zeta) = a^2/2 + 50\zeta^2$. Directly,

$$\begin{aligned} E(h,\zeta/9) - E(a,\zeta) + hv &= -\tfrac{1}{2}h^2 - \tfrac{1}{2}a^2 + \tfrac{5}{8}ha - 3h\zeta - \tfrac{4000}{81}\zeta^2 \\ &\le -\tfrac{3}{32}h^2 - \tfrac{3}{16}a^2 - \tfrac{2056}{81}\zeta^2. \end{aligned}$$

The last line uses $5|ha|/8 \le 5(h^2+a^2)/16$ and $3|h\zeta| \le 3h^2/32 + 24\zeta^2$. The coordinate change is invertible, so this gives positive storage and strict full-state dissipation in the original coordinates too.

The coercive lift and actual-orbit dissipation. Choose one such storage for each of the finitely many eigenvalues of A and define, in that orthogonal dual basis,

$$\mathcal{H}(p,t) = \tfrac{1}{2}\|Sp\|^2 + \sum_{j=1}^{m} E_j((Bp)_j, t_j). \tag{513}$$

If $m_P = \min_j \lambda_{\min}(P_j) > 0$ and $M_P = \max_j \lambda_{\max}(P_j)$, then $\|Sp\|^2 + \|Bp\|^2 = \|p\|^2$ gives

$$\min\{1/2, m_P\}(\|p\|^2 + \|t\|^2) \le \mathcal{H}(p,t) \le \max\{1/2, M_P\}(\|p\|^2 + \|t\|^2). \tag{514}$$

The physical coordinate w differs from t by a fixed positive scale, so this is a fixed coercive quadratic metric on (p, w). No simultaneous diagonalization with L is used. The remaining proximal-point supply satisfies the exact identity

$$\tfrac{1}{2}\|Sp_n\|^2 - \tfrac{1}{2}\|Sp_{n-1}\|^2 + \langle Sp_n, S(p_{n-1} - p_n)\rangle = -\tfrac{1}{2}\|S(p_n - p_{n-1})\|^2.$$

Adding the scalar inequalities to this identity and using (508) yields, for some fixed $\epsilon > 0$,

$$\begin{aligned}&\mathcal{H}(p_n, t_n) - \mathcal{H}(p_{n-1}, t_{n-1}) + \langle p_n, v_n\rangle \\ &\quad\le -\tfrac{1}{2}\|S(p_n - p_{n-1})\|^2 - \epsilon\big(\|u_{n-1}\|^2 + \|t_{n-1}\|^2 + \|u_n\|^2\big).\end{aligned} \tag{515}$$

The supply is nonnegative at each sample by (505). This is an actual finite-horizon Lyapunov inequality, not a soft IQC on an orbit whose summability remains to be proved.

From boundedness to full convergence. Coercivity and descent prove boundedness, square summability of u, t, and square summability of $S(p_n - p_{n-1})$. Also $\sum_n \|B(p_n - p_{n-1})\|^2 = \sum_n \|u_n - u_{n-1}\|^2 < \infty$. Since $S^*S + B^*B = I$, the primal increments are square summable and tend to zero. The recurrence then gives $v_n \to 0$, while $w_n \to 0$. Choose a convergent subsequence $p_{n_k} \to \bar{p}$ in finite dimensions. We have $L\bar{p} = 0$ since $Bp_{n_k} \to 0$. Closedness of the convex subdifferential graph, applied to (505), gives $0 \in \partial F(\bar{p})$. Therefore $(x^* + \bar{p}, y^*)$ is a KKT pair. The same quadratic form $\mathcal{H}$ works about every KKT pair, since its coefficients depend only on the linear parameters and the anchored supply is valid at every pair. Its distance to the cluster pair is nonincreasing and tends to zero along the chosen subsequence. It must tend to zero along the whole orbit. Coercivity proves the stated paired norm convergence and completes the proof. ∎

Both parameter equalities are included. The conclusion permits a nonunique primal minimizer and controls $L(x_n - x^*)$, not the full primal error sum. This static proof does not extend its quadratic-metric monotonicity claim to unrestricted $\sigma Q \succ 0$; general convergence follows instead from Theorem 16. It makes no additional claim for semidefinite Q, a nonquadratic dual, or products above 3. If one space has dimension zero, omit its storage block: the remaining argument is the proximal-point identity or the stable affine dual recursion. For $L = 0$ both are independent, and any $Q \succ 0$ suffices for finite-dimensional convergence even without the scaled lower bound.

D.2 Explicit soft IQC gains and the sharper identity estimate

For the remainder of these soft-IQC estimates, specialize the common problem to finite-dimensional $X = \mathbb{R}^d$, $Y = \mathbb{R}^k$ and $Q \succ 0$. Their role is quantitative: they give explicit constants without selecting a graph-storage certificate. The arbitrary-coupling branch assumes composite growth, while identity coupling has a sharper separator. Both use the same literal map and KKT equations already specified above.

Theorem 18 (Rectangular coupling and composite curvature)**.** *Let $f : X \to (-\infty, +\infty]$ be proper lower-semicontinuous convex, and suppose the composite* (1) *is globally μ-strongly convex for a specified $\mu > 0$. Let $\tau, \sigma > 0$, $\theta = 1/8$, and $\tau\sigma\|L\|^2 \le 3$. Use the literal iteration* (3)*. There is a unique KKT pair satisfying* (2)*. For every initialization in $X \times Y$ and every restart $j \ge 0$,*

$$\sum_{n=j}^{\infty} \left(\|x_n - x^*\|^2 + \|y_n - y^*\|^2\right) \le C\left(\|x_j - x^*\|^2 + \|y_j - y^*\|^2\right), \tag{516}$$

and consequently

$$\|(x_n - x^*, y_n - y^*)\|^2 \le C(1 - C^{-1})^n \|(x_0 - x^*, y_0 - y^*)\|^2. \tag{517}$$

Here $C > 1$ depends only on τ, σ, Q, L, μ. One explicit choice is obtained by setting $\ell = \|L\|$, $d_s = \tau\sigma$, and

$$\begin{aligned} A = (I + \sigma Q)^{-1}, \qquad q_- = \lambda_{\min}(A), \qquad \rho = \lambda_{\max}(A), \qquad \kappa = q_-(1 - \rho), \\ D_1 = \frac{7\rho}{4(1-\rho)^2} + \frac{11\rho^2}{2(1-\rho)^4}, \qquad B = 1 + d_s \ell^2 D_1, \\ t_0 = \min\left\{1, \frac{\tau\mu}{2B}\right\}, \qquad c_0 = \frac{7\kappa}{8}, \qquad H_{\max} = \tau\mu + 2B, \\ \eta = \frac{c_0 t_0^3}{2(1 + H_{\max})}, \qquad \delta = \min\left\{\frac{\eta\tau\mu}{2}, \frac{c_0 t_0^3}{2}\right\}, \qquad K = \frac{20 + \eta}{\delta}, \\ E = 1 + \tau^2\ell^2 + \frac{\tau^2\ell^2\rho^2}{1 - \rho^2}\left(1 + \frac{\sigma^2\ell^2}{64}\right), \qquad U_0 = 1 + K^2 E. \end{aligned} \tag{518}$$

Then

$$C = \left(1 + \frac{25\sigma^2\ell^2\rho^2}{8(1-\rho)^2}\right) U_0 + \frac{2}{1 - \rho^2}. \tag{519}$$

All constants are finite and $\delta > 0$, including when $L = 0$. Neither injectivity nor surjectivity of L, commutation of Q with LL^*, strong convexity of f alone, finite $f(x_0)$, nor an initial subgradient is required.

For clarity, arbitrary steps can also be reduced to unit steps without changing the rectangular geometry. Put

$$\bar{x} = x/\sqrt{\tau}, \quad \bar{y} = y/\sqrt{\sigma}, \quad \bar{L} = \sqrt{\tau\sigma}\, L, \quad \bar{Q} = \sigma Q, \quad \bar{c} = \sqrt{\sigma}\, c, \quad \bar{f}(\bar{x}) = f(\sqrt{\tau}\,\bar{x}). \tag{520}$$

Equation (3) becomes the same primal-first map with unit steps and data $(\bar f, \bar L, \bar Q, \bar c)$, and $\|\bar L\|^2 \le 3$. Its composite objective is $\bar\Phi(\bar x) = \Phi(\sqrt{\tau}\,\bar x)$, with strong-convexity modulus $\tau\mu$. The estimates above and the proof below use the original, unscaled paired-state Euclidean norm.

Identity coupling admits a stronger statement with no additional growth hypothesis. We retain its quantitative constants, rather than obtaining them by specializing the more conservative general formula.

Theorem 19 (Positive definite quadratic dual curvature)**.** *Let Q be a real symmetric positive definite $d \times d$ matrix, let $f : \mathbb{R}^d \to (-\infty, +\infty]$ be proper lower-semicontinuous convex, and let $c \in \mathbb{R}^d$. With identity coupling and parameters $\theta = 1/8$, $\tau = r$, $\sigma = 1$, suppose $0 < r \le 3$ and consider the literal primal-first iteration*

$$\begin{aligned} x_n &= \operatorname{prox}_{rf}(x_{n-1} - r y_{n-1}), \\ y_n &= (I+Q)^{-1}\left(y_{n-1} + \frac{9x_n - x_{n-1}}{8} - c\right), \qquad n \ge 1. \end{aligned} \tag{521}$$

There is a unique KKT pair (x^*, y^*), characterized by

$$-y^* \in \partial f(x^*), \qquad x^* = Qy^* + c. \tag{522}$$

For every initialization $(x_0, y_0) \in \mathbb{R}^d \times \mathbb{R}^d$ and every restart $m \ge 0$,

$$\sum_{n=m}^{\infty} \left(\|x_n - x^*\|^2 + \|y_n - y^*\|^2\right) \le C_{r,Q}\left(\|x_m - x^*\|^2 + \|y_m - y^*\|^2\right), \tag{523}$$

where $C_{r,Q} > 1$ depends only on r and Q. In particular,

$$\|(x_n - x^*, y_n - y^*)\|^2 \le C_{r,Q}(1 - C_{r,Q}^{-1})^n \|(x_0 - x^*, y_0 - y^*)\|^2, \qquad n \ge 0. \tag{524}$$

One explicit choice is as follows. Set

$$\begin{gathered} A = (I+Q)^{-1}, \qquad q_- = \lambda_{\min}(A), \qquad q_+ = \lambda_{\max}(A), \qquad \rho = q_+, \\ \kappa = q_-(1-q_+), \qquad \eta = r\kappa, \qquad \gamma = \frac{r^2\kappa^2}{4}, \qquad K = \frac{20+\eta}{\gamma}, \\ E = 1 + r^2 + \frac{65 r^2 \rho^2}{64(1-\rho^2)}. \end{gathered} \tag{525}$$

Then the bound in (523) *holds with*

$$C_{r,Q} = \left(1 + \frac{25\rho^2}{8(1-\rho)^2}\right)(1 + K^2 E) + \frac{2}{1-\rho^2}. \tag{526}$$

These constants are independent of f, c, and the initialization.

The symbol $C_{r,Q}$ denotes the bound from the present matrix argument; it is distinct from the scalar-curvature bound $C_{r,\mu}$ used earlier. Even when $Q = \mu I$, the two formulas need not agree.

D.2.1 The shared anchored IQC argument

Proof of Theorems 18 and 19.

The anchor and centered normalization. For Theorem 18, the proper closed convex function $\Phi - \mu\|\cdot\|^2/2$ has an affine minorant. Hence Φ is coercive and attains a unique minimum x^* in finite dimensions. The smooth sum rule gives $y^* = Q^{-1}(Lx^* - c)$ and $-L^*y^* \in \partial f(x^*)$. The dual variable is uniquely determined even if L has a kernel or a nontrivial cokernel. For Theorem 19, the same existence argument applies to

$$x \longmapsto f(x) + \frac{1}{2}\langle Q^{-1}(x-c), x-c\rangle,$$

which is automatically strongly convex and coercive because $Q \succ 0$. Thus (522) requires no additional assumption on f. These arguments also apply to extended-valued f with lower-dimensional domain.

In the general parameters write

$$\begin{gathered} u_n = x_n - x^*, \qquad w_n = y_n - y^*, \qquad \xi_n = (u_n, w_n), \\ \widehat{f}(u) = f(x^*+u) - f(x^*) + \langle L^*y^*, u\rangle, \qquad F = \tau\widehat{f}, \qquad P = \operatorname{prox}_F. \end{gathered} \tag{527}$$

The KKT equations imply $\widehat{f} \geq 0$, $\widehat{f}(0) = 0$ and $0 \in \partial\widehat{f}(0)$. Thus $P(0) = 0$, and P is everywhere defined and 1-Lipschitz. For $L = I$, $\tau = r$, $\sigma = 1$, this is precisely the previous normalization

$$F(u) = r\big(f(x^*+u) - f(x^*) + \langle y^*, u\rangle\big), \qquad P = \operatorname{prox}_F. \tag{528}$$

For arbitrary rectangular L, the centered iteration is

$$u_n = P(u_{n-1} - \tau L^* w_{n-1}), \qquad w_n = A\left(w_{n-1} + \frac{\sigma L(9u_n - u_{n-1})}{8}\right), \tag{529}$$

where $A = (I + \sigma Q)^{-1}$ and $0 < q_- \leq \rho < 1$. Its identity specialization is

$$u_n = P(u_{n-1} - r w_{n-1}), \qquad w_n = A\left(w_{n-1} + \frac{9u_n - u_{n-1}}{8}\right). \tag{530}$$

We use one anchored matrix R to distinguish the two branches:

$$R = \tau(\mu I - W), \quad W = L^*Q^{-1}L \quad \text{in Theorem 18}, \qquad R = 0 \quad \text{in Theorem 19}. \tag{531}$$

In the general case, $\widehat{f}(u) + \langle Wu, u\rangle/2 = \Phi(x^*+u) - \Phi(x^*)$ is μ-strongly convex and minimized at zero. Strong monotonicity of its subdifferential against the anchor gives

$$\langle p, v\rangle \geq \langle p, Rp\rangle, \qquad \langle p, v\rangle \geq 0, \qquad v \in \partial F(p). \tag{532}$$

The second inequality follows separately from ordinary convexity of F. Both are also valid in the identity branch with $R = 0$. The first is an anchored inequality only: R may be indefinite, and we do not assert convexity after subtracting its quadratic, spatially weighted cyclic monotonicity, or positivity of $I + R$.

The IQC theorem and one stable plant. We use [3, Theorem 2] in its ordinary discrete-time form. For a stable proper real-rational plant G and a bounded causal operator Δ, the hypotheses are causal well-posedness of the feedback with $\alpha\Delta$ for every $\alpha \in [0,1]$, a common Hermitian frequency IQC for all these homotopies, and uniform strict negative separation of the plant graph on the unit circle. The conclusion is a finite ℓ_2 input-output gain with additive inputs at both ports. A causal multiplier or a finite-horizon hard IQC is not required. We use this theorem first to establish actual square summability; the quantitative state estimate and rate then follow from the same supply and an autonomous restart argument.

Index the feedback signals by $n \geq 1$, and put $p_n = u_n$, $s_n = u_{n-1} - \tau L^* w_{n-1}$. The shared open linear realization is

$$\begin{gathered} \xi_n = B_0\xi_{n-1} + B_1 p_n, \qquad s_n = C_0\xi_{n-1}, \\ B_0 = \begin{pmatrix} 0 & 0 \\ -\sigma AL/8 & A \end{pmatrix}, \qquad B_1 = \begin{pmatrix} I \\ 9\sigma AL/8 \end{pmatrix}, \qquad C_0 = \begin{pmatrix} I & -\tau L^* \end{pmatrix}. \end{gathered} \tag{533}$$

Its state eigenvalues are zero and those of A, so the plant is stable and strictly causal. Throughout, z denotes a one-step delay. The transfer is

$$\begin{gathered} G(z) = I + V(z), \qquad V(z) = (z-1)I - d_s z L^* A \frac{9I - zI}{8}(I - Az)^{-1}L, \\ s = Gp + e, \qquad p = P(s), \qquad d_s = \tau\sigma. \end{gathered} \tag{534}$$

With $L = I$, $\tau = r$, $\sigma = 1$, this becomes exactly

$$G(z) = zI - rzA\left(\frac{9}{8}I - \frac{z}{8}I\right)(I - Az)^{-1}, \qquad s = Gp + e, \quad p = P(s). \tag{535}$$

The change $z = \lambda^{-1}$ gives the conventional proper real-rational transfer variable of the cited theorem.

The map P is memoryless and bounded on one-sided ℓ_2, since $\|P(s_n)\| \leq \|s_n\|$, and it is defined on extended ℓ_2 (signals with square-summable finite truncations). For arbitrary additive inputs a, b and every $\alpha \in [0,1]$, the interconnection $s = Gp + a$, $p = \alpha P(s) + b$ is uniquely solved in time order: s_n depends only on past plant inputs and a_n, after which p_n is determined. This constructs its causal inverse on extended ℓ_2 without presupposing closed-loop square summability.

One soft IQC for the entire homotopy. For $0 < \alpha \leq 1$, define

$$F_\alpha(p) = \alpha F(p/\alpha) + \frac{1-\alpha}{2\alpha}\|p\|^2. \tag{536}$$

This function is proper closed convex, nonnegative, and zero at zero. Moreover,

$$s - p \in \partial F_\alpha(p) \quad \Longleftrightarrow \quad s - p/\alpha \in \partial F(p/\alpha) \quad \Longleftrightarrow \quad p = \alpha P(s).$$

The strengthened anchored supply also survives this homotopy. Indeed, set $b = P(s)$, $g = s - b$, $p = \alpha b$, $v = s - p$. Then

$$\begin{aligned}\langle p, v\rangle - \langle p, Rp\rangle &= \alpha^2\big(\langle b, g\rangle - \langle b, Rb\rangle\big) \\ &\quad + \alpha(1-\alpha)\big(\langle b, g\rangle + \|b\|^2\big) \geq 0.\end{aligned} \tag{537}$$

The two terms are nonnegative by the two separate inequalities in (532); no sign condition on R is needed. For $\alpha = 0$ the nonlinear output is zero and all supplies vanish.

Take an arbitrary $s \in \ell_2$, put $p = \alpha P(s)$, $v = s - p$, and extend the signals by zero to all indices $n \leq 0$. For $\alpha > 0$, both $p, v \in \ell_2$ and the added pair $(0, 0)$ belongs to the same subgradient graph. Thus, for every integer n,

$$v_n \in \partial F_\alpha(p_n), \qquad 0 \leq F_\alpha(p_n) \leq \langle p_n, v_n\rangle. \tag{538}$$

Cauchy–Schwarz makes the potential values summable. For any integer k, sum the ordinary Euclidean subgradient inequalities

$$F_\alpha(p_{n+k}) \geq F_\alpha(p_n) + \langle v_n, p_{n+k} - p_n\rangle$$

over $n \in \mathbb{Z}$. The absolutely convergent potential sums cancel under the shift; all cross pairings are absolutely summable. Hence

$$\sum_{n\in\mathbb{Z}} \langle p_n - p_{n+k}, v_n\rangle \geq 0 \qquad (k \in \mathbb{Z}). \tag{539}$$

For $\eta > 0$, use the single temporal multiplier

$$(M_\eta p)_n = (10 + \eta)p_n - 7p_{n+1} - 2p_{n-2} - p_{n-3}. \tag{540}$$

Combine shifts $1, -2, -3$ with weights $7, 2, 1$ and add η times (537). The common supply is

$$\sum_{n\in\mathbb{Z}} [\langle (M_\eta p)_n, s_n - p_n\rangle - \eta\langle p_n, Rp_n\rangle] \geq 0, \qquad p = \alpha P(s), \quad s \in \ell_2. \tag{541}$$

For $R = 0$ this is the identity-coupling soft IQC. The forward shift makes the multiplier generally noncausal; its output at $n = 0$ causes no boundary term because $s_n - p_n = 0$ there. No finite-horizon hard IQC is asserted, and no spatial weighting of convexity was used.

With $z = e^{-it}$, its scalar symbol is $M_\eta(z) = 10 + \eta - 7z^{-1} - 2z^2 - z^3$. In signal order (s, p), let

$$\Pi_{\eta,R}(z) = \begin{pmatrix} 0 & M_\eta(z) I_d/2 \\ \overline{M_\eta(z)} I_d/2 & -(M_\eta(z) + \overline{M_\eta(z)}) I_d/2 - \eta R \end{pmatrix}, \qquad \Pi_\eta = \Pi_{\eta,0}. \tag{542}$$

It is Hermitian, continuous and bounded on the circle. For the convention $\widehat{a}(t) = \sum_{n\in\mathbb{Z}} a_n e^{-int}$, Parseval gives

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \begin{pmatrix} \widehat{s}(t) \\ \widehat{p}(t) \end{pmatrix}^* \Pi_{\eta,R}(e^{-it}) \begin{pmatrix} \widehat{s}(t) \\ \widehat{p}(t) \end{pmatrix} dt \geq 0. \tag{543}$$

This is a common IQC for every scalar homotopy of the coupled proximal map. Its plant restriction is

$$\begin{pmatrix} G \\ I \end{pmatrix}^* \Pi_{\eta,R} \begin{pmatrix} G \\ I \end{pmatrix} = \mathrm{Re}(\overline{M_0} V) + \eta\big(\mathrm{Re}\, V - R\big), \tag{544}$$

where $\mathrm{Re}\, T = (T + T^*)/2$ denotes the Hermitian part for matrices.

D.2.2 A common factorization and two strict separators

The scalar identity used in both branches. For a scalar dual eigenvalue $q \in (0, 1)$ and $0 \le r \le 3$, define

$$V_{r,q}(z) = -\frac{1 + (9qr/8 - 1 - q)z + q(1 - r/8)z^2}{1 - qz}. \tag{545}$$

In the identity branch these are the eigenvalues of $G - I$; otherwise we use them only to derive a scalar bound before congruence with L. Let $a = \cos t$, $h_t = 1 - a$, $D = 1 - 2qa + q^2$, and

$$S(q, t) = \mathrm{Re}\big(\overline{M_0(z)} V_{r,q}(z)\big), \qquad \overline{M_0(z)} = 10 - 7z - 2z^{-2} - z^{-3}.$$

The common exact factorization is

$$\begin{aligned} DS &= (1 - r/3)N_0 + (r/3)N_3, \\ N_0 &= -4(1 - a)^3 D, \\ N_3 &= -(1 - a)\Big[4(1 - q)(1 - a)^2 + 2q(1 - a)(2a + 1)^2 \\ &\qquad + \frac{q(1 - q)}{2}(19a^2 + 58a + 67)\Big]. \end{aligned} \tag{546}$$

Since $19a^2 + 58a + 67 - 28 = (a + 1)(19a + 39) \ge 0$ on $[-1, 1]$, $S(q, t) < 0$ away from zero frequency for $0 < r \le 3$, but $S(q, 0) = 0$. Only the linear spectral calculus is used; F is never decomposed coordinatewise.

General coupling: congruence and composite DC repair. Write $b_0 = -4h_t^3$ and

$$b_1(q, a) = \mathrm{Re}\left(-\overline{M_0(z)} \frac{zq(9 - z)}{8(1 - qz)}\right).$$

The $r = 3$ endpoint of (546) says $b_0 + 3b_1 = N_3/D$. Since $D \le 4$ and $q(1 - q) \ge \kappa = q_-(1 - \rho)$ for $q \in [q_-, \rho]$, it implies

$$b_1(q, a) \le -\frac{b_0}{3} - \frac{7\kappa}{6} h_t. \tag{547}$$

Spectral calculus for A, followed by congruence with L, gives

$$\begin{aligned} \mathrm{Re}(\overline{M_0} V) &= b_0 I + d_s L^* b_1(A, a) L \\ &\preceq b_0 I + \left(-\frac{b_0}{3} - \frac{7\kappa h_t}{6}\right) T, \qquad T = d_s L^* L, \quad 0 \preceq T \preceq 3I. \end{aligned} \tag{548}$$

For an eigenvalue $\lambda \in [0,3]$ of T, the last expression is

$$-4h_t^3(1-\lambda/3) - \frac{7\kappa}{6}h_t\lambda \le -h_t^3\left[\frac{7\kappa}{8}(1-\lambda/3) + \frac{7\kappa}{24}\lambda\right] = -\frac{7\kappa}{8}h_t^3.$$

Here $h_t^2 \le 4$ and $\kappa < 1$. Thus

$$\operatorname{Re}(\overline{M_0}V) \preceq -c_0 h_t^3 I. \tag{549}$$

This holds on the original primal Euclidean space, including $\ker L$; no simultaneous diagonalization of Q and LL^* is needed.

At zero frequency, $\sigma A(I-A)^{-1} = Q^{-1}$, so

$$V(1) = -\tau W, \qquad H(1) := \operatorname{Re} V(1) - R = -\tau\mu I. \tag{550}$$

To make this repair uniform, the scalar real part of the dual factor is

$$R_q(a) = \operatorname{Re}\frac{zq(9-z)}{8(1-qz)} = \frac{q[-2a^2 + (9+q)a + 1 - 9q]}{8(1-2qa+q^2)}. \tag{551}$$

For $q \in [q_-, \rho]$ and $a \in [-1,1]$, the numerator bracket and its a-derivative have absolute values at most 22 and 14, respectively, while $D \ge (1-\rho)^2$. The quotient rule yields $|R_q'(a)| \le D_1$ with D_1 in (518). Spectral calculus and congruence then give

$$\begin{aligned} \|\operatorname{Re} V(a) - \operatorname{Re} V(1)\| &\le Bh_t, \\ H(a) := \operatorname{Re} V(a) - R \preceq (-\tau\mu + Bh_t)I, &\qquad \|H(a)\| \le H_{\max}. \end{aligned} \tag{552}$$

If $h_t \le t_0$, this is $H(a) \preceq -\tau\mu I/2$. Combining (549) with the parameters in (518) yields the two regions

$$\operatorname{Re}(\overline{M_0}V) + \eta H(a) \preceq \begin{cases} -\eta\tau\mu I/2, & h_t \le t_0, \\ (-c_0 t_0^3 + \eta H_{\max})I \preceq -c_0 t_0^3 I/2, & h_t \ge t_0. \end{cases}$$

Consequently the full-circle separator, including zero frequency, is

$$\begin{pmatrix} G(z) \\ I \end{pmatrix}^* \Pi_{\eta,R}(z) \begin{pmatrix} G(z) \\ I \end{pmatrix} \preceq -\delta I_d, \qquad |z| = 1. \tag{553}$$

Identity coupling: the sharper anchored separation. Here $L = I$, $\tau = r$, $\sigma = 1$, $R = 0$, and all constants in this paragraph are those of (525). Put $q_+ = \rho$. An explicit anchored margin controls zero frequency as well. Let

$$\beta = 9qr/8 - 1 - q, \qquad \chi = q(1 - r/8) > 0, \qquad W(q,a) = D\operatorname{Re} V_{r,q}(z).$$

Another exact expansion yields

$$\begin{aligned} W(q,a) &= -1 + \beta q + \chi + (-\beta + \chi q + q)a - 2\chi a^2, \\ W(q,1) &= -rq(1-q), \\ W(q,a) - W(q,1) &= (1-a)\big[\beta - \chi q - q + 2\chi(a+1)\big]. \end{aligned} \tag{554}$$

For $a \le 1$ and $0 < r \le 3$, the bracket is bounded above by $9/4$. The following decomposition verifies the bound with nonnegative terms:

$$\begin{aligned}\frac{9}{4} - \big[\beta - \chi q - q + 2\chi(a+1)\big] &= 2\chi(1-a) + (1-q)^2 \\ &\quad + \frac{6(3-r) + r(1-q)(q+6)}{8} \ge 0.\end{aligned} \tag{555}$$

For $q \in [q_-, q_+]$,

$$q(1-q) - \kappa = (q - q_-)(1-q) + q_-(q_+ - q) \ge 0.$$

It follows from (554) and (546) that

$$W(q,a) \le -r\kappa + \frac{9}{4}(1-a), \qquad DS(q,t) \le -\frac{14r}{3}\kappa(1-a). \tag{556}$$

Choose η and γ as in (525). Since $M_\eta = M_0 + \eta$,

$$\begin{aligned}D\,\mathrm{Re}\big(\overline{M_\eta(z)}V_{r,q}(z)\big) &= DS + \eta W \\ &\le -\frac{29r\kappa}{12}(1-a) - r^2\kappa^2 \le -r^2\kappa^2.\end{aligned} \tag{557}$$

Because $0 < D \le 4$, this proves the uniform full-circle bound

$$\mathrm{Re}\big(\overline{M_\eta(z)}V_{r,q}(z)\big) \le -\gamma, \qquad |z| = 1, \quad q \in [q_-, q_+]. \tag{558}$$

Orthogonal diagonalization of A now gives the full matrix separation

$$\begin{aligned}\begin{pmatrix} G(z) \\ I \end{pmatrix}^* \Pi_\eta(z) \begin{pmatrix} G(z) \\ I \end{pmatrix} &= \frac{1}{2}\left[\overline{M_\eta(z)}(G(z) - I) + M_\eta(z)(G(z) - I)^*\right] \\ &\preceq -\gamma I_d, \qquad |z| = 1.\end{aligned} \tag{559}$$

D.2.3 One forcing estimate, gain bound and restart argument

All plant, bounded-operator, well-posedness, homotopy-IQC and strict separation hypotheses of [3, Theorem 2] have now been checked. We use margin δ in the general branch and margin γ in the identity branch, with their respective η and K.

Actual square summability precedes the orbit IQC. The natural response of (533) is exactly

$$e_1 = u_0 - \tau L^* w_0, \qquad e_n = -\tau L^* A^{n-1}(w_0 - \sigma L u_0/8), \qquad n \ge 2. \tag{560}$$

In the identity case this is

$$e_1 = u_0 - r w_0, \qquad e_n = -r A^{n-1}(w_0 - u_0/8), \qquad n \ge 2. \tag{561}$$

The squared row-operator bounds $1+\tau^2\ell^2$ and $1+\sigma^2\ell^2/64$, together with the geometric tail, give

$$\begin{aligned}\|e\|_{\ell_2}^2 &\le \left[1+\tau^2\ell^2+\tau^2\ell^2\sum_{n=2}^{\infty}\rho^{2(n-1)}\left(1+\frac{\sigma^2\ell^2}{64}\right)\right]\|\xi_0\|^2 \\ &= E\|\xi_0\|^2.\end{aligned} \tag{562}$$

For $(L,\tau,\sigma)=(I,r,1)$ this is exactly $E=1+r^2+65r^2\rho^2/[64(1-\rho^2)]$ in (525); $65/64$ is the squared norm of $(u,w)\mapsto w-u/8$. Thus $e\in\ell_2$ for every initial state.

More generally, for *any* external $e\in\ell_2$ in $s=Gp+e$, $p=P(s)$, the cited IQC theorem first establishes $s,p\in\ell_2$. Only after that conclusion do we apply (541) to these actual signals. Parseval, the appropriate strict separator, and $\|M_\eta\|_{\ell_2\to\ell_2}\le 20+\eta$ give

$$0\le -\delta_*\|p\|_{\ell_2}^2+(20+\eta)\|p\|_{\ell_2}\|e\|_{\ell_2},\qquad \delta_*=\begin{cases}\delta, & \text{general branch},\\ \gamma, & \text{identity branch}.\end{cases}$$

The extra term $-\eta\langle p,Rp\rangle$ produces no forcing cross term. Hence the common arbitrary-input gain is

$$\|p\|_{\ell_2}\le K\|e\|_{\ell_2},\qquad K=(20+\eta)/\delta_*. \tag{563}$$

This includes $p=0$ without division. Specializing to the natural response proves

$$\|p\|_{\ell_2}\le K\|e\|_{\ell_2},\qquad U^2:=\sum_{n=0}^{\infty}\|u_n\|^2\le(1+K^2E)\|\xi_0\|^2. \tag{564}$$

There is no circular use of an infinite-horizon IQC: square summability was supplied by the stability theorem before the orbit supply was summed.

Recovering the dual state. The same stable filter in both branches is

$$w_n=A^nw_0+\sigma\sum_{j=1}^{n}A^{n-j+1}L\frac{9u_j-u_{j-1}}{8}.$$

The forcing sequence $L(9u_j-u_{j-1})/8$, $j\ge1$, has ℓ_2 norm at most $5\ell U/4$, including u_0. Since $\sum_{k\ge1}\|A^k\|\le\rho/(1-\rho)$, the ℓ_1–ℓ_2 convolution inequality gives

$$\left(\sum_{n=0}^{\infty}\|w_n\|^2\right)^{1/2}\le\frac{\|w_0\|}{\sqrt{1-\rho^2}}+\frac{5\sigma\ell\rho}{4(1-\rho)}U. \tag{565}$$

The identity estimate is exactly

$$\left(\sum_{n=0}^{\infty}\|w_n\|^2\right)^{1/2}\le\frac{\|w_0\|}{\sqrt{1-\rho^2}}+\frac{5\rho}{4(1-\rho)}U. \tag{566}$$

Squaring with $(a+b)^2\le2a^2+2b^2$ proves the general state bound

$$\sum_{n=0}^{\infty}\|\xi_n\|^2\le\left[\left(1+\frac{25\sigma^2\ell^2\rho^2}{8(1-\rho)^2}\right)(1+K^2E)+\frac{2}{1-\rho^2}\right]\|\xi_0\|^2, \tag{567}$$

and in the identity branch recovers exactly

$$\sum_{n=0}^{\infty}\|\xi_n\|^2 \le \left[\left(1+\frac{25\rho^2}{8(1-\rho)^2}\right)(1+K^2E)+\frac{2}{1-\rho^2}\right]\|\xi_0\|^2. \tag{568}$$

These are (516) and (523) at zero, with the stated constants $C > 1$ and $C_{r,Q} > 1$. No initial potential or subgradient has been used.

Autonomous restarts and geometric decay. The KKT anchor, centered function, plant, multiplier and constants are unchanged at every restart. Thus the same state estimate holds from every index $m \ge 0$. Write C_* for the applicable constant and $R_m = \sum_{n=m}^{\infty}\|\xi_n\|^2$. These finite tails satisfy

$$R_m \le C_*\|\xi_m\|^2, \qquad R_{m+1} = R_m - \|\xi_m\|^2 \le (1-C_*^{-1})R_m.$$

Consequently

$$\|\xi_n\|^2 \le R_n \le (1-C_*^{-1})^n R_0 \le C_*(1-C_*^{-1})^n\|\xi_0\|^2.$$

This proves (517) and (524), without an exponential-IQC theorem or positivity of a KYP storage. ∎

D.2.4 Global anchored growth and scope

Corollary D.1 (Attained global single-anchor secant and growth conditions)**.** *In Theorem 18, replace global strong convexity of Φ by the following assumptions: Φ has an attained minimizer x^*, and for a specified $m_* > 0$,*

$$\langle x-x^*, g\rangle \ge m_*\|x-x^*\|^2 \quad \textit{for every } x \in X \textit{ and every } g \in \partial\Phi(x). \tag{569}$$

Then the minimizer and KKT pair are unique, and all conclusions and constants of that theorem hold with m_* in place of μ. In particular, this applies if x^* is an attained minimizer and the global single-anchor quadratic-growth condition

$$\Phi(x) \ge \Phi(x^*) + \frac{\nu}{2}\|x-x^*\|^2 \quad \textit{for every } x \in X, \qquad \nu > 0, \tag{570}$$

holds: one may use $m_* = \nu/2$ in all constants.

Proof. At another minimizer x, take $g = 0$ in (569) to obtain $x = x^*$. Attainment and the smooth sum rule give the unique KKT pair. After translation, the secant condition gives precisely (532) with $R = \tau(m_*I - W)$. This is the only use of strong convexity in the shared IQC proof after existence; all its remaining steps apply unchanged. Under (570), the convex subgradient inequality at x gives, for every $g \in \partial\Phi(x)$,

$$\langle x-x^*, g\rangle \ge \Phi(x) - \Phi(x^*) \ge \frac{\nu}{2}\|x-x^*\|^2.$$

Thus the valid secant coefficient is $\nu/2$. ∎

Attainment is explicitly assumed in Corollary D.1; it is not inferred from a potentially vacuous subgradient condition. Both conditions are global and centered on one minimizer. Local growth, a local residual error bound, or growth relative to a non-singleton solution set is not a hypothesis of this proof. No equivalence between quadratic growth and strong convexity is claimed.

Corollary D.2 (Finite-cycle exclusion)**.** *Under the hypotheses of Theorem 19, every finite periodic orbit of* (521) *is the constant orbit* (x^*, y^*)*. The same conclusion holds for* (3) *under Theorem 18 or Corollary D.1.*

Proof. Every such orbit converges to its unique KKT pair by the corresponding geometric estimate, and a convergent periodic sequence is constant. ∎

For a concrete rank-deficient noncommuting example, let

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad c = 0, \qquad f(x) = |x_1 + x_2 + x_3| + x_3^2/2.$$

Then f is not strongly convex and L has a kernel, but Φ is μ-strongly convex with $\mu = 1 - 1/\sqrt{2}$; indeed,

$$W = L^*Q^{-1}L = \begin{pmatrix} 3/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad QLL^* - LL^*Q = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Taking $\sigma = 1$ and $\tau = 6/(3 + \sqrt{5})$ gives $\tau\sigma\|L\|^2 = 3$. Zero coupling is also covered whenever f supplies the stated composite curvature; no division by $\|L\|$ occurs.

Strict positive definiteness makes the open plant stable. These explicit constants are conservative and may diverge in singular limits. The soft-IQC estimates themselves are finite-dimensional and the general-coupling rate uses growth. The broader qualitative Hilbert conclusion has already been proved in Theorem 16; it is not inferred from these estimates. The identity theorem requires only proper closed convex f, and the soft-IQC branches make no finite-memory storage assertion.

D.3 Small smooth departures from a quadratic dual

The quadratic estimates imply quantitative nonlinear extensions through a fixed tangent at the actual KKT pair. The isotropic identity-coupling argument uses finite blocks on arbitrary real Hilbert spaces. In finite dimensions, both identity coupling and rectangular coupling use the same forced-baseline and small-gain proof. We first retain the sharper normalized identity-coupling constants; the rectangular extension uses its own constants and requires global strong convexity of the original quadratic composite.

Theorem 20 (Two quadratic baselines for robustness)**.** *Fix $0 < r \le 3$. Let f be proper lower-semicontinuous convex, let $c \in \mathcal{H}$, and let $\psi : \mathcal{H} \to \mathbb{R}$ be convex and continuously Fréchet differentiable with globally ε-Lipschitz gradient. Set $h(y) = \frac{1}{2}\langle y, Qy\rangle + \langle c, y\rangle + \psi(y)$ and use the literal map*

$$x_{n+1} = \operatorname{prox}_{rf}(x_n - ry_n), \qquad y_{n+1} = \operatorname{prox}_h\big(y_n + (9x_{n+1} - x_n)/8\big). \tag{571}$$

The following two cases give a unique KKT pair $z^* = (x^*, y^*)$ for every finite $\varepsilon \ge 0$, and global geometric convergence from every initial pair under their respective smallness conditions.

Hilbert isotropic case. *Let $\mathcal{H}$ be any real Hilbert space, $Q = \mu I$ with $\mu > 0$, $q_b = (1+\mu)^{-1}$, and $C = C_{r,\mu} > 1$ from Theorem 10. Define*

$$\begin{aligned} K_b &= \sqrt{1+r^2}, \quad B_b = \tfrac{9}{8}(1+K_b), \quad L_b = \sqrt{K_b^2 + q_b^2 B_b^2} > 1, \\ \rho_b &= 1 - C^{-1}, \quad m = \lceil 4C^2 \rceil, \qquad \varepsilon_0 = \frac{1}{4q_b^2 B_b m L_b^{m-1}} > 0. \end{aligned} \tag{572}$$

If $0 \le \varepsilon \le \varepsilon_0$, then for $n = km + s$, $k \ge 0$, $0 \le s < m$,

$$\|z_n - z^*\| \le L_b^s (3/4)^k \|z_0 - z^*\| \le A_b \alpha_b^n \|z_0 - z^*\|, \qquad \alpha_b = (3/4)^{1/m}, \quad A_b = (L_b/\alpha_b)^{m-1}. \tag{573}$$

Any positive integer m with $C\rho_b^m \le 1/4$ is also admissible, with the corresponding tolerance in (572)*.*

Finite-dimensional positive-definite case. *Let $\mathcal{H} = \mathbb{R}^d$ and $Q = Q^{\mathsf{T}} \succ 0$. Use A, ρ, K, E from* (525)*, with frequency margin γ. Define*

$$\begin{aligned} J &= \frac{r\rho}{1-\rho}, \qquad a = \frac{5\rho}{4(1-\rho)}, \qquad b = \frac{\rho}{1-\rho}, \quad t = 1 + K\sqrt{E}, \\ A_w &= \frac{1}{\sqrt{1-\rho^2}} + at, \quad B_w = aKJ + b, \quad \varepsilon_{\mathrm{alg}} = \frac{1}{2B_w}. \end{aligned} \tag{574}$$

If $0 \le \varepsilon \le \varepsilon_{\mathrm{alg}}$, set

$$C_\varepsilon = (t + 2KJ\varepsilon A_w)^2 + 4A_w^2 > 1. \tag{575}$$

Then, for every restart $m \ge 0$,

$$\sum_{n=m}^{\infty} \|z_n - z^*\|^2 \le C_\varepsilon \|z_m - z^*\|^2, \qquad \|z_n - z^*\|^2 \le C_\varepsilon (1 - C_\varepsilon^{-1})^n \|z_0 - z^*\|^2. \tag{576}$$

Replacing ε in (575) *by $\varepsilon_{\mathrm{alg}}$ gives one constant uniform over the whole admitted perturbation class. Neither f nor ψ need be separable. Both cases give square-summable errors and increments, without any initial objective value or initial subgradient requirement.*

Theorem 21 (Rectangular coupling with a smooth convex dual perturbation)**.** *Let $X = \mathbb{R}^d$, $Y = \mathbb{R}^m$, $Q = Q^{\mathsf{T}} \succ 0$ on Y, and let $L : X \to Y$ be arbitrary. Let $f : X \to (-\infty, +\infty]$*

be proper lower-semicontinuous convex, $c \in Y$, and suppose that, for a specified $\mu > 0$, the original quadratic composite

$$\Phi_0(x) = f(x) + h_0^*(Lx), \qquad h_0(y) = \tfrac{1}{2}\langle y, Qy\rangle + \langle c, y\rangle, \qquad h_0^*(v) = \tfrac{1}{2}\langle v - c, Q^{-1}(v - c)\rangle \tag{577}$$

is globally μ-strongly convex. Let $\psi : Y \to \mathbb{R}$ be convex and C^1 with globally ε-Lipschitz gradient, and set $h = h_0 + \psi$. For $\tau, \sigma > 0$ with $\tau\sigma\|L\|^2 \le 3$, use the literal primal-first map

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^* y_{n-1}), \\ y_n &= \operatorname{prox}_{\sigma h}\left(y_{n-1} + \sigma L(9x_n - x_{n-1})/8\right), \qquad n \ge 1. \end{aligned} \tag{578}$$

For every finite $\varepsilon \ge 0$, the actual objective $f + h^* \circ L$ has a unique minimizer, and the unique KKT pair satisfies

$$-L^* y^* \in \partial f(x^*), \qquad Lx^* = Qy^* + c + \nabla\psi(y^*). \tag{579}$$

This existence assertion requires no perturbation smallness.

For convergence, put $\ell = \|L\|$ and take A, ρ, K, E from (518)*, using the original curvature μ. In particular, $A = (I + \sigma Q)^{-1}$ and the positive separator margin there is δ, distinct from ε. Define*

$$\begin{aligned} J &= \frac{\tau\sigma\ell\rho}{1-\rho}, \qquad a = \frac{5\sigma\ell\rho}{4(1-\rho)}, \qquad b = \frac{\sigma\rho}{1-\rho}, \quad t = 1 + K\sqrt{E}, \\ A_w &= \frac{1}{\sqrt{1-\rho^2}} + at, \quad B_w = aKJ + b, \quad \varepsilon_{\mathrm{alg}} = \frac{1}{2B_w}. \end{aligned} \tag{580}$$

If $0 \le \varepsilon \le \varepsilon_{\mathrm{alg}}$, the constant C_ε defined by (575)*, now with* (580)*, satisfies both all-restart and geometric estimates in* (576)*, with $z_n = (x_n, y_n) \in X \times Y$. The bounds hold for every finite initialization, including $x_0 \notin \operatorname{dom} f$. They give square-summable errors and increments. Substituting $\varepsilon_{\mathrm{alg}}$ for ε in* (575) *gives a uniform constant over this perturbation class. No rank condition on L, separability, or commutation of Q and LL^* is required.*

The original global strong-convexity hypothesis in Theorem 21 is essential to the anchor transfer below. The separately stated anchored-secant and single-anchor quadratic-growth consequences of the quadratic theorem are not being extended to an unknown perturbed anchor. For $L = I$, $\tau = r$ and $\sigma = 1$, use the sharper constants of Theorem 20; the general constants do not replace them.

D.3.1 Actual KKT existence and a fixed tangent baseline

We first record the conjugate facts common to all three cases. Set $\underline{q} = \mu$, $M = \mu + \varepsilon$ in the Hilbert isotropic case, and $\underline{q} = \lambda_{\min}(Q)$, $M = \lambda_{\max}(Q) + \varepsilon$ in finite dimensions. Then h is $\underline{q}$-strongly convex with globally M-Lipschitz gradient. For each v in the dual space, $h(y) - \langle v, y\rangle$ is strongly convex, weakly lower-semicontinuous and coercive, so it has

a unique minimizer $y(v)$ with $\nabla h(y(v)) = v$. In a Hilbert space the direct method uses weak sequential compactness of bounded sequences. Strong monotonicity gives $\|y(v + d) - y(v)\| \le \|d\|/\underline{q}$. The maximizing characterization of the conjugate then gives

$$0 \le h^*(v + d) - h^*(v) - \langle y(v), d\rangle \le \langle y(v + d) - y(v), d\rangle \le \|d\|^2/\underline{q}.$$

Thus h^* is finite and continuously Fréchet differentiable, with $\nabla h^*(v) = y(v)$. Integrating the Lipschitz gradient bound for h and testing the supremum for $h^*(v')$ at $y(v)+(v'-v)/M$ proves

$$h^*(v') \ge h^*(v) + \langle y(v), v' - v\rangle + \|v' - v\|^2/(2M). \tag{581}$$

Identity coupling. The function f has an affine minorant: $p = \operatorname{prox}_f(0)$ satisfies $-p \in \partial f(p)$. Consequently $f+h^*$ is proper, strongly convex, weakly lower-semicontinuous and coercive, and has a unique minimizer x^*. Set $y^* = \nabla h^*(x^*)$. For $z \in \operatorname{dom} f$ and $0 < t < 1$, minimality and convexity yield

$$0 \le t[f(z) - f(x^*)] + h^*(x^* + t(z - x^*)) - h^*(x^*).$$

Dividing by t and passing to zero proves $-y^* \in \partial f(x^*)$; also $\nabla h(y^*) = x^*$. Monotonicity of ∂f and strong monotonicity of ∇h prove uniqueness of the pair.

Rectangular coupling. Here define

$$\beta = \frac{\lambda_{\min}(Q)}{\lambda_{\max}(Q) + \varepsilon} \in (0, 1]. \tag{582}$$

Equation (581) says that $h^* - \|\cdot\|^2/(2M)$ is convex. Since $Q^{-1} \preceq I/\underline{q}$ and $\beta = \underline{q}/M$, $M^{-1}I - \beta Q^{-1} \succeq 0$. Adding this positive-semidefinite quadratic and an affine function proves that $h^* - \beta h_0^*$ is finite, continuous and convex. For $0 < \beta < 1$ the exact extended-valued decomposition is

$$f + h^* \circ L = \beta\Phi_0 + (1 - \beta)f + (h^* - \beta h_0^*) \circ L. \tag{583}$$

It proves $\beta\mu$-strong convexity, properness and lower semicontinuity of the actual objective. If $\beta = 1$, omit the term $(1 - \beta)f$ entirely rather than multiply $+\infty$ by zero; the same conclusion follows from $\Phi_0 + (h^* - h_0^*) \circ L$. The strongly convex objective minus $\beta\mu\|\cdot\|^2/2$ is proper lower-semicontinuous convex and has an affine minorant. Hence the objective is coercive and has a unique minimizer x^*. The finite smooth sum rule gives $-L^*\nabla h^*(Lx^*) \in \partial f(x^*)$. With $y^* = \nabla h^*(Lx^*)$ this proves (579) and uniqueness of the pair, including when L has a kernel or cokernel and when $\operatorname{dom} f$ is lower-dimensional. No smallness condition was used in either existence argument.

One fixed tangent and normalization. For all three cases fix $g^* = \nabla\psi(y^*)$ once and define

$$h_{\tan}(y) = \tfrac{1}{2}\langle y, Qy\rangle + \langle c, y\rangle + \psi(y^*) + \langle g^*, y - y^*\rangle. \tag{584}$$

This quadratic has affine coefficient $c + g^*$ and has the actual (x^*, y^*) as its KKT pair. In the rectangular case, ignoring the irrelevant scalar constant in $h^*_{\tan}$, its composite is

$$\Phi_{\tan}(x) = f(x) + \tfrac{1}{2}\langle Lx - c - g^*, Q^{-1}(Lx - c - g^*)\rangle.$$

The exact difference is

$$\Phi_{\tan}(x) - \Phi_0(x) = -\langle Lx - c, Q^{-1}g^*\rangle + \tfrac{1}{2}\langle g^*, Q^{-1}g^*\rangle. \tag{585}$$

It is affine in x, so $\Phi_{\tan}$ retains the original strong convexity modulus μ, not merely $\beta\mu$. This is the reason Theorem 18 and its constants apply at the actual perturbed anchor.

In identity coupling use F from (416). More generally, with $\tau = r$, $\sigma = 1$ and $L = I$ understood in the two normalized cases, put

$$F(u) = \tau\big(f(x^* + u) - f(x^*) + \langle L^*y^*, u\rangle\big), \qquad P = \operatorname{prox}_F. \tag{586}$$

The KKT equations imply that F is proper lower-semicontinuous convex, $F \geq 0$, $F(0) = 0$, and $P(0) = 0$. The tangent, the KKT pair and this normalization remain fixed at every restart.

D.3.2 The Hilbert isotropic finite-block argument

In this subsection $Q = \mu I$. Let T, T_0 be the maps in (571) with $h, h_{\tan}$ respectively, using the fixed tangent (584). Both fix z^*, and the quadratic baseline gives, for every z and $n \geq 0$,

$$\|T_0^n(z) - z^*\| \leq \sqrt{C}\,\rho_b^{n/2}\|z - z^*\|. \tag{587}$$

The tangent, anchor and comparison map remain fixed at every restart.

A relative perturbation at the actual proximal output. Subtracting the proximal optimality equations for $a = \operatorname{prox}_h(v)$ and $b = \operatorname{prox}_{h_{\tan}}(v)$ gives the exact identity

$$a - b = -(I + Q)^{-1}[\nabla\psi(a) - \nabla\psi(y^*)]. \tag{588}$$

It uses the actual nonlinear output a and no commutation with Q. Strong monotonicity proves that both dual proximal maps are q_b-Lipschitz. Their common input anchor is $v^* = y^* + x^*$, so

$$\|a - b\| \leq q_b\varepsilon\|a - y^*\| \leq \varepsilon q_b^2\|v - v^*\|. \tag{589}$$

For $z = (x, y)$ put $p(z) = \operatorname{prox}_{rf}(x - ry)$ and $v(z) = y + (9p(z) - x)/8$. Nonexpansiveness and Cauchy–Schwarz give

$$\|p(z) - p(z')\| \leq K_b\|z - z'\|, \qquad \|v(z) - v(z')\| \leq B_b\|z - z'\|.$$

Both T and T_0 are therefore L_b-Lipschitz. Their first components agree, and (589) implies

$$\|T(z) - T_0(z)\| \le d_b \|z - z^*\|, \qquad d_b = \varepsilon q_b^2 B_b. \tag{590}$$

This estimate is global and has no additive anchor-dependent error.

Finite comparison and repeated blocks. Fix z, write $D_z = \|z - z^*\|$, and compare $u_j = T^j(z)$ with $w_j = T_0^j(z)$. Since $\|u_j - z^*\| \le L_b^j D_z$, inserting $T_0(u_j)$ gives

$$e_{j+1} \le d_b L_b^j D_z + L_b e_j, \qquad e_0 = 0, \qquad e_j := \|u_j - w_j\|.$$

Induction yields $e_j \le d_b j L_b^{j-1} D_z$ for $j \ge 1$. Bernoulli's inequality gives

$$\rho_b^{-m} \ge 1 + m/C, \qquad C\rho_b^m \le \frac{C^2}{C+m} \le \frac{C}{1+4C} < \frac{1}{4}$$

when $m = \lceil 4C^2 \rceil$. Hence the baseline error at step m is at most $D_z/2$, and the perturbation error is at most $D_z/4$ under (572). Thus

$$\|T^m(z) - z^*\| \le \tfrac{3}{4}\|z - z^*\| \quad \text{for every } z.$$

Repeating this estimate for the same autonomous map and using L_b for the remaining s steps proves (573). Error square summability follows geometrically; increment summability follows from $\|e_{n+1} - e_n\|^2 \le 2\|e_{n+1}\|^2 + 2\|e_n\|^2$. This is contraction of distance to a fixed equilibrium, not pairwise contraction of T^m. No finite initial objective or initial subgradient, bounded-orbit premise, or compactness argument for the convergence step is required.

D.3.3 A shared finite-dimensional small-gain proof

For identity coupling take $\tau = r$, $\sigma = 1$, $L = I$ and use A, ρ, K, E from (525) and the constants (574). For rectangular coupling use (518) and (580) instead. These are two choices of the quadratic input gain for the same argument; in particular, the stronger normalized identity estimate is retained. Put $u_n = x_n - x^*$, $w_n = y_n - y^*$ and $d_n = \nabla\psi(y^* + w_n) - \nabla\psi(y^*)$ for $n \ge 1$. Centering the actual proximal optimality equations gives

$$\begin{aligned} u_n &= P(u_{n-1} - \tau L^* w_{n-1}), \\ w_n &= A\left(w_{n-1} + \sigma L \frac{9u_n - u_{n-1}}{8} - \sigma d_n\right), \qquad \|d_n\| \le \varepsilon \|w_n\|. \end{aligned} \tag{591}$$

In the normalized identity case these equations read

$$u_n = P(u_{n-1} - r w_{n-1}), \qquad w_n = A\left(w_{n-1} + \frac{9u_n - u_{n-1}}{8} - d_n\right), \quad \|d_n\| \le \varepsilon\|w_n\|. \tag{592}$$

Although d_n depends on the current dual state, the actual map is the everywhere-defined proximal update (578). No d_0, finite initial objective value, or initial subgradient is needed.

An exogenous disturbance and the quadratic input gain. First replace d by any exogenous sequence in $\ell^2(Y)$. The resulting recursion is explicit and causal. With signals indexed by $n \geq 1$, $p_n = u_n$ and $s_n = u_{n-1} - \tau L^* w_{n-1}$ obey

$$s = Gp + e + \mathcal{L}d, \qquad \mathcal{L}(z) = \tau\sigma z L^* A(I - Az)^{-1}, \qquad \|\mathcal{L}\|_{\ell^2 \to \ell^2} \leq J. \tag{593}$$

Here $G : X \to X$ is the plant in (534) and $\mathcal{L} : Y \to X$. The natural response from (560) is

$$e_1 = u_0 - \tau L^* w_0, \qquad e_n = -\tau L^* A^{n-1}(w_0 - \sigma L u_0/8), \quad n \geq 2.$$

Only A and its own resolvent commute in the elimination; no commutation between Q and LL^* is used. The sign of the disturbance transfer is positive because d_n enters w_n negatively and the next proximal input contains $-\tau L^* w_n$. Its first contribution is $+\tau\sigma L^* A d_1$ in s_2, with no feedthrough into s_1. The impulse coefficient at delay $k \geq 1$ is $\tau\sigma L^* A^k$, whose norm sum is at most $\tau\sigma\ell\rho/(1-\rho) = J$. For normalized identity coupling this reduces exactly to

$$s = Gp + e + \mathcal{L}d, \qquad \mathcal{L}(z) = rzA(I - Az)^{-1}, \qquad \|\mathcal{L}\|_{\ell^2 \to \ell^2} \leq J, \tag{594}$$

with the original identity plant and natural response. Writing $H = \|(u_0, w_0)\|$, the corresponding initial estimate in either branch is $\|e\|_2 \leq \sqrt{E}H$.

Apply the verified soft-IQC feedback theorem of Section 6 to the tangent baseline with additive input $e + \mathcal{L}d$. Its full-homotopy well-posedness and strict full-circle separator first establish $s, p \in \ell^2$. Only then may the soft IQC and its margin be applied to obtain the arbitrary-input gain, as in (563):

$$\|p\|_2 \leq K\|e + \mathcal{L}d\|_2 \leq K\sqrt{E}H + KJ\|d\|_2. \tag{595}$$

This uses the proved external-input conclusion, not an inference from a zero-input state estimate. The normalized P meets all the IQC hypotheses: in the general branch the fixed tangent retains global μ-strong composite curvature by (585); in the identity branch ordinary convexity suffices for its sharper separator.

Recovering both state components. For the full state sequence norms $U = \|u\|_2$ and $W = \|w\|_2$, including index zero, the triangle inequality yields

$$U \leq \|u_0\| + \|p\|_2 \leq tH + KJ\|d\|_2. \tag{596}$$

The exact dual convolution, including the current disturbance, is

$$w_n = A^n w_0 + \sigma \sum_{j=1}^{n} A^{n-j+1} \left(L\frac{9u_j - u_{j-1}}{8} - d_j \right). \tag{597}$$

The primal forcing has sequence norm at most $5\ell U/4$ and the filter kernel has ℓ^1 norm at most $\rho/(1-\rho)$. With the appropriate branch's a, b, A_w, B_w this gives

$$W \le \frac{H}{\sqrt{1-\rho^2}} + aU + b\|d\|_2 \le A_w H + B_w \|d\|_2. \tag{598}$$

For $L = I$, $\sigma = 1$ this is precisely the earlier $5U/4$ forcing bound and (574). These estimates hold for every exogenous ℓ^2 disturbance. They do not yet assume square summability of the actual nonlinear orbit.

Finite-prefix absorption and every restart. To close that gap, fix $N \ge 1$ and prescribe the exogenous sequence $d_n^{[N]} = d_n$ for $1 \le n \le N$, and zero afterward, using the finite prefix of the actual orbit. This sequence belongs to ℓ^2. The forced trajectory with the same initialization agrees with the actual trajectory for every $0 \le n \le N$: the primal update agrees inductively from the previous state, and the explicit dual update then uses the same supplied current d_n. In particular, the truncation must retain sample N, not only samples through $N-1$. No agreement after N is asserted.

Let $U_N^2 = \sum_{n=0}^{N} \|u_n\|^2$ and $W_N^2 = \sum_{n=0}^{N} \|w_n\|^2$. Prefix agreement gives $\|d^{[N]}\|_2 \le \varepsilon W_N$, so (598) yields

$$W_N \le A_w H + B_w \varepsilon W_N \le A_w H + \tfrac{1}{2} W_N.$$

Absorption in this finite inequality proves $W_N \le 2A_w H$. Equation (596) then gives

$$U_N \le tH + KJ\varepsilon W_N \le (t + 2KJ\varepsilon A_w)H.$$

Both estimates are independent of N. Monotone convergence of the nonnegative partial sums proves the first actual infinite-horizon square-summability assertion, with the constant in (575). Every restart uses the same P, Q, L, ψ, tangent and KKT pair, and only its initial paired norm. No disturbance sample at the restart itself is required. Thus the first inequality of (576) holds at every restart. The tail argument of Section C.10 proves the second. Increment summability follows from the same elementary error-difference inequality used above. This is standard small gain applied with a causal truncation argument, not a hard finite-horizon IQC or an assumed positive storage.

The exact finite absorption also gives the slightly smaller valid constant

$$\left(t + \frac{KJ\varepsilon A_w}{1 - \varepsilon B_w}\right)^2 + \left(\frac{A_w}{1 - \varepsilon B_w}\right)^2.$$

The displayed choice (575) uses $\varepsilon B_w \le 1/2$ for a simpler uniform half-gain bound. For $L = 0$ all general formulas remain valid: $J = a = 0$, $E = 1$, $B_w = b > 0$, $A_w = (1-\rho^2)^{-1/2}$, and $C_\varepsilon = t^2 + 4/(1-\rho^2) > 1$. No division by $\|L\|$ occurs. The hypothesis then says that f itself is μ-strongly convex. The tolerance is sufficient, with no optimality claim even in this decoupled case.

A normalized example and the scope of the extension. For a unit vector e, the example $\psi(y) = a_0 \log\cosh(\langle e, y\rangle)$ has globally a_0-Lipschitz gradient and is genuinely nonquadratic when $a_0 > 0$. Choose a_0 below the appropriate tolerance. A concrete coupled example has $r = 3$, $Q = \operatorname{diag}(1, 2)$, $c = 0$, $f(x) = |x_1 + x_2|$ and $e = (1, 1)/\sqrt{2}$. Then $q_- = 1/3$, $\rho = 1/2$, $\kappa = 1/6$, $\eta = 1/2$, $\gamma = 1/16$, $K = 328$, $J = 3$, $a = 5/4$ and $b = 1$. Consequently $B_w = 1231$ and $\varepsilon_{\mathrm{alg}} = 1/2462$, so every $0 < a_0 \le 1/2462$ yields the claimed convergence for a coupled nonsmooth primal and genuinely nonlinear dual at product 3.

The algebraic tolerance avoids the block-length exponential of the Hilbert argument, but is not asserted to dominate every optimized finite-block bound pointwise. Both tolerances are conservative and may degenerate in singular or stiff curvature limits. There is no claim for arbitrary strongly convex nonlinear duals, local smoothness alone, indefinite perturbations, positive-semidefinite rather than positive-definite Q, or products beyond 3. The rectangular extension is finite-dimensional and requires the stated global strong convexity of Φ_0; it does not transport original-anchor quadratic growth or assert convergence for a merely convex composite. The only infinite-dimensional assertion here remains the isotropic identity-coupling finite-block theorem. No positive finite-memory storage or hard finite-horizon IQC is claimed.

E Periodic constructions and scalar order

E.1 Proof of the scalar period three classification

Exact elimination and normalization. All indices below are modulo three. Write

$$D = 8(\mu^2 + 3\mu + 3), \qquad p_i = \frac{x_{i-1} - x_i}{r} - y_{i-1} \in \partial f(x_i).$$

The dual cyclic system

$$(1 + \mu)y_i - y_{i-1} = \frac{9}{8}x_i - \frac{1}{8}x_{i-1} - c$$

has a unique solution because $(1 + \mu)^3 - 1 > 0$. In particular, a constant primal triple forces a constant dual triple and hence a fixed point. For a nonconstant primal triple, cyclic rotation and simultaneous sign reversal of both coordinates, followed by translation and common nonzero rescaling, reduce all six strict cyclic orders to

$$(x_0, x_1, x_2) = (0, t, 1), \qquad 0 < t < 1.$$

With exactly two equal primal values, sign reversal makes the repeated value the lower one, and rotation gives the same form with $t = 0$. The transformations preserve the primal monotonicity products. They preserve μ and the linear dual difference equations; they may change c, which is irrelevant to those differences. Thus no assertion that normalization preserves the original objective is needed.

For this normalized triple, solving the cyclic system gives, with an additive $-c/\mu$ in each entry,

$$y_0 = \frac{-\mu^2 - \mu t + 7\mu + 8t + 8}{\mu D} - \frac{c}{\mu},$$
$$y_1 = \frac{9\mu^2 t + 18\mu t - \mu + 8t + 8}{\mu D} - \frac{c}{\mu},$$
$$y_2 = \frac{-\mu^2 t + 9\mu^2 + 7\mu t + 18\mu + 8t + 8}{\mu D} - \frac{c}{\mu}.$$

Consequently

$$y_1 - y_0 = \frac{U}{D}, \quad y_2 - y_0 = \frac{V}{D}, \qquad U = (9\mu+19)t+\mu-8, \quad V = (8-\mu)t+10\mu+11. \tag{599}$$

Notice that $V > 0$ for $0 \le t \le 1$ and $\mu > 0$.

Three distinct primal values. Scalar monotonicity requires $p_0 \le p_1 \le p_2$, equivalently

$$rV \ge D(1+t), \qquad rU \le D(2t-1). \tag{600}$$

If $U \ge 0$, the second inequality forces $t \ge 1/2$. The case $U = 0$ is impossible, since $t \ge 1/2$ implies $U \ge (11\mu+3)/2 > 0$. For $U > 0$, the two inequalities would imply $(1+t)U \le (2t-1)V$, contradicting the exact identity

$$(1+t)U - (2t-1)V = (11\mu+3)(t^2 - t + 1) > 0. \tag{601}$$

Thus $U < 0$. This requires $\mu < 8$ and

$$0 < t < \frac{8-\mu}{9\mu+19} < \frac{1}{2}.$$

The second inequality in (600) now yields

$$r \ge B_\mu(t) := \frac{D(1-2t)}{8-\mu-(9\mu+19)t} = R(\mu) + \frac{D(11\mu+3)t}{(8-\mu)[8-\mu-(9\mu+19)t]} > R(\mu). \tag{602}$$

The first inequality is automatically strict under this bound: identity (601) shows $B_\mu(t) > D(1+t)/V$. This proves exclusion for all strict orders and also describes their necessary and sufficient monotone interpolation conditions.

Two equal primal values. Now $t = 0$ and primal monotonicity imposes precisely $p_2 \ge p_1$ and $p_2 \ge p_0$, with no comparison required between p_0 and p_1. In terms of (599), these conditions are

$$r(\mu-8) \le -D, \qquad r(9\mu+19) \ge 2D.$$

The first requires $\mu < 8$ and $r \ge R(\mu)$. It implies the second strictly, since $9\mu + 19 > 2(8-\mu)$ for $\mu > 0$. This handles every primal tie. Dual ties require no extra cases: their exact values were solved from the recurrence rather than ordered by a necessary relaxation. Together with the constant-primal case, these arguments establish the claimed exclusion without any KKT assumption.

Fixed-objective realization at every admissible product. Let $0 < \mu < 8$ and $r \geq R(\mu)$. Set

$$a_0 = \frac{(8-\mu)(\mu+1)}{\mu D}, \qquad a_1 = \frac{8-\mu}{\mu D}, \qquad a_2 = \frac{9\mu^2+18\mu+8}{\mu D}, \qquad y_i = a_i - \frac{c}{\mu}.$$

Choose the fixed finite continuous convex function

$$\alpha = \frac{1}{r} - y_2, \qquad \beta = -\frac{1}{r} - y_1, \qquad f(x) = \max\{\alpha x, \beta x\}.$$

For the proposed triple $x = (0, 0, 1)$ the exact primal residuals are

$$(p_0, p_1, p_2) = (\alpha, -y_0, \beta).$$

Indeed

$$\begin{aligned} p_1 - \alpha &= \frac{10\mu+11}{D} - \frac{1}{r} > 0, \\ \beta - p_1 &= \frac{8-\mu}{D} - \frac{1}{r} \geq 0, \\ \beta - \alpha &= \frac{9\mu+19}{D} - \frac{2}{r} > 0. \end{aligned}$$

Hence $p_0, p_1 \in \partial f(0) = [\alpha, \beta]$ and $p_2 \in \partial f(1) = \{\beta\}$. These are all three primal proximal inclusions. The exact dual residuals, either by the elimination above or direct substitution, satisfy

$$y_{i-1} - y_i + \frac{9}{8}x_i - \frac{1}{8}x_{i-1} = \mu y_i + c \quad (i = 0, 1, 2),$$

which are all three dual proximal inclusions. Proximal minimizations are strongly convex, so these inclusions give unique updates. Therefore

$$(0, y_0) \longmapsto (0, y_1) \longmapsto (1, y_2) \longmapsto (0, y_0)$$

is an actual orbit of the same fixed objectives. Since $y_1 - y_0 = (\mu - 8)/D < 0$, the first two states differ, and the third has a different primal coordinate. Its least period is three.

There is also the explicit KKT pair

$$y_* = -\beta = \frac{1}{r} + a_1 - \frac{c}{\mu}, \qquad x_* = \mu y_* + c = \frac{\mu}{r} + \frac{8-\mu}{D} > 0.$$

At this point $-y_* = \beta \in \partial f(x_*)$ and $x_* = \mu y_* + c \in \partial h(y_*)$.

Optional exact shape refinement. For $r \geq R(\mu)$ define

$$t_{\max}(\mu, r) = \frac{r(8-\mu) - D}{r(9\mu+19) - 2D}.$$

The denominator is positive. The admissible normalized primal shapes are exactly $0 \leq t \leq t_{\max}(\mu, r)$: the endpoint $t = 0$ is the tie case; for $t > 0$ this is precisely (602).

Moreover $t_{\max} < (8-\mu)/(9\mu+19) < 1/2$. Sufficiency for strict shapes follows without further constraints: the selected slopes satisfy $p_0 < p_1 \le p_2$, so the fixed function

$$f(x) = \begin{cases} p_0 x, & x \le 0, \\ p_1 x, & 0 \le x \le t, \\ p_1 t + p_2(x-t), & x \ge t \end{cases}$$

is convex, continuous, and contains each required residual in its subdifferential. The solved dual recurrence then supplies the actual cycle. This also has a KKT pair: the continuous coercive strongly convex function $f(x) + (x-c)^2/(2\mu)$ has a minimizer, whose first-order condition is exactly the KKT system. The refinement is not needed for the onset claim, but makes the strict-order interpolation sufficiency explicit.

Attribution and exact relation to the published phase criterion. The sufficient region above is exactly the period-three, unit-slope specialization of Seiler–Carrasco's existing Lur'e-system construction: [4, Proposition 1]: P. Seiler and J. Carrasco, *Construction of Periodic Counterexamples to the Discrete-Time Kalman Conjecture*, IEEE Control Systems Letters **5**(4), 1291–1296 (2021), DOI 10.1109/LCSYS.2020.3033443; arXiv:2009.02468v1 (5 September 2020), Proposition 1, equation (21). The directly checked source section is `https://arxiv.org/html/2009.02468#S5.SS2`. This attribution concerns the exact sufficient threshold, not merely an analogy between methods.

Here is the parameter identification, with the same $D = 8(\mu^2+3\mu+3)$ as above. It is enough to take $c = 0$: translating y to $\widetilde{y} = y + c/\mu$ and tilting f to $\widetilde{f}(x) = f(x) - (c/\mu)x$ removes c and preserves convexity. Put $a = 1+\mu$, $P = \mathrm{prox}_{rf}$ and $w_n = x_n - r y_n$. The exact recurrence, using $x_n = P(w_{n-1})$, is

$$w_{n+1} = \frac{w_n}{a} + \frac{8a-9r}{8a} P(w_n) + \frac{r-8}{8a} P(w_{n-1}).$$

With the negative-feedback convention $u_n = -P(w_n)$, the transfer function from u to w is

$$G(z) = \frac{(9r-8a)z + 8 - r}{8z(az-1)}.$$

This real rational plant is stable: its only possible poles are 0 and $1/a$. The scalar proximal map is monotone and 1-Lipschitz. Conversely a scalar monotone 1-Lipschitz map is firmly nonexpansive, hence the resolvent of a scalar maximal monotone graph, which is a convex subdifferential. Thus there is no missing multidimensional cyclic-monotonicity assumption in using the scalar slope-restricted construction here.

At $\zeta = e^{2\pi \mathrm{i}/3}$,

$$\operatorname{Re} G(\zeta) + 1 = \frac{3}{2} - \frac{r(8\mu+27)}{2D}, \qquad \operatorname{Im} G(\zeta) = \sqrt{3}\left(\frac{1}{2} - \frac{r(10\mu+11)}{2D}\right).$$

Proposition 1 with its parameters $\alpha = 2$, $\beta = 3$, slope bound $k = 1$, and period $T = 3$, requires

$$\mathrm{Re}(G(\zeta) + 1) < 0, \qquad |\operatorname{Im} G(\zeta)| \le -\sqrt{3}\,\mathrm{Re}(G(\zeta) + 1).$$

The two sides of the absolute-value inequality reduce exactly to

$$r(8 - \mu) \ge D, \qquad r(9\mu + 19) \ge 2D.$$

For $\mu > 0$ their solution is precisely $0 < \mu < 8$, $r \ge R(\mu)$. At $r = R(\mu)$ the real part in the strict condition is $-(11\mu + 3)/(2(8 - \mu)) < 0$, and it decreases with r, so equality at onset is included by the published criterion.

Accordingly, the result should be used as an exact scalar Chambolle–Pock specialization, with an elementary all-orders necessity proof and an explicit CP proximal/KKT witness. The cited proposition is a sufficient test; the necessity proof above is given directly rather than attributed to it. Neither that distinction nor the targeted source check establishes novelty of the necessity argument, the threshold, or the general interpolation/cycle mechanism. The source already provides finite monotone interpolation, including degenerate samples. No new general cycle-construction claim is intended.

E.2 Sharpness of the curvature-independent universal cap

Corollary E.1 (Universal cap with prescribed composite curvature)**.** *Fix any $m_0 > 0$. Over all finite-dimensional instances with $Q \succ 0$, arbitrary rectangular L, proper lower-semicontinuous convex f, and composite Φ globally m_0-strongly convex, the largest cap C for which every initialization converges whenever $\tau\sigma\|L\|^2 \le C$ is $C = 3$. The convergence at and below this cap is geometric for each instance; no rate uniform over all curvatures is asserted.*

Proof. The lower bound, including equality, is Theorem 18. For any $r > 3$, take $X = Y = \mathbb{R}$, $L = 1$, $\sigma = 1$, $\tau = r$, and choose the positive dual curvature

$$\mu = \min\{1, (r - 3)/5, 1/m_0\}.$$

In particular $0 < \mu \le 1 < 8$. The exact identities

$$R(\mu) - 3 = \frac{\mu(8\mu + 27)}{8 - \mu}, \qquad 5\mu - [R(\mu) - 3] = \frac{13\mu(1 - \mu)}{8 - \mu} \ge 0$$

give $R(\mu) \le 3 + 5\mu \le r$. Use the fixed two-slope objective and the initialization $(0, y_0)$ constructed above, with $c = 0$. They produce the actual least-period-three orbit $(0, y_0), (0, y_1), (1, y_2)$, including equality at onset. The composite $f(x) + x^2/(2\mu)$ is at least m_0-strongly convex because $1/\mu \ge m_0$. Thus this is an instance in the stated class with a unique KKT pair and a nonconvergent initialization. Taking $r = C$ excludes every proposed cap $C > 3$. ∎

The quantifiers are essential: for every prescribed $m_0 > 0$ and every $r > 3$ we choose a positive $Q = \mu$ depending on both. This is a direct consequence of the attributed Seiler–Carrasco construction, not a new cycle mechanism or a publication-priority claim. It establishes neither sharpness for a fixed Q nor sharpness when a fixed positive lower bound on dual curvature is imposed. Near $r = 3$ the selected curvature tends to zero; the endpoint itself is covered by the positive-curvature theorem. The primal hinge objective is not itself strongly convex.

E.3 Fourier necessity and normal-cone realizations

Indices in this appendix are read modulo the indicated period N. Let S be the backward temporal shift, $(Sx)_i = x_{i-1}$. The literal cyclic dual recurrence has the unique solution

$$y = B_N x, \qquad B_N = \frac{q}{8(1-q^N)} \sum_{a=0}^{N-1} q^a S^a (9I - S), \qquad v = (S - I - rSB_N)x. \tag{603}$$

The last coordinate is exactly $v_i = x_{i-1} - x_i - r y_{i-1} \in r\partial f(x_i)$. All values $f(x_i)$ on an actual periodic orbit are finite, being attained proximal outputs. Summing the subgradient inequality for the same f therefore gives, for $1 \le j < N$,

$$\Gamma_j = \sum_i \langle v_i, x_i - x_{i-j} \rangle \ge r \sum_i (f(x_i) - f(x_{i-j})) = 0. \tag{604}$$

This uses neither a KKT anchor nor a Lyapunov function.

E.3.1 Exact Hilbert-valued Fourier reduction and converse

For $\omega_k = 2\pi k/N$, $z = e^{-\mathrm{i}\omega_k}$, define

$$C_{jk} = \Re\{V_r(z)(1 - \overline{z}^{\,j})\}. \tag{605}$$

One may avoid complex conventions altogether: if $S - I - rSB_N = \sum_a m_a S^a$, then $C_{jk} = \sum_a m_a[\cos(a\omega_k) - \cos((a-j)\omega_k)]$. Real Fourier orthogonality yields

$$\Gamma_j / N = \sum_{k=1}^{\lfloor N/2 \rfloor} E_k C_{jk}, \qquad E_k \ge 0. \tag{606}$$

For a non-Nyquist mode $a_k \cos(i\omega_k) + b_k \sin(i\omega_k)$, $E_k = (\|a_k\|^2 + \|b_k\|^2)/2$; for an alternating mode its energy is the squared norm of its coefficient. Summation over time cancels the cross terms even if these Hilbert-valued coefficients are not orthogonal. Equivalently, the real temporal orthogonal projectors P_k satisfy

$$\tfrac{1}{2}\{M^\top (I - S^j) + (I - (S^\top)^j) M\} = \sum_k C_{jk} P_k, \qquad M = S - I - rSB_N,$$

with zero DC coefficient, and $E_k = \|P_k x\|^2_{\mathcal{H}^N}/N$. All matrices act only on the finite temporal index, so the identity holds on every real Hilbert space.

Conversely, suppose a nonzero $E \geq 0$ satisfies $\sum_k E_k C_{jk} \geq 0$ for every j. Choose mutually orthogonal real planes and set

$$X_i = \bigoplus_{k<N/2} \sqrt{E_k}(\cos(i\omega_k), \sin(i\omega_k)), \qquad Y = B_N X, \tag{607}$$

adding the coordinate $\sqrt{E_{N/2}}(-1)^i$ when needed. Let $\mathcal{C} = \operatorname{conv}\{X_0, \dots, X_{N-1}\}$ and $f = \iota_{\mathcal{C}}$. Direct real inner products give at every phase

$$\left\langle \frac{X_{i-1} - X_i}{r} - Y_{i-1}, X_i - X_{i-j} \right\rangle = \frac{1}{r} \sum_k E_k C_{jk} \geq 0. \tag{608}$$

These are all vertex normal-cone inequalities, so the residual belongs to $N_{\mathcal{C}}(X_i)$. Consequently $X_i = \operatorname{proj}_{\mathcal{C}}(X_{i-1} - rY_{i-1})$, the unique actual primal step, while (603) is the actual dual step. The finite convex hull is compact and its indicator is proper closed convex. The zero centroid gives $0 \in \mathcal{C}$, hence $(0, 0)$ is a KKT pair. This proves a converse over the Hilbert-space class, using dimension at most $N - 1$; it does not claim that every energy vector is realizable by a scalar graph. Least period is checked separately below.

For explicit arithmetic let $t = \cos\omega$ and $\Delta = 1 + q^2 - 2qt$. Then, with the Chebyshev polynomials T_j, U_j,

$$\begin{aligned} A(t) &= t - 1 - \frac{rq((9+q)t - 2t^2 + 1 - 9q)}{8\Delta}, \\ B(t) &= -1 + \frac{rq(9 + q - 2t)}{8\Delta}, \\ C_j(\omega) &= A(t)(1 - T_j(t)) + B(t)(1 - t^2)U_{j-1}(t). \end{aligned} \tag{609}$$

This formula includes $t = -1$ by its polynomial expression.

E.3.2 Proof of the four-period theorem

For $N = 4$, the nonconstant frequencies are $\pi/2$ and π. The third row in (609) is

$$C_{31} = \frac{rq(4q - 5)}{4(1 + q^2)} < 0, \qquad C_{32} = \frac{5qr - 8q - 8}{2(1 + q)}. \tag{610}$$

For $r < r_{\text{lin}}$ both are negative, so (604)–(606) force both energies to vanish. A constant primal sequence forces a constant dual sequence. At $r = r_{\text{lin}}$, only the quarter-turn energy must vanish; DC and alternating modes remain, and the shift-commuting B_N preserves period dividing two. This proves the endpoint exclusion correctly without excluding the known two-cycle.

For $r > r_{\mathrm{lin}}$ set $E_1 = 1$, $E_2 = -C_{31}/C_{32} > 0$. The third normal row is zero. The other rows satisfy

$$\begin{aligned} C_{11}(r_{\mathrm{lin}}) &= \frac{2(9q-1)}{5(1+q^2)}, \quad C_{21}(r_{\mathrm{lin}}) = \frac{4(2q^2+4q-3)}{5(1+q^2)}, \\ \partial_r C_{11} &= \frac{q(5q+4)}{4(1+q^2)}, \qquad \partial_r C_{21} = \frac{q(9q-1)}{4(1+q^2)}, \\ C_{12} &= C_{32} > 0, \qquad C_{22} = 0. \end{aligned} \tag{611}$$

For the stated $q_\star < q < 1$, all needed signs are strict: $q_\star > 3/5$ and $2(3/5)^2 + 4(3/5) - 3 = 3/25 > 0$. Both remaining row gaps are positive. Thus (607) becomes $X_i = (\cos(\pi i/2), \sin(\pi i/2), \sqrt{E_2}(-1)^i)$ in $\mathbb{R}^3$. Its first plane has four distinct points, so its least period is four. The verified normal products realize one fixed indicator throughout each orbit and prove Theorem 14 in exactly its stated band.

E.3.3 The two five-period radicals and their signs

For $N = 5$ take $t_1 = (\sqrt{5}-1)/4$ and $t_2 = -(\sqrt{5}+1)/4$ in (609). The determinant of rows two and four is

$$\begin{aligned} D(r) &= C_{21}C_{42} - C_{22}C_{41} = \frac{5\sqrt{5}H_q(r)}{256(1+q+q^2+q^3+q^4)}, \\ H_q(r) &= q^2(90q^2+71q-9)r^2 - 8q(28q^3+25q^2+25q+7)r + 128(1+q+q^2+q^3+q^4). \end{aligned} \tag{612}$$

Clearing positive denominators at $q = 9/10$ and $19/20$ gives respectively

$$258795r^2 - 1262916r + 1310432, \qquad 10084535r^2 - 47009952r + 46329664.$$

The larger roots are exactly (118); substitution gives zero and a positive derivative of the upward quadratic fixes the branch. Write $R = R_5(q)$, use a star for evaluation there, and put

$$a = C_{21}^*, \quad b = C_{22}^*, \quad \lambda = -C_{41}^*/a, \qquad E = (-b, a). \tag{613}$$

The following common coarse rational bounds hold at both selected q:

$$\begin{aligned} -b > 1/10, \qquad a > 2, \qquad -C_{41}^* - a/2 > 1/250, \\ C_1^* \cdot E > 3/4, \qquad C_3^* \cdot E > 1/2, \qquad C_j(0) \cdot E < -3/2 \quad (1 \le j \le 4). \end{aligned} \tag{614}$$

Here is an exact radical certificate for these strict signs. For each integer N appearing below, put $d = 10^{18}$ and $m = \lfloor \sqrt{Nd^2} \rfloor$:

N	m
5	2236067977499789696
183969001	13563517279820894599102
919845005	30328946651672556892067
3690736609	60751432978984125358477
18453683045	135844333871531056950434

Integer squaring gives $m^2 \le Nd^2 < (m+1)^2$, so $m/d \le \sqrt{N} < (m+1)/d$. Substitute these intervals into the explicit formulas (609), (118), and (613), using rational outward addition, multiplication, and inversion (all denominators are separated from zero). The lower or upper endpoints give precisely the weaker bounds (614); thus no floating-point sign oracle is required. The extra radicals $919845005 = 5\cdot 183969001$ and $18453683045 = 5\cdot 3690736609$ allow the expanded expressions to be checked directly. The source data retain the much tighter rational endpoints. The same calculation gives $R_5(9/10) - 152/45 > 1/200$ and $R_5(19/20) - 9128/2869 > 3/50$.

E.3.4 Exclusion below the five-period onset

By $D(R) = 0$ and $a > 0$, the row $\lambda C_2(R) + C_4(R)$ is zero. Also $\lambda > 1/2$ by (614). At $r = 0$,

$$C_2(0) = \left(-\frac{5+\sqrt{5}}{4}, -\frac{5-\sqrt{5}}{4}\right), \qquad C_4(0) = \left(\frac{3\sqrt{5}-5}{4}, -\frac{5+3\sqrt{5}}{4}\right).$$

Both entries of $\lambda C_2(0) + C_4(0)$ are negative: its first is less than $(-15+5\sqrt{5})/8 < 0$, and its second is a sum of negative terms. Since every row is affine in r,

$$\lambda C_2(r) + C_4(r) = (1 - r/R)\{\lambda C_2(0) + C_4(0)\}$$

has two strictly negative entries throughout $0 < r < R$. But an actual convex-graph orbit must have $\lambda\Gamma_2 + \Gamma_4 \ge 0$. The Fourier reduction forces both energies to vanish, hence a constant paired orbit. This excludes every Hilbert-valued shape and tie, not just single-harmonic or scalar candidates, and also excludes any possibility at the smaller determinant root.

E.3.5 Attainment and persistence at every larger product

For $E = (-b, a)$, the second row gap is zero by definition, the fourth is $D(R) = 0$, and the first and third are positive. The construction is

$$\begin{aligned} X_i &= (\sqrt{E_1}\cos(2\pi i/5), \sqrt{E_1}\sin(2\pi i/5), \sqrt{E_2}\cos(4\pi i/5), \sqrt{E_2}\sin(4\pi i/5)), \\ Y_i &= \frac{q}{8(1-q^5)} \sum_{a=0}^{4} q^a (9X_{i-a} - X_{i-a-1}), \qquad f = \iota_{\mathrm{conv}\{X_0,\dots,X_4\}}. \end{aligned} \tag{615}$$

Equation (608) verifies all twenty nontrivial normal products, hence all original proximal updates at the endpoint. Keep the same E, X, Y, f for $r \ge R$. Affinity gives

$$C_j(r) \cdot E = C_j(R) \cdot E + (r/R - 1)\{C_j(R) \cdot E - C_j(0) \cdot E\} \ge 0,$$

by (614). Thus all normal inclusions remain true with the residual divided by the new r; projection onto the fixed convex hull is its proximal map for every $r > 0$. Since $E_1 > 0$, all five primal points are distinct. Their least period is five. This proves the exact closed rays in Theorem 15.

E.3.6 An entirely rational original-iteration witness

Independently of the irrational endpoint recipe, set $q = 9/10$, $r = 17/5$, and in $\mathbb{R}^5$ take

$$X_0 = (-42, 25, -4, -4, 25), \qquad Y_0 = \frac{(-267417, 235233, -41139, -3438, 76761)}{10568}. \quad (616)$$

Let T be the right coordinate shift and $X_i = T^i X_0$, $Y_i = T^i Y_0$. Use $f = \iota_{\mathrm{conv}\{X_0,\dots,X_4\}}$ and $h(y) = \|y\|^2/18$. Substitution verifies $Y_i = (9/10)(Y_{i-1} + (9X_i - X_{i-1})/8)$ at every phase. At phase zero, $p_0 = (X_4 - X_0)/r - Y_4$ has the four normal products

$$\langle p_0, X_0 - X_j\rangle, \quad j = 1,2,3,4: \qquad \frac{2394355}{179656}, \quad \frac{10556005}{44914}, \quad \frac{220105}{179656}, \quad \frac{33684325}{89828}.$$

They are positive; cyclic orthogonal equivariance proves the other phases. The centroid is zero and all five vertices are distinct. This is a genuine least-period-five projection orbit with KKT pair $(0, 0)$, for one fixed primal and dual objective. It lies above the earlier failure envelope and is an original-recurrence validation, not an improved upper bound.

E.4 Scalar order rigidity at the distinct threshold r_*

This appendix concerns the different parameter $r_0 = r_*$ from (4) and two arbitrary scalar monotone graphs. It gives local kernel information only. In particular, it does not extend the general scalar convergence range above r_0.

Let

$$r_0 = \frac{88 + 16\sqrt{91}}{81}, \qquad C = \frac{9r_0}{16} - \frac{1}{2}, \qquad D_0 = \frac{11}{16} + \frac{3}{2r_0}, \qquad A = \begin{pmatrix} C & -r_0 \\ D_0 & -1 - C \end{pmatrix}. \quad (617)$$

One checks $A^2 + A + I = 0$, and hence $A^3 = I$.

Theorem 22 (Four-output order rigidity)**.** *Take arbitrary $a, b, v \in \mathbb{R}^2$ and set*

$$z_i = (x_i, y_i) = a + ib + A^i v, \qquad i = -1, 0, 1, 2, 3.$$

At the four output indices $i = 0, 1, 2, 3$, define the actual-form residuals

$$p_i = (x_{i-1} - x_i)/r_0 - y_{i-1}, \qquad q_i = y_{i-1} - y_i + (9x_i - x_{i-1})/8.$$

If

$$(x_i - x_j)(p_i - p_j) \geq 0, \qquad (y_i - y_j)(q_i - q_j) \geq 0 \quad (i, j \in \{0, 1, 2, 3\}), \quad (618)$$

then $v = 0$ and $b_x b_y = 0$. Conversely, every history with $v = 0$ and $b_x b_y = 0$ satisfies (618).

Proof. The center a contributes a species-wide constant to each of the four lists x_i, y_i, p_i, q_i. It disappears from every difference, so set $a = 0$ and write $\omega = (b_x, b_y, v_x, v_y)$. Each graph coordinate is a linear form in ω with coefficients in $\mathbb{Q}(\sqrt{91})$.

A finite scalar set of graph pairs (x_i, p_i) is monotone if and only if some permutation orders both coordinate lists nondecreasingly. To see necessity, sort by x and sort p within each tied-x block; monotonicity orders p between the strict-x blocks. Sufficiency follows immediately from the two nondecreasing lists. This treats all ties without dividing by a coordinate difference.

For each pair (π, ρ) of permutations of $\{0, 1, 2, 3\}$, form the 12 by 4 matrix $M_{\pi,\rho}$ whose rows, in order, are the three consecutive differences of x along π, of p along π, of y along ρ, and of q along ρ. The feasible set in (618) is exactly the union of the $24^2 = 576$ cones

$$\mathcal{C}_{\pi,\rho} = \{\omega : M_{\pi,\rho}\omega \geq 0\}.$$

The supplied finite certificate gives, for each cone, each $j \in \{3, 4\}$, and each $\varepsilon \in \{-1, 1\}$, nonnegative weights satisfying

$$M_{\pi,\rho}^{\mathsf{T}}\lambda = \varepsilon e_j, \qquad \lambda \geq 0. \tag{619}$$

The certificate is stored as exact rational expressions in $\sqrt{91}$, with at most four nonzero weights per identity. The standalone checker reconstructs every matrix from (617) and the displayed residual formulas, verifies all 2304 identities (619), and checks every coefficient sign by exact rational arithmetic. For a coefficient $\alpha + \beta\sqrt{91}$, opposite signs of the rational coefficients are resolved by comparing α^2 with $91\beta^2$. No floating-point sign test or optimization solver is part of verification.

For $\omega \in \mathcal{C}_{\pi,\rho}$, (619) implies $\varepsilon\omega_j = \lambda^{\mathsf{T}} M_{\pi,\rho}\omega \geq 0$ for both signs. Thus $v_x = v_y = 0$ on every cone. Once $v = 0$, direct residual differences give, for $i \neq j$,

$$(x_i - x_j)(p_i - p_j) = -(i-j)^2 b_x b_y, \qquad (y_i - y_j)(q_i - q_j) = (i-j)^2 b_x b_y.$$

Both inequalities hold exactly when $b_x b_y = 0$, proving necessity and the converse. ∎

E.4.1 Surviving ramps and a second-difference consequence

The nonzero axis ramps in Theorem 22 can occur as finite actual histories even with a KKT pair. For example, fix $s > 0$, let

$$f = \iota_{\{0\}}, \qquad h(y) = -sy + \iota_{[0,4s]}(y), \qquad (x_0, y_0) = (0, 0).$$

Then $x_n = 0$ and $y_n = \min\{ns, 4s\}$. The first five states form a nonzero dual-axis ramp, while $(0, 4s)$ is KKT. Removing the critical period-three mode therefore cannot, by itself, make endpoint dissipation coercive in first differences or positions.

Corollary E.2 (Conditional second-difference control)**.** *Let $\mathcal{D}$ be any fixed real positive-semidefinite* 10 *by* 10 *matrix whose kernel is exactly*

$$\ker \mathcal{D} = \{(a + ib + A^i v)_{i=0}^4 : a, b, v \in \mathbb{R}^2\}.$$

For a five-state window $W = (z_0, \ldots, z_4)$, define

$$SW = (z_2 - 2z_1 + z_0,\ z_3 - 2z_2 + z_1,\ z_4 - 2z_3 + z_2).$$

There is a finite constant $K_{\mathcal{D}}$ such that every window with scalar-monotone actual output graph pairs at indices $1, 2, 3, 4$ *satisfies*

$$\|SW\|_2^2 \le K_{\mathcal{D}}\, W^{\mathsf{T}} \mathcal{D} W. \tag{620}$$

The constant is existential; no numerical value or uniform continuation above r_0 is asserted.

Proof. As before, the scalar-order conditions define a union of finitely many polyhedral cones $\mathcal{C}_j$ in $\mathbb{R}^{10}$. Let L be a square root of $\mathcal{D}$. On each cone,

$$\mathcal{C}_j \cap \ker L \subseteq \ker S.$$

Indeed, a vector in $\ker L$ has the displayed kernel form. Shift indices by one to put its four outputs into the convention of Theorem 22; its critical component becomes Av, which vanishes by that theorem. Since A is invertible, $v = 0$, so the window is affine and is annihilated by S.

The image

$$\mathcal{B}_j = \{(LW, SW) : W \in \mathcal{C}_j\}$$

is a polyhedral cone and is closed. This follows by eliminating the variables W from its finitely many homogeneous linear equalities and inequalities. If no finite constant c_j satisfied $\|SW\| \le c_j \|LW\|$ on the cone, choose $W_k \in \mathcal{C}_j$ with $\|SW_k\| > k\|LW_k\|$ and divide by $\|SW_k\|$. The resulting image pairs have second coordinate of norm one and first coordinate tending to zero. A convergent subsequence has limit $(0, s)$ with $\|s\| = 1$. Closedness supplies some $W \in \mathcal{C}_j$ with $LW = 0$ and $SW = s$, contradicting the kernel inclusion. Take $K_{\mathcal{D}} = \max_j c_j^2$. ∎

Whenever an independently available decreasing storage supplies summability of $W_n^{\mathsf{T}} \mathcal{D} W_n$ along actual windows at this fixed endpoint, (620) supplies square summability of their second differences. The conclusion remains compatible with nonzero first differences along surviving finite ramps. This appendix supplies no first-difference bound, position bound, or convergence theorem on $(r_0, 3)$ for two arbitrary convex objectives.

F Multiplier comparisons and alternative certificates

F.1 Static and one delay comparison proofs

The results of Section 5.2 are proved here.

Proof of Proposition 5.2. Direct expansion of the literal port gives

$$8\Delta_q(t)\,\mathrm{Re}\,V = 8(t-1)\Delta_q(t) + rqB_q(t). \tag{621}$$

Since $B_q(-1) = 10(q+1) > 0$ and $B_q(1) = 8(q-1) < 0$, its positive domain is nonempty and separated from $t = 1$. At a boundary approached from $B_q > 0$ where $B_q \to 0$, the ratio in (77) tends to $+\infty$. Compactness away from these boundaries proves attainment and finiteness. Where $B_q \le 0$, (621) is strictly negative for $r > 0$: for $t < 1$ its first term is negative, and at $t = 1$ its second term is negative. Where $B_q > 0$, strict negativity is equivalent to r being below the ratio in (77). Consequently $\mathrm{Re}\,V < 0$ on the entire unit circle exactly for $0 < r < r_{\mathrm{static}}(q)$.

To pass from this frequency test to a positive ordinary quadratic storage, let $s = (p_{n-1}, R_{n-1})$ and let the free input be $u = p_n$. The scalar realization is

$$F_0 = \begin{pmatrix} 0 & 0 \\ -1/8 & q \end{pmatrix}, \qquad b_0 = \begin{pmatrix} 1 \\ 9/8 \end{pmatrix}, \qquad c_0 = \begin{pmatrix} 1 \\ -rq \end{pmatrix}, \qquad s^+ = F_0 s + b_0 u, \tag{622}$$

with $v_n = c_0^{\mathsf{T}} s - u$. Completing the square gives

$$u(c_0^{\mathsf{T}} s - u) = y^2 - e^2, \qquad y = \tfrac{1}{2} c_0^{\mathsf{T}} s, \qquad e = u - y. \tag{623}$$

The completed plant has state matrix $A_0 = F_0 + b_0 c_0^{\mathsf{T}}/2$, input b_0 and output $c_0^{\mathsf{T}} s/2$. Its full characteristic polynomial, multiplied by 16, and its transfer numerator in the same normalization are

$$\begin{aligned} D_0(\lambda) &= 16\lambda^2 + (9qr - 16q - 8)\lambda - qr + 8q, \\ N_0(\lambda) &= (8 - 9qr)\lambda + qr - 8q. \end{aligned} \tag{624}$$

On $|\lambda| = 1$, with $\zeta = \lambda^{-1}$,

$$|D_0(\lambda)|^2 - |N_0(\lambda)|^2 = -256\Delta_q(t)\,\mathrm{Re}\,V(\zeta) > 0. \tag{625}$$

At $r = 0$, $D_0 = 8(\lambda - q)(2\lambda - 1)$, so the actual state matrix is Schur. For every positive r below the static radius, (625) excludes a unit-circle root. Root continuation from $r = 0$, with fixed nonzero leading coefficient, proves Schur stability throughout this interval. This uses the full characteristic polynomial, including at nonminimal parameters such as $q = 1/9$. The gain N_0/D_0 is strictly less than one in modulus on the unit circle. Lemma 4 therefore yields $P_0 \succ 0$ and a strict state/input margin. The invertible change $(s, u) \mapsto (s, e)$ gives a $\delta > 0$ such that

$$\langle s^+, P_0 s^+ \rangle - \langle s, P_0 s \rangle + \langle u, c_0^{\mathsf{T}} s - u \rangle \le -\delta(\|s\|^2 + \|u\|^2). \tag{626}$$

The same scalar matrix inequality holds for Hilbert-valued states. On actual outputs, $\langle p_n, v_n\rangle \geq F_n \geq 0$. Starting with the finite state (p_0, R_0) and telescoping (626) proves the stated summability and convergence, without evaluating $F(p_0)$. Conversely, testing any strict unrestricted-state storage inequality on a nonzero harmonic state cancels its storage difference and forces $\operatorname{Re} V < 0$. This proves exactness for the stated supply.

At the linear cap, the numerator factors as

$$8\Delta_q(t)\operatorname{Re} V\big|_{r=r_{\mathrm{lin}}} = \frac{8}{5}(t+1)\big[(2-8q)t - 6 + 8q + 4q^2\big]. \tag{627}$$

The affine bracket has endpoint values $4(q^2-1) < 0$ at $t = 1$ and $4(q^2+4q-2)$ at $t = -1$. It is nonpositive on $[-1, 1]$ exactly when $q \leq q_\circ$. In that range, interpolation in r between $r = 0$ and $r = r_{\mathrm{lin}}$ gives strict negativity at every frequency for $0 < r < r_{\mathrm{lin}}$: the zeros at the two endpoints in r do not coincide. Also the ratio in (77) at $t = -1$ equals r_{lin}. Thus $r_{\mathrm{static}} = r_{\mathrm{lin}}$ there. For $q > q_\circ$, (627) is positive at frequencies just above $t = -1$, so $r_{\mathrm{static}} < r_{\mathrm{lin}}$.

Proposition 5.1 gives $\mathcal{R} = r_{\mathrm{lin}}$ for $q \leq q_\star$ and $\mathcal{R} < r_{\mathrm{lin}}$ for $q > q_\star$. This already proves (78) through $q = q_\star$. For $q > q_\star$, a graph-radius minimizing frequency t_* lies in $(-1, 1)$ and has $A_q(t_*) > 0$, where $A_q(t) = (9q+7)t^2 + (14q-34)t + 25q - 21$. Write $\rho_q(t) = 8(1-t)^2\Delta_q(t)/(qA_q(t))$. Substitution into (621) yields

$$\Delta_q(t)\operatorname{Re} V|_{r=\rho_q(t)} = \frac{2(1-t)(t+1)(t+2)(5-4q-t)\Delta_q(t)}{A_q(t)} > 0 \tag{628}$$

for $-1 < t < 1$ with $A_q(t) > 0$. At $r = 0$ the same real part is negative. Its numerator is affine in r, so at t_* its static crossing lies strictly below $\rho_q(t_*) = \mathcal{R}(q)$. The minimum static crossing is no larger, proving the strict ordering. ∎

Example 1 (The static baseline at $q = 1/2$)**.** In this case

$$\begin{aligned} r_{\mathrm{static}}(1/2) &= \min_{\substack{-1\leq t\leq 1 \\ 4t^2-19t+7>0}} \frac{8(t-1)(4t-5)}{4t^2-19t+7} \\ &= \frac{600+128\sqrt{21}}{249} \simeq 4.7653401163 < \mathcal{R}(1/2) = \frac{24}{5}. \end{aligned} \tag{629}$$

The derivative of the ratio is $-64(5t^2-2t-4)/(4t^2-19t+7)^2$. Its only relevant zero is $t = (1-\sqrt{21})/5$; the other zero exceeds one. On the positive-denominator domain the derivative changes from negative to positive there, and the ratio tends to infinity at the right boundary. The endpoint value at $t = -1$ is $24/5$. Thus the comparison is with the variable-product static radius, not merely with the older sufficient bound $r \leq 3$.

Proof. For a normalized one-delay coefficient $a \geq 0$ write

$$E(q, r, a, t) = \Delta_q(t)\operatorname{Re}\big[(1-a\zeta)V(\zeta)\big]. \tag{630}$$

It is affine in r with q, a, t fixed.

Small q. For $0 < q \le 1/3$, Proposition 5.2 already gives the static test throughout $0 < r < r_{\mathrm{lin}}$, since $1/3 < q_\circ$. Explicitly, at the cap

$$E(q, r_{\mathrm{lin}}, 0, t) = \frac{t+1}{5}\big[(2-8q)t - 6 + 8q + 4q^2\big] \le 0.$$

The bracket is negative at both endpoints of $[-1, 1]$ in this q range. At $r = 0$ one has $E(q, 0, 0, t) = (t-1)\Delta_q(t) \le 0$. Their sole zeros occur at $t = -1$ and $t = 1$, respectively, so strict affine interpolation proves the whole open interval.

Low r. The static test at $r = 2$ has

$$8E(q, 2, 0, t) = -12qt^2 + (8 + 6q^2 - 2q)t + 10q^2 - 2q - 8 =: S_q(t). \tag{631}$$

For each $t \in [-1, 1]$ this is convex in q, and

$$S_0(t) = 8(t-1) \le 0, \qquad S_{3/4}(t) = \frac{-72t^2 + 79t - 31}{8} < 0.$$

The latter quadratic has negative discriminant and negative leading coefficient. Convex interpolation gives $S_q(t) < 0$ for $0 < q \le 3/4$. Interpolating in r from the nonpositive expression at $r = 0$ proves the static strict FDI for all $0 < q \le 3/4$, $0 < r \le 2$. This includes the required low-product region $1/3 \le q \le q_\star$.

High r. First use the limiting coefficient $a = 1$. At the linear cap,

$$\begin{aligned} E(q, r_{\mathrm{lin}}, 1, t) &= -\frac{2}{5}(t-1)(t+1)L_q(t), \\ L_q(t) &= 4q^2 - 8qt - q + 2t - 5. \end{aligned} \tag{632}$$

The affine function L_q has endpoint values $L_q(-1) = 4q^2 + 7q - 7$ and $L_q(1) = 4q^2 - 9q - 3$. The first is nonpositive for $q \le q_\star$, and the second is strictly negative there. Hence (632) is nonpositive on $[-1, 1]$ for the entire stated curvature band. At $r = 2$,

$$\begin{aligned} E(q, 2, 1, t) &= \frac{t-1}{4}C_q(t), \\ C_q(t) &= -(6t+8)q^2 + (12t^2 + 16t + 10)q - 8t. \end{aligned} \tag{633}$$

For fixed $t \in [-1, 1]$, $C_q(t)$ is concave in q, and

$$C_{1/3}(t) = \frac{2(18t^2 - 15t + 11)}{9} > 0, \qquad C_{3/4}(t) = \frac{72t^2 + 5t + 24}{8} > 0. \tag{634}$$

Both inequalities follow from positive leading coefficients and negative discriminants. Thus $C_q(t) > 0$ for $1/3 \le q \le 3/4$. Since $q_\star < 3/4$, interpolation in r proves

$$E(q, r, 1, t) < 0 \quad (t < 1,\ 1/3 \le q \le q_\star,\ 2 \le r < r_{\mathrm{lin}}).$$

DC repair. Fix such a pair (q, r) and replace the limiting multiplier by $M = 1 + \eta - \zeta$ with $\eta > 0$. Its scaled real form is $E(q, r, 1, t) + \eta E(q, r, 0, t)$. At $t = 1$ the additional term is

$$\eta E(q, r, 0, 1) = -\eta r q(1 - q) < 0. \tag{635}$$

The static term stays negative in a neighborhood of $t = 1$. On the compact complement the unperturbed form has a strictly negative maximum. Boundedness of the static term there allows $\eta > 0$ small enough to preserve strict negativity. This proves a strict FDI with $m = 1 + \eta > \gamma = 1$ for the whole high-r region. No single η uniform up to the excluded product boundary is claimed. Together with the preceding two regions this covers every pair in (83).

The simpler fixed multiplier at $q = 1/2$. For $a = 1/3$ exact factorization gives

$$\begin{aligned} E(1/2, 0, 1/3, t) &= -\frac{2}{3}(t - 1)^2(5/4 - t) \le 0, \\ E(1/2, 24/5, 1/3, t) &= \frac{(t + 1)(4t^2 + t - 11)}{15} \le 0. \end{aligned} \tag{636}$$

The quadratic in the second line is convex with endpoint values $-8, -6$, hence is negative throughout $[-1, 1]$. The first expression vanishes only at $t = 1$ and the second only at $t = -1$. Strict affine interpolation proves the asserted full open interval. ∎

For comparison with the direct source specialization, its convenient choice $M = 1 - \zeta/2$ at $q = 1/2$ gives

$$\begin{aligned} E(1/2, 2, 1/2, t) &= \frac{(4t - 5)(6t^2 - 3t + 5)}{32} < 0, \\ E(1/2, 24/5, 1/2, t) &= \frac{(t + 1)(t + 2)(4t - 5)}{10} \le 0. \end{aligned} \tag{637}$$

It covers $2 \le r < 24/5$ by interpolation, while the static $r = 2$ value $(-12t^2 + 17t - 13)/16 < 0$ and its nonpositive $r = 0$ value cover $0 < r \le 2$. Thus that source-matched route also has no missing lower-product interval. The refinement (636) uses one fixed multiplier instead.

Proof of Proposition 5.4. The exact one-delay specialization. For convergence, apply Lemma 10 with $K = 1$, $\gamma_1 = \gamma$ and $\alpha = m - \gamma > 0$. Its port, objective and physical normalization are exactly (74)–(76). The result starts from the first actual graph output, so it needs neither $F(p_0)$ nor v_0. We retain below the distinct positive-storage formulas in the original three-slot coordinates. These coordinates need not be invertibly related to the minimal port-history state at $r = 8$; the full state argument below covers that value as well.

Literal state and square completion. Let $s_{n-1} = (p_{n-1}, p_{n-2}, R_{n-1})$ and take the free input $u = p_n$. Use the following scalar matrices:

$$\begin{gathered} F_0 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ -1/8 & 0 & q \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 0 \\ 9/8 \end{pmatrix}, \\ c_{\mathrm{new}} = \begin{pmatrix} 1 \\ 0 \\ -rq \end{pmatrix}, \qquad c_{\mathrm{old}} = \begin{pmatrix} 9r/8 - 1 \\ 1 - r/8 \\ -r \end{pmatrix}, \qquad d = mc_{\mathrm{new}} - \gamma c_{\mathrm{old}}. \end{gathered} \tag{638}$$

Then $s_n = F_0 s_{n-1} + bu$, $v_n = c_{\mathrm{new}}^{\mathsf{T}} s_{n-1} - u$, and $v_{n-1} = c_{\mathrm{old}}^{\mathsf{T}} s_{n-1}$ on actual histories. The last identity follows by eliminating w_{n-2} with (75); it is valid at the actual first graph point, so it can be used for $n \geq 2$ without inventing v_0. The exact causal supply and its completion are

$$\begin{aligned} Q_n &= \langle p_n, mv_n - \gamma v_{n-1} \rangle, \\ Q(s,u) &= ud^{\mathsf{T}} s - mu^2 = m(y^2 - e^2), \\ y &= \frac{d^{\mathsf{T}} s}{2m}, \qquad e = u - y. \end{aligned} \tag{639}$$

For Hilbert-valued states the middle line uses inner products and squared norms. The full determinant of $\lambda I - F_0$ is $\lambda^2(\lambda - q)$, so F_0 is Schur. Define

$$g(\lambda) = d^{\mathsf{T}} (\lambda I - F_0)^{-1} b. \tag{640}$$

The realization gives the exact identity $g - m = (m - \gamma\zeta)V$. Thus the FDI is precisely $\operatorname{Re} g < m$ on the unit circle.

Schur stability of the actual completed plant. The completed equations are

$$s^+ = \left(F_0 + \frac{bd^{\mathsf{T}}}{2m} \right) s + be, \qquad y = \frac{d^{\mathsf{T}} s}{2m}.$$

For every $0 \leq \beta \leq 1/(2m)$ the determinant identity is

$$\det(\lambda I - F_0 - \beta b d^{\mathsf{T}}) = \det(\lambda I - F_0)\,[1 - \beta g(\lambda)]. \tag{641}$$

The continuation argument of Lemma 10 applies to this full state: a unit-circle crossing would require $g = 1/\beta \geq 2m$, contradicting $\operatorname{Re} g < m$. Thus $F_0 + bd^{\mathsf{T}}/(2m)$ is Schur, including its nonminimal cases. The completed transfer is $g/(2m - g)$ and satisfies

$$|2m - g|^2 - |g|^2 = 4m(m - \operatorname{Re} g) > 0. \tag{642}$$

Lemma 4, which requires no minimality, gives $Z \succ 0$ and $\epsilon > 0$ such that

$$\langle s^+, Zs^+ \rangle - \langle s, Zs \rangle + \|y\|^2 - \|e\|^2 \leq -\epsilon(\|s\|^2 + \|e\|^2).$$

Put $P = mZ \succ 0$. Since $(s,u) \mapsto (s,e)$ is an invertible linear change, for some $\delta > 0$ this implies

$$\langle s^+, Ps^+\rangle - \langle s, Ps\rangle + Q(s,u) \le -\delta(\|s\|^2 + \|u\|^2). \tag{643}$$

The finite scalar matrix inequalities lift to Hilbert-vector slots by factoring their positive semidefinite differences.

Convex potential and the actual startup index. At each actual output let

$$A_n = \gamma\big[\langle p_n, v_n\rangle - F_n\big] = \gamma F^*(v_n) \ge 0, \qquad V_n = \langle s_n, Ps_n\rangle + A_n, \qquad n \ge 1. \tag{644}$$

Fenchel equality makes $F^*(v_n)$ finite, and $F(0) = 0$ implies $F^*(v_n) \ge 0$. Direct subtraction gives

$$\begin{aligned} Q_n - (A_n - A_{n-1}) &= (m-\gamma)\langle p_n, v_n\rangle \\ &\quad + \gamma\big[F_n - F_{n-1} - \langle v_{n-1}, p_n - p_{n-1}\rangle\big]. \end{aligned} \tag{645}$$

The bracket is nonnegative by the subgradient inequality at p_{n-1}, and $\langle p_n, v_n\rangle \ge F_n$. Hence

$$\begin{aligned} V_n &\ge \lambda_{\min}(P)\|s_n\|^2, \\ V_n - V_{n-1} &\le -\delta(\|s_{n-1}\|^2 + \|p_n\|^2) - (m-\gamma)F_n, \qquad n \ge 2. \end{aligned} \tag{646}$$

Crucially, $s_1 = (p_1, p_0, R_1)$ is finite, and A_1 uses only the actual graph pair (p_1, v_1) and F_1. Thus $V_1 < \infty$ even if $f(x_0) = +\infty$. There is neither a fictitious graph point at index zero nor an appeal to $F(p_0)$.

The exact $K = 1$ potential and descent in Lemma 10 therefore also hold for this positive raw-state storage. Its proof gives all square-summability and strong-convergence assertions, with $w_n = qR_n$ and $v_n = p_{n-1} - p_n - rqR_{n-1}$; finite prefixes are harmless. For the normalized composite objective, completion at the KKT pair also gives

$$\Phi(x_n) - \Phi(x^*) = \frac{F_n}{r} + \frac{\|p_n\|^2}{2\mu}, \qquad n \ge 1, \tag{647}$$

where an additive constant from the dual conjugate cancels. Thus its positive-output gaps are summable. This proof uses a finite first-output energy; it does not assert a uniform quadratic bound for that energy or a pointwise rate. ∎

Combining Propositions 5.3 and 5.4 yields the arbitrary-real-Hilbert identity-coupled sharp-band conclusion by a static or one-delay potential. Sharpness is supplied by the actual endpoint and divergent orbits in Proposition 3.1. That proposition is the single linear endpoint proof used by all the class-wide comparisons; it is logically distinct from exactness of any certificate ansatz.

Proof. Take either unit-circle point with $t = \operatorname{Re}\zeta = -11/20$. Exact evaluation gives

$$\begin{aligned} \operatorname{Re} V &= \frac{15369965}{99100832} > 0, \\ \operatorname{Re}[(1-\zeta)V] &= \frac{1405447}{991008320} > 0, \\ \operatorname{Re}[(1-\zeta^{-1})V] &= \frac{118765867}{247752080} > 0. \end{aligned} \tag{648}$$

For every pair (h_+, h_-) in (84),

$$\begin{aligned} \operatorname{Re}(MV) = (1-h_+-h_-)\operatorname{Re} V + h_+ \operatorname{Re}[(1-\zeta)V] \\ + h_- \operatorname{Re}[(1-\zeta^{-1})V] \geq \frac{1405447}{991008320} > 0. \end{aligned} \tag{649}$$

This one-frequency necessary test excludes the whole simplex, including its causal and anticausal boundary cases. Positive rescaling cannot change the sign, and the zero multiplier cannot be strict.

For orientation, the zero-η three-history multiplier at the same point gives the exact values

$$\begin{aligned} \operatorname{Re}[(10-7\zeta-2\zeta^{-2}-\zeta^{-3})V] &= -\frac{1512361691}{49550416000} < 0, \\ W_{q,3}(-11/20) &= \frac{341501027}{6250000} > 0. \end{aligned} \tag{650}$$

This favorable single frequency is not a feasibility proof. Global strict feasibility with some $\eta > 0$ follows instead from Theorem 7 and Proposition 5.1, since $3 < \mathcal{R}(q)$ for every $0 < q < 1$. Corollary 3.2 gives actual convergence at the stated point. ∎

The proposition excludes only the displayed nonnegative first-neighbor frequency-multiplier class. It is not an algorithmic counterexample, a necessity theorem for all longer histories, or an exclusion of signed multipliers, other storages, or nonlinear proofs. It complements the strong-curvature result: one delay covers that sharp band, whereas this limited first-neighbor family does not subsume the three-history certificate throughout weak curvature.

F.1.1 The residual PPA sign and general coupling

The identity-coupled proof has no residual block when its normalization gives $B = I$. For a general coupling the normalized splitting uses $B = \sqrt{\tau\sigma/\kappa_{\mathrm{cert}}}\, L$ and $C = (I - B^*B)^{1/2}$; the coupled and residual directions still come from one convex function. Separate nonlinearities on those directions cannot be assumed. Thus an identity-coupled FDI alone is insufficient to establish a common graph supply and a noncommuting Hilbert lift.

The exact limitation for an unchanged nonnegative one-delay test is particularly simple. With $V_{\mathrm{PPA}}(\zeta) = \zeta - 1$ and $a \geq 0$,

$$\begin{aligned} \operatorname{Re}[(1-a\zeta)V_{\mathrm{PPA}}] &= (t-1)[1-a(2t+1)] \\ &= (1-t)(2at+a-1). \end{aligned} \tag{651}$$

It is nonpositive for every $t \in [-1,1]$ exactly when $0 \le a \le 1/3$. Indeed, in this range the last bracket has maximum $3a-1 \le 0$. If $a > 1/3$, choose $\max\{-1,(1-a)/(2a)\} < t < 1$ to make both factors strictly positive. In particular, the otherwise convenient $a = 1/2$ identity multiplier has value $1/16 > 0$ at $t = 3/4$. Similarly the small-DC-repair choice gives

$$\mathrm{Re}[(1+\eta-\zeta)V_{\mathrm{PPA}}] = (t-1)(\eta-2t), \tag{652}$$

which is positive near $t = 1$ when $0 < \eta < 2$.

Neither calculation proves impossibility of a one-delay general-coupling method. In fact

$$\mathrm{Re}[(1-\zeta/3)V_{\mathrm{PPA}}] = -\frac{2}{3}(1-t)^2 \le 0, \tag{653}$$

while the same multiplier certifies the full $q = 1/2$ identity-coupled interval in (636). This compatible residual sign does not by itself give a common multiplier across a spectral band, graph coercivity, operator normalization, a noncommuting lift, or arbitrary-start and restart estimates. Those are separate tasks addressed for the specified three-history certificate in the general Hilbert proof.

F.1.2 Scope, attribution, and unresolved comparison

The generic dissipativity and graph-potential mechanisms are established prior art. The strict discrete-time KYP and soft-IQC machinery in Boczar–Lessard–Packard–Recht [3], the convex-conjugate causal and anticausal potentials in Scherer [9], and the graph-consistent finite-history framework of Upadhyaya–Banert–Taylor–Giselsson [6] provide the relevant methodological context. In particular, Scherer's inspected arXiv:2207.01363v1 uses the function and its conjugate in Theorem 5 and Corollary 6, and quadratic-plus-terminal-cost positivity in Theorem 9 and equation (26). Our graph correction can be written, by Fenchel equality, as

$$7F^*(v_n) + 3F(p_n) + 3F(p_{n-1}) + F(p_{n-2}).$$

This is substantive overlap in the type of potential. Neither that ancestry nor the explicit parameter calculations is a claim of a new general graph-potential principle.

The exact anchored map is also representable by the established one-oracle linear framework. Put $\varphi(p) = f(x^*+p) - f(x^*) + \langle y^*, p\rangle$ and let $a \in \partial\varphi(p^+)$. With $s = (p,w)$,

$$\begin{gathered} s^+ = As + Da, \qquad p^+ = Cs + Ja, \\ A = \begin{pmatrix} 1 & -r \\ q & q(1-9r/8) \end{pmatrix}, \quad D = \begin{pmatrix} -r \\ -9qr/8 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & -r \end{pmatrix}, \quad J = -r. \end{gathered} \tag{654}$$

The known quadratic dual has been absorbed exactly into the linear state equation, leaving one arbitrary convex oracle. The representation and finite-history graph terms are within the framework of [6]; representability is not an exhibited software certificate or

a statement that a particular default history attains the radius. Nor does an unreported or failed SDP imply instability.

There is also qualitative convergence overlap at smaller products. The literal small-extrapolation Hilbert theorem of Upadhyaya [5], Theorem 1.2 of the inspected v2, gives $r < 120/113$ at $\theta = 1/8$. Its anchored-gap and increment summability estimates imply, in the present positive scalar-curvature identity subclass, $w \in \ell^2$ because the dual anchored gap is $\mu\|w\|^2/2$. The exact identity

$$p_n = \mu w_n + (w_n - w_{n-1}) - \frac{p_n - p_{n-1}}{8}$$

then gives $p \in \ell^2$ from those same increment estimates. Strong paired convergence in this subclass is therefore already a derived consequence in that smaller range; it is not by itself a basis for a priority claim.

Wu–Ma [12] remains *unresolved at theorem level*. The verified journal record is *Numerical Algorithms* **102** (2026), 2485–2523, DOI 10.1007/s11075-025-02207-5. The complete algorithm and convergence theorem were not available in the inspected sources. Whole-article pagination is not a theorem pin. Without the exact update, allowable parameters, admissible starts, and any coordinate-equivalence map, neither coverage nor noncoverage of the literal fixed-step result is established. Two-sided extrapolation or a general Peaceman–Rachford correspondence alone does not settle that question. The comparison consequently makes no novelty assertion.

Finally, beyond $q_\star$, $\mathcal{R}(q)$ remains an exact *certificate* radius. It is not a proved full algorithmic threshold. The actual scalar period-three failures already established in Section 7.2, with the sufficient construction attributed there to Seiler–Carrasco [4], must also be included when describing the region above the sufficient radius. They cannot be replaced by a blanket description of the entire interval up to r_{lin} as undecided. Their complete classification and endpoint construction are kept in that separate section.

The finite-dimensional general-coupling, product-three positive-metric result, Theorem 8, has a distinct conclusion: monotonicity of a fixed positive quadratic distance to every KKT anchor under its own curvature condition. Neither the scalar formula (77) nor the more general graph-corrected convergence theorem substitutes for that ordinary-metric statement.

F.2 Causal finite-delay storage and weak-curvature calculations

We use the nonminimal strict bounded-real storage lemma (Lemma 4) once for all causal finite-delay multipliers. Its proof retains stable invisible modes by adding a small state-energy term to the one-sided ℓ_2 supremum. It requires neither controllability nor observability.

Lemma 10 (Causal FIR storage with actual startup)**.** *Use the literal recurrence* (38) *and the graph normalization* (40)*, with any fixed* $0 < q < 1$*,* $r > 0$*, and an integer* $K \geq 1$*. Suppose*

$$M(z) = m - \sum_{k=1}^{K} \gamma_k z^k, \qquad \gamma_k \geq 0, \quad \alpha = m - \sum_{k=1}^{K} \gamma_k > 0, \quad \Re(M(z)V_r(z)) < 0 \quad (|z| = 1). \tag{655}$$

Then the primal and dual anchor errors and the exact anchored port are square summable, the attained F_n *are summable, and every actual orbit converges strongly to the unique KKT pair. A positive quadratic storage exists on all* $K+2$ *raw state coordinates, with first finite energy at index* K *and first descent at index* $K+1$*.*

Proof. The KKT pair and all genuine Fenchel values are those in (40)–(122). Keep the full before-step state

$$s = (p_{n-1}, R_{n-1}, v_{n-1}, \ldots, v_{n-K}), \qquad h = p_n.$$

Write $s^+ = F_0 s + bh$ for the literal raw update

$$\begin{aligned} p^+ &= h, & R^+ &= qR - p/8 + 9h/8, \\ v^+ &= p - rqR - h, & v_j^+ &= v_{j-1} \quad (2 \leq j \leq K). \end{aligned} \tag{656}$$

Thus, with the same coordinate order, put

$$\begin{aligned} b &= (1, 9/8, -1, 0, \ldots, 0)^\top, \\ c &= (1, -rq, 0, \ldots, 0)^\top, \\ d &= mc - (0, 0, \gamma_1, \ldots, \gamma_K)^\top. \end{aligned}$$

The raw F_0 is lower triangular with diagonal $(0, q, 0, \ldots, 0)$, so its full characteristic polynomial is $\lambda^{K+1}(\lambda - q)$. Every raw mode is Schur, including hidden delay modes. Its exact supply is

$$Q(s, h) = hd^\top s - mh^2 = \langle p_n, mv_n - \sum_{k=1}^{K} \gamma_k v_{n-k} \rangle. \tag{657}$$

Before any pole-zero cancellation, the transfer $g(\lambda) = d^\top(\lambda I - F_0)^{-1} b$ satisfies

$$g(\lambda) - m = M(\lambda^{-1}) V_r(\lambda^{-1}).$$

Thus the hypothesis gives $\Re g < m$ on the entire unit circle.

Complete the square by putting $y = d^\top s/(2m)$ and $e = h - y$. Then $Q = m(y^2 - e^2)$ and the completed plant is

$$s^+ = (F_0 + bd^\top/(2m))s + be, \qquad y = d^\top s/(2m).$$

Its stability must be established on the full state before applying a gain lemma. For $0 \le \beta \le 1/(2m)$ the full determinant identity is

$$\det(\lambda I - F_0 - \beta b d^\top) = \det(\lambda I - F_0)[1 - \beta g(\lambda)]. \tag{658}$$

There is no raw pole on $|\lambda| = 1$. A unit-circle eigenvalue at $\beta > 0$ would imply $g = 1/\beta \ge 2m$, contradicting $\Re g < m$. Continuation from the Schur F_0, along a bounded matrix path with a fixed monic characteristic degree $K + 2$, proves that the completed full-state matrix is Schur. This argument does not infer stability from a cancelled transfer function. The completed transfer is $g/(2m - g)$, and

$$|2m - g|^2 - |g|^2 = 4m(m - \Re g) > 0.$$

Compactness of the unit circle gives a strictly subunit gain. Lemma 4 therefore supplies $Z \succ 0$, $\epsilon_0 > 0$ with

$$(s^+)^\top Z s^+ - s^\top Z s + y^2 - e^2 \le -\epsilon_0(\|s\|^2 + |e|^2).$$

Taking $P = mZ \succ 0$ and using the invertible finite-dimensional change $(s, h) \mapsto (s, e)$ yields some $\delta > 0$ such that

$$(s^+)^\top P s^+ - s^\top P s + Q(s, h) \le -\delta(\|s\|^2 + |h|^2). \tag{659}$$

Orthogonal diagonalization of finite real scalar matrices lifts this inequality to arbitrary Hilbert-valued coordinates with the same constants.

For actual $n \ge K$ define the nonnegative conjugate potential and energy

$$\begin{aligned} \mathcal{A}_n &= \sum_{k=1}^{K} \gamma_k \sum_{j=0}^{k-1} B_{n-j}, \\ s_n &= (p_n, R_n, v_n, \dots, v_{n-K+1}), \qquad E_n = \langle s_n, P s_n \rangle + \mathcal{A}_n. \end{aligned} \tag{660}$$

For $n \ge K + 1$, every index in $C_{n,k} = B_{n-k} - B_n - \langle p_n, v_{n-k} - v_n \rangle \ge 0$ is positive. The finite potential telescopes exactly, giving

$$\begin{aligned} \mathcal{A}_n - \mathcal{A}_{n-1} &= \sum_k \gamma_k (B_n - B_{n-k}), \\ Q_n - (\mathcal{A}_n - \mathcal{A}_{n-1}) &= \alpha \langle p_n, v_n \rangle + \sum_k \gamma_k C_{n,k} \ge \alpha F_n. \end{aligned} \tag{661}$$

Because $P \succ 0$ and $\mathcal{A}_n \ge 0$, the total energy is coercive: $E_n \ge \lambda_{\min}(P)\|s_n\|^2 \ge 0$. Combining the last two identities yields

$$E_n - E_{n-1} \le -\delta(\|s_{n-1}\|^2 + \|p_n\|^2) - \alpha F_n, \qquad n \ge K + 1. \tag{662}$$

The initial E_K uses precisely the actual graph values $v_1, \dots, v_K$ and $B_1, \dots, B_K$. It is finite even if $f(x_0) = +\infty$; the oldest point at the first descent is index one. Telescoping

proves square summability of the full state, hence of p, R, v, and summability of F_n. Finite attained prefixes do not change these claims. Strong primal and dual convergence follows. No convergence rate or uniform quadratic restart estimate is inferred. ∎

For $K = 1$, the lemma has state $(p_{n-1}, R_{n-1}, v_{n-1})$ and potential $\gamma_1 B_n$, so its actual startup is exactly E_1. The corresponding one-delay history coordinates satisfy

$$v_{n-1} = (9r/8 - 1)p_{n-1} - rR_{n-1} + (1 - r/8)p_{n-2}.$$

This identifies the original one-delay actual history and its conjugate potential. The raw-state proof remains valid without any minimality assumption on that realization. In particular it does not provide the mixed anticausal energy in Section A.1.

F.2.1 The global minimizing branch near weak curvature

To justify that the expansion in (85) concerns the actual minimum, write the exact radius formula as

$$\begin{aligned}
\varrho_q(t) &= \frac{8(1-t)^2\Delta_q(t)}{qA_q(t)},\\
A_q(t) &= (9q+7)t^2 + (14q-34)t + 25q - 21,\\
\mathcal{R}(q) &= \min_{[-1,1]\cap\{A_q>0\}} \varrho_q(t).
\end{aligned}$$

For $q < 1$, $A_q(-1) > 0$, $A_q(1) < 0$, and strict convexity gives the positive domain $[-1, \alpha(q))$, where $\alpha(q) \to 1/4$ as $q \uparrow 1$. The objective diverges at its excluded boundary. At $q = 1$ the limiting formula is

$$\varrho_1(t) = 3 + \frac{(2t+1)^2}{1-4t}, \qquad -1 \le t < 1/4,$$

with unique minimum at $t = -1/2$. Every minimizing sequence for $q \uparrow 1$ has value at most $\varrho_q(-1/2) \to 3$. A limit at $1/4$ would make the positive numerator tend to $27/4$ while the denominator tends to zero; any other limit must equal $-1/2$ by the displayed unique minimum. Thus all global minimizers enter an interior neighborhood of $-1/2$.

There ϱ is analytic, with $\partial_t\varrho = 0$, $\partial_t^2\varrho = 8/3 > 0$ and $\partial_q\partial_t\varrho = -91/12$ at $(1, -1/2)$. The analytic implicit-function theorem gives a unique local stationary branch, which the preceding localization makes the global minimizing branch for all q sufficiently near one. Writing $t = -1/2 + u$ gives

$$\begin{aligned}
\varrho_{1-\varepsilon}(-1/2+u) = 3 &+ \tfrac{27}{8}\varepsilon + \tfrac{307}{64}\varepsilon^2 + \tfrac{3275}{512}\varepsilon^3 + \tfrac{91}{12}\varepsilon u + \tfrac{851}{48}\varepsilon^2 u\\
&+ \tfrac{4}{3}u^2 + \tfrac{91}{6}\varepsilon u^2 + \tfrac{16}{9}u^3 + O((|\varepsilon| + |u|)^4).
\end{aligned} \tag{663}$$

Stationarity gives $u = -(91/32)\varepsilon + (4877/512)\varepsilon^2 + O(\varepsilon^3)$. Substitution proves the $\mathcal{R}$ expansion. The cubic position correction does not affect the cubic value because the first variation vanishes. Expansion of the rational upper bounds gives

$$r_3 = 3 + \tfrac{27}{8}\varepsilon + \tfrac{307}{64}\varepsilon^2 + \tfrac{3275}{512}\varepsilon^3 + O(\varepsilon^4), \qquad r_{\mathrm{lin}} = \tfrac{16}{5} + \tfrac{8}{5}\varepsilon + \tfrac{8}{5}\varepsilon^2 + O(\varepsilon^3),$$

and subtraction proves all of (85). The squared leading gap $91^2/768$ is the decrease obtained by minimizing $(91/12)\varepsilon u + (4/3)u^2$. In fact, for every $1/9 < q < 1$,

$$\varrho_q(-1/2) = r_3(q), \qquad \partial_t \varrho_q(-1/2) = \frac{16(1-q)(64q^2+121q+88)}{9q(9q-1)^2} > 0.$$

A small move left within the positive domain proves $\mathcal{R}(q) < r_3(q)$. This is a limitation of the specified ansatz, not a nonlinear orbit at its minimizing frequency.

F.2.2 Exact frequency certificates for the two causal witnesses

For either witness put $B_{q,r}(t) = -\Delta_q(t)\Re(M(z)V_r(z))$. At the upper products in (87) and (86), respectively,

$$\begin{aligned} 400000B_{9/10,10/3}(t) = &-840000t^5 - 802000t^4 + 551000t^3 \\ &+ 800920t^2 + 251041t + 39159. \end{aligned} \tag{664}$$

For the second witness write $\widetilde{B} = 2400000000000B_{99/100,30337/10000}$. Its exact coefficients are

$$\begin{aligned} \widetilde{B}(t) = &-125865907200000t^8 - 7990779353600t^7 \\ &+ 27092646611200t^6 + 16518263868800t^5 \\ &- 18612436156800t^4 - 15666565374400t^3 \\ &+ 3683313053800t^2 + 6132068334111t + 1430151817601. \end{aligned}$$

Use (145), now in t, on each of the four intervals $[-1,-1/2],[-1/2,0],[0,1/2],[1/2,1]$. All degree-five coefficients of the first polynomial are at least $3/10000$; those at $r = 3$ are at least $27/100000$. All degree-eight coefficients of the second are at least $3003363/100000000000$; those at $r = 3$ are at least $297/10000000$. The $r = 3$ polynomials are obtained directly by replacing r in (139); this specifies every coefficient without an optimizer or numerical grid. Their full 120 rational Bernstein coefficients and exact polynomial reconstructions are in the source data. Affinity in r gives, on the whole respective closed product intervals,

$$B_{9/10,r} \geq 27/100000, \qquad B_{99/100,r} \geq 297/10000000.$$

Thus (655) holds, with $\alpha = 1/1000$, $K = 3$ and $K = 6$ respectively. Lemma 10 proves the asserted Hilbert conclusions there; $\mathcal{R}(q) > 3$ and the old theorem cover all smaller positive products. Finally the exact tests $t = -7/10$ at $q = 9/10$ and $t = -21/40$ at $q = 99/100$ lie in $A_q > 0$ and give respectively $709784/214911 < 10/3$ and the first comparison in (88). This proves strict improvement over the old ansatz without rounding its minimizing frequency.

F.3 A fixed-q alternate certificate

The sharp statement at $q = 9/10$, $0 < r < 152/45$, is a direct specialization of Theorem 3. There is also a useful alternate exact storage using powers of the first-difference numerator. It is retained as a separate certificate rather than identified entrywise with the variable-q construction.

Keep $a = 45/71$, $b = 26/71$, the same actual state $s = (p, R, p_2)$, transition (127), supply $\mathcal{Q}_r = hv - ahv^- - bp_2v$, and mixed potential $aB_n + b(F_n + F_{n-1})$. At $q = 9/10$ the rows (128) are

$$\begin{aligned} w_1 &= h + p/71 - (36/355)R, \\ w_2 &= h - (79/71)p + (4/355)R, \\ w_3 &= h - (149/71)p + (10/9)p_2 - (4/3195)R, \\ w_e &= h + (81/71)p - (76/355)R. \end{aligned}$$

Replace the W-basis dissipation in (129) by

$$\widehat{D}_3 = \frac{1217}{56800}w_1^2 + \frac{539}{113600}w_2^2 + \frac{3060}{227200}w_3^2, \qquad \widehat{D}_\ell = \frac{3}{500}w_e^2. \tag{665}$$

For each endpoint, the Hessian/Lyapunov recipe (130), with these $\widehat{D}_j$, reconstructs $\widehat{P}_3, \widehat{P}_\ell$ uniquely. Interpolate using $\rho = (45r - 135)/17$. The resulting unrestricted-state identity is

$$\langle s^+, \widehat{P}s^+\rangle - \langle s, \widehat{P}s\rangle + \mathcal{Q}_r = -(1-\rho)\widehat{D}_3 - \rho\widehat{D}_\ell.$$

Thus the alternate matrices can be reconstructed without a long list of entries. The exact matrices and coefficient data are retained in `author_result.json` in the directory `mixed_identity/checkers/q09/`. The separate `check_independent.py` in that directory reconstructs them by solving all unrestricted quadratic coefficient equations, without importing the author matrices.

For a short check of the distinct graph certificate, its correction weights are

$$\widehat{\lambda} = \frac{17588493r + 1353256}{60844870r}, \qquad \widehat{\nu} = \frac{149571r - 152}{453690r}.$$

Their four graph budgets in (133) are nonnegative on the closed $[3, 152/45]$, and the corrected matrix has zero $12, 23$ entries and

$$\begin{aligned} \widehat{T}_{11} &= \frac{42879537r^2 + 197401280r - 10826048}{486758960r}, \\ \widehat{T}_{22} &= \frac{1030527r - 1499480}{1369009575}, \\ \widehat{T}_{33} &= \frac{-309321r^2 + 1736320r - 1216}{3629520r}, \\ \widehat{T}_{13} &= \frac{1097428743r^2 - 2128108320r + 49451072}{4380830640r}. \end{aligned} \tag{666}$$

For example, the determinant of its $(1,3)$ block is

$$\frac{N(r)}{2398959637045351200r^2},$$

where

$$\begin{aligned} N(r) = & -168553961839685925r^4 + 602046792584136000r^3 \\ & - 109782450211664448r^2 + 458732053739520r - 287800354611200. \end{aligned}$$

Its degree-four Bernstein coefficients on $[3, 152/45]$ are

$$\begin{gathered} 1615438834658739403, \qquad 6846205072787554344/5, \\ 697151014642992140576/675, \qquad 132078788922138877696/225, \\ 231539723500240896/25. \end{gathered}$$

All are positive. The same exact conversion verifies $\widehat{T}_{11} > 0$, $\widehat{T}_{22} > 0$, and the budgets; their short coefficient lists are included in the cited exact data. The graph identity, actual E_2 startup, and DC bridge (137), now with $k = 9r$, apply verbatim to this separately defined storage. It therefore gives an alternate proof on $3 \le r < 152/45$, with the same lower-product bridge. Its positive endpoint storage does not remove the product restriction: the $(1-\rho)$ coefficient of the needed dissipation still vanishes at $152/45$.

F.4 The three tap common frequency and graph interface

We prove Proposition 5.7 and all the strict-budget claims of Section 5.4. In this subsection $q = 19/20$, $R = 9128/2869$, and V_r, Δ have the definitions in (89). All polynomial statements below hold on the entire interval $-1 \le t \le 1$.

F.4.1 Necessity and sufficiency for the complete family

Write

$$C_r(t) = -\Delta(t)\,\mathrm{Re}\,V_r(z), \qquad B_r(t) = -\Delta(t)\,\mathrm{Re}\big[M(z)V_r(z)\big].$$

At the period-three point one has the exact identities

$$\begin{aligned} C_R(-1/2) &= -\frac{19397}{30200} < 0, \\ -\Delta\,\mathrm{Re}\big[(1-z^j)V_R\big] &= 0 \quad \text{when } t = -1/2, \quad j \in \{-3,-2,1\}. \end{aligned} \tag{667}$$

If $m = \sum_j h_j \le 1$, then $M = (1-m) + \sum_j h_j(1-z^j)$. Hence $B_R(-1/2) \ge 0$ forces $m = 1$. Set $h_{-3} = x$, $h_{-2} = y$, $h_1 = 1-x-y$. Direct Laurent expansion gives $B_R(-1/2) = 0$ and

$$B_R'(-1/2) = \frac{3}{15100}\big[19397x - 14345y + 5052(1-x-y)\big]. \tag{668}$$

A differentiable nonnegative polynomial with an interior zero has zero first derivative there. Thus

$$y = \frac{14345x + 5052}{19397}, \qquad h_1 = \frac{14345 - 33742x}{19397}. \tag{669}$$

Substitution gives the full-circle factorizations

$$\begin{aligned} B_R(t) &= (1-t)(2t+1)^2 K_x(t), \\ -\operatorname{Re}[MV_0] &= (1-t)H_x(t), \\ K_x(t) &= \frac{19}{193970}(1794 - 21191x - 19397tx), \\ H_x(t) &= \frac{2}{19397}\big[38794xt^2 + (67484x - 9293)t + 14345x + 5052\big]. \end{aligned} \tag{670}$$

Their relevant endpoint values are

$$\begin{aligned} K_x(-1) &= \frac{17043(1-x)}{96985}, \qquad K_x(1) = \frac{19(897 - 20294x)}{96985}, \\ H_x(1) &= \frac{2(120623x - 4241)}{19397}. \end{aligned} \tag{671}$$

If either of the right-hand DC values is negative, the corresponding frequency expression is negative immediately to the left of $t = 1$. Therefore feasibility requires

$$L_0 := \frac{4241}{120623} \le x \le U_0 := \frac{897}{20294}.$$

Conversely, throughout this interval $x > 0$, $y > 0$, and h_1 is decreasing with $h_1(U_0) = 6724/10147 > 0$. The affine polynomial K_x is nonnegative at both endpoints, hence on the whole interval. Moreover,

$$H_x''(t) = 8x > 0, \qquad H_x'(t) \le H_x'(1) \le H_{U_0}'(1) = -\frac{3014}{10147} < 0. \tag{672}$$

Consequently $H_x(t) \ge H_x(1) \ge 0$. These implications prove both necessity and sufficiency, including the two parameter endpoints.

For $x = 1/25$ the strict interior gaps are

$$\frac{1}{25} - L_0 = \frac{14598}{3015575} > 0, \qquad U_0 - \frac{1}{25} = \frac{2131}{507350} > 0. \tag{673}$$

The resulting selected polynomials and their exact lower bounds are

$$\begin{aligned} H(t) &= \frac{4}{25}t^2 - \frac{329682}{484925}t + \frac{281290}{484925}, \\ K(t) &= \frac{449521}{4849250} - \frac{19}{250}t, \\ H(t) &\ge h_0 := \frac{29196}{484925} > 0, \qquad K(t) \ge \kappa_0 := \frac{40489}{2424625} > 0. \end{aligned} \tag{674}$$

For example, the whole-interval sign is also exhibited by

$$\begin{aligned} H(t) - h_0 &= (1-t)\left(\frac{252094}{484925} - \frac{4}{25}t\right), \\ K(t) - \kappa_0 &= \frac{19}{250}(1-t), \\ \Delta(t) - \frac{1}{400} &= \frac{19}{10}(1-t), \qquad \frac{1521}{400} - \Delta(t) = \frac{19}{10}(1+t). \end{aligned} \tag{675}$$

The parenthesis in the first line is positive at $t = 1$ and decreasing in t, so it is positive everywhere on the interval. The residual expression has exactly its DC zero $z = 1$. The coupled endpoint has, in addition, the two primitive cube-root zeros, because $|1 - z^3|^2 = 2(1 - t)(2t + 1)^2$ and $K(t) > 0$.

F.4.2 One diagonal repair for every effective product

From now on $M = M_0$ in B_r. Affinity of the plant gives

$$B_r = (1 - r/R)\Delta(1 - t)H + (r/R)(1 - t)(2t + 1)^2 K. \tag{676}$$

For $0 < k < R$ put $\gamma = (1 - k/R)h_0/400$. Then $B_k \geq \gamma(1 - t)$; the exact nonnegative remainder is

$$\begin{aligned} B_k - \gamma(1 - t) = (1 - k/R)(1 - t)\left[(\Delta - 1/400)H + \frac{H - h_0}{400}\right] \\ + (k/R)(1 - t)(2t + 1)^2 K. \end{aligned} \tag{677}$$

To add a positive diagonal, direct calculation gives

$$\begin{aligned} C_r(t) &= (1 - t)\Delta + \frac{rq}{8}[-2t^2 + (9 + q)t + 1 - 9q], \\ C_r(1) &= \frac{19r}{400}, \qquad C_r(t) = A_r t^2 + D_r t + E_r, \\ A_r &= \frac{19}{10} - \frac{19r}{80}, \qquad D_r = \frac{3781r}{3200} - \frac{1521}{400}, \qquad E_r = \frac{761}{400} - \frac{2869r}{3200}. \end{aligned} \tag{678}$$

For $0 \leq r \leq R$, $A_r > 0$ and

$$2A_r + D_r = \frac{2261r - 8}{3200} \leq \frac{33907}{15100} < 3.$$

The useful one-sided bound follows without a frequency grid:

$$\begin{aligned} C_r(t) - C_r(1) = (t - 1)[A_r(t + 1) + D_r] \geq -3(1 - t), \\ C_r(t) - C_r(1) + 3(1 - t) = (1 - t)[3 - 2A_r - D_r + A_r(1 - t)] \geq 0. \end{aligned} \tag{679}$$

Take any $0 < \eta \leq \gamma/6$ and set $M_\eta = M_0 + \eta$. Equations (677)–(679) yield

$$-\Delta \operatorname{Re}(M_\eta V_k) = B_k + \eta C_k \geq \frac{\gamma}{2}(1 - t) + \frac{19\eta k}{400}. \tag{680}$$

Indeed, subtracting this lower bound gives the sum of $B_k - \gamma(1 - t)$, $\eta[C_k - 19k/400 + 3(1 - t)]$, and $(\gamma/2 - 3\eta)(1 - t)$, each nonnegative. Since $\Delta \leq 1521/400$, it follows that

$$\operatorname{Re}(M_\eta V_k) \leq -\epsilon, \qquad \epsilon = \frac{19\eta k}{1521} > 0. \tag{681}$$

One explicit rational choice for rational k is $\eta = \gamma/6$; $\eta = \gamma/16$ is another sufficient choice. At product zero the same symbol satisfies the exact identity

$$-\operatorname{Re}(M_\eta V_0) = (1 - t)(H + \eta) \geq (h_0 + \eta)(1 - t). \tag{682}$$

Interpolate (680) with Δ times (682). The inequality

$$\frac{h_0}{400} - \frac{\gamma}{2} = \frac{h_0(1 + k/R)}{800} > 0$$

proves (93) for every $0 \le r \le k$, with the same η and all three taps fixed. The lower bound at $t = 1$ is exactly $19\eta r/400$ before division by Δ, so strictness cannot be uniform as $r \downarrow 0$. Dividing the whole symbol by $1 + \eta$ puts it in the usual normalized O'Shea–Zames–Falb form and scales every margin by $1/(1 + \eta)$.

The strict-budget restriction matters for this repair. At $r = R$,

$$-\Delta \operatorname{Re}(M_\eta V_R)\big|_{t=-1/2} = -\frac{19397}{30200}\eta < 0 \qquad (\eta > 0). \tag{683}$$

Thus a positive diagonal cannot repair this mass-one endpoint family. This statement concerns the specified multiplier; it supplies no additional actual-algorithm counterexample.

F.4.3 Bounded operator congruence and a single graph

Let $L : X \to Y$ be bounded, $Q = I/19$, $\sigma = 1$, $\tau > 0$, and $\tau\|L\|^2 \le k < R$. Introduce

$$\mathsf{B} = \sqrt{\tau/k}\, L, \qquad \mathsf{C} = (I - \mathsf{B}^*\mathsf{B})^{1/2}, \qquad \mathsf{C}^*\mathsf{C} + \mathsf{B}^*\mathsf{B} = I.$$

The harmonic primal plant is the exact operator congruence

$$\begin{aligned} V_L(z) &= (z-1)I - \frac{\tau q z(9-z)}{8(1-qz)} L^*L \\ &= \mathsf{C}^* V_0(z)\mathsf{C} + \mathsf{B}^* V_k(z)\mathsf{B}. \end{aligned} \tag{684}$$

Real parts here are Hermitian parts on the complexified Hilbert spaces. Congruencing (680) and (682), then using (675), gives

$$\begin{aligned} -\Delta \operatorname{Re}(M_\eta V_L) &\succeq \frac{\gamma}{2}(1-t)I + \frac{19\eta k}{400}\mathsf{B}^*\mathsf{B}, \\ -\operatorname{Re}(M_\eta V_L) &\succeq \frac{200\gamma}{1521}(1-t)I + \frac{19\eta\tau}{1521}L^*L. \end{aligned} \tag{685}$$

At DC, $V_L(1) = -19\tau L^*L$. The required degeneracy on $\ker L$ is therefore retained. The congruence uses only scalar curvature; it neither simultaneously diagonalizes a nonlinear graph nor requires injectivity, surjectivity, closed range, compactness or finite dimension.

At an existing KKT anchor let $p_n = x_n - x^*$ and $d_n = \sqrt{\tau/k}(y_n - y^*)$, and set

$$\begin{aligned} u_n &= \mathsf{B}p_n, & r_n &= \mathsf{C}p_n, \\ d_n &= q[d_{n-1} + (9u_n - u_{n-1})/8], & & \\ g_n &= u_{n-1} - u_n - k d_{n-1}, & z_n &= r_{n-1} - r_n, \\ v_n &= \mathsf{C}^* z_n + \mathsf{B}^* g_n. & & \end{aligned} \tag{686}$$

For every pair of actual indices,

$$\langle p_i, v_j\rangle = \langle r_i, z_j\rangle + \langle u_i, g_j\rangle. \tag{687}$$

This identity splits the supply and any common scalar graph correction, but does not put the two ports on independent subdifferential graphs.

For completeness, the original-space graph potential attached to this interface is

$$\begin{aligned} v_n \in \partial F(p_n), \quad F_n = F(p_n), \quad B_n = F^*(v_n) = \langle p_n, v_n\rangle - F_n \quad (n \ge 1),\\ \mathcal{A}_n = aB_n + (b+c)(F_n + F_{n-1}) + cF_{n-2} \quad (n \ge 3),\\ \mathcal{Q}_n^\eta = (1+\eta)\langle p_n, v_n\rangle - a\langle p_n, v_{n-1}\rangle - b\langle p_{n-2}, v_n\rangle - c\langle p_{n-3}, v_n\rangle \quad (n \ge 4). \end{aligned} \tag{688}$$

Here F is one proper lower-semicontinuous convex function on X with $F(0) = \min F = 0$. Lemma 5, which is an algebraic convexity lemma independent of any storage, gives the full three-gap identity and the lower bound $\mathcal{Q}_n^\eta - (\mathcal{A}_n - \mathcal{A}_{n-1}) \ge \eta(F_n + B_n)$. It also gives the common scalar correction for any triple satisfying

$$\begin{aligned} \lambda_1, \lambda_2 \ge 0, \quad a - \lambda_0 - \lambda_1 - \lambda_2 \ge 0,\\ b + c - \lambda_0 \ge 0, \quad b + c - \lambda_1 \ge 0, \quad c - \lambda_2 \ge 0. \end{aligned} \tag{689}$$

There is no separate nonnegativity requirement on λ_0. All three scalars must be the same for both linear ports and the original graph. The term cF_{n-2} is essential: the first finite potential is $\mathcal{A}_3$, involving only attained graph indices $1, 2, 3$, and its first allowed difference is at $n = 4$. Neither F_0 nor v_0 occurs.

These frequency and potential calculations alone do not establish a coercive state storage. Theorem 6 provides that extra step for the fixed budget $159/50$, using its own four verified matrix inequalities and actual residual/DC argument. The larger interval $159/50 < \tau\|L\|^2 < R$ is not given an actual operator convergence conclusion by the present interface.

G Polynomial signs for the common anisotropic multiplier

The signs in Lemma 1 reduce to a few univariate polynomials. This appendix records the normalization so that the finite exact verification is reproducible independently of any solver. Put $S = 1 + w + w^2$ and

$$\begin{aligned} A_+(w) &= 16 + 3w + 13w^2 - 12w^3,\\ B_+(w) &= 44 + 9w + 39w^2 - 32w^3, \qquad D_+ = A_+B_+,\\ A(w) &= 160w^6 - 556w^5 + 353w^4 - 306w^3 + 775w^2 + 106w + 308,\\ B(w) &= 384w^6 - 272w^5 - 467w^4 - 890w^3 + 413w^2 + 368w + 704, \end{aligned}$$

$$
\begin{aligned}
C(w) &= -160w^6 - 56w^5 + 417w^4 + 390w^3 + 35w^2 - 198w - 308,\\
E(w) &= -1152w^5 - 1144w^4 + 1386w^3 + 4118w^2 + 3744w + 2288,\\
H(w) &= 2144w^7 + 12120w^6 - 14706w^5 - 22867w^4\\
&\qquad - 14040w^3 + 10101w^2 + 20168w + 11880.
\end{aligned}
$$

Direct expansion of (13) and (11) gives

$$
\begin{aligned}
a = A/D_+, \quad b = B/D_+, \quad c_3 = C/D_+,&\\
\frac{K_0(1)}{1-w} = \frac{8S}{A_+}, \quad \frac{-K_0'(1)}{1-w} = \frac{E}{D_+}, \quad K_0'' = \frac{8C}{D_+},&\\
\frac{J_w(1)}{1-w} = \frac{8wS}{B_+}, \quad -J_w' = \frac{2wC}{D_+},&\\
-K_2(-1) = \frac{4(4w^2-6w-5)(34w^3-39w^2-9w-46)}{D_+},&\\
-K_2'(-1) = \frac{2H}{w(9w-1)D_+}, \quad -K_2'' = \frac{8(10w+9)C}{w(9w-1)D_+}.&
\end{aligned}
\tag{690}
$$

The two factors in the numerator of $-K_2(-1)$ are negative on $47/50 \le w \le 1$: $4w^2 - 6w - 5 \le 4 - 6(47/50) - 5 < 0$, and $34w^3 - 39w^2 - 9w - 46 \le 34 - 46 < 0$. All remaining signs follow from positivity of the seven displayed polynomials. Here is a compact exact check of that positivity.

For any degree-d polynomial $P(w)$, expand $P((47+3s)/50) = \sum_{j=0}^{d} c_j s^j$ and put

$$
\beta_i(P) = \sum_{j=0}^{i} c_j \frac{\binom{i}{j}}{\binom{d}{j}}, \qquad P((47+3s)/50) = \sum_{i=0}^{d} \beta_i(P) \binom{d}{i} s^i (1-s)^{d-i}. \tag{691}
$$

For the seven polynomials above, rational arithmetic gives the following lower bounds on every β_i:

P	A_+	B_+	A	B	C	E	H
d	3	3	6	6	6	5	7
$\min_i \beta_i(P) \ \ge$	20	60	816	240	34	8858	4800

(692)

Since the Bernstein basis is nonnegative and sums to one on $[0,1]$, these integer bounds establish strict positivity on the entire closed parameter interval. Equations (690) then give all endpoint signs and derivative signs used in the proof: K_0 is decreasing convex; J_w is decreasing positive; and $-K_2$ is increasing positive convex. The factors $1-w$ explain the strict restriction $w < 1$ where needed.

The source file `common_curvature/check_printed.py` verifies (692), the displayed polynomials, the graph-weight example and the restart constant algebra. The author reconstruction `check_family.py` and the separate advance-variable `check_parameter_band.py` independently derive the complete frequency numerator, critical double-zero condition and interpolation identity. `check_classwide.py` checks the inherited scalar hinge and (10). Full rational coefficients remain ancillary in

`exact_certificate.json`. These finite checks certify their stated algebra; the soft inequality, causal continuation, actual startup and geometric conclusion are proved in Section 2. The current source adds the common-curvature theorem and a full v7-to-v8 preservation ledger for all 802 baseline labels, complete source contents, and physical PDF destinations. The older ledgers described below retain their historical scope; the new theorem does not replace their results.

Reproducibility

The source contains this manuscript and its reference style, a current result map, the semantic correspondence for all 637 pre-existing labels and their complete proof and unnumbered mathematical content, portable exact checkers with separate independent reconstructions, and rational certificate data. The prerequisite universal-threshold certificate [1] is included unchanged in `dependencies/report30/`. The storage-existence, convex-graph, functional-calculus, actual-startup, quantitative-restart and whole-sequence arguments are proved in the manuscript. Exact arithmetic supports those proofs and reproduces the displayed certificate branches; it does not replace readable mathematics. The README gives one build command, one complete verification command and the public file manifest.

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